OPT Plus TSEK

Operation Timbang Plus

Tamang Sukat, Eksaktong Kalidad!

MANUAL OF OPERATIONS AND DATA QUALITY CHECK PROTOCOL

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Made in partnership with

National Nutrition Council
Department of Health
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Institute of Human Nutrition and Food
College of Human Ecology
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This publication was prepared by the OPT Plus National Project Team:

University of the Philippines Los Baños
Prof. Leila S. Africa, *Project Leader*
Ms. Ma. Charina P. Maneja, *Project Manager*
Dr. Nancy Tandang, *Statistician*
Asst. Prof. Reanne Len C. Arlan, *Statistician*
Asst. Prof. Aiza Kris M. Bernardo, *Study Leader*
Asst. Prof. Joyce P. Parco, *Study Leader*
Dr. Raymundo Celestino Habito Jr., *Consultant*
Ms. Katrina F. Baluyot, *Research Assistant*
Ms. Danica Mae V. Inoceno, *Research Assistant*

UNICEF
Ms. Alice Nkoroi
Dr. Rene Gerard Galera
Mr. Frederich Christian Tan

National Nutrition Council
Dr. Azucena M. Dayanghirang, MD, MCH, CESO III, Assistant Secretary & Executive Director IV
Ms. Ellen Ruth F. Abella, RND, MPM
Ms. Maria Cynthia B. Vengco, RND, MPS
Mr. Daniel G. Salunga, RND, MSPH, *Project Coordinator*
Ms. Christine Jane B. Almira, RND, MAHPS, *Asst. Project Coordinator*

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1630 Taguig City
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Email Address: info@nnc.gov.ph

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A Message of Support from

Dr. Azucena M. Dayanghirang

The conduct of Operation Timbang (OPT), which started in the early 70s, is regularly done by local government units (LGUs) through their local nutrition committees. Every one of us in the nutrition community put a premium on the quality of data, high quality data, to be precise. Sound and reliable information is the foundation of sound decision making. Thus, the quality of data is of prime importance for accurate, reliable, and valid results. High-quality data results in better information, which redounds to better decision-making, and better health of the population.

In this very significant project, we dealt with the quality of the Operation Timbang Plus results. The quality of the data collected from the OPT has long been an issue in the past, and at present, and may pose a potential problem in the future if not addressed. Some of you may be familiar with studies done by Dr. Barba, Dr. Habito, Myra Bondad and Director Alicia Ramos in the 1990s on the implementation of the OPT which highlighted the need to improve the capacity of data collectors to ensure quality of OPT results. This was addressed in part by NNC and other partners by providing the tools, the weight and measuring tools; as well as in the development of the electronic OPT Plus to minimize the errors in computation of age in months and determination of the child's nutritional status. Operations research done by NNC in 2019 showed that there were still gaps that were consistent with the previous study (Salunga et al.), the research recommendations lead to the development of data quality check protocol and the revision of the OPT Plus implementing guidelines, which we attained from the output of this project.

The results of the OPT Plus is the only source of information of the nutritional situation of the barangay, without which, the barangay nutrition committee cannot formulate their nutrition action plans that will identify appropriate interventions. Without the results of the OPT, there is no way to measure the effectiveness, efficiency, impact, and sustainability of the nutrition programs implemented at the local level. A good measure of nutrition progress depends on quality data. Similarly, without quality data, we cannot measure the contribution of the local level to the PPAN, the Philippine Development Plan, the *Ambisyon Natin 2040*, and the Sustainable Development Goals.

We express our gratitude and appreciation to the United Nations Children's Fund (UNICEF) and the Korea International Cooperation Agency (KOICA) for the continuous assistance and strong support in our goal to improve the OPT Plus. We extend our thanks and appreciation too for the hard work, dedication and professionalism of the team from the University of the Philippines Los Baños who helped in developing the Manual of Operations, Data Quality Check Protocol and other OPT

Plus information products. The synergy and efforts of the National Project Team of NNC, UNICEF and UPLB, and all the stakeholders consulted from the barangay up to the national level led to the success of this project.

Our data quality initiatives do not end with this project. In addition to this project, NNC has several initiatives lined up to improve data quality: along policy, tools and capacity. In this day and age, quality data is the new oil. Let us use this to fuel our calibrated and responsive nutrition policies, plans, projects and activities.

Through this OPT Plus Manual of Operations and Data Quality Check Protocol, we can do **OPT Plus TSEK: Tamang Sukat, Eksaktong Kalidad!** Let us all work together to build a structure to safeguard correct measurements to produce exact quality of data for the OPT Plus, exact meaning - nothing more, nothing less, *walang dagdag bawas.*

Together, we can do things better, i-TSEK na natin ang OPT Plus at sama sama nating itaguyod ang nutrisyong sapat, para sa lahat!

DR. AZUCENA M. DAYANGHIRANG, MD, MCH, CESO III
Assistant Secretary and Executive Director IV
National Nutrition Council, Department of Health

A Message of Support from

Dr. Enrico P. Supangco

Greetings!

This revised and improved Operation Timbang Plus (OPT Plus) publication which includes manuals and infographics was completed in partnership with UNICEF, the National Nutrition Council (NNC) and the Institute of Human Nutrition, College of Human Ecology, UPLB through the UPLB Foundation Incorporated. This publication is the output of two projects namely the "Technical Support to NNC in the Development of Data Quality Audit Protocol for the Operation Timbang Plus (OPT Plus)" and the "Updating of Implementing Guidelines". It incorporates the revised and improved OPT Plus guidelines and should help ensure the high quality of nutrition data in the country and integrity of its use as a basis for local and national nutrition program planning.

We would like to congratulate Dr. Leila S. Africa, the project Leader, and all the members of her research team for their commitment and dedication in the successful implementation of the two projects and the publication of the Operation Timbang Plus (OPT Plus).

DR. ENRICO P. SUPANGCO, PhD
Executive Director, UPLBFI

A Message of Support from

Dr. Jose V. Camacho, Jr.

On behalf of the University of the Philippines Los Baños (UPLB), I extend my warmest greetings and congratulations to the College of Human Ecology's Institute of Human Nutrition and Food and to Dr. Leila S. Africa and her team for publishing a manual for our country's revised and improved Operation Timbang (OPT) Plus.

This manual, which is a product of the National Nutrition Council and United Nation's Children's Fund-commissioned project *Development of Data Quality Audit (DQA) Protocol for the Operation Timbang (OPT) Plus and Updating of Implementing Guidelines (IG)*, aims to improve the Philippines' OPT Plus system, a routine child growth assessment for all children aged 0 to 59 months. Conducted annually in all barangays nationwide, OPT Plus serves as a baseline for assessing quality nutrition programs and interventions that are essential for local nutrition action planning and determining priority areas for nutrition programming. This manual includes the revised and improved OPT Plus Guidelines designed to better integrate with the Monitoring and Evaluation of Local Level Plan Implementation (MELLPI) Protocol Tool.

Apart from serving as a valuable reference for local government units in keeping their communities healthy and thriving, this publication is also a testament to the power of collaboration and teamwork between the University and its project partners towards achieving its mutual goals for the communities and people we serve.

I commend Dr. Africa and her team for their work has resulted in a successful project and the betterment of communities all over our country. You should be incredibly proud of this project's impact and the lives it will improve. You have set a shining example of excellence and raised the bar for future projects. Keep up the excellent work and continue to inspire others with your dedication and passion for making a difference.

Mabuhay kayong lahat!

DR. JOSE V. CAMACHO, JR.
Chancellor, University of the Philippines Los Baños

A Message of Support from

Oyunsaikhan Dendevnorov

Children have a fundamental right to optimal nutrition and health, and proper nourishment enables them to grow, learn, play, and develop resilience, even in difficult circumstances. Tragically, many Filipino children are currently deprived of the basic needs, essential services, and nurturing environment they require to reach their full potential.

Accurate data and information are crucial for informed decision-making, policy development, and effective implementation. The Operation Timbang (OPT) program provides valuable data on the nutritional status of children aged 0-59 months, allowing for targeted service delivery, nutrition program management, and evaluation of local programs. This tool is essential for local government units to collect and analyze accurate anthropometric data on malnutrition trends, analyse and report emerging trends and ultimately improve nutrition programs for children. I believe this new approach will significantly enhance the quality of data collected from the OPT and OPT Plus.

UNICEF is proud to support the National Nutrition Council (NNC) in its efforts to improve the existing OPT Plus standards and tools in accordance with global standards. Working with our institutional partner, University of the Philippines Los Baños Foundation Incoporated, we have updated the guidelines, training materials, and tools used by local government units to incorporate data quality checks and automatic functions. These improved standards and tools aim to enhance the capacities of frontline workers and health professionals to collect and report quality nutrition data in a timely manner, enabling prompt and appropriate interventions.

Nutrition has always been a central focus of UNICEF's work for children. We remain committed to supporting the National Government and our priority local government units in scaling up nutrition interventions to reach the SDG 2 targets and the national strategic plan. I commend the NNC for its continued leadership and dedication to improving data quality and systems, with the ultimate goal of enhancing nutrition for all Filipino children.

OYUNSAIKHAN DENDEVNOROV
Representative, UNICEF Philippine Country Office

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Section One

OPERATION TIMBANG PLUS (OPT PLUS) OVERVIEW

OPERATION TIMBANG PLUS (OPT PLUS) OVERVIEW

OVERVIEW

The general objective of this section is to provide information on the basic concepts and procedures of the OPT Plus.

After reading this section, one should be able to:

1. Describe the OPT Plus, its purposes, and processes;
2. Identify who are involved in OPT Plus and their roles;
3. Enumerate the forms used in OPT Plus;
4. Identify the common anthropometric measurements used in OPT Plus; and
5. Explain the mechanics in the OPT Plus implementation.

This overview covers the basic definitions and principles of anthropometry, nutrition assessment, and OPT Plus used throughout the succeeding sections.

LIST OF FIGURES

LIST OF ACRONYMS

AIP	Annual Investment Plan
BARMM	Bangsomoro Autonomous Region of Muslim Mindanao
BHC	Barangay Health Center
BHW	Barangay Health Worker
BHS	Barangay Health Station
BNAP	Barangay Nutrition Action Plan
BNC	Barangay Nutrition Committee
BNS	Barangay Nutrition Scholar
C/MNAO	City/Municipal Nutrition Action Officer
C/MNAP	City/Municipal Nutrition Action Plan
CHAT	Community Health Action Team
cm	centimeter
CGS	Child Growth Standard
CNAO	City Nutrition Action Officer
CNAP	City Nutrition Action Plan
CNC	City Nutrition Committee
DCW	Day Care Worker
DQC	Data Quality Check
ECCD	Early Child Care and Development
eOPT	Electronic Operation Timbang
FANTA	Food and Nutrition Technical Assistance
GB	Governing Board
LGU	Local Government Unit
LNAP	Local Nutrition Action Plan
MAM	Moderate Acute Malnutrition
MNAO	Municipal Nutrition Action Officer
MNAP	Municipal Nutrition Action Plan
MNC	Municipal Nutrition Committee
MOP	Manual of Operation

MSt	Moderately Stunted
MUAC	Mid-Upper Arm Circumference
MUW	Moderately Underweight
MW	Moderately Wasted
NCP	Nutrition Center of the Philippines
NNC	National Nutrition Council
Ob	Obese
OPT Plus	Operation Timbang Plus
Ow	Overweight
P/C/M/BNAP	Provincial/City/Municipal/Barangay Nutrition Action Plan
PNAO	Provincial Nutrition Action Officer
PNC	Provincial Nutrition Committee
PPA	Programs, projects, and activities
RA	Republic Act
RNC	Regional Nutrition Council
RHM	Rural Health Midwife
RHU	Rural Health Unit
RNAP	Regional Nutrition Action Plan
RNPC	Regional Nutrition Program Coordinator
RSDC	Regional Social Development Committee
RTWG	Regional Technical Working Group
SAM	Severe Acute Malnutrition
SD	Standard Deviation
SK	Sangguniang Kabataan
SW	Severely Wasted
SSt	Severely Stunted
St	Stunted
SUW	Severely Underweight
UNICEF	United Nations Children's Fund
WHO	World Health Organization
WHO-CGS	World Health Organization Child Growth Standards

DEFINITION OF TERMS

Anthropometric Measurements	Non-invasive and quantitative measurements of the body mass, stature, and size [i.e., weight, length, height, and mid-upper arm circumference (MUAC)].
Certified OPT Plus Measurer	A member of the OPT Plus team who has completed the anthropometry training and has passed both the standardization exercise and post test.
Early Childhood Care and Development Card	The card used in regular monitoring of growth and health indicators among 0-to-59-month-old children.
Height	The stature measurement of 24-to-59-month-old children collected while standing up.
Length	The stature of 0-to-23-month-old children measured while in a recumbent or lying position.
Length/Height-forAge (L/HFA)	An indicator used to classify stunting among 0-to-59-month-old children.
Local Nutrition Action Plan	Three-year action plan to address prevailing nutritional problems based on administrative level i.e., Provincial, City, Municipal, Barangay Nutrition Action Plan (P/C/M/BNAP) and is annually updated.
Malnutrition	Deficiencies, excesses, or imbalances in a person's intake of protein, energy (carbohydrates and fats), and/or nutrients covering both undernutrition which includes suboptimal breastfeeding, stunting, wasting or thinness, underweight, and micronutrient deficiencies or insufficiencies, as well as overnutrition, which includes overweight and obesity (RA 11148 definition).
Mid-Upper Arm Circumference	Circumference of the left arm taken at the midpoint between the elbow and shoulder that is used in determining the presence of acute malnutrition among 6-to-59-month-old children.
Moderately Stunted (MSt)	Nutritional status classification of 0-to-59-month-old children whose height/length-for-age is below -2 to -3 standard deviations (SD) of the World Health Organization (WHO) Child Growth Standards (CGS) median.

Moderately Wasted/ Moderate Acute Malnutrition (MW/MAM)	Nutritional status classification of 0-to-59-month-old children whose weight-for-length/height Z-score is less than –2 SD to –3 SD from the median of the WHO growth standards or a MUAC measurement of <12.5 cm and ≥11.5 cm.
Nutritional Status	Condition of the body resulting from intake, absorption, and use of nutrients, as well as the influence of disease-related factors.
Operation Timbang Plus	Regular growth assessment of all 0-to-59-month-old children in the barangay, done by the trained team and involves planning, follow-up, and data quality check.
Overnutrition	Includes overweight and obesity.
Obese (Ob)	Nutritional status classification of 0-to-59-month-old children whose weight-for-length/height Z-score is above three (3) SD from the median of the WHO Child Growth Standards.
Overweight (OW)	Nutritional status classification of 0-to-59-month-old children whose weight-for-length/height Z-score is above 2 to 3 SD from the median of the WHO Child Growth Standards.
Severely Stunted (SSt)	Nutritional status classification of 0-to-59-month-old children whose length/height-for-age Z-score is less than –3 SD from the median of the WHO Child Growth Standards.
Severely Wasted/Severe Acute Malnutrition (SW/SAM)	Nutritional status classification of 0-to-59-month-old children whose weight-for-length/height Z-score is below –3 SD of the median of the WHO Growth Standards; characterized by visible severe wasting, or by the presence of bilateral pitting edema, or a MUAC measurement of less than 115 mm or 11.5 cm (RA 11148 definition).
Stunted (St)	A form of malnutrition that results from chronic or long-term undernutrition and can be classified using the length/height-for-age indicator. It is a general term that refers to both moderate and severe stunting.
Undernutrition	Includes suboptimal breastfeeding, stunting, wasting or thinness, underweight, and micronutrient deficiencies or insufficiencies.
Underweight	A form of malnutrition classified using the weight-for-age indicator. It is also a general term referring to both moderately and severely underweight.

Wasted	A form of malnutrition that results from acute onset of undernutrition classified using the weight-for-length/height indicator or MUAC. It is also the term referring to both moderately and severely wasted.
Weight	The measurement of body mass or heaviness of a child.
Weight-for-Age (WFA)	An indicator used to classify underweight among 0-to-59-month-old children.
Weight-for-Length (WFL)	Indicator used to classify wasting, overweight, and obesity among 0-to-23-month-old children.
Weight-for-Height (WFH)	Indicator used to classify wasting, overweight, and obesity among 24-to-59-month-old children.
WHO Child Growth Standards	Represents how children should grow through globally accepted prescriptive measures considered normal growth for all 0-to-59-month-old children.

Section 1.1

OPERATION TIMBANG PLUS (OPT PLUS) OVERVIEW

What is OPT Plus?

Operation Timbang Plus (OPT Plus) is the regular growth assessment for all 0-to-59-month-old children in the barangay done by a trained team. It involves planning, follow-up, and data quality check.

What is the purpose of OPT Plus?

The annual OPT Plus done at the start of the year will serve as the basis for close and regular follow-up of 0-to-23-month-old children and children at-risk of malnutrition.

Weight, length, and height are the anthropometric measurements collected during the OPT Plus to generate data on the nutritional status of children as inputs to community nutrition assessment, nutrition program planning, and monitoring and evaluation of the local nutrition action plan (LNAP).

GENERAL OBJECTIVES

The general objectives of the OPT Plus are to generate data on the nutritional status of children in a barangay; locate under- or over-nourished children; and guide local government units (LGU) in nutrition program management.

SPECIFIC OBJECTIVES

Community Level

1. Assess the nutritional status of the 0-to-59-month-old children;
2. Locate families with children who are underweight, stunted, wasted, overweight, and obese;
3. Determine priority areas and individuals for program planning;
4. Serve as the basis for the preparation and evaluation of the nutrition action plan;
5. Monitor the nutrition trend and assess the effects of nutrition interventions implemented.

Family Level

1. Provide parents/caregivers information on the nutritional status of their child/children to prevent malnutrition;
2. Early detection and immediate referral of undernutrition to prevent severely wasted/severe acute malnutrition (SW/SAM) and moderately wasted/moderate acute malnutrition (MW/MAM) cases;
3. Guide the parents/caregivers on the growth and development of their children.

Who are the members of the OPT Plus Team?

The OPT Plus Team is composed of at least 7 members. The team is headed by the Rural Health Midwife (RHM), assisted by the Barangay Nutrition Scholars (BNS) and Barangay Health Workers (BHWs).

Other members of the OPT Plus Team can be the Committee Chair on Health and Nutrition, other barangay officials and staff, Sangguniang Kabataan (SK) officials, Day Care Workers (DCW), and teachers-in-charge.

The team may be assisted by the purok or mother leaders, other community leaders, and representatives from civic organizations. The roles and responsibilities of the OPT Plus Team and the supervisors are enumerated in Annex 1.1.

In the Bangsamoro Autonomous Region of Muslim Mindanao (BARMM), some of the members of the community health action team (CHAT) may be tapped to join the OPT Plus Team.

Who are the target groups of OPT Plus?

All 0-to-59-month-old children shall have their MUAC or weight, length/height measured regularly.

When is the OPT Plus conducted?

The OPT Plus preparation at the municipal/city level should start every August to prepare for the OPT Plus activities for the following year. Figure 1.1 describes the schedule of activities for the whole year.

Figure 1.1. OPT Plus Timeline of Activities

Where is the OPT Plus conducted?

OPT Plus activities can be conducted in the health facility, weighing post, barangay hall, day care center, house-to-house, or other conducive areas making sure that all 0-to-59-month-old children are covered.

What are the OPT Plus follow-up activities?

- Monthly weight and length measurement of all 0-to-23-month-old children and plotting them in the growth chart in the Early Childhood Care and Development (ECCD) Card or Mother-baby handbook;
- Monthly weight and height measurements of 24-to-59-month-old children who are malnourished from February to December; and
- Semestral weight and height measurements of 24-to-59-month-old children.

NOTE 1.1.

For OPT Plus, collection of MUAC shall be imposed if the situation does not allow the collection of weight, length, and height such as in case of emergencies (NNC, 2020).

What anthropometric measurements are used for the OPT Plus?

WEIGHT refers to the measurement of body mass or heaviness of a child.

LENGTH refers to the stature of a 0-to-23-month-old child measured while in a recumbent or lying position.

HEIGHT refers to the stature measurement of a 24-to-59-month-old child collected while standing up.

MID-UPPER ARM CIRCUMFERENCE (MUAC) refers to the circumference of the left arm taken at the midpoint between the elbow and shoulder that is used in determining the presence of acute malnutrition among 6-to-59-month-old children (FANTA, 2018). MUAC identifies children at high risk for mortality (UNICEF, 2015).

NOTE 1.2. MUAC

Why is MUAC not recommended to assess acute malnutrition in children <6 months?

There is insufficient evidence for recommending MUAC to determine acute malnutrition among <6 months old children (FANTA, 2018; WHO, 2013). The weight-for-length index is used instead.

What are the recommended tools and equipment for weighing and measurement of length or height?

WEIGHING SCALES

Several weighing tools can be purchased but only recommended weighing scales can be used in assessing nutritional status to get accurate data. The National Nutrition Committee (NNC) issued NNC Governing Board (GB) Resolution No. _____ Approving the Guidelines on the selection of weighing scales (Annex 1.2).

The following are the recommended weighing scales that can be used:

1. **Hanging Infant Weighing Scale** (Figure 1.2) is a type of mechanical scale designed to be used for children less than five years old weighing less than 25 kg. The reading scale should only be in grams with a gradation of 0.1 kg or 100 grams. It comes in a set composed of the scale and hanging pants where the child sits during weighing.

Figure 1.2. Hanging Infant Weighing Scale

2. **Digital Scales**

   a. **Hanging Digital Scale** (Figure 1.3) is another hanging scale for children with a maximum weight capacity of 25 kg. The parts are similar to the Mechanical Hanging Infant Weighing Scale, except that this scale is battery-powered and features a digital monitor where the weight can easily be read (NCP, n.d.).

Figure 1.3. Hanging Digital Scale

   b. **Digital Platform Scale** (Figure 1.4) is a battery-powered digital scale with a maximum capacity load of 150 kg and can be used by all age groups. It has no buttons and automatically displays the weight, which makes it easy to use and operate while maintaining the accuracy and reliability of weight measurement (Tanita, n.d.).

Figure 1.4. Digital Platform Scale

c. **Digital Taring Scale** (Figure 1.5) is a digital scale specifically designed for mobile use and for all age groups. It is lightweight but can measure weight up to a maximum of 200 kg. It is simple and features a dual screen where both the measurer and the person being measured can simultaneously view the reading. It also simplifies weighing a child with their parent/ caregiver (SECA, n.d.).

Figure 1.5. Digital Taring Scale

3. **Mechanical Column Scale** (Figure 1.6) is a mechanical scale that makes use of an eye-level beam balance to obtain the weight. This type of scale is usually made from heavy duty steel which makes it durable but not portable. It can weigh up to 200 kg.

Figure 1.6. Mechanical Column Scale

NOTE 1.3.
Weighing Scales

Why are regular bathroom scales not recommended for OPT Plus?

Regular bathroom scales, especially the mechanical or spring-type kind, may not maintain accurate readings with repeated use (UW, n.d.). Although cheaper in price, these types of scales are not meant to last. Recommended ones, on the other hand, ensure accurate and precise readings and are designed to last.

LENGTH/HEIGHT MEASUREMENT TOOLS

While there are many available length and height measurement tools in the market, not all are recommended for nutritional status assessment due to inaccuracies that may result during use. The NNC issued NNC Governing Board (GB) Resolution No. 3 s.2012 Approving the Guidelines on the Fabrication, Verification, and Maintenance of Wooden Height Boards (Annex 1.3) and NNC GB Resolution No. 3 s.2018 Approving the Guidelines on the selection of non-wood height and length measuring tool (Annex 1.4).

1. **Length/Height Board** (Figure 1.7) is a measurement tool that can be set up horizontally against a flat surface for measuring the length and vertically against a wall for children's height measurement. Generally, length/height boards can measure up to 130 cm and have a precision of 0.1 cm (FANTA, 2018). A sliding piece acts as a foot piece when taking length or a headpiece when taking height. Wooden length/height board is a commonly used type in the country as recommended by the WHO (NNC, 2012). Metal, although durable, is not recommended as it can become too hot in areas with high temperatures and might cause burns (FANTA, 2018).

Figure 1.7. Length/Height Board

2. **Stadiometer** (Figure 1.8) is a height measurement tool which consists of a vertical ruler and a sliding paddle which rests on top of the head when height is being taken. It is lightweight, convenient, and portable. It features a level which assures that the tool is completely perpendicular to the floor. It is similar to a length/height board in form with its graduated body and sliding piece. However, a stadiometer can only be used vertically to measure the height of children.

Figure 1.8. Stadiometer

3. **Microtoise** (Figure 1.9) is a height measurement tool that is mounted to a flat wall. The bottom part of the tool can be slid down to the top of the child's head and the height measurement can be seen through a window. It is practical to use, lightweight, and portable.

Figure 1.9. Microtoise

NOTE 1.3.
Mobile Measuring Mat

Mobile measuring mat (Figure 1.10) is a light-weight and portable length measurement tool composed of a fixed headpiece and a sliding foot positioner. This tool is recommended only during emergencies and if the length board is not available due to its gradation.

Figure 1.10. Mobile Measuring Mat

NOTE 1.4.
Length/Height Board

Wooden length/height boards have been widely used in the country since the adoption of the WHO recommendation for its use. But more recently, use of non-wood materials in fabricating length/height measurement tools has been approved as a response to the total log ban policy. See Annex 1.4 NNC GB Resolution No. 3, Series 2018 Approving the Guidelines on the Selection of Non-Wood Height and Length Measuring Tool.

Figure 1.11. Length/Height Boards

What are the forms to be used in OPT Plus?

The OPT Plus uses different forms to record the gathered data. Below are the list of forms to be used at the Barangay Level:

1. **OPT Plus Form 1A**: List of 0-to-59-month-old children measured which will be used for recording of weight and length/height in the next OPT Plus
2. **List of 0-23 mos**: List of 0-to-23-month-old children
3. **List_MUAC Status**: List of 6-to-59-month-old children by MUAC status
4. **List_MW (MAM)**: List of 0-to-59-month-old children who are moderately wasted/ moderate acute malnutrition
5. **List_SW (SAM)**: List of 0-to-59-month-old children who are severely wasted/severe acute malnutrition
6. **List_MSt&SSt**: List of 0-to-59-month-old children who are moderately stunted or severely stunted
7. **List_OW&Ob**: List of 0-to-59-month-old children who are overweight and obese
8. **List_MUW, SUW,MSt&SSt**: List of 0-to-59-month-old children who are both stunted and underweight. The supervisor's role is to ensure that the OPT Plus protocols are followed (Section 8)
9. **List_MSt,SSt,MW&SW**: List of 0-to-59-month-old children who are both stunted and wasted
10. **List_MSt,SSt,OW&Ob**: List of 0-to-59-month-old children who are both stunted and overweight/obese

What are the uses of OPT Plus?

- Identify and quantify the 0-to-59-month-old children who are malnourished.
- Locate families with 0-to-59-month-old children who are malnourished.
- Prevent growth faltering among 0-to-23-month-old children.
- Determine priority areas and individuals for nutrition interventions.
- Provide a basis for assessing local nutrition programs, projects, and activities.

What are the units used in OPT Plus?

1. **Weight-for-Age**: An indicator used to classify underweight among 0-to-59-month-old children.
2. **Length/Height-for-Age**: An indicator used to classify stunting among 0-to-59-month-old children.
3. **Weight-for-Length/Height**: An indicator used to classify wasting, overweight, and obesity among 0-to-59-month-old children.

How is OPT Plus implemented?

The OPT Plus protocol comprises five (5) major activities. As a protocol, the OPT Plus Team should strictly follow the different activities to generate timely, reliable, accurate, and precise OPT Plus data. The barangays will be clustered and assigned a supervisor. Figure 1.12 presents OPT's five major activities. Figure 1.13 indicates which sections of the Manual of Operations (MOP) the OPT Plus activities are discussed in detail.

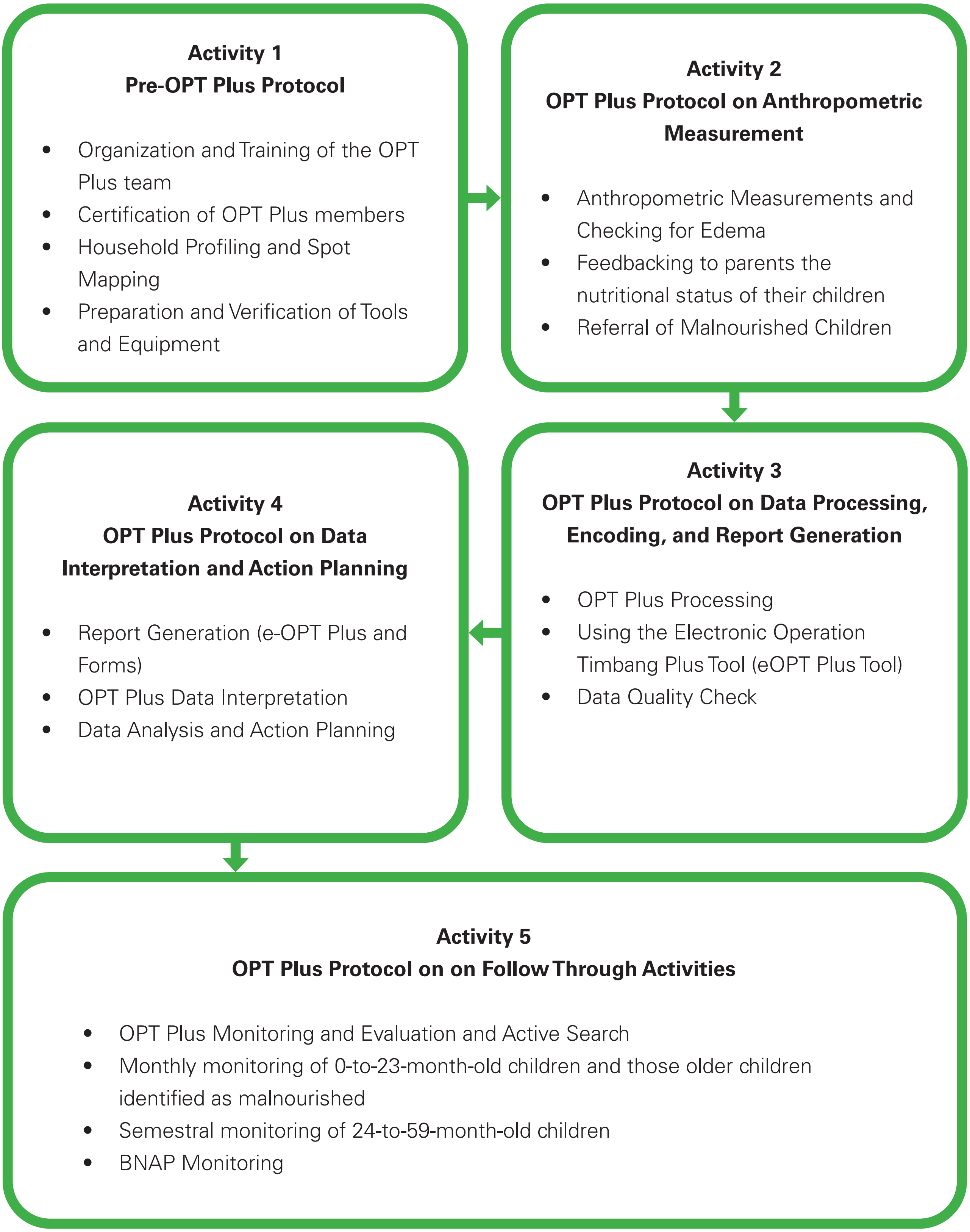

Figure 1.12. OPT Plus activities flowchart

Activity 1
Pre-OPT Plus Protocol

Section 2.1. Organizing the OPT Plus team
Section 2.2. Certification of OPT Plus members
Section 2.3. Household Profiling
Section 2.4. Spot Mapping
Section 2.5. Calibration and Verification of Weighing Tools and Equipment
Section 2.6. Verification of Length/Height Tools and Equipment
Section 2.7. Standardization Exercise

Activity 2
OPT Plus Protocol on Anthropometric Measurement

Section 3.1. Roles and Responsibilities of OPT Plus Team and Supervisors
Section 3.2. General Reminders for Taking Anthropometric Measurements
Section 3.3. Instruction on How to Weigh the 0-to -59-month-old Children
Section 3.4. Instructions on Measuring the Length of the 0-to-23-month-old Children
Section 3.5. Instructions on Measuring the Height of the 24-to-59-month-old Children
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Section 3.8. OPT Plus Referral
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Activity 3
OPT Plus Protocol on Data Processing, Encoding, and Report Generation

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Section 5.3. Report Generation (eOPT Plus and Forms)
Section 5.4. Data Analysis and Interpretation
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Section 6.1. Roles and Responsibilities of the Supervisors and OPT Plus Team in the Follow Through Activities
Section 6.2. Child Growth Monitoring and Active Search
Section 6.3. BNAP Monitoring

Figure 1.13. Flowchart of Operation Timbang Plus Activities and MOP Sections

Activity 1. Pre-OPT Plus Protocol

- OPT Plus Team should be organized and trained accordingly (Section 2.1). The pre-OPT Plus planning phase should start by August, allowing enough time to prepare for the conduct of OPT Plus. Measurement of OPT Plus will start in the first quarter of the succeeding year.
- The measurer needs to be a Certified OPT Plus Measurer (Sections 2.2 and 2.7).
- The master listing of 0-to-59-month-old children (Section 2.3) and spot-mapping (Section 2.4) should be conducted.
- Tools and equipment for weighing and measuring should also be calibrated and verified (Sections 2.5 and 2.6).

Activity 2. OPT Plus Protocol on Anthropometric Measurement

- This activity involves measuring the length or height, weight, and checking for edema of 0-to-59-month-old children using calibrated and verified equipment and tools and the correct OPT Plus data collection protocol (Sections 3.1 to 3.6).
- The weight and length/height of children are recorded in OPT Plus Form 1A with one decimal place (0.0). This is also an opportune time to actively search for malnourished children with apparent manifestations of malnutrition to immediately refer them to appropriate offices for proper intervention (i.e., health center, social welfare office, agriculture office, etc.) (Section 3.7).
- The parents or caregivers should be given immediate feedback on the child's nutritional status using the weight-for-length/height indicator (Section 3.7). This can easily be done by using the Child Growth Standards (CGS) weight-for-length/height Growth Chart in the Early Childhood Care and Development (ECCD) Card (Section 3.8) or through the National Nutrition Information System (NNIS) mobile application.

Activity 3. OPT Plus Protocol on Nutritional Status Assessment

- The main activity is processing the OPT Plus data using the eOPT Plus Tool. This tool was developed by the National Nutrition Council (NNC) to make data collection, calculation, and reporting of OPT Plus data more efficient and accurate (Section 4.2).
- Data Quality Checks (DQC) are included in the eOPT Plus Tool to generate reliable, accurate, and precise OPT Plus data (Section 7).
- In case there is no access to a computer, a smartphone may be used by downloading the National Nutrition Information System (NNIS) mobile application to automatically determine the nutritional status, and fill out the required OPT Plus Forms (Section 4.3).

Activity 4. OPT Plus Protocol on Data Interpretation and Action Planning

- After generating reliable, accurate, and precise OPT Plus data, its utility should be promoted at the barangay (Section 5.1). The OPT Plus Team's efforts will be useless if the Barangay Nutrition Committee (BNC) will not use the OPT Plus results for the identification of nutrition actions.
- The data should be presented, analyzed, and interpreted (Sections 5.2 and 5.3). The data presented should be simple and easy to understand. Then, the BNC will review the results if there are inconsistencies and will interpret the trends or changes in the prevalence of malnutrition in the community. What does the data mean? What story is the data telling?
- After data interpretation, the OPT Plus Team should be ready to plan and/or update the Barangay Nutrition Action Plan (BNAP) (Section 5.4).

Activity 5. OPT Plus Protocol on Follow Through Activities

- Monitoring and evaluation are important components of the OPT Plus process. These aim to enhance the continuity and implementation of nutrition programs, projects, and activities (PPA). The main activity is the OPT Plus monitoring of the nutritional status of children (Section 6.3).
- Monthly monitoring of 0-to-23-month-old and those older children identified as malnourished (i.e., underweight, stunted, wasted, overweight, and with obesity), and semestral monitoring of 24-to-59-month-old children (Section 6.2). The eOPT Plus Tool can generate these lists (Section 6.3).
- The spot map should also be updated (Section 2.4). Lastly, the OPT Plus follow through activities provide a chance to monitor and evaluate the accomplishments in the BNAP and assess its contribution to the improvement of malnutrition in the community (Section 6.4).

What resources are needed in the OPT Plus Implementation?

The OPT Plus Team, appropriate weighing scale, length/height board, MUAC tape, CGS tables, growth chart for male and female, computer with Excel application, OPT Plus forms, pens/pencils, clipboard and calibration tools such as:

- Calibrated test weights
- Calibrated steel rule
- L-shaped ruler

Thus, the OPT Plus activities should have a budget appropriation in the City/Municipal Nutrition Action Plan (C/MNAP), Barangay Nutrition Action Plan (BNAP), and Annual Investment Program (AIP).

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LIST OF ANNEXES

ANNEX 1.1 Roles and responsibilities of the OPT Plus Team and supervisors

Activity	National Nutrition Council Central Office	National Nutrition Council Regional Office/ Regional Nutrition Program Coordinator (RNPC)	Provincial Nutrition Office/ Provincial Nutrition Action Officer (PNAO)	Supervisors: City/Municipal Nutrition Action Officers, District/ City/Municipal Coordinators, Midwives, Barangay Nutrition Action Officers among others as applicable	City/Municipal Nutrition Action Officer (C/MNAO)	OPT Plus Team members	Barangay Nutrition Committee
Activity 1. Pre-OPT Plus Protocol	1. Issue national guidance and updates as necessary on the conduct of OPT Plus. 2. Issue updated electronic OPT Plus tool. 3. Provide technical assistance as needed to stakeholders.	1. Issue letter on the conduct of OPT Plus starting August with reminders on the following: • Calibration/ Verification • Household/ Family Profiling • OPT Plus team Certification • Refresher Training 2. Provide technical support during the training of the OPT Plus team.	1. Disseminate memorandum on the conduct of OPT Plus issued by the RNPC. 2. Assist in the calibration/ verification of tools and equipment. 3. Provide technical support during the training of the OPT Plus team. 4. Allocate funds for OPT Plus.	1. Assist in organizing the OPT Plus training, standardization and refreshers. 2. Facilitate the organization of the OPT Plus Team in assigned cluster/ area. 3. Check if the weighing scales are calibrated and if length/ height boards are verified. If not, initiate or lead the calibration or verification of the tools and equipment.	1. Organize the OPT Plus Training and Standardization and Refresher Training. 2. Remind the barangay OPT Plus Team of the scheduled conduct of the OPT Plus every January to March of the current year. 3. Oversee the conduct of family profiling and spot-mapping. 4. Allocate funds for the purchase of OPT Plus.	1. Attend the Training and Standardization and Refresher Training on the OPT Plus. 2. Assist the supervisor in facilitating the organization of the OPT Plus Team. 3. Check if the weighing scales are calibrated and verify the length/ height board. 4. Conduct household profiling and spot-mapping of target households.	1. Provide resources to the OPT Plus Team members to attend the Training and Standardization or Refresher Training on the OPT Plus. 2. Organize the OPT Plus Team. 3. Assist the OPT Plus Team members in calibrating/ verifying the weighing scales and verifying the length/height board. 4. Assist in conducting household profiling and spot-mapping of target households. 5. Allocate funds needed for the OPT Plus.

Activity	**National Nutrition Council Central Office**	**National Nutrition Council Regional Office/ Regional Nutrition Program Coordinator (RNPC)**	**Provincial Nutrition Office/ Provincial Nutrition Action Officer (PNAO)**	**Supervisors: City/Municipal Nutrition Action Officers, District/ City/Municipal Coordinators, Midwives, Barangay Nutrition Action Officers among others as applicable**	**City/Municipal Nutrition Action Officer (C/MNAO)**	**OPT Plus Team members**	**Barangay Nutrition Committee**
				4. For highly urbanized cities, the City/Municipal Health Officer (C/MHO) will prepare a letter of communication to be signed by the Local Chief Executive (LCE) to allow the conduct of OPT Plus in exclusive subdivisions and villages.			
Activity 2. OPT Plus Anthropometric Measurement	Monitor the status of the OPT Plus activity at the national level	1. Monitor the status of the OPT Plus activity 2. Provide mentoring and supportive supervision visits to LGUs/ barangays needing assistance	1. Monitor the status of the OPT Plus activity 2. Provide mentoring and supportive supervision visits to barangays needing assistance	1. Check the master list of 0-to-59-month-old children 2. Oversee the actual collection of length/height measurement and weighing 3. Review the recording of the sex, date of birth, date of weighing, length/height, and weight	Monitor the status of the conduct of the OPT in each Barangay Health Station (BHS)	1. Conduct monthly OPT Plus monitoring of MW/MAM, SW/SAM, underweight, and 0-to-23-month-old children, and quarterly monitoring of 24-to-59-month old children.	1. Assist in the preparation of the master list of 0- to 59-month-old children. 2. Mobilize the under-five children to the weighing post. 3. Provide logistic support, i.e., transportation and food allowance to the OPT Plus Team.

				4. Refer to the Barangay Health Centers (BHC)/ Rural Health Unit (RHU) the children with edema and obviously manifesting malnutrition (i.e. extreme thinness, hollow cheeks, edematous feet, and hands, etc.) 5. Feedback the child's nutritional status immediately after measurement using the weight-for-length/height index		2. Participate in the bi-annual review of the BNAP and present BNAP accomplishments during the Barangay General Assembly/BNC meetings. 3. Prepare the master list of 0-to-59-month-old children. 4. Conduct the actual collection of length/height measurement and weighing with the supervisor, if feasible. 5. Record the sex, date of birth, date of weighing, length/height, and weight. 6. Refer to the BHC/ RHU the children with edema and obviously manifest malnutrition if the supervisor is unavailable. 7. Feedback the child's nutritional status immediately after measuring using the weight-for-length/height CGS Growth Chart in the ECCD Card, if the	4. Provide transportation and other services to the family with malnourished children to visit the BHC/RHU. 5. Assist coordination with the DCW for the children who have not been measured at home or center-based.

Activity	National Nutrition Council Central Office	National Nutrition Council Regional Office/ Regional Nutrition Program Coordinator (RNPC)	Provincial Nutrition Office/ Provincial Nutrition Action Officer (PNAO)	Supervisors: City/Municipal Nutrition Action Officers, District/ City/Municipal Coordinators, Midwives, Barangay Nutrition Action Officers among others as applicable	City/Municipal Nutrition Action Officer (C/MNAO)	OPT Plus Team members	Barangay Nutrition Committee
						supervisor is unavailable. 8. Coordinate with the DCW to measure the children who were not covered during the center-based OPT Plus or home visit.	
Activity 3. OPT Plus Data Processing, Encoding and Report Generation	1. Provide technical assistance at the national level in using the eOPT Plus Tool 2. Document issues and concerns encountered in the tool for future improvements	1. Provide technical assistance at the regional level in using the eOPT Plus Tool 2. Submit reports to the NNC CO on issues and concerns encountered in the tool for future improvements	Provide capacity building in using the eOPT Plus Tool	1. Assist in encoding the anthropometric measurements in the eOPT Plus Tool. 2. Review the outputs of the BNS for possible errors in data recording such as duplicate children, too big or too small measurements, etc. 3. Endorse the OPT Plus reports to the barangay.	Conduct a data quality check of the eOPT Plus Tool.	1. If the team has access to a computer or laptop, use the eOPT Plus Tool to compute children's age in months and nutritional status automatically. Make sure to get a soft copy of the eOPT Plus Tool.	Review the submitted OPT Plus reports and endorse them to the C/MNAO.

						2. In case there is no access to a computer, a smartphone may be used by downloading the NNIS mobile application to automatically determine the nutritional status. 3. Submit the OPT Plus reports to the C/MNAO and Midwife.	
Activity 4. OPT Plus Data Interpretation and Action Planning	1. Presentation of the OPT Plus trend during National BNS Conference and to the NNC Technical Committee, Technical Working Group, and other meetings 2. Consolidate the OPT Plus results 3. Include OPT Plus results in the Philippine Plan of Action for Nutrition specifically areas not covered by National Nutrition Survey to address nutrition problems	1. Presentation of the OPT Plus trend during BNS Conference and to the Regional Nutrition Council (RNC), Regional Technical Working Group (RTWG), Regional Social Development Committee (RSDC) among other fora. 2. Consolidate the OPT Plus results 3. Include OPT Plus results in the Regional Nutrition Action Plan (RNAP) to address nutrition problems	1. Presentation of the OPT Plus trend to the Provincial Nutrition Committee (PNC) and Provincial Board (Sangguniang Panlalawigan) 2. Consolidate the OPT Plus results 3. Include OPT Plus results in the PNAP to address nutrition problems 4. Provision of technical assistance in data analysis and interpretation	Participate in the preparation of the LNAP	1. Organize the presentation of OPT Plus results to Municipal Nutrition Committee (C/MNC) 2. Review and consolidate the updated BNAP 3. Facilitate the preparation of the C/MNAP 4. Present the results of the OPT Plus to the Sangguniang Panlungsod/Bayan for the budget appropriation of the C/MNAP 5. Facilitate approval of resolutions at the barangay, municipal, city levels concerning OPT Plus activities	1. Organize the presentation of OPT Plus results to BNC. 2. Facilitate the preparation and updating of the BNAP to the BNC. 3. Present the results of the OPT Plus to the Sangguniang Barangay for the budget appropriation of the BNAP. 4. Facilitate approval of resolutions at the barangay level concerning OPT Plus activities.	1. Participate in the presentation of OPT Plus. 2. Participate in preparing and updating the BNAP. 3. Allocate funds to implement the programs/ projects/ activities of the BNAP. 4. Review and approve resolutions concerning OPT Plus activities.

Activity	National Nutrition Council Central Office	National Nutrition Council Regional Office/ Regional Nutrition Program Coordinator (RNPC)	Provincial Nutrition Office/ Provincial Nutrition Action Officer (PNAO)	Supervisors: City/Municipal Nutrition Action Officers, District/ City/Municipal Coordinators, Midwives, Barangay Nutrition Action Officers among others as applicable	City/Municipal Nutrition Action Officer (C/MNAO)	OPT Plus Team members	Barangay Nutrition Committee
Activity 5. OPT Plus Protocol on Follow Through Activities	1. Conduct mid-year and end-year meetings within the Nutrition Surveillance Division staff regarding the conduct of OPT Plus 2. Conduct national activities related to data quality check	1. Conduct end-year meetings with P/C/MNAOs to discuss the implemented programs and projects and updates on the nutritional status of children 2. Conduct regional activities related to data quality check	1. Conduct mid-year meetings with C/MNAOs to discuss the implemented programs and projects and updates on the nutritional status of children 2. Conduct provincial activities related to data quality check	1. Assist in reviewing the monthly and quarterly OPT Plus monitoring coverage if all the children weighed and measured in the first quarter were followed up 2. Participate in the bi-annual review of the BNAP and assist in presenting BNAP accomplishments during the Barangay General Assembly.	1. Remind the BNC of the schedules of the monthly OPT Plus monitoring of SW/MAM and MW/MAM, underweight and 0-to-23-month-old, and quarterly monitoring of 24-to-59-month old children 2. Review the monthly and quarterly OPT Plus monitoring coverage if all the children weighed and measured in the first quarter were all followed up OR Assist in reviewing the monthly and quarterly OPT Plus monitoring coverage if all the children measured were covered during child growth monitoring.	1. Remind the BNC of the schedules of the monthly OPT Plus monitoring of SW/MAM and MW/MAM, underweight and 0-to-23-month-old, and quarterly monitoring of 24-to-59-month old children 2. Review the monthly and quarterly OPT Plus monitoring coverage if all the children weighed in the first quarter were all weighed. 3. Organize a bi-annual review of the BNAP and assist in presenting BNAP accomplishments during the Barangay General Assembly.	1. Assists in conducting monthly OPT Plus monitoring of SW/MAM and MW/MAM, underweight and 0-to23-month-old, and quarterly monitoring of 24-to-59-month-old children. 2. Participate in the bi-annual review of the BNAP and present BNAP accomplishments during the Barangay General Assembly. 3. Conduct barangay activities related to data quality check

					3. Conduct a bi-annual review of the BNAP and present BNAP accomplishments during the Barangay General Assembly. 4. Follow-up to the BNS on the status of the referred malnourished children. 5. Conduct city/municipal activities related to data quality check		

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Section Two

PRE-OPT PLUS PROTOCOL

PRE-OPT PLUS PROTOCOL

OVERVIEW

This section aims to cover an in-depth discussion of the pre-OPT Plus protocol to instill integrity in conducting the OPT Plus.

After reading this section, one should be able to:

1. Describe how to organize the OPT Plus team;
2. Explain the process of OPT Plus members;
3. Collect household profiles;
4. Prepare and update spot maps;
5. Calibrate and verifiy weighing tools and equipment;
6. Verify length and height tools and equipment; and
7. Accurately and precisely measure the weight, length, height, and MUAC.

The general objective of the pre-OPT Plus protocol is to help prepare all the resources needed for the conduct of the OPT Plus. Specifically, the pre-OPT Plus protocol aims to provide knowledge and skills in the following subtopics:

Section 2.1	Organizing the OPT Plus Team
Section 2.2	Certification of OPT Plus Members
Section 2.3	Household Profiling
Section 2.4	Spot Mapping
Section 2.5	Calibration and Verification of Weighing Tools and Equipment
Section 2.6	Verification of Length/Height Tools and Equipment
Section 2.7	Standardization Exercise

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LIST OF ACRONYMS

AIP	Annual Investment Plan
BGA	Barangay General Assembly
BHW	Barangay Health Worker
BNAO	Barangay Nutrition Action Officer
BNC	Barangay Nutrition Committee
BNS	Barangay Nutrition Scholar
C/MHO	City/Municipal Health Officer
C/MPDO	City/Municipal Planning Development Office
cm	centimeter
CMV	Conventional Mass Value
D/C/MNPC	District/City/Municipal Nutrition Program Coordinator
DCW	Day Care Worker
DOST	Department of Science and Technology
g	gram
HRH	Human Resource for Health
kg	kilogram
LCE	Local Chief Executive
LGU	Local Government Unit
mm	millimeter
MSt	Moderately Stunted
MUW	Moderately Underweight
NDDP	Nutritionist-Dietitian Deployment Program
NHTS	National Housing Targeting System
NNC	National Nutrition Council
OPT Plus	Operation Timbang Plus
P/C/MNAO	Provincial/City/Municipal Nutrition Action Officer
PNAO	Provincial Nutrition Action Officer
R	Coefficient of Reliability
RHM	Rural Health Midwife

RNPC	Regional Nutrition Program Coordinator
SK	Sangguniang Kabataan
SSt	Severely Stunted
St	Stunted
SUW	Severely Underweight
TEM	Technical Error of Measurement
UW	Underweight

DEFINITION OF TERMS

Anthropometry Training	Training sessions that strengthen the measurers' knowledge and skills on the use, maintenance, calibration, and verification of tools and equipment; procedures and precautions when taking the weight, length/height, and MUAC measurement; and roles of measurers and assistants.
Calibration	Process of checking and adjusting of the weighing scale within the possible and allowable error.
Household Profile	Summary of demographic, health, and nutrition information of households living in the barangay.
Spot Map	Visual representation of the barangay, where information such as the geographic and demographic characteristics of the barangay are indicated.
Standardization Exercise	An activity usually conducted after the general training that enables the trainers/facilitators to directly observe the trainee's measurement techniques and correct any error that may lead to an inaccurate measurement.
Verification	Process of checking if the length/height board and weighing scale calibration results have not changed significantly since it had its last calibration.

Section 2.1

ORGANIZING THE OPT PLUS TEAM

Through the help of the City/Municipal Nutrition Action Officer (C/MNAO), two (2) sets of Operation Timbang (OPT) Plus Teams will be organized: one at the city/municipal and another at the barangay level. At the municipal level, the OPT Plus Supervisors are composed of the C/MNAO, Rural Health Midwife (RHM), nurses, District/City/Municipal Nutrition Program Coordinator (D/C/MNPC), and Human Resource for Health (HRH), i.e., Nutritionist-Dietitian Deployment Program (NDDP).

Meanwhile, the members of the Barangay OPT Plus Team can include the RHM, Barangay Nutrition Action Officer (BNAO), Committee Chair on Health and Nutrition, Barangay Nutrition Scholar (BNS), Sangguniang Kabataan (SK) officials, other Barangay Health Workers (BHW), Day Care Worker (DCW), and teacher-in-charge, and other barangay officials and staff.

The roles of the different persons at the different Local Government Units (LGU) levels are defined in Annex 1.1 while Table 2.1 shows the roles and functions of the different persons involved during the Pre-OPT Plus activities. Meetings should be done to discuss the organization of the OPT Plus Team with the Barangay Nutrition Committee (BNC).

Table 2.1. Roles and Functions of the different persons involved in the OPT Plus during the Pre-OPT Plus Protocol.

ORGANIZATION/ PERSON INVOLVED	FUNCTION
National Nutrition Council Central Office (NNC CO)	1. Issue national guidance and updates as necessary on the conduct of OPT Plus. 2. Issue updated electronic OPT Plus tool. 3. Provide technical assistance to stakeholders as needed.
Regional Nutrition Program Coordinator (RNPC)	1. Issue letter on the conduct of OPT Plus starting August with reminders on the following: i. Calibration/Verification ii. Household Profiling iii. OPT Plus Certification iv. Refresher Training 2. Provide technical support during the training of the OPT Plus team.
Provincial Nutrition Action Officer (PNAO)	1. Disseminate memorandum on the conduct of OPT Plus issued by the RNPC. 2. Assist in the calibration/verification of tools and equipment. 3. Provide technical support during the training of the OPT Plus team. 4. Allocate funds for OPT Plus.
Supervisors	1. Assist in organizing the OPT Plus Training and Standardization and Refresher Training. 2. Facilitate the organization of the OPT Plus Team in the assigned cluster. 3. Check if the weighing scales are calibrated and if length/height boards are verified. If not, initiate or lead the calibration or verification of the tools and equipment. 4. For highly urbanized cities, the City/Municipal Health Officer (C/MHO) will prepare a letter of communication to be signed by the Local Chief Executive (LCE) to allow the conduct of OPT Plus in exclusive subdivisions and villages.
City/Municipal Nutrition Action Officer (C/MNAO)	1. Organize the OPT Plus Training and Standardization and Refresher Training. 2. Remind the barangay OPT Plus Team of the scheduled conduct of the OPT Plus every January to March of the current year. 3. Oversee the conduct of family profiling and spot-mapping. 4. Allocate funds for the purchase of tools and equipment.

ORGANIZATION/ PERSON INVOLVED	FUNCTION
OPT Plus Team members	1. Attend the OPT Plus Training and Standardization and Refresher Training. 2. Assist the supervisor in facilitating the organization of the OPT Plus Team. 3. Check if the weighing scales are calibrated and verify the length/height board. 4. Conduct household profiling and spot-mapping of target households.
Barangay Nutrition Committee (BNC)	1. Provide resources to the BNS to attend the OPT Plus Training and Standardization and Refresher Training. 2. Organize the OPT Plus Team. 3. Assist the BNS in calibrating the weighing scales and verifying the length/ height board. 4. Assist in conducting household profiling and spot-mapping of target households. 5. Allocate funds needed for the OPT Plus.

SUPERVISOR'S BOX 2.1.
Data Quality Check

The details of the data quality check per activity are discussed in Section 7. Data Quality Check.

Section 2.2

CERTIFICATION OF OPT PLUS MEMBERS

Certifying the OPT Plus members after the Anthropometry Training is vital to OPT Plus activities. The certification will ensure that the OPT Plus Team has the knowledge, skills, and confidence in generating quality data. The details of the Anthropometry Training can be found in the OPT Plus Training Manual.

The basis for the level of certification includes the completion of the training, results of the standardization exercise, and post-test scores of the participating OPT Plus member. The certification is classified into five: Level 1, 2, 3, 4, and 5. Table 2.2 details the criteria and implication of each level.

The minimum requirement for an OPT Plus team member to get a certification of any level includes completing the Anthropometry Training and passing the post-test by obtaining at least 60% of the total score. Level 1 certification is given to those who were not able to pass any component of the standardization exercise. A Level 2 certification is given to those who passed one (1) out of the three (3) standardization tests. A Level 3 Certification will be given to those who passed the MUAC component and either weight or height component of the standardization test. If the OPT Plus team member passed both weight and height, they will be given a Level 4 certification. If they pass all the components of the standardization exercise, they will be given a Level 5 certification. A Level 5 certification makes an OPT Plus team member eligible to be a resource person in the succeeding OPT Plus training that will be conducted. Lastly, participants who completed the training, but failed the post-test and all components of standardization exercises will be given a certificate of attendance.

All certified OPT Plus measurers will be recorded in the Capacity Map database.

Table 2.2. Level of Classification of OPT Plus Team after completing the standardization exercise.

CRITERIA	LEVELS					
	LEVEL 1	LEVEL 2	LEVEL 3	LEVEL 4	LEVEL 5*	CERTIFICATE OF ATTENDANCE
Training**	Completed the training and passed the post-test (60%)	Completed the training and passed the post-test (60%)	Completed the training and passed the post-test (60%)	Completed the training and passed the post-test (60%)	Completed the training and passed the post-test (60%)	Completed the training but did not pass the post-test
Standardization***	Did not pass the standardization	Passed 1 out of 3 standardization test	Passed MUAC plus either weight or height standardization test	Passed both weight and height standardization test	Passed MUAC, weight and height standardization test	Did not pass any standardization test
Implication	Eligibility to measure depends on the passed measurement (both precision and accuracy). Those who failed could still help in assisting and recording the measurement.					

**Can be tapped as resource persons for the succeeding OPT Plus training.*
*** Participants who failed the post-test can take the removal test.*
****If ≥20% failed, those who failed will undergo restandardization. If <20% failed, re-standardization will not be necessary.*

The Regional Nutrition Program Coordinator (RNPC) shall issue the certification upon endorsement of the PNAO. The certified OPT Plus Team member is eligible for weighing and measurement of 0-59-month-old children in the barangay while those who passed the Level 5 can be tapped as resource persons for the succeeding OPT Plus training.

If an OPT Plus member did not pass the post-test, they are allowed to take a removal test and they should get 60% and above to pass.

On the other hand, failing the standardization test requires the measurer to repeat the anthropometry training and standardization exercise. If during the standardization test, ≥20% failed, those who failed may participate in the re-standardization.

In case no OPT Plus member was certified, the trained team can conduct the OPT Plus with close supervision of the trained supervisors or certified OPT Plus member from other LGUs or villages.

What are Anthropometry Training and Standardization Exercises?

ANTHROPOMETRY TRAINING is an activity that aims to strengthen the OPT Plus Team's knowledge and skills on the use, maintenance, and calibration of tools and equipment; procedures and precautions when taking weight and length/height measurements; and roles of measurers and assistants. The technical staff from the regional office, P/C/MNAOs, and D/C/MNPCs will support and coordinate the conduct of the Anthropometry Training provided by duly recognized training institutions.

The **STANDARDIZATION EXERCISE** is one of the components of the Anthropometry Training which will enable the trainers/facilitators to directly observe the trained OPT Plus Team members' measurement techniques and correct any error that may lead to an inaccurate measurement. The recorded measurements of the OPT Plus members will be subjected to accuracy, precision, and reliability analysis which will be used as one of the bases for the certification of participants.

SUPERVISOR'S BOX 2.2.
Training, Standardization, and Certification

As supervisors, here are some reminders regarding training, standardization, and certification of the OPT Plus Team:

1. In the second semester of the year, do an inventory of OPT Plus team members that do not have a certification. Make sure that the OPT team has a certification before the conduct of the OPT Plus. If all members have already received their certification, conduct a refresher training on measuring the weight, length, height, and MUAC.
2. Allot budget for regular training and standardization in the Annual Investment Plan (AIP).
3. The certification is valid for five (5) years with regular refresher course for updating as necessary.

Section 2.3

HOUSEHOLD PROFILING

Household Profile

A household profile records the health and nutrition information of families in the barangay and is similar to the Household Profile which can be accessed through the City/Municipal Health Office (C/MHO). Household profiling is routinely conducted before or during the OPT Plus Anthropometric Measurement.

To accomplish the household profile, the nutrition worker should use the Household Profile form (Annex 2.1). The nutrition worker may use the Household Profile as a reference for completing this form. Table 2.3 shows the definition of terms used in the household profile.

Table 2.3. Definition of terms used in the household profile.

TERM	DEFINITION
Adolescent	Household member ages 10-19 years old.
Caregiver	Household member who functions as a head or guardian of children living in the household in the absence of the mother and/or father.
Family	A group of persons usually living together and composed of the head and other persons related to the head by blood, marriage, or adoption. It includes both the nuclear and extended family (PSA, n.d.).
Father	Male head of the family with biological or adoptive relationship to the children.
Household	A social unit consisting of a person living alone or a group of persons who sleep in the same housing unit and have a common arrangement in the preparation and consumption of food (PSA).
Indigenous People (IP)	A group of people or homogenous societies identified by self-ascription and ascription by others, who have continuously lived as organized community on communally bounded and defined territory, and who have, under claims of ownership since time immemorial, occupied, possessed customs, tradition and other distinctive cultural traits, or who have, through resistance to political, social and cultural inroads of colonization, non-indigenous religions and culture, become historically differentiated from the majority of Filipinos (PSA).
Infant	Children ages 29 days to 11 months old.
Mother	Female head of the family with biological or adoptive relationship to the children.
Newborn	Children ages 0-28 months old.
National Housing Targeting System (NHTS)	An information management system that identifies who and where the poor households in the Philippines.
Pantawid Pamilyang Pilipino Program (4Ps)	Poverty alleviation program that aims to provide social assistance through cash grants and social development through investing in the health, nutrition, and education of children from poor families (Official Gazette, n.d.).
Postpartum (PP)	A woman who is in the postpartum period or the period not exceeding 6 weeks or 90 days after giving birth.
Person with Disability	Children affected by developmental, neurologic, and genetic conditions, which may affect growth patterns and overall growth potential.
Senior Citizen	Household member 60 years old and above.
Under-five	Children 0-59 months old.

The household profile form has 34 columns that will be filled out based on the number of household members based on age group, father and mother's name, occupation, and educational attainment. It also identifies pregnant mothers, couples practicing family planning, feeding status of <6-month-old children, toilet type, water source, food production activities, and utilization of iodized salt, iron-fortified rice, and micronutrient powder. and should be filled out completely.

C1 - HH No.

Write the control number of each household that will be interviewed. A household is defined as an individual or group of people living under one housing unit.

C2 - No. of families living in the house

Write the number of families living within a household. A family is a group of people living together and are related by blood, marriage, or adoption. A family can be nuclear composed of the father, mother, and children living together or extended with the inclusion of kin in the direct or indirect line of one member or members of the nuclear family.

C3 - No. of HH member

Write the total number of household members.

C4 - NHTS Household

Write the code that corresponds to the household's inclusion in the NHTS list using the following criteria:

CODE	CRITERIA	DEFINITION
1	NHTS 4Ps	Household listed in the NHTS and a member of 4Ps
2	NHTS Non-4Ps	Household listed in the NHTS but not a member of 4Ps
3	Non-NHTS	Not listed in the NHTS

C5 - Indigenous Group

Write the code that corresponds to the household's membership to an IP group using the following criteria:

CODE	TERM	DEFINITION
1	IP	Household belonging to an IP group
2	Non-IP	Household not belonging to an IP group

C6-C7 - Newborn (0-28 days)	Write the total number of male (C6) and female (C7) newborn babies 0-28 days old within a family.
C8-C9 - Infant (29 days-11 months)	Write the total number of male (C8) and female (C9) infants 29 days to 11 months old within a family.
C10-C11 - Under-five (1-4 y.o.)	Write the total number of male (C10) and female (C11) children 1-4 years old within a family.
C12-C13 - Children (5-9 y.o.)	Write the total number of male (C12) and female (C13) children 5-9 years old within a family.
C14-C15 - Adolescence (10-19 y.o.)	Write the total number of male (C14) and female (C15) adolescents 10-19 years old within a family.
C16 - Pregnant	Write the total number of pregnant women in the family.
C17 - Adolescent Pregnant	Write the number of pregnant women 10-19 years old in the family.
C18 - Postpartum (PP)	Write the number of women in the family who are in the postpartum period or the period not exceeding 6 weeks after giving birth.
C19 - 15-49 y.o. not pregnant & non-PP	Write the number of women in the family who are 15-49 years old and are not pregnant nor in postpartum period.
C20-C21 - Adult (20-59 y.o.)	Write the number of male (C20) and female (C21) adults 20-59 years old in the family.
C22-C23 - Senior Citizen	Write the number of male (C22) and female (C23) senior citizens 20-59 years old in the family.
C24-C25 - Person with Disability	Write the number of male (C24) and female (C25) individuals in the family who has a disability.
C26 - Name of Father (Fa) and Mother (Mo); Caregiver (Ca)	Write the name of the mother, father, and/or caregiver in the family.

C27 - Occupation Write the code of the type of occupation that the mother, father, and/or caregiver has using the following:

CODE	CRITERIA	DEFINITION
1	Manager	Workers in this group plan, direct, coordinate and evaluate the overall activities of enterprises, governments and other organizations, or of organizational units within them, and formulate and review their policies, laws, rules and regulations.
2	Professional	Workers in this group increase the existing stock of knowledge, apply scientisfic or artistic concepts and theories, teach about the foregoing in a systematic manner, or engage in any combination of these activities.
3	Technician & associate professionals	Workers in this group perform mostly technical and related tasks connected with research and the application of scientific or artistic concepts and operational methods, and government or business regulations.
4	Clerical support workers	Workers in this group record, organize, store, compute and retrieve information related, and perform a number of clerical duties in connection with money-handling operations, travel arrangements, requests for information, and appointments.
5	Service and Sales Worker	Workers in this group provide personal and protective services related to travel, housekeeping, catering, personal care, or protection against fire and unlawful acts, or demonstrate and sell goods in wholesale or retail shops and similar establishments, as well as at stalls and on markets.
6	Skilled Agricultural, forestry, and fishery workers	Workers in this group grow and harvest field or tree and shrub crops, gather wild fruits and plants, breed, tend or hunt animals, produce a variety of animal husbandry products, cultivate, conserve and exploit forests, breed or catch fish and cultivate or gather other forms of aquatic life in order to provide food, shelter and income for themselves and their households.
7	Craft and related trade workers	Workers in this group apply specific knowledge and skills in the fields to construct and maintain buildings, form metal, erect metal structures, set machine tools, or make, fit, maintain and repair machinery, equipment or tools, carry out printing work, produce or process foodstuffs, textiles, or wooden, metal and other articles, including handicraft goods.

CODE	CRITERIA	DEFINITION
8	Plant and machine operators and assemblers	Workers in this group operate and monitor industrial and agricultural machinery equipment on the spot or by remote control, drive and operate trains, motor vehicles and mobile machinery and equipment, or assemble products from component parts according to strict specifications and procedures.
9	Elementary occupations	Occupations in this group involve the performance of simple and routine tasks which may require the use of handheld tools and considerable physical effort.
10	Armed forces occupations	This major group includes all jobs held by members of the armed forces. Members of the armed forces are those personnel who are currently serving in the armed forces, including auxiliary services, whether on a voluntary or compulsory basis, and who are not free to accept civilian employment and are subject to military discipline. Included are members of the army, navy, air force and other military services, as well as conscripts enrolled for military training or other service for a specified period.
11	None	

C28 - Educational Attainment

CODE	CRITERIA	DEFINITION
N	None	Have not attended school
EU	Elementary undergraduate	Did not graduate from elementary level
EG	Elementary graduate	Graduated from elementary level
HU	High School Undergraduate	Did not graduate from high school level
HG	Highschool graduate	Graduated from high school level
CU	College undergraduate	Did not graduate from college
CG	College graduate	Graduated from college
V	Vocational	Graduated from vocational
PG	Post-Graduate Studies	College graduate and undertook further study

C29 - Couple Practicing Family Planning

Put a check mark to answer "Yes" or "No".

C30 - Toilet Type

Write the code that corresponds to the type of toilet the household has.

CODE	CRITERIA	DEFINITION
1	Improved Sanitation	Flush/pour flush to a piped sewer system Flush/pour flush to septic tank Flush/pour flush to a pit latrine Ventilated improved pit (VIP) latrine Pit latrine with slab Composting toilet
2	Shared Facility	Flush/pour flush to a piped sewer system Flush/pour flush to septic tank Flush/pour flush to a pit latrine Ventilated improved pit (VIP) latrine Pit latrine with slab Composting toilet Public toilet
3	Unimproved Facility	Flush/pour flush not to sewer/septic tank/pit latrine Pit latrine without slab/open pit Bucket Hanging toilet/hanging latrine Other
4	Open Defecation	No facility Bush Field

C31 - Water Source (P, W, S) Write the code that corresponds to the water source the household has.

CODE	CRITERIA	DEFINITION
1	Improved Source	• Piped into dwelling/yard/plot • Piped to neighbor • Public tap/standpipe • Tube well/borehole • Protected dug well • Protected spring • Rainwater • Bottled water/refilling station, improved source for cooking/handwashing (with certification)
2	Unimproved Source	• Unprotected dug well • Unprotected spring • Tanker truck/cart with small tank • Surface water • Bottled water/refilling station, unimproved source for cooking/handwashing

C32 - Food Production Activity (VG/PL/FP) Write the code of the applicable means of food production of the family. Multiple answers may be selected.

C33 - HH using Iodized Salt Put a check mark to answer "Yes" or "No" depending on the use of iodized salt.

C34 - HH using Iron-Fortified Rice Put a check mark to answer "Yes" or "No" depending on the use of iron-fortified rice.

Section 2.4

SPOT MAPPING

Spot Map

A spot map is a visual presentation of the barangay, where information such as the geographic, demographic, health, and nutrition characteristics of the barangay are indicated. Ideally, the updating of the spot map is conducted during the anthropometric measurement, however, this can also be done after the anthropometric measurement. It helps the nutrition worker to locate target households with 0-59 month old children for OPT Plus. A spot map is created through sketching the barangay, downloading the map on the internet (Google Maps), or asking for the Geographic Information System (GIS) barangay map from the City/Municipal Planning Development Office (C/MPDO). The spot map can be used during presentations with the Barangay Nutrition Committee (BNC) and Barangay General Assembly (BGA).

As mentioned, there are several ways to create a spot map – sketch the barangay, download the map through Google Maps, or ask for the GIS.

A. SKETCH THE SPOT MAP

Here are the steps in sketching a spot map:

1. Draw the streets, pathway, and the creek/canal/drainage of the barangay by purok.
2. Estimate and draw the location of the houses, schools, churches, health center, barangay hall, daycare centers, water pumps, water tanks, public toilets, water system, basketball court, business establishments, fishpond, poultry, piggery, rest house, resort, small market, and other infrastructure in the barangay.
3. Merge the maps by purok, then draw the whole barangay on clean paper.
4. After generating the map, you need to put the codes or legends on the needs and problems of the families in the houses on the spot map similar to what is indicated in Figure 2.1. Use the color/legend prescribed by NNC in Memorandum No. 2016-015 Dissemination and Application of Revised Color Coding/Legend in the Preparation of Spot Maps Among Local Government Units (Table 3). The details of the memo are shown in Annex 2.3.

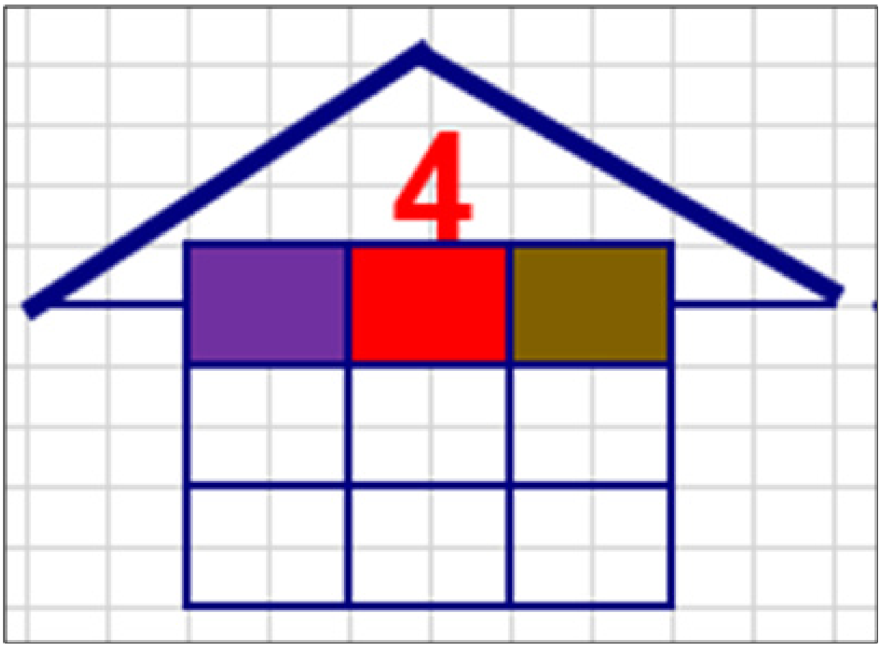


Figure 2.1. Sample coded house.

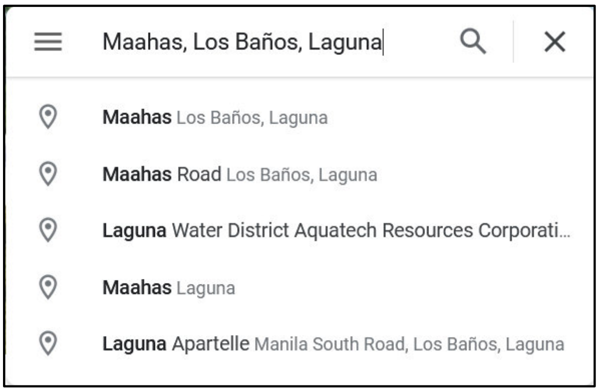
Table 2.4. Color/Legend for the preparation of the barangay spot-map.

COLOR		INTERPRETATION
	Green	Children with 'normal weight/length/height'
	Yellow	Children with 'underweight/stunted/wasted' Mark with "UW" for underweight, "St" for stunted, "W" for wasted
	Red	Children with "severely underweight/stunted/wasted" Mark with "SUW" for severely underweight, "SSt" for severely stunted, "SW" for severely wasted
	Orange	Children with "overweight/obese" Mark with "OW" for overweight, "Ob" for obese
	Pink	With pregnant women
	Violet	With lactating women
	Maroon	Infants who were exclusively breastfed for 6 months
	Purple	Infants who were given complimentary food in addition to breastmilk (starting at 6 months old)
	Brown	With infants 0-11 months
	Blue	With large family size (households with greater than 5 family members)
	Black	Without potable water source
	Gray	Without sanitary toilets

B. DOWNLOAD THE MAP FROM GOOGLE MAPS

Here are the steps in creating a spot map through downloading a map of the barangay from Google Maps:

1. On your computer, sign in to My Maps or https://maps.google.com/.
2. Open or create a map.
3. In the search bar, type the name or address of a place.



4. Click one of the results on the map.
5. Screenshot 

or use the snipping tool 

to copy the map. Then paste (ctrl +v) and save (ctrl+s) it in the word file. The downloaded file already shows the roads and bodies of water like rivers and creeks.
6. Identify the boundaries of each purok.
7. On the map, draw the pathway and other roads not captured by the downloaded map.
8. Some structures are shown on the map. Identify the structures like schools, churches, health center, barangay hall, daycare centers, water pumps, water tanks, public toilets, water system, basketball court, business establishments, fishpond, poultry, piggery, rest house, resort, small market, and other infrastructure in the barangay.
9. Identify the households on the map and add the houses not included on the map.
10. Merge the maps by purok, then draw the whole barangay on clean paper.
11. After generating the map, you need to put the codes or legends on the needs and problems of the families in the houses on the spot map similar to what is indicated in Figure 2.1. Use the color/legend prescribed by NNC in Memorandum No. 2016-015 Dissemination and Application of Revised Color Coding/Legend in the Preparation of Spot Maps Among Local Government Units (Table 2.4).

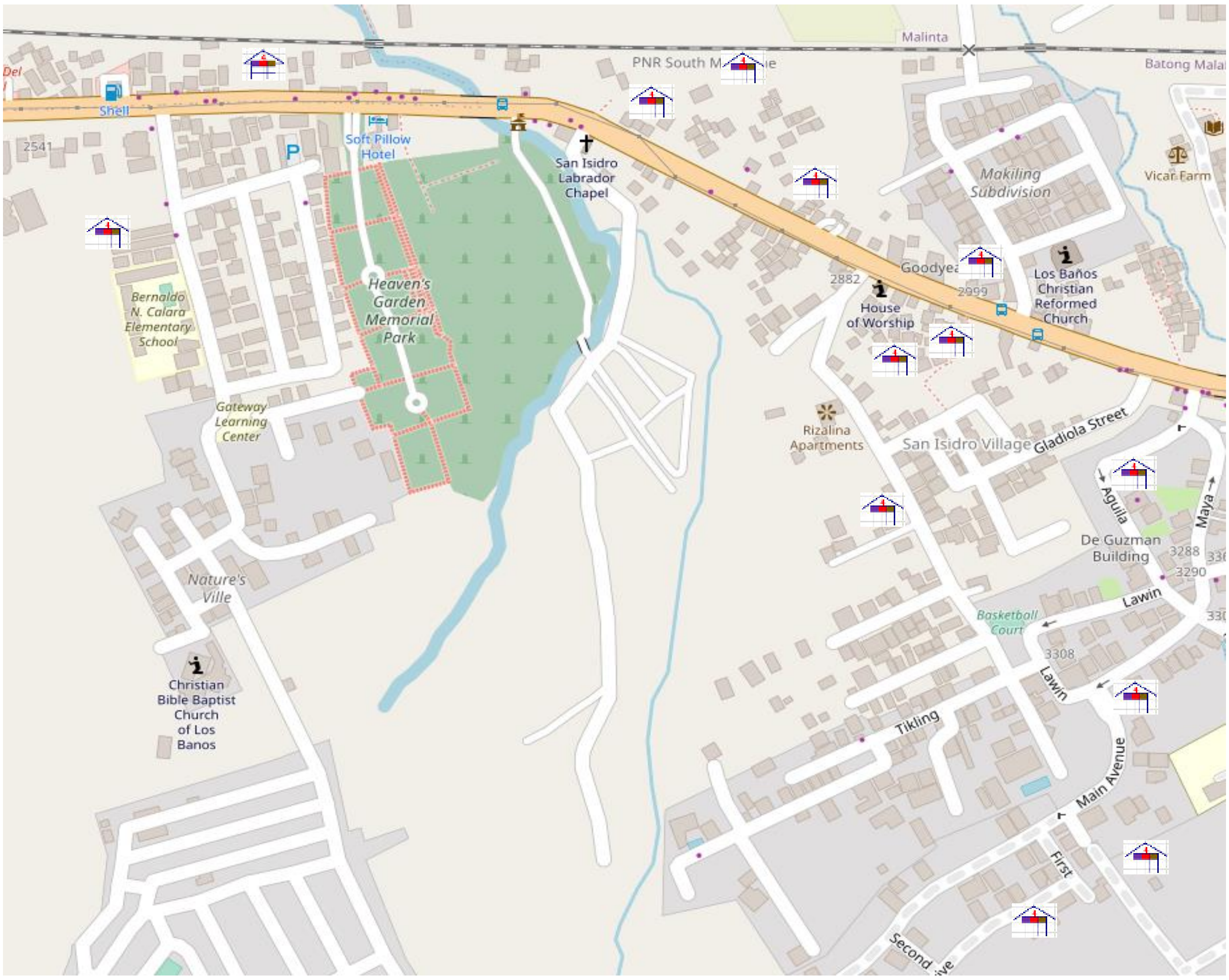


Figure 2.2. Aerial map downloaded from Open Street Map.

C. GEOGRAPHIC INFORMATION SYSTEM (GIS)

Here are the steps in creating a spot map through the GIS barangay map:

1. Ask the C/MPDO if they have an available GIS barangay map. Secure a copy if they have it on file. The maps in the municipality are usually used for disaster risk mapping.
2. To locate the houses, get the coordinates of the houses (latitude and longitude) called geotagging. Geotagging is the process of appending geographic coordinates to media based on the location of a mobile device. It can be done through the GPS or by downloading it on a smartphone.
3. Merge the coordinates with the barangay shape file and encode the household and children's profiles in the table attributes.
4. Generate the map per indicator as prescribed in Table 1.

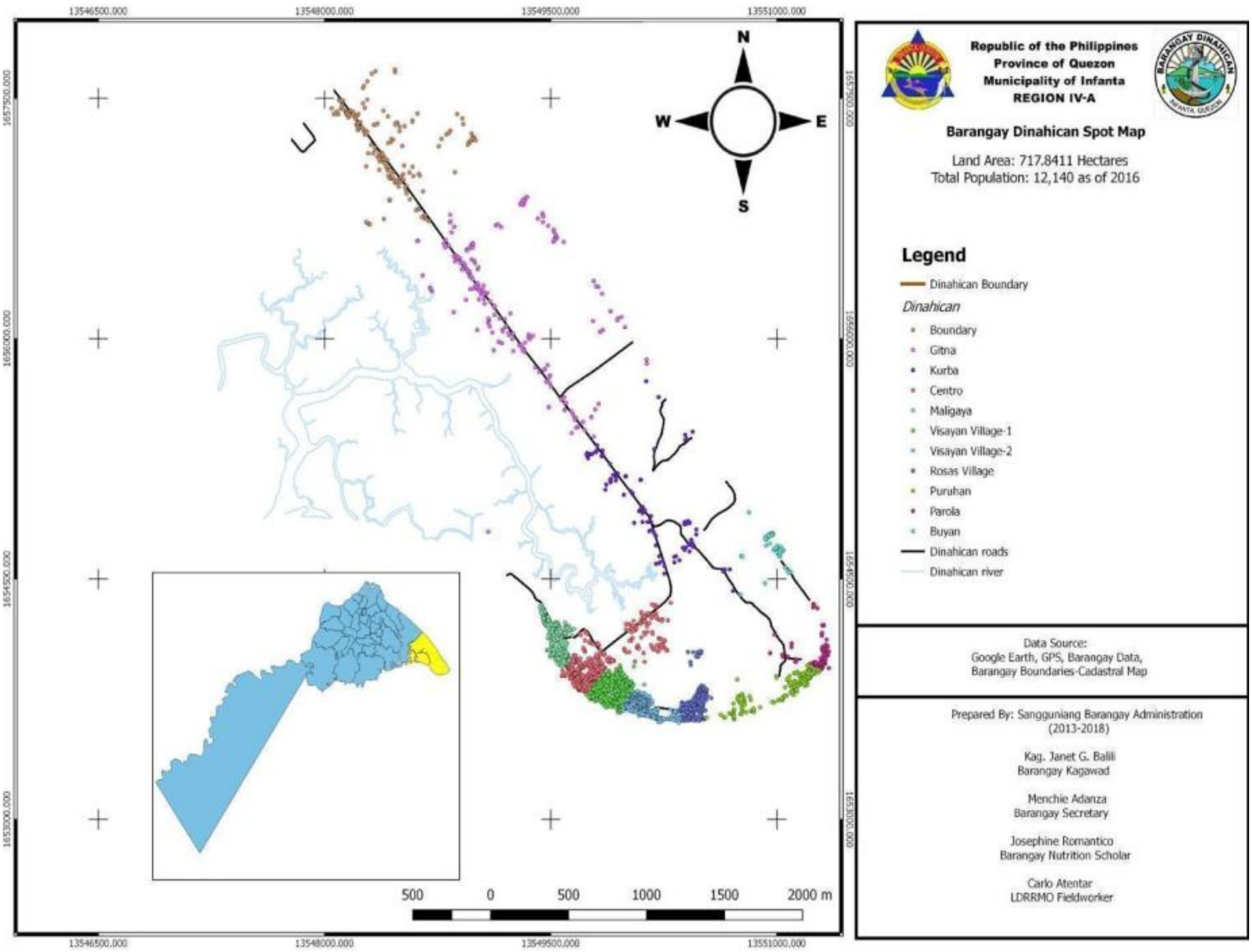


Figure 2.3. Sample GIS barangay map from C/MPDO.

Section 2.5

CALIBRATION AND VERIFICATION OF WEIGHING TOOLS AND EQUIPMENT

What is Calibration?

Calibration is the comparison of two instruments or measuring devices, one of which is a standard of known accuracy traceable to national standards, to detect, correlate, or eliminate by adjustment any discrepancy in accuracy of the instrument or measuring device being compared with the standard (NIST, USA). A weighing scale's performance eventually degrades as it ages and undergoes mechanical stress. Once this happens, the weight measurement that can be obtained may be inaccurate. For OPT Plus, calibration is recommended every three years to ensure that the weighing scale remains functional and reliable.

MATERIALS NEEDED FOR WEIGHING SCALE CALIBRATION OR VERIFICATION

1. One (1) 5 kg and two (2) 10 kg calibrated standard weights
2. Weighing Scale Calibration and Verification worksheet (Annex 2.8)

Figure 2.4. Calibrated Standard Weights.

How to calibrate weighing scales?

TESTS FOR THE CALIBRATION OF WEIGHING SCALES

1. Departure from Nominal Value Test. Tests whether the weighing scale reading is consistent and accurate at different points (i.e., 5, 10, 15, 20, and 25 kg).
2. Repeatability Test. Ensures that the weighing scale provides consistent and accurate readings after repeated measurements.
3. Effect Off-Center Loading/Eccentric Test. Determines if the weighing scale reading is consistent and accurate if test weight is placed on different points in the weighing scale. This test is applicable with platform scales only.

STEPS IN CALIBRATING WEIGHING SCALES

Step 1. Fill out the information part of the calibration/verification form for documentation.

Using the Weighing Scale Calibration and Verification Form Record (Annex 2.8), record all the pertinent information about the weighing scale to be verified or calibrated located at the upper section of the form. Once filled out, proceed to the next step.

Weighing Scale Calibration and Verification Form

Type of weighing scale:		Date of Calibration:	
Manufacturer's Name:		Place of Calibration:	
Serial number:		Resolution:	
Max capacity:		Requesting Agency:	
		Calibration Validity:	

Step 2. Conduct Pre-loading

Before starting the calibration, pre-load the weighing scale by placing or hanging a 20-kg standard weight for three (3) minutes. This procedure "warms up" the instrument before the actual calibration. After removing the test weight, check if the dial has returned to zero. Otherwise, the weighing scale is suggested to be repaired or replaced.

Figure 2.5. Pre-loading.

Figure 2.6. Testing if the scale goes back to zero.

Step 3. Conduct Departure from Nominal Value Test

1. Tare the weighing scale to 0.0 kg.
2. Prepare 5, 10, 15, 20, and 25 kg standard weights
3. Get the Conventional Mass Value (CMV) found in the Calibration Certificate of the standard weights. The CMV in the sample shown in Figure 2.7 is 120 mg.

CALIBRATION CERTIFICATE

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Sample : Single Test Weight
Serial No. : 10024175
Date Received : June 27, 2022
Date and Place of calibration : June 27, 2022 / Calibration Lab. 2

Conventional Mass Value	Uncertainty of Measurement	k interval	Remarks
10 kg +120 mg	160 mg	2.00	-

Figure 2.7. Sample calibration certificate with CMV.

4. Start weighing the 5 kg standard weight, followed by 10 kg until 25 kg. This is Trial 1.
5. Repeat weighing until Trial 3.

Figure 2.8. Hanging of 5 kg, 10 kg, 15 kg, 20 kg and 25kg weights for Departure from Nominal Value Test.

Step 4. Conduct Repeatability Test.

1. Tare the weighing scale to 0.0 kg.

   For hanging weighing scale, ensure that the rope that will be used to hang the 20 kg standard weight is attached to the hook. Tie the rope securely around the handle of the standard weight.

2. Load the 20 kg standard weight and record the weight measurement in the verification form. This is Trial 1.

3. Tare the scale to 0.0 kg before proceeding to Trial 2.

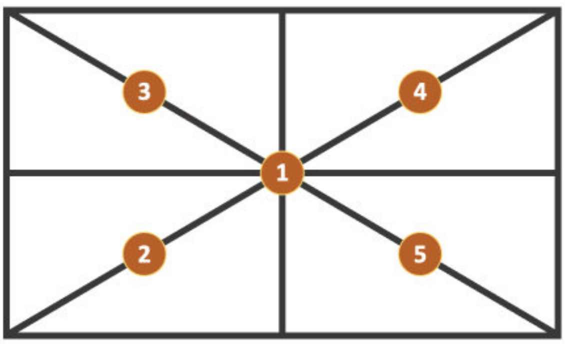
4. Load then unload the standard weight until Trial 10.

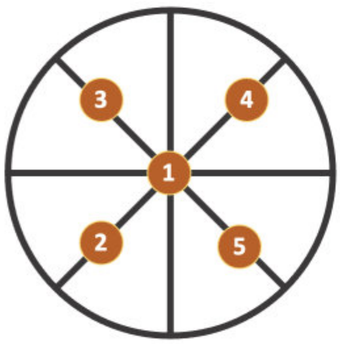
Repeatability Test using 20 kg test weight	
Resolution of weighing scale (kg)	
Trial 1	
Trial 2	
Trial 3	
Trial 4	
Trial 5	
Trial 6	
Trial 7	
Trial 8	
Trial 9	
Trial 10	
Standard Deviation	
Criteria	

For platform scales, follow steps 1 to 5.

Step 5. Conduct Effect Off-Center Loading/Eccentric Test.

1. Tare the weighing scale to 0.0 kg.
2. Apply the 10 kg test weight starting at position 1 and record in the verification sheet.
3. Proceed to position 2 until position 5.
4. Repeat weight measurement at position 1.





Step 6. Encode the collected data in the Weighing Scale Verification Tool and interpret results.

The Weighing Scale Verification Tool will automatically generate the calibration results. To determine if the weighing scale can be considered "Calibrated" or fit for use, check the Correction column found in the Departure from Nominal Value Test. If any of the rows appear in red, the weighing scale should be replaced or repaired. Otherwise, the weighing scale is fit for use.

Fit for use:

Departure From Nominal Value Test							
Test Weights (kg)	Conventional Mass Value (CMV) found in Calibration Certificate of test weights (mg)	Nominal Value (kg)	Trial 1 (kg)	Trial 2 (kg)	Trial 3 (kg)	Average	Correction (should be within -0.1 kg to 0.1 kg)
5	25	5.00003	5.00	4.95	5.00	4.983333333	0.017
10	120	10.00012	9.90	9.95	9.95	9.933333333	0.067
15	145	15.00015	15.00	15.00	15.00	15	0.000
20	240	20.00024	20.00	20.00	20.00	20	0.000
25	265	25.00027	25.00	24.95	25.00	24.98333333	0.017

For replacement or repair:

Departure From Nominal Value Test							
Test Weights (kg)	Conventional Mass Value (CMV) found in Calibration Certificate of test weights (mg)	Nominal Value (kg)	Trial 1 (kg)	Trial 2 (kg)	Trial 3 (kg)	Average	Correction (should be within -0.1 kg to 0.1 kg)
5	25	5.00003	5.00	5.00	5.05	5.016666667	-0.017
10	120	10.00012	9.95	10.00	10.00	9.983333333	0.017
15	145	15.00015	15.00	15.00	15.00	15	0.000
20	240	20.00024	20.00	19.95	18	19.31666667	0.684
25	265	25.00027	25.00	25	25	25	0.000

Step 7. Attach Calibration Sticker.

Fill out the calibration sticker (Figure 2.9) and attach it at the back of the weighing scale (Figure 2.10). Calibration is valid for three (3) years.

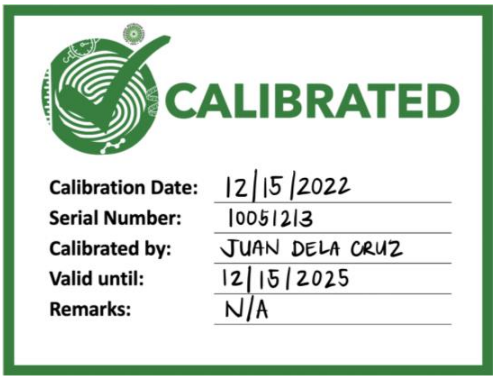


Figure 2.9. Sample Calibration sticker.

Figure 2.10. Calibration sticker attached at the back of the weighing scale.

What is Verification?

Verification or intermediate check is done in between regular calibration and is done more frequently. For OPT Plus, verification of weighing scales is recommended every year before the next calibration. This process ensures that operational stability and calibration status is maintained at tolerable limits. To perform verification, calibration records and information of the weighing scale should be available. If none is available, an initial calibration should first be conducted. The verification process follows the Scale Value Check Procedure (Source: NATA Technical Note 13: User Checks and Maintenance of Laboratory Balances).

VERIFICATION PROCESS

1. Tare the weighing scale to 0.0 kg (z1)
2. Load the 20 kg standard weight onto the scale (M) and record the weight reading (m1).
3. Remove the standard weight. Do not tare the weighing scale to 0.0 kg.
4. Load the standard weight again and record the weight reading (m2).
5. Unload the standard weight and record the weight reading (z2).
6. Encode the weight measurements in the Weighing Scale Calibration and Verification Tool and interpret the results.

Verification	
Single Point Value Check (using 20 kg test weight)	
M (Nominal value of 20 kg)	
z_1	
m_1	
m_2	
z_2	
C_1	
C_2	
User's correction	
Change in correction	
Basis of decision	
Decision *(fit for use if Change in Correction is less than 3x the Criteria)*	

The Weighing Scale Verification Tool will automatically generate the verification results. To determine if the weighing scale can be considered as functional or fit for use, check the Decision found in the Verification column which could state "fit for use" or "not fit for use."

Fit for use:

Verification	
Single Point Value Check (using 20 kg test weight)	
M (Nominal value of 20 kg)	20.000240
z_1	0
m_1	20
m_2	20
z_2	0
C_1	0.000240
C_2	0.000240
User's correction	0.000240
Change in correction	0.000000
Basis of decision	Standard Deviation
Decision *(fit for use if Change in Correction is less than 3x the Criteria)*	fit for use

Not fit for use:

Verification	
Single Point Value Check (using 20 kg test weight)	
M (Nominal value of 20 kg)	20.000240
z_1	0
m_1	20
m_2	19.8
z_2	0
C_1	0.000240
C_2	0.200240
User's correction	0.100240
Change in correction	0.100000
Basis of decision	Standard Deviation
Decision *(fit for use if Change in Correction is less than 3x the Criteria)*	not fit for use

7. Attach the Verification Sticker. Fill the sticker with pertinent information. Verification should be done annually.

Figure 2.11. Sample Verification sticker.

Figure 2.12. Verification sticker attached at the back of the weighing scale.

NOTE: Weighing Scale Calibration and Verification Form/Tool

The Weighing Scale Calibration and Verification Form/Tool should be used when performing calibration and verification. Sample of the form/tool can be seen Annex 2.8. It can also be accessed at the NNC website (https://nnc.gov.ph/).

NOTE: Standard Weights

The OPT Plus Team may request the purchase of standard weights. If this is not possible, the OPT Plus Team may request the use of standard weights from the City/ Municipal Treasurer's Office.

Section 2.6

VERIFICATION OF LENGTH/ HEIGHT TOOLS AND EQUIPMENT

VERIFICATION OF LENGTH AND HEIGHT BOARDS

The purpose of this section is to provide maintenance and verification guidelines that can be used to ensure the reliability and long-term use of height/length measuring equipment. Before a length or height board can be used, verification should be done by trained personnel (passed the Competency Training on Verification). The points to be checked should include:

1. Correct placement and attachment of the steel rule;
2. Zero placement of the steel rule at the base;
3. Dimension of the board and parts;
4. Quality of materials used;
5. Perpendicularity of base/footboard and headpiece against the beam;
6. Functionality and integrity of the unit; and
7. Correction factor, if any.

The height board should be verified upon installation and yearly thereafter in a well-lit and controlled temperature room. A verified length/height board is one with a sticker of verification issued and posted at the upper right corner of the board. The certificate shall contain the verification date, name, and signature of the person who verified and agency/institution, steel rule serial number (which will serve as unit number), and correction factor, if any.

Materials needed for height board verification

1. 1.5M calibrated steel ruler
2. Paper
3. L-square angle ruler
4. Leveling tool
5. Magnifying glass
6. Length/Height Board Verification Sheet (Annex 2.7)

Figure 2.13. A calibrated steel ruler.

Figure 2.14. An L-square angle ruler.

Figure 2.15. A magnifying glass.

NOTE: Calibration

Before verification, the length/height board should already be calibrated by the Department of Science and Technology (DOST) and has a visible calibration sticker on its back or side.

Care for measuring equipment

1. Practice hand hygiene whenever conducting measurement activities.
2. Clean and disinfect measuring equipment before and after use/transporting.
3. Store equipment at normal indoor temperature protected from extreme temperatures and humidity.

STEPS IN LENGTH AND HEIGHT BOARD VERIFICATIONS

Step 1. Fill out the information part of the verification sheet for documentation (Annex 2.7).

a. Date and place of verification
b. Name of the verifier
c. Type of height/length board, model, date of acquisition, serial/code number

Step 2. Start with a visual inspection of the equipment.

a. Check that the joints are tight and straight. If not, tighten or straighten them.
b. Check that the board can stand on its own and stay straight when rested on the wall or laid on a flat surface. You may also use a leveling tool to check if the board is flat.
c. Check that the lines of the steel ruler can still be read. Replace it if it is too worn out to be read.
d. Check for other damages–chipped parts, needs repainting, immovable or loose head board, etc. If it has any, it must be repaired or replaced immediately.

Figure 2.16. Visual inspection of length/height board.

Step 3. Measure in three trials the width and thickness of the different parts of the length/height board using the calibrated steel ruler.

a. Headboard – left corner, right corner, center
b. Foot board – left corner, right corner, center
c. Body – top, center, bottom

Step 4. Check the squareness of the headboard and footboard.

Using the L-square angle ruler, measure the space using a calibrated steel ruler. In assessing the squareness, the space should not be >0.2 cm.

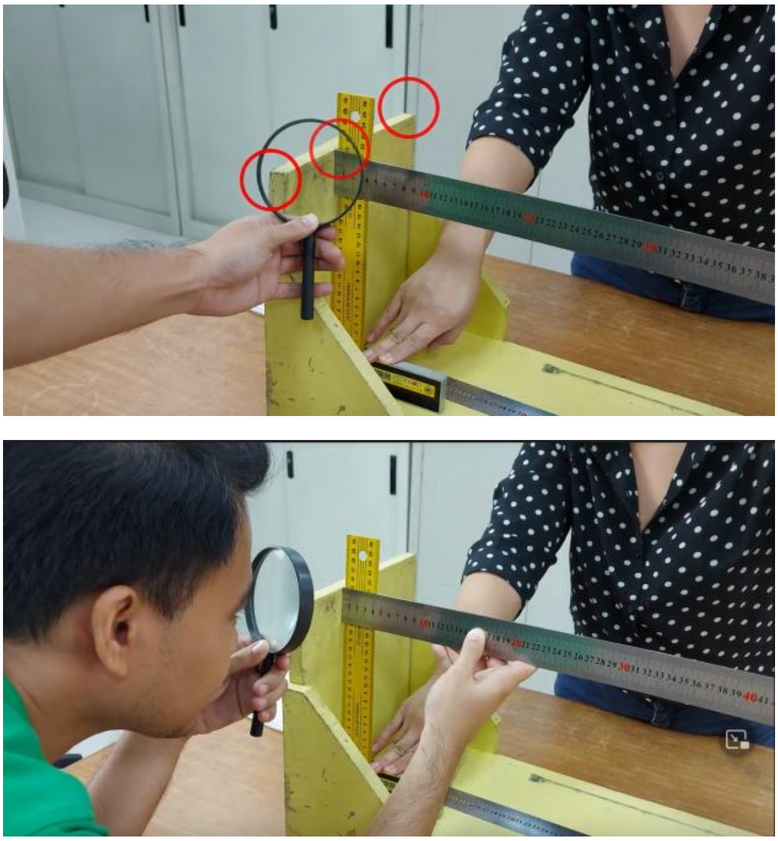


Figure 2.17. Checking the squareness of headboard and footboard using L-square angle ruler.

Step 5. Check calibration of the 10 verification points.

Check calibration of the 10 verification points (every 10 cm) of the measuring steel rod of the board, do this for three trials. The 10 verification points of Trial 1 should be done first before Trial 2, and so on.

a. Lay the length/height board on a stable and flat table. Place the calibrated steel ruler in line with the measuring steel rod of the board resting firmly on the foot board and should not be moved until the next trial.

b. Looking at the measuring steel rod of the board, read the measurement that aligns with the verification point in the calibrated steel ruler, select the nearest 0.1 cm. Record the measurement in the worksheet under Trial 1.

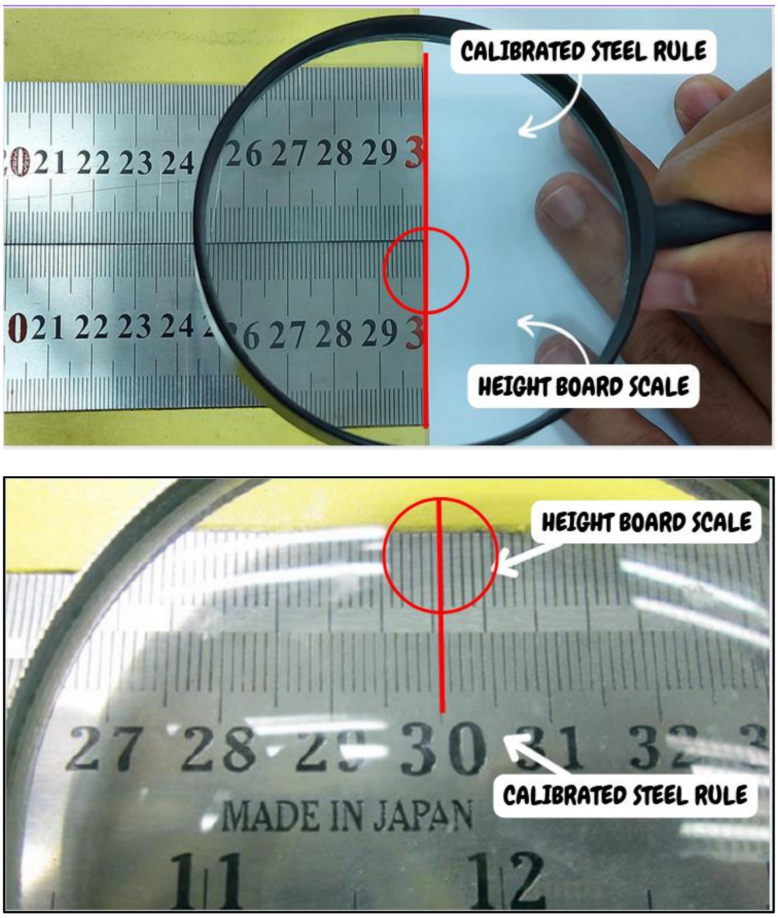


Figure 2.18. Conducting check calibration of the 10 verification points.

c. Use magnifying glass and flashlight to aid in reading the measurements.

Figure 2.19. Using magnifying glass and flashlight while reading measurements.

d. Remove the calibrated steel ruler only after the 10th verification point of Trial 1 and repeat the steps until the third trial.

e. Get the difference between the verification point and the average of the three readings. The allowable error is +/- 2mm or +/- 0.2 cm.

f. If the height/length board reads outside the acceptable allowable error, adjust measurements accordingly when using the board until fixed or replaced.

Step 6. Encode the collected measurements in the Length/Height Board Verification Tool and interpret results.

Once all the measurements have been encoded, the Length/Height Board Verification Tool will automatically display the results. If at least one of the rows under the Error column displays "Not Acceptable," then the length/height board is recommended for repair or replacement. Otherwise, it is fit for use.

Fit for use:

NATIONAL NUTRITION COUNCIL

Length/Height Board Verification Form

Manufacturer's Name:	HDSI	Model:	Marine wood w/ steel rule, straight
Serial number:	NNC-2022-0003	RA's date of acquisition:	5-Dec-22
Requesting Agency (RA):	NNC Central Office	Date of Verification:	6-Dec-22
Type of length/height board:	Marine wood w/ steel rule, straight	Place of Verification:	Taguig City
		Verification Validity:	5-Dec-23

Preliminary Evaluation

Scale/Graduations	☑ Readable ☐ Unsatisfactory	**Missing/broken parts**	☑ None ☐
Visual inspection shows that the general condition and workmanship of the instrument are:	level surface, foot/head piece slides smoothly, clear gradation		
Headboard Squareness at 30cm	☑ Acceptable ☐ Not Acceptable	**Footboard Squareness at 30cm**	☑ Acceptable ☐ Not Acceptable

Verification of Instrumental Error

Verification Point (cm)	Length/Height Board Scale Reading (cm) Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)	Average of 3 trials (cm)	Difference *(Verification Point - Average)*	Error *Acceptable if absolute value of difference is less than or equal to 0.2 cm*
30	30	30	29.9	29.97	0.03	Acceptable
40	40	40	40.00000	40.00	0.00	Acceptable
50	49.9	49.9	49.90000	49.90	0.10	Acceptable
60	60	59.9	59.90000	59.93	0.07	Acceptable
70	69.9	70	70.00000	69.97	0.03	Acceptable
80	80	80	80.00000	80.00	0.00	Acceptable
90	90	90	90.00000	90.00	0.00	Acceptable
100	100	100	100.00000	100.00	0.00	Acceptable
110	110	110	110.00000	110.00	0.00	Acceptable
120	120	120	120	120.00	0.00	Acceptable

Overall Decision ☑ Fit for use ☐ For replacement Remarks: ____________

Prepared by: ____________ Date: ____________

Checked by: ____________ Date: ____________

Not fit for use:

NATIONAL NUTRITION COUNCIL

Length/Height Board Verification Form

Manufacturer's Name:	HDSI	Model:	Marine wood w/ steel rule, straight
Serial number:	NNC-2022-0003	RA's date of acquisition:	5-Dec-22
Requesting Agency (RA):	NNC Central Office	Date of Verification:	6-Dec-22
Type of length/height board:	Marine wood w/ steel rule, straight	Place of Verification:	Taguig City
		Verification Validity:	5-Dec-23

Preliminary Evaluation

Scale/Graduations	☑ Readable ☐ Unsatisfactory	**Missing/broken parts**	☑ None ☐
Visual inspection shows that the general condition and workmanship of the instrument are:	level surface, foot/head piece slides smoothly, clear gradation		
Headboard Squareness at 30cm	☑ Acceptable ☐ Not Acceptable	**Footboard Squareness at 30cm**	☑ Acceptable ☐ Not Acceptable

Verification of Instrumental Error

Verification Point (cm)	Length/Height Board Scale Reading (cm) Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)	Average of 3 trials (cm)	Difference *(Verification Point - Average)*	Error *Acceptable if absolute value of difference is less than or equal to 0.2 cm*
30	29.7	29.8	29.8	29.77	0.23	Not Acceptable
40	39.8	39.8	39.70000	39.77	0.23	Not Acceptable
50	49.9	49.8	49.90000	49.87	0.13	Acceptable
60	60	59.9	59.90000	59.93	0.07	Acceptable
70	69.9	70	70.00000	69.97	0.03	Acceptable
80	80	80	79.80000	79.93	0.07	Acceptable
90	90	90	90.00000	90.00	0.00	Acceptable
100	99.7	99.7	99.80000	99.73	0.27	Not Acceptable
110	110	110	110.00000	110.00	0.00	Acceptable
120	120	120	120	120.00	0.00	Acceptable

Overall Decision ☐ Fit for use ☑ For replacement Remarks: ____________

Prepared by: ____________ Date: ____________

Checked by: ____________ Date: ____________

Step 7. Attach a Verification sticker to the length/height board.

Write down the information needed on the verification sticker and attach it at the back of the board. Verification should be repeated annually.

Figure 2.20. Verification sticker attached at the back of the height board.

NOTE: Calibration and Verification

Use instructional video available at NNC website (https://nnc.gov.ph/). Only trained personnel are allowed to conduct calibration and verification of measuring tools and equipment.

Section 2.7

STANDARDIZATION EXERCISE

To assure that all OPT Plus Teams are accurate, precise, and reliable in measuring weight, length, height, and MUAC standardization test will be conducted. The purpose of the standardization test is to evaluate the accuracy, precision, and reliability of the measurements taken by each pair of measurers.

ACCURACY is the degree to which the measurement is close to the "true" value. The measurements made by the supervisors can be considered as the true value.

PRECISION refers to how close the different weight measurements are to each other when weighing a subject. The size of the variation between repeated measurements is calculated to assess precision.

RELIABILITY is the consistency or the ability to repeat measurement under the same condition.

PREPARATION OF EQUIPMENT, SUPPLIES AND FORMS

Equipment needed:

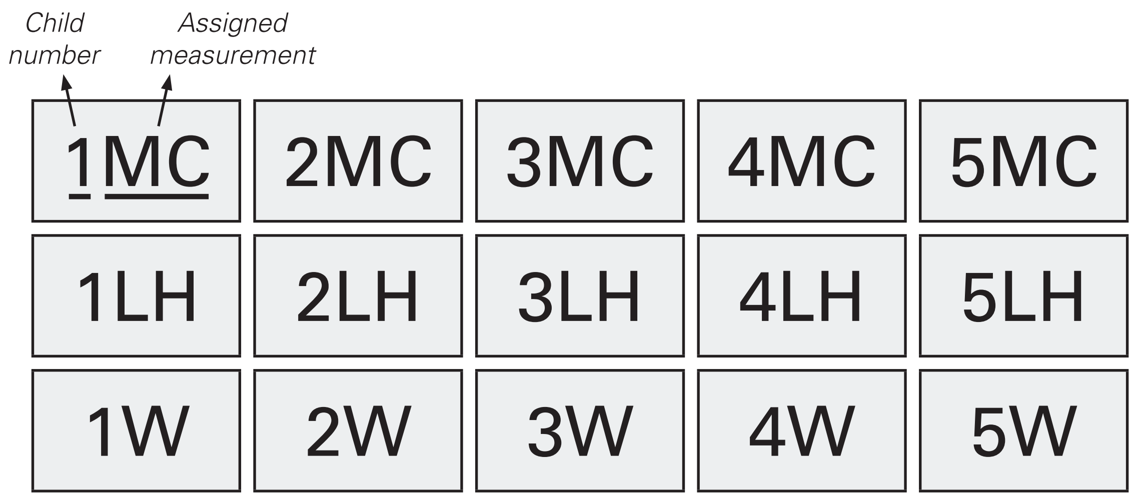
- 5 verified hanging weighing scale
- 5 verified length/height board
- 5 MUAC tape
- 1 computer/laptop with ENA SMART software or application (can be downloaded from this site: https://smartmethodology.org/survey-planning-tools/smart-emergency-nutrition-assessment/?doing_wp_cron=1657333343.8602769374847412109375)

Supplies needed:

- 1 scissor
- 1 permanent marker
- 10 pencils
- 10 small clipboard

Forms needed:

- Sheets of sticker paper cut into four (4) for the tags of the 15 children



MC - MUAC *LH - Length/Height* *W - Weight*

- Sheets of sticker paper cut into 4 for the tags of participants (depending on the number of participants)

SRM	SRA	1M	1A	2M	2A
3M	3A	4M	4A	5M	5A

SRM - Supervisor Measurer
SRA - Supervisor Assistant

M - Measurer
A - Assistant

Number of pair trainer

Classification of trainer (measurer or assistant)

- 5 pieces of Pink Form 1 for all Measurers (i.e., 1M, 2M, 3M, etc.) and 5 pieces of Pink Form 2 for all Assistants (i.e., 1A, 2A, 3A, etc.)

Pink Form 1 (Weight):

OPT DQC Standardization 1
Measurers' Code: ____________________
Child's Code: ____________________
Weight - HS: ____________________

Pink Form 2 (Weight):

OPT DQC Standardization 2
Measurers' Code: ____________________
Child's Code: ____________________
Weight - HS: ____________________

- 5 pieces of Yellow Form 2 for all Assistants (i.e., 1A, 2A, 3A, etc.) and 5 pieces of Yellow Form 2 for all Assistants (i.e., 1A, 2A, 3A, etc.)

Yellow Form 1 (Lt/Ht):

OPT DQC Standardization 1
Measurers' Code: ____________________
Child's Code: ____________________
Length/Height: ____________________

Yellow Form 2 (Lt/Ht):

OPT DQC Standardization 2
Measurers' Code: ____________________
Child's Code: ____________________
Length/Height: ____________________

- 5 pieces of Blue Form 1 for all Measurers (i.e., 1M, 2M, 3M, etc.) and 5 pieces of Blue Form 2 for all Assistants (i.e., 1A, 2A, 3A, etc.)

Blue Form 1: (MUAC)

OPT DQC Standardization 1
Measurers' Code: ____________________
Child's Code: ____________________
MUAC: ____________________

Blue Form 2: (MUAC)

OPT DQC Standardization 2
Measurers' Code: ____________________
Child's Code: ____________________
MUAC: ____________________

- 5 pieces of White Form 1 for all Measurers (i.e., 1M, 2M, 3M, etc.) and 5 pieces of White Form 2 for all Assistants (i.e., 1A, 2A, 3A, etc.)

White Form 1: (edema)

OPT DQC STANDARDIZATION EXERCISE **1**
Measurer's Code: ____________________
Child's Code: ____________________
Presence of Edema: ☐ **Yes** ☐ **No** **Severity:** ______

White Form 2: (edema)

OPT DQC STANDARDIZATION EXERCISE **2**
Measurer's Code: ____________________
Child's Code: ____________________
Presence of Edema: ☐ **Yes** ☐ **No** **Severity:** ______

- In case there is no available colored paper, use forms with different shapes to avoid confusion.

OPT DQC Standardization 1
Measurers' Code: ______________________
Child's Code: ______________________
Weight - HS: ______________________

OPT DQC Standardization 1
Measurers' Code: ______________________
Child's Code: ______________________
Length/Height: ______________________

OPT DQC Standardization 2
Measurers' Code: ______________________
Child's Code: ______________________
Weight - HS: ______________________

OPT DQC Standardization 2
Measurers' Code: ______________________
Child's Code: ______________________
Length/Height: ______________________

OPT DQC Standardization 1
Measurers' Code: ______________________
Child's Code: ______________________
MUAC: ______________________

OPT DQC STANDARDIZATION EXERCISE 1
Measurer's Code: ______________________
Child's Code: ______________________
Presence of Edema: ☐ Yes ☐ No Severity: ________

OPT DQC Standardization 2
Measurers' Code: ______________________
Child's Code: ______________________
MUAC: ______________________

OPT DQC STANDARDIZATION EXERCISE 2
Measurer's Code: ______________________
Child's Code: ______________________
Presence of Edema: ☐ Yes ☐ No Severity: ________

- 15 pieces informed consent form (Attachment 6)

METHODOLOGY

1. Invite children to participate in the standardization exercise.

The total number of children needed for the standardization exercise is 15. There should be six (6) 6-23-month-old and nine (9) 24-59-month-old children as shown in Table 1. Do not forget to inform the parents that their child will be measured at least 20 times (weight, length/height, or MUAC).

Table 2.5. Number of children needed per age group, nutrition assessment type, and coding procedure.

AGE GROUP	WEIGHT	LENGTH/HEIGHT	MUAC	TOTAL
6-23 months	2 Code: 1W, 2W	2 1LH, 2LH	2 1MC, 2MC	**6**
24-59 months	3 Code: 3W, 4W, 5W	3 3LH, 4LH, 5LH	3 3MC, 4MC, 5MC	**9**
TOTAL	**5**	**5**	**5**	**15**

2. Setting up the training venue

Conduct the standardization test in a covered multi-purpose hall/covered court to ensure social distancing.

3. Obtaining of informed consent

Following the principles of informed consent, each guardian will be briefed on the purpose, benefits, risk, and confidentiality of the exercise prior to obtaining their consent to participate (Annex 2.2).

4. Ensuring the household profile of each guardian is available.

The household profile, which summarizes the demographic, health, and nutrition information of the household, will be confidential as stipulated in the Data Privacy Act of 2012.

5. Ensuring set-up for the standardization activity (as presented in Figure 2.21).

There will be three stations and two tables:

- Station 1 for hanging weighing scale (weight-HS);
- Station 2 for length board/infantometer (0-23 months) or height board (24-59 months)
- Station 3 for MUAC and edema
- Table 1 for registration of children with their parents/caregivers
- Table 2 for the computer/laptop where the encoder will be stationed.

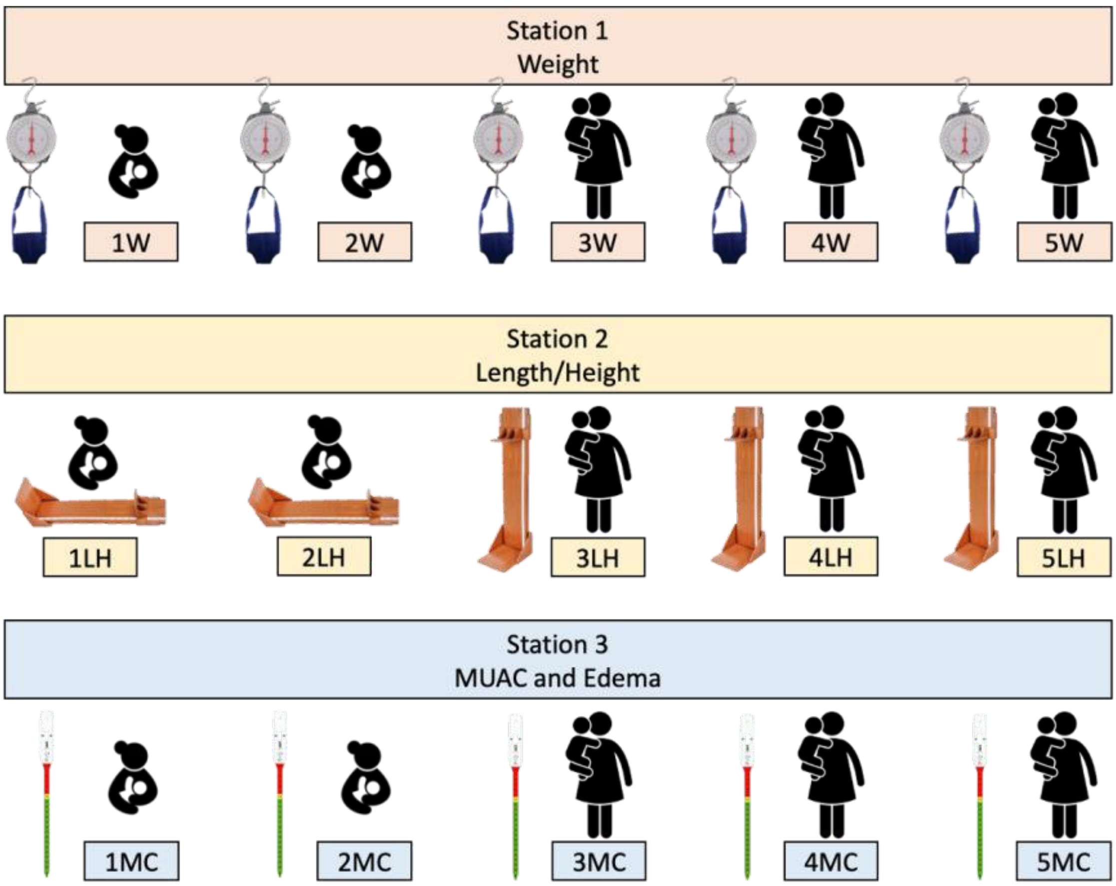


Figure 2.21. Initial set–up and flow of the standardization activity.

5. Coding of trainees and children

Trainee

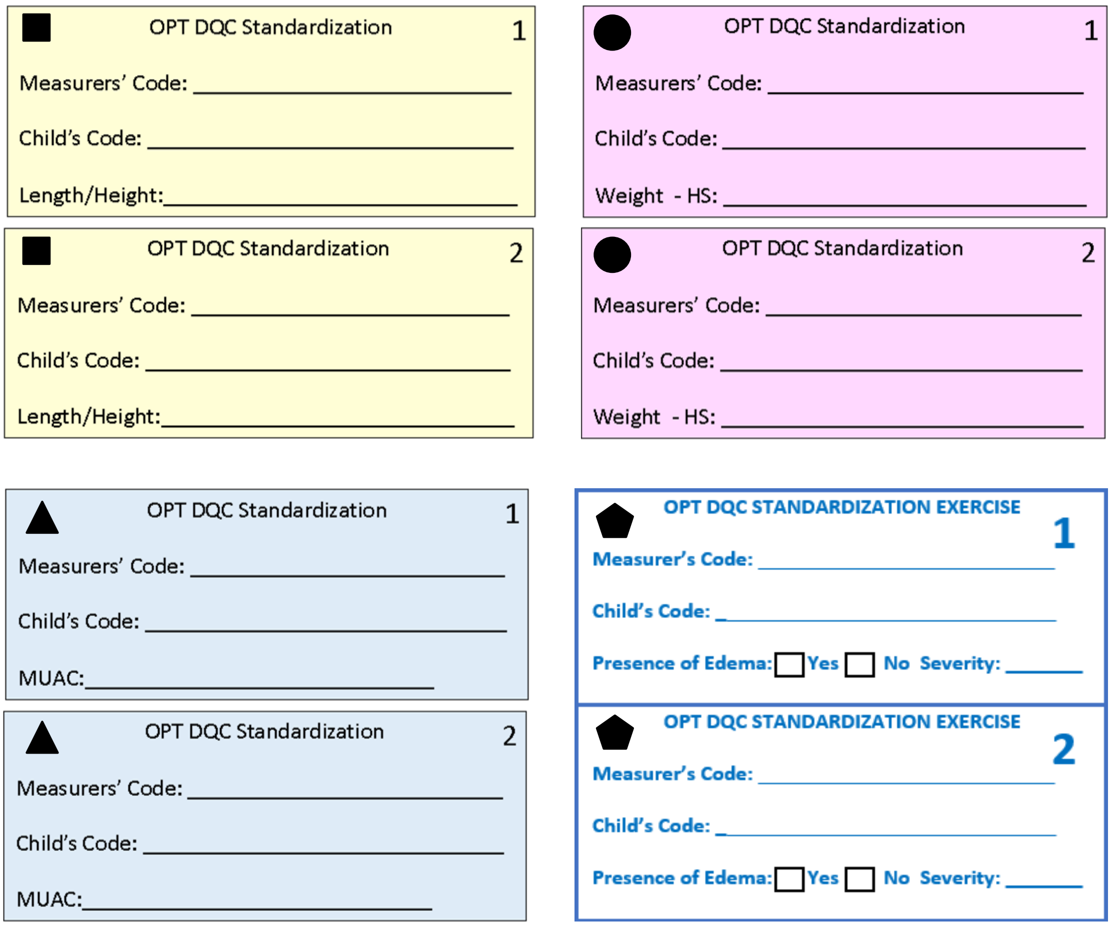
- Participants will be paired, and each pair will be coded as SR, 1, 2, 3, 4, 5, and so on, depending on the number of trainees.
- The code on the sticker will be used to identify the pair of measurers. A document with their names and corresponding code should be set aside.

Children

- The parent/caregiver of the children will sign the attendance sheet with the assigned code. The children's code is indicated in Table 2.5.
- The stickers with written codes should be attached to the children's clothes at the back. Make sure that the code on the sticker is the same on the attendance sheet.

6. Actual measurements

- Each pair will carry out the measurements in turn. In the first round, one will be assigned as measurer (Measurer M) while the other will be assigned as a recorder or assistant measurer (Measurer A).
- The supervisor and each pair of measurers will be assigned first to a station in the initial set-up (Figure 2.21). Likewise, each of the five (5) children will be assigned to station 1.
- The supervisor and each pair of measurers will be given color-coded form based on their assigned stations to record the measurers' code, child's code, and measurement (weight-HS; length/height-LH; and MUAC-MC). Since two (2) measurements per child will be needed for the activity, each piece of form was numbered 1 or 2. Form 1 is for the measurements of Measurer M while form 2 is for the measurements of Measurer A. Each measurer will need five (5) pieces of colored-coded form 1, while the assistants will need five (5) pieces color-coded form 2 (Figure 2.22).


OPT DQC Standardization 1
Measurers' Code: ______
Child's Code: ______
Length/Height: ______

OPT DQC Standardization 1
Measurers' Code: ______
Child's Code: ______
Weight - HS: ______

OPT DQC Standardization 2
Measurers' Code: ______
Child's Code: ______
Length/Height: ______

OPT DQC Standardization 2
Measurers' Code: ______
Child's Code: ______
Weight - HS: ______

OPT DQC Standardization 1
Measurers' Code: ______
Child's Code: ______
MUAC: ______

OPT DQC STANDARDIZATION EXERCISE 1
Measurer's Code: ______
Child's Code: ______
Presence of Edema: ☐ Yes ☐ No Severity: ______

OPT DQC Standardization 2
Measurers' Code: ______
Child's Code: ______
MUAC: ______

OPT DQC STANDARDIZATION EXERCISE 2
Measurer's Code: ______
Child's Code: ______
Presence of Edema: ☐ Yes ☐ No Severity: ______

Figure 2.22. Color-coded forms are used to record the measurements of children.

- The activity will commence when the supervisor pair, measurer pair 1, measurer pair 2, and measurer pair 3 measured the child on their assigned station and recorded the measurement in the assigned form.
- The recorder will submit all the accomplished forms to the encoder.
- All the measurers will then transfer to the next station assigned to them until they complete the first measurement of each of the five (5) children's weight, length or height, and MUAC.

  Note: These procedures will be repeated to get the second measurements of these 15 children. For this round, the recorder or assistant measurer (Measurer A) will become the measurer, while the original measurer (Measurer M) will become the recorder or assistant measurer.

Data Entry and Analysis

The data for each measurement will be encoded in the Training Screen of the SMART ENA software version 2020. The data will be analyzed in this software following the methods defined by Ulijaszek and Kerr (1999) and used in the training of the anthropometrists for the WHO Multicentre Growth Reference Study (MGRS) for Child Growth Standards.

The analyses are more rigorous compared with the older procedures which used acceptable measures based on two thresholds:

1. **PRECISION:** <2 times the precision value of the supervisor.
2. **ACCURACY:** <3 times the precision value of the supervisor. This older procedure is not the standard method used by anthropometrists and the results are not presented in recognizable units. In addition, the interpretation of the standardization test results is compromised if the supervisor is not a competent anthropometrist. The results of the standardization test were carried out on survey enumerators to assess their capacity to take accurate and precise measurements.

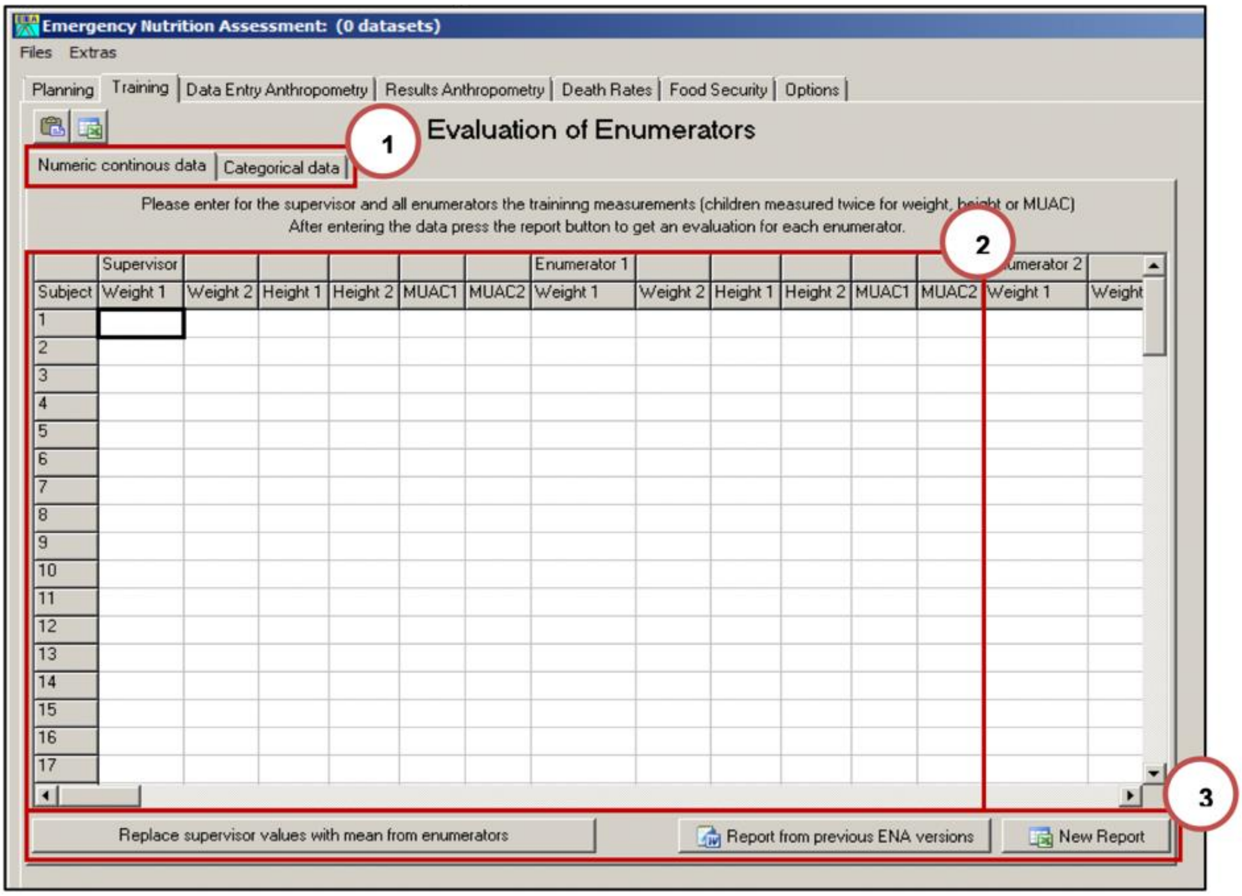


Figure 2.23. Screenshots of Training Screen.

As shown in Figure 2.23 (number 1), both numeric continuous data (i.e., weight, height, and MUAC) and categorical data (e.g., yes/no, green/yellow/red for MUAC, etc.) can be entered and analyzed. Different analyses are performed and different types of reports are generated depending on the type of data.

Data collected during the standardization should be entered in the ENA training tab. The measurement values of 100 subjects from 20 enumerators and 1 supervisor can be entered at a time.

The height and weight data are entered in 0.1 cm and 0.1 kg respectively, While MUAC data must always be entered in millimeters (mm). Make sure you entered the right values and unit of measurement to avoid generating wrong data.

The standardization test report consists of 1) descriptive results; 2) precision report; and 3) accuracy report (Table 2.6). The descriptive results consist of: a) mean value of all measurements of 10 children measured twice; b) standard deviation of each pair of measurer; and c) the largest difference between two measurements (i.e., Max). There are three (3) indicators for precision including: a) Technical Error of Measurement (TEM); b) % TEM; and c) R or Coefficient of Reliability. Lastly, the two (2) outputs of accuracy include bias from supervisor and bias from median.

Table 2.6. Components of standardization test report.

REPORT	OUTPUT
1. Descriptive	• Mean • Standard Deviation • Max
2. Precision	• Technical Error of Measurement (TEM) • % TEM • R or Coefficient of Reliability.
3. Accuracy	• Bias from supervisor • Bias from median

ANALYSIS OF PRECISION

a. **Technical Error of Measurement (TEM).** This reflects the mistakes that the pairs made between their 1st and 2nd measurements. It is computed using the following formula:

$$TEM = \sqrt{(\Sigma D^2)/2N}$$

where D is the difference between the 1st and 2nd measurements.

The analysis will be done at each pair and on the team level and the following cut-off points were used for the acceptability of measurements (Table 2.7).

Table 2.7. Suggested cut-off points for acceptability of measurements using TEM.

PARAMETER	CATEGORIES	WEIGHT, KG	PARAMETER	CATEGORIES	WEIGHT, KG
Pair	good	<0.04	**Team TEM (intra+inter) and Total**	good	<0.10
TEM	acceptable	<0.10		acceptable	<0.21
(intra)	poor	<0.21		poor	<0.24
	reject	>0.21		reject	>0.24

In the results of the standardization test, enumerator (ENUM) as used in the standardization test is also called a pair. There were five (5) results using TEM; 1) pair inter 1st; 2) pair inter 2nd; 3) inter pair + sup; 4) TOTAL intra+inter; and 5) TOTAL + sup.

In pair inter 1st, the measurements between enumerators in their 1st round of measurements were compared while in pair inter 2nd the comparison between enumerators was on their 2nd round of measurements. In inter pair + sup, the measurements between everyone who measured including the supervisor (pair + supervisor) were compared. In TOTAL intra+inter, the comparison was between each enumerator/pair to themself, plus comparison between enumerators or pairs. Lastly, in TOTAL + sup, this is similar with TOTAL intra+inter but this time the supervisor was included. Individual TEM (pair inter 1st or pair inter 2nd) is the most important measure of precision but the team TEM analysis (inter pair + sup, TOTAL + sup, or TOTAL intra+inter) gives an indication of consistency of measurements within and between all the enumerators/pairs.

b. **% TEM.** This is a unit-less measurement which shows the relative mistake as a percent of the whole measurement size and appropriate for comparing error across populations. This is computed by using the following formula:

(TEM/mean measurement) x 100%

c. **Coefficient of Reliability (R).** This indicator shows the % of natural variation between measured children (of different sizes) in recognition that not all variation in measurements is due to enumerator mistakes. This is the most widely used measure of precision in population studies. To interpret: using the scale of 0-1, where 1=perfect, if R=0.70, then this means that 70% of variance in measurement is due to actual differences in size of children, but 30% of variance in measurement is due to enumerator measurement error. The cut-off points used for acceptable measurements are as follow:

CATEGORIES	%
Good	>99
Acceptable	>95
Poor	>90
Reject	<90

ANALYSIS OF ACCURACY

As previously mentioned, the first output of accuracy is the bias from supervisor or the difference between the mean of supervisor and the mean of enumerator/pair. On the other hand, bias from median is the difference from median of all measurers and mean of enumerator/pair or supervisor.

If the supervisor has good or acceptable precision as measured by TEM, then bias from supervisor is used. This means that individual/pair results are compared to the supervisor. However, if the supervisor's TEM is poor, then the bias from median should be used. When this is used, the individual/pair results are compared to the median of the group. The cut-off points for acceptable measurements are as follows:

CATEGORIES	WEIGHT, KG
Good	<0.04
Acceptable	<0.10
Poor	<0.21
Reject	>0.21

Tables 2.8a and 2.8b show two examples of the results of standardization test for weight measurement. Using the digital flat scale, all measurers passed the test in terms of accuracy and precision. However, using the hanging weighing scale, the supervisor's TEM was poor hence the outcome from median should be used for the analysis of accuracy. The results showed that all measurers were precise based on coefficient of reliability but measurers pair 1 and pair 6 were precise but not accurate.

What do we want?

The aim of the standardization exercises is to produce skilled OPT Plus measurers capable of obtaining and recording accurate, precise, and reliable anthropometric measurements of children.

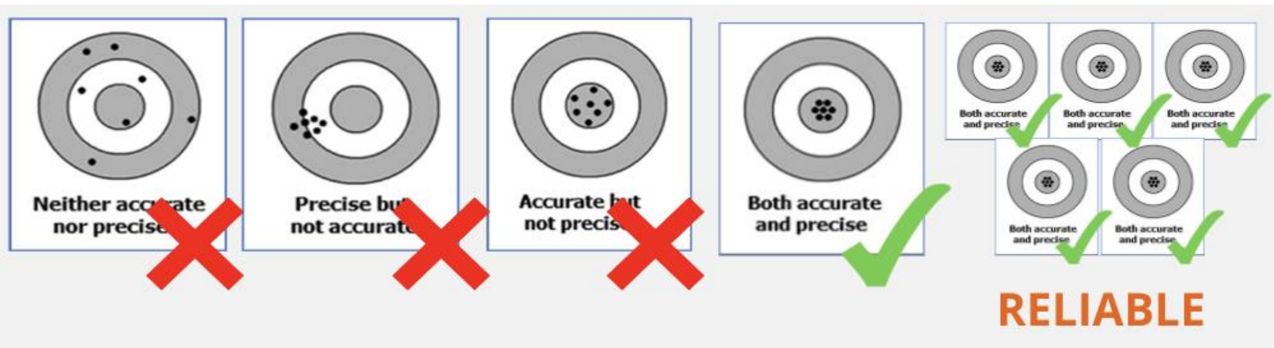


Table 2.8a. Standardization Test of weight measurement using digital flat scale.

Tanita Digital bathroom scale HD-662				Precision				Accuracy		OUTCOME			
Measurer	subjects	mean	SD	max	Technical error	TEM/mean	Coef of reliability	Bias from superv	Bias from median	TEM	R value	From	From
	#	kg	kg	kg	TEM (kg)	TEM (%)	R (%)	Bias (kg)	Bias (kg)			Supervisor	Median
Supervisor	8	11.7	1.4	0.1	0.04	0.3	99.9	0	0.01	TEM good	R value good	Bias good	Bias good
Pair 1	8	11.7	1.4	0.3	0.09	0.7	99.6	0.05	0.05	TEM acceptable	R value good	Bias acceptable	Bias acceptable
Pair 2	8	11.7	1.4	0.2	0.05	0.4	99.9	0.04	0.04	TEM acceptable	R value good	Bias acceptable	Bias acceptable
Pair 3	8	11.7	1.4	0.1	0.04	0.3	99.9	0.03	0.02	TEM good	R value good	Bias good	Bias good
Pair 4	8	11.7	1.4	0.1	0.05	0.4	99.9	0.03	0.03	TEM acceptable	R value good	Bias good	Bias good
Pair 5	8	11.7	1.4	0.2	0.07	0.6	99.7	0.03	0.02	TEM acceptable	R value good	Bias good	Bias good
Pair 6	8	11.7	1.4	0.1	0.04	0.4	99.9	0.04	0.02	TEM acceptable	R value good	Bias good	Bias good
pair inter 1st	6x8	11.7	1.4	-	0.05	0.4	99.9	-	-	TEM good	R value good		
pair inter 2nd	6x8	11.7	1.4	-	0.05	0.5	99.8	-	-	TEM good	R value good		
inter pair + sup	7x8	11.7	1.4	-	0.05	0.4	99.9	-	-	TEM good	R value good		
TOTAL intra+inter	6x8	-	-	-	0.08	0.7	99.7	-	-	TEM good	R value good		
TOTAL+ sup	7x8	-	-	-	0.08	0.7	99.7	-	-	TEM good	R value good		

Table 2.8b. Initial standardization test of weight measurement using Hanging Weighing Scale.

Hanging Infant Weighing Scale				Precision				Accuracy		OUTCOME			
Measurer	subjects	mean	SD	max	Technical error	TEM/mean	Coef of reliability	Bias from superv	Bias from median	TEM	R value	From	From
	#	kg	kg	kg	TEM (kg)	TEM (%)	R (%)	Bias (kg)	Bias (kg)			Supervisor	Median
Supervisor	8	11.8	1.4	0.4	0.15	1.3	98.8	0	0.08	TEM poor	R value acceptable	Bias good	Bias acceptable
Pair 1	8	11.8	1.4	0.4	0.12	1	99.3	0.08	0.1	TEM poor	R value good	Bias acceptable	Bias poor
Pair 2	8	11.8	1.4	0.4	0.17	1.4	98.6	0.07	0.06	TEM poor	R value acceptable	Bias acceptable	Bias acceptable
Pair 3	8	11.8	1.4	0.4	0.12	1	99.3	0.05	0.06	TEM poor	R value good	Bias acceptable	Bias acceptable
Pair 4	8	11.8	1.4	0.3	0.11	0.9	99.4	0.06	0.07	TEM poor	R value good	Bias acceptable	Bias acceptable
Pair 5	8	11.8	1.4	0.4	0.14	1.2	99	0.05	0.06	TEM poor	R value acceptable	Bias acceptable	Bias acceptable
Pair 6	8	11.9	1.4	1	0.28	2.3	95.8	0.09	0.11	TEM reject	R value acceptable	Bias acceptable	Bias poor
pair inter 1st	6x8	11.8	1.3	-	0.15	1.3	98.7	-	-	TEM acceptable	R value acceptable		
pair inter 2nd	6x8	11.8	1.4	-	0.08	0.7	99.7	-	-	TEM good	R value good		
inter pair + sup	7x8	11.8	1.4	-	0.11	0.9	99.3	-	-	TEM acceptable	R value good		
TOTAL intra+inter	6x8	-	-	-	0.2	1.7	97.8	-	-	TEM acceptable	R value acceptable		
TOTAL+ sup	7x8	-	-	-	0.2	1.7	97.9	-	-	TEM acceptable	R value acceptable		

Reference:

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LIST OF ANNEXES

ANNEX 2.1. Household Profile Format

BNS Form No. 1A
Philippine Plan of Action for Nutrition

HOUSEHOLD PROFILE

Purok/Sitio:	Barangay:	Municipality/City:	Province:

HH No.	No. of family living in the house	Number of HH members	NHTS Household 1-NHTS 4Ps 2-NHTS Non-4Ps 3-Non-NHTS	Indigenous Group 1-IP 2-Non-IP	Number of Family Members by Age Classification / Health Risk Group																				Name of Father (F_a) and Mother (M_o); Caregiver (Ca)	Occupation	Educational Attainment	Couple Practicing Family Planning	Fill in:			Check If	
					Newborn (0-28 days)		Infant (29 days-11 months)		Under-five (1-4 y.o.)		Children 5 to 9 y.o.		Adolescence (10-19 y.o.)		Pregnant	Adolescent Pregnant	Post-Partum (PP)	15-49 y. o. not pregnant & non-PP	Adult 20-59 y. o.		Senior Citizens		Person With Disability						Toilet Type (WS, OP, O, N)	Water Source (P, W, S)	Food Production Activity (VG/PL/FP)	HH using Iodized Salt	HH using Iron-Fortified Rice
					M	F	M	F	M	F	M	F	M	F					M	F	M	F	M	F									
C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20	C21	C22	C23	C24	C25	C26	C27	C28	C29	C30	C31	C32	C33	C34
																									(F_a) (M_o) (C_a)			☐ YES ☐ NO				☐ YES ☐ NO	☐ YES ☐ NO
																									(F_a) (M_o) (C_a)			☐ YES ☐ NO				☐ YES ☐ NO	☐ YES ☐ NO
																									(F_a) (M_o) (C_a)			☐ YES ☐ NO				☐ YES ☐ NO	☐ YES ☐ NO
																									(F_a) (M_o) (C_a)			☐ YES ☐ NO				☐ YES ☐ NO	☐ YES ☐ NO
																									(F_a) (M_o) (C_a)			☐ YES ☐ NO				☐ YES ☐ NO	☐ YES ☐ NO
																									(F_a) (M_o) (C_a)			☐ YES ☐ NO				☐ YES ☐ NO	☐ YES ☐ NO
																									(F_a) (M_o) (C_a)			☐ YES ☐ NO				☐ YES ☐ NO	☐ YES ☐ NO
																									(F_a) (M_o) (C_a)			☐ YES ☐ NO				☐ YES ☐ NO	☐ YES ☐ NO

Abbreviations
HH – household M – male F_a – father
No – number F – female M_o – mother
Ca - Caregiver
NHTS – National Household Targeting System
4P – Pantawid Pamilyang Pilipino Program
IP – indigenous people y. o. – year-old
PP – postpartum

Occupation
1 – Manager
2 – Professional
3 – Technician & Associate Professionals
4 – Clerical Support Workers
5 – Service and Sales Worker
6 – Skilled agricultural, forestry, and fishery workers
7 – Craft and related trade workers
8 – Plant and machine operators and assemblers
9 – Elementary occupations
10 – Armed Forces Occupations
11 - none

Educational Attainment
N – none
EU – Elementary undergraduate
EG – Elementary graduate
HU – high school undergraduate
HG – high school graduate
CU – College undergraduate
CG – College graduate
V – Vocational
PG – post-graduate studies

Toilet
1 -Improved sanitation
2 - shared facility
3 - Unimproved facility
4 - Open defecation (no facility/bush/field)
(Refer to the definition in the OPT Plus MOP

Water Source
1 Improved
2 Unimproved
(Refer to the definition in the OPT Plus MOP

Food Production
VG – Vegetable Garden
FT - Fruit
PL – Poultry Livestock
FP – Fish pond

ANNEX 2.2. Standardization Exercise Informed Consent

**Standardization Exercise Informed Consent Form
for Children and Parent/Caregiver**

This informed consent form is for the Standardization Exercises which is a prerequisite for the certification of the OPT Plus Team.

PART I: INFORMATION SHEET

1. Introduction

I am ___________, a member of the OPT Plus Team. As part of the training, members of the OPT Plus Team including the supervisors, BNS, BHW etc. are expected to participate in the standardization exercise as measurers and assistants in the measurement of the weight, length, height, MUAC, and checking of the severity of the bilateral pitting edema. The data gathered through this activity will be used to check the proper techniques, correct errors that may lead to inaccurate measurement, and ensure that the data collected during the OPT Plus are accurate, precise, and reliable. We would like to invite you to participate in this activity. I will accommodate and answer all your questions or clarifications anytime as I introduce you to the activity. There may be some words that you do not understand. Please ask me to stop as we go through the information, and I will take time to explain.

2. Purpose

This activity aims to: 1) standardize the skills of the OPT team, including the BNSs, BHWs, SK members, and Barangay Kagawad; and 2) ensure that the data collected by the OPT team are accurate, precise, and reliable.

3. Selection of Participants

Participants of the standardization exercise are 6-59-month-old children living in the barangay or nearby barangay who will be joined by, at least, a parent/caregiver. They shall be selected and invited based on the organizers of the training. A total of 15 pairs of parents and children - six (6) pairs of parents and 6–23-month-old children, and nine (9) pairs of parents and 24–59-month-old children were selected for this sampling.

4. Voluntary Participation

Participation is entirely voluntary. Willingness and approval to participate in the activity is expressed by signing this Informed Consent Form. You may ask any question regarding the activity. Should you opt not to participate for any reason, we assure you that you will not be affected in any way.

5. Procedure

You will be requested to attend the standardization exercise according to schedule indicated by the organizers. The activity will start by a registration and a brief interview. You and your child will be assigned a code and a station (e.g. weight, length/height, or MUAC station). Your child will be measured multiple times depending on the number of participants. The activity will be supervised by the training organizers and facilitators.

6. Duration

The whole activity may take up to four (4) hours.

7. Risks and Discomforts

The activity may cause the child to feel irritated or discomfort due to repeated collection of anthropometric measurement. Although the participants were briefed on safety protocols, injuries due to accidents may also happen. Should you find some of the activities uncomfortable for you or your child, you may inform the organizers and facilitators.

8. Benefits

Your participation in this study will be treated as a considerable contribution to the scant body of knowledge on child growth monitoring and nutrition assessment in the country.

9. Confidentiality

In compliance with the Data Privacy Act of 2012 (R.A. 10173), any personal information and responses obtained from the activity that can be identified with you will remain strictly confidential.

10. Right to Refuse of Withdraw

You may decline or withdraw participation at any given time without any obligation or penalty. In this case, please do inform the training facilitator before leaving the venue.

11. Contact Person

If you have further clarifications regarding the procedure, you may contact the following:

Name: ____
Address: ______________________________________
Contact Information: _____________________________

PART II: CERTIFICATE OF CONSENT

Certificate of Consent

I have read and understood the procedure and implications of participating in the standardization exercise. The questions and clarification that I have raised were answered adequately. I hereby declare consent to participate in this activity.

____

Signature over Printed Name of Participant

____

Date

Statement of the researcher:

I have read and discussed all the details of the research and procedures to the participant. I have also adequately answered any questions that were raised to the best of my knowledge. I confirm that the participants were given the opportunity to ask and clarify any questions and that the participant was not forced to participate. Participation is entirely voluntary. A copy of the Informed Consent Form was provided to the participant.

____

Signature over Printed Name of Facilitator

____

Date

ANNEX 2.3. NNC Memorandum No. 2016-015, Series 2016 Revised Color Coding/Legend for the Preparation of Spot Map

Republika ng Pilipinas
KAGAWARAN NG KALUSUGAN
PAMBANSANG SANGGUNIAN SA NUTRISYON
(NATIONAL NUTRITION COUNCIL)
Nutrition Building, 2332 Chino Roces Avenue Extension
Taguig City, Philippines

Revised Spot Map Color Coding/Legend Scheme
Series November 2016

Revised color coding/legend for the preparation of barangay spot maps:

Color	Interpretation
Green	Children with "normal weight/length/height"
Yellow	Children "underweight/stunted/wasted" Mark with "UW" for underweight, "St" for stunted, "W" for wasted
Red	Children "severely underweight/stunted/wasted" Mark with "SUW" for severely underweight, "SSt" for severely stunted, "SW" for severely wasted
Orange	Children "overweight and obese" Mark with "OW" for overweight, "Ob" for obese
Pink	With pregnant women
Violet	With lactating women
Maroon	Infants who were exclusively breastfed for 6 months
Purple	Infants who were given complementary food in addition to breastmilk (starting at 6 months old)
Brown	With infants 0-11 months
Blue	With large family size (households with greater than 5 family members)
Black	Without potable water source
Gray	Without sanitary toilets

**Children means "age 0-71 months"*

Reference: NNC Governing Board Resolution No. 2, Series 2012: Implementing Guidelines on Operation Timbang Plus (OPT +)

Approved:

Assistant Secretary of Health Maria-Bernardita T. Flores, CESO II
Executive Director IV

"First 1000 days ni baby, pahalagahan para sa malusog na kinabukasan!"

P.O. Box 2490
Makati Central Post Office
Makati City
Tel. Nos. (63-2) 843-0412 • 843-5824 • 843-1337
(63-2) 843-5856 • 818-5834 • 816-4239
(63-2) 843-5868 • 818-7398 • 892-4271
Fax No. (63-2) 843-5818
www.nnc.gov.ph info@nnc.gov.ph www.facebook.com/nncofficial www.youtube.com/user/NNC1974

Wastong Nutrisyon
Alamin, Gawin at Palaganap

ANNEX 2.4. NNC GB Resolution No. ______ Approving the Guidelines on the selection of weighing scales

ANNEX 2.5. NNC GB Resolution No. 3 s.2012 Approving the Guidelines on the Fabrication, Verification, and Maintenance of Wooden Height Boards

Republic of the Philippines
NATIONAL NUTRITION COUNCIL

NNC GOVERNING BOARD
Resolution No. 3, Series of 2012

Approving the Guidelines on the Fabrication, Verification, and Maintenance of Wooden Height Boards

WHEREAS, the Philippines adopted the WHO Child Growth Standards for determining the weight and height status of children under five years old;

WHEREAS, triggered by the adoption of the WHO Child Growth Standards, the annual weighing of all preschool children or Operation Timbang (OPT) has been expanded to include the measurement of height (for children 2 years and older) or length (for children less than 2 years old) into what is now called OPT Plus;

WHEREAS, the measurement of height and length would require an affordable instrument;

WHEREAS, the World Health Organization has recommended the use of wooden height board, for which it has drawn up technical specifications for fabrication;

WHEREAS, there is a need to ensure the integrity of measurements using the wooden height board through clear guidelines on fabrication, calibration, verification, and maintenance;

NOW THEREFORE, BE IT RESOLVED AS IT IS HEREBY RESOLVED, in consideration of the above premises, the National Nutrition Council Governing Board hereby approves the guidelines on the fabrication, verification, and maintenance of wooden height boards.

RESOLVED FURTHER that all agencies, local nutrition committees of local government units, or organizations procuring, fabricating or marketing, and using wooden height boards should use these guidelines in preparing technical specifications for procurement of height boards, in manufacturing, in inspecting deliveries, and in maintaining the boards

RESOLVED FURTHER that the National Nutrition Council Secretariat shall ensure the widest dissemination of the guidelines and monitor its implementation.

Approved this 12th day of January 2012.

ENRIQUE T. ONA, MD
Secretary of Health and Chairperson
National Nutrition Council Governing Board

Attested:

Assistant Secretary Maria-Bernardita T. Flores, CESO II
Council Secretary and Executive Director IV
National Nutrition Council

NNC GOVERNING BOARD
Resolution No. 3, Series of 2012

Approving the Guidelines on Fabrication, Verification, and Maintenance of Wooden Height Boards

CONFORME:

HON. PROCESO J. ALCALA
Secretary of Agriculture
Vice-Chairperson, NNC Governing Board

HON. JESSE M. ROBREDO
Secretary of the Interior and Local Government
Vice-Chairperson, NNC Governing Board

HON. FLORENCIO B. ABAD
Secretary of Budget and Management
Member, NNC Governing Board

HON. CORAZON JULIANO-SOLIMAN
Secretary of Social Welfare and Development
Member, NNC Governing Board

HON. CAYETANO W. PADERANGA, JR.
Secretary of Socio-Economic Planning
Director General
Member, NNC Governing Board

HON. ARMIN A. LUISTRO FSC
Secretary of Education
Member, NNC Governing Board

HON. GREGORY T. DOMINGO
Secretary of Trade and Industry
Member, NNC Governing Board

HON. MARIO G. MONTEJO
Secretary of Science and Technology
Member, NNC Governing Board

HON. ROSALINDA DIMAPILIS-BALDOZ
Secretary of Labor and Employment
Member, NNC Governing Board

DR. IRENE B. MARAMAG
President of RIC Philippines
Member, NNC Governing Board

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Republic of the Philippines
Department of Health
NATIONAL NUTRITION COUNCIL

Guidelines on the Fabrication, Verification and Maintenance of Wooden Height Boards

Rationale

This guidepost is being issued to LGU functionaries, relevant government agencies and business entities with interest in fabrication, marketing, verification and maintenance of wooden height boards used in measuring the length of 0-23 months old and height of 24-71 months old children. This document serves as a guide in ensuring the integrity of all height boards for field use.

Coverage

This guidepost applies to wooden length/height boards, which is the length and height measuring equipment being recommended, by the Technical Working Group on Child Growth Standards (TWG-CGS), for use in both facility- and non-facility based child growth assessment. The TWG-CGS considered the WHO's recommendation on wooden height boards, an affordable measuring equipment, for which it has drawn up technical specifications for fabrication, and from which the TWG based its recommendation.

The guidepost is likewise provided for length boards for those who want to fabricate a separate length board for younger children. Illustrations of the length and height boards are attached for reference.

Materials

- Wood - Kiln dried, water-proofed light weight but sturdy wood, 1.5 cm thick with *Supplemental treatment using FPA Registered Wood Preservatives*
- Steel rule - At least 2.0 cm wide steel rule, 1.5 meter long, with engraved millimeter markings; major tick should be in centimeter and font of every 10 centimeter marks should be bigger and/or colored
- Wood screws for attaching the steel rule and pieces
- Rubber or equivalent material as stopper for the headpiece
- Handle
- Bag (optional)

Additional note:

Advantages of properly kiln dried wood (15%-18% moisture) include:

a. Light in weight
b. Prevents dimensional changes and improves machining properties.
c. Provides good adhesion of glues and paint
d. Improves treatability and nail holding capacity
e. Resistant to decay but not to termites and beetles. Dry wood termites and other species of powder-post beetles prefer to attack wood materials even at the lowest MC of 10.0%.

Dimension (in centimeters)

<u>For height boards</u>

Assembled height of the height board: 146.5 cm

A. Head board (HHB) – movable; consists of 8 pieces of wood measuring as follows (*See illustration of head board*)

Label	**Quantity**	**Height** cm	**Length** cm	**Width** cm	**Diagonal/ Slant** (approximate)
HHB_A (back)	1	10.0	25.0	1.5	-
HHB_B [1/] (flat front)	1	25.0	25.0	1.5	-
HHB_C_1C_2 (outer side)	2	10.0 (at the back)	6.5 (on top)	1.5	12.0
		1.5 (in front)	15.0 (at the bottom)		
HHB_D_1D_2 (inner side)	2	8.5	3.5 (on top)	1.5	12.0
			12.0 (at the bottom)		
HHB_E [2/] (middle/handle)	1	8.5	3.5 (on top)	1.5	18.5
			20.0 (at the bottom)		
HHB_F (stopper)	1	3.5	15.0	2.5	-

1/ provide a thin slot measuring 4.5 cm x 0.5 cm, see illustration for the verification procedure

2/ provide a hole measuring 8 cm x 4cm which will serve as a handle (can be circular or rectangular in shape) of the headboard, see illustration

B. Back board (HBB) – consists of 3 pieces of wood measuring as follows
(*See illustration for back board*)

Label	Quantity	Height	Length	Width
HBB_A (backboard)	1	146.5	25.0	1.5
HBB_B (back top)	1	3.5	25.0	3.5
HBB_C (back bottom)	1	3.5	28.0	3.5

C. Foot board (HFB) – fixed; consists of 3 pieces of wood measuring as follows
(*See illustration of foot board*)

Label	Quantity	Height	Length	Width	Diagonal/ Slant
HFB_A (main foot board)	1	25.0	25.0	1.5	-
HFB_B$_1$B$_2$ (side)	2	15.0 (at the back)	6.0 (on top)	1.5	17
		3.0 (in front)	18.0 (at the bottom)		

For length boards

Assembled length of the length board: 116.5 cm

D. Head board (LHB) –fixed; consists of 3 pieces of wood measuring as follows
(*See illustration of head board*)

Label	Quantity	Height	Length	Width	Diagonal/ Slant
LHB_A (Head board)	1	20.0	20.0	1.5	-
LHB_B$_1$B$_2$ (side)	2	15.0 (at the back)	6.0 (on top)	1.5	17.0
		3.0 (in front)	18.0 (at the bottom)		

E. Back board (LBB) – consists of 3 pieces of wood measuring as follows
(*See illustration for back board*)

Label	Quantity	Height	Length	Width
LBB_A (backboard)	1	116.5	20.0	1.5
LBB_B (back top)	1	3.5	20.0	2.5
LBB_C	1	3.5	23.0	2.5

(back bottom)				

F. Foot board (LFB) – movable; consists of 8 pieces of wood measuring as follows (*See illustration of foot board*)

Label	Quantity	Height	Length	Width	Diagonal/ Slant (approximate)
LFB_A (back)	1	10.0	20.0	1.5	-
LFB_B [1/] (flat front)	1	20.0	20.0	1.5	-
LFB_C_1C_2 (outer side)	2	10.0 (at the back)	4.5 (on top)	1.5	12.0
		1.5 (in front)	13.0 (at the bottom)		
LFB_D_1D_2 (inner side)	2	8.5	1.5 (on top)	1.5	12.0
			10.0 (at the bottom)		
LFB_E [2/] (middle/handle)	1	8.5	1.5 (on top)	1.5	18.5
			18.0 (at the bottom)		
LFB_F (stopper)	1	3.5	15.0	2.5	-

1/ provide a thin slot measuring 4.5 cm x 0.5 cm, see illustration for the verification procedure

2/ provide a hole measuring 8 cm x 4 cm which will serve as a handle of the foot board, see illustration

Other specifications

1. Steel rule should be placed in the middle and a little bit embedded
2. Steel rule should be screwed to keep it in place
3. Headboard should have a slot (4.5 cm x 0.5 cm) through which the calibrated steel rule can be placed during the height board verification
4. Headpiece and footboard should be at 90^o angle with the back board
5. For models with detachable pieces (2-pc boards), care should be done that the pieces are properly attached and gaps in the joints are considered so as not to affect the accuracy of reading
6. Last reading should be at least 130.0 cm but not to exceed 1.0 cm for height boards.
7. Last reading should be at least 95.0 cm but not to exceed 1.0 cm for length boards.

Verification

Before a length or height board can be used, verification should be done by a DOST-trained personnel (passed the Competency Training on Verification). The points to be checked should include:

1. Correct placement and attachment of the steel rule
2. Zero placement of the steel rule at the base
3. Dimension of the board and parts
4. Quality of materials used
5. Perpendicularity of base/footboard and headpiece against the beam
6. Functionality and integrity of the unit
7. Correction factor, if any

A verified length/height board is one with a sticker of verification issued and posted at the upper right corner of the board. The certificate shall contain the date of verification, name and signature of person who verified and agency/institution, steel rule serial number (which will serve as unit number) and correction factor, if any.

Verification shall be repeated every 6 months by DOST-trained personnel.

Maintenance and Handling of the Height Board

Lifespan of wooden length/height board is 3-5 years, depending on the frequency of use and storage practices.

To maximize service life of the length/height board:

1. Use the length/height board properly according to its specific purpose.
2. Conduct regular tightness checks of all screws and bolts for safety.
3. Conduct regular visual inspection of the length/height board to check the board quality and avoid potential risk to children while in-service.
4. Do not drag during transport or transfer the board from one place to another.
5. Replace parts when necessary following the standard specifications in the fabrication of the length/height board.
6. Re-apply finishing when the length/height board in service needs supplemental finishes to retain its appearance. Excessive use can cause damage to both the finish and wood itself.
7. Let the kids stand on the board barefoot not only to obtain accurate measurements but also to avoid damage on the surface of the foot board.
8. Regularly clean with a soft, clean and dry cloth to keep the length/height board looking good.
9. Keep and store the length/height board in adequately dry area. Do not expose to water, excessive heat and direct sunlight.

The wooden board should be replaced if already showing warping or advanced deterioration. The rule should be replaced if the markings are already not legible.

Notes:

1. *Use of other materials in the fabrication of height board is still subject to review by the TWG-CGS and approval by the NNC Technical Committee.*

2. *The length/height boards with tape installed on the right side are acceptable only if these came from donations of UN organizations like UNICEF, WFP and Save the Children which were produced under the strictest guidance.*

ANNEX 2.6. NNC GB Resolution No. 3 s.2018 Approving the Guidelines on the selection of non-wood height and length measuring tool

Republic of the Philippines
Department of Health
NATIONAL NUTRITION COUNCIL

National Nutrition Council Governing Board
Resolution No. 3, Series 2018

APPROVING THE GUIDELINES ON THE SELECTION OF NON-WOOD HEIGHT AND LENGTH MEASURING TOOL

WHEREAS, the Philippines adopted the WHO Child Growth Standards (WHO-CGS) for determining the nutritional status of Filipino children under five years old;

WHEREAS, the WHO-CGS includes measurement of length and height of children to more accurately allow comprehensive child growth assessment;

WHEREAS, the NNC adopted the use of wooden height board in the country based on the recommendations of the World Health Organization where the first prototype of the wooden height board's specifications had been drawn and adapted;

WHEREAS, the NNC Governing Board on motion of the Department of Agriculture as Vice-Chair, instructed the Secretariat to explore other materials apart from wood for the fabrication of height board in view of the country's policy on total log ban;

WHEREAS, there is a need to field test height boards fabricated by various interest groups using other materials such as aluminum, acrylic, and plastics, a Field Trial of Alternative Height Measuring Equipment: Allenstick, Aluminum-Acrylic Height Board, Stadiometer vis-à-vis Wooden Height Board was conducted to provide evidence on accuracy, reliability, and ease of use of the other height board models; possibly replacing the wooden height board in the long-term in consideration of its weight and harm to the environment;

WHEREAS, the results of the field trial attested to the feasibility and practicability of the three (3) height boards tested made from other materials, i.e. aluminum-acrylic height board, fiberglass, and plastic, as acceptable measuring equipment for length/height of children in addition to the wooden height board with improvements as indicated in the study;

NOW THEREFORE, BE IT RESOLVED AS IT IS HEREBY RESOLVED, in consideration of the above premises, the National Nutrition Council Governing Board hereby approves the Guidelines on the Selection of Acceptable Non-Wood Height and Length Measuring Tool and use of height boards made of aluminum-acrylic, fiberglass, and plastic as acceptable measuring equipment in addition to the recommended wooden height board for assessing growth of children in the Philippines.

RESOLVED FURTHER, that the National Nutrition Council Secretariat shall ensure the dissemination of this Resolution and shall provide guidance to local government units and other agencies and organizations producing height boards, conducting length/height assessment of children and purchasing length/height measuring equipment.

APPROVED this 24th day of April Two Thousand and Eighteen.

FRANCISCO T. DUQUE III, MD, MSc
Secretary of Health and Chairperson
National Nutrition Council Governing Board

Attested by:

Assistant Secretary of Health Maria-Bernardita T. Flores, CESO II
Council Secretary and Executive Director IV
National Nutrition Council

CONFORME:

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ANNEX 2.7. Length/Height Board Verification Form

NATIONAL NUTRITION COUNCIL

Length/Height Board Verification Form

Manufacturer's Name:		Model:	
Serial number:		RA's date of acquisition:	
Requesting Agency (RA):		Date of Verification:	
Type of length/height board:		Place of Verification:	
		Verification Validity:	

Preliminary Evaluation			
Scale/Graduations	☐ Readable ☐ Unsatisfactory	Missing/broken parts	☐ None ☐ ____
Visual inspection shows that the general condition and workmanship of the instrument are:	____		
Headboard Squareness at 30cm	☐ Acceptabl☐ Not Acceptable	Footboard Squareness at 30cm	☐ Acceptable ☐ Not Acceptable

Verification of Instrumental Error						
Verification Point (cm)	Length/Height Board Scale Reading (cm)			Average of 3 trials (cm)	Difference *(Verification Point - Average)*	Error *Acceptable if absolute value of difference is less than or equal to 0.2 cm*
	Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)			
30						
40						
50						
60						
70						
80						
90						
100						
110						
120						

Overall Decision	☐ Fit for use☐ For replacement Remarks: ____

Prepared by: ____ Date: ____

Checked by: ____ Date: ____

ANNEX 2.8. Weighing Scale Calibration and Verification Form

NATIONAL NUTRITION COUNCIL

Weighing Scale Calibration and Verification Form

Type of weighing scale:		Date of Calibration:	
Manufacturer's Name:		Place of Calibration:	
Serial number:		Resolution:	
Max capacity:		Requesting Agency:	
		Calibration Validity:	

Departure From Nominal Value Test							
Test Weights (kg)	Conventional Mass Value (CMV) found in Calibration Certificate of test weights (mg)	Nominal Value (kg)	Trial 1 (kg)	Trial 2 (kg)	Trial 3 (kg)	Average	Correction (should be within -0.1 kg to 0.1 kg)
5							
10							
15							
20							
25							

Repeatability Test using 20 kg test weight	
Resolution of weighing scale (kg)	
Trial 1	
Trial 2	
Trial 3	
Trial 4	
Trial 5	
Trial 6	
Trial 7	
Trial 8	
Trial 9	
Trial 10	
Standard Deviation	
Criteria	

Eccentricity Test (using 10 kg test weight)		
Position	Weight	Offcenter Error
1		
2		
3		
4		
5		
1 (last)		
Reference		

Overall decision

☐ Fit for use:

0 - ____ kg

☐ For replacement

Verification	
Single Point Value Check (using 20 kg test weight)	
M (Nominal value of 20 kg)	
z_1	
m_1	
m_2	
z_2	
C_1	
C_2	
User's correction	
Change in correction	
Basis of decision	
Decision *(fit for use if Change in Correction is less than 3x the Criteria)*	

Prepared by: ____________________

Checked by: ____________________

OPT Plus TSEK

Operation Timbang Plus

Tamang Sukat, Eksaktong Kalidad!

Section Three

OPT PLUS PROTOCOL ON ANTHROPOMETRIC MEASUREMENT

OPT PLUS PROTOCOL ON ANTHROPOMETRIC MEASUREMENT

OVERVIEW

The general objective of this section is to provide information on the proper measurement of weight, length, height, and MUAC.

After reading this section, one should be able to:

1. Properly measure the weight of 0-to-59-month-old children using the appropriate weighing scale;
2. Properly measure the recumbent length of the 0-to-23-month-old children;
3. Properly measure the height of 24-to-59-month-old children;
4. Properly measure the MUAC of 6-to-59-month-old children;
5. Properly check for and classify severity of bilateral pitting edema;
6. Properly record the weight, length, height, and MUAC measurements; and
7. Provide immediate feedback to parents/caregivers and refer malnourished children to relevant rural health offices for appropriate interventions.

Section 3 covers how to get the weight and height using the recommended anthropometric tools. This section includes the following:

Section 3.1	Roles and Responsibilities of OPT Plus Team and Supervisors
Section 3.2	General Reminders in Taking Anthropometric Measurements
Section 3.3	Instruction on How to Weigh the 0-to-59-month-old Children
Section 3.4	Instructions on Measuring the Length of the 0-to-23-month-old Children
Section 3.5	Instructions on Measuring the Height of the 24-to-59-month-old Children
Section 3.6	Instruction on How to Measure MUAC of 6-to-59-month-old Children
Section 3.7	Instructions on How to Check and Classify Severity of Bilateral Pitting Edema
Section 3.8	OPT Plus Referral
Section 3.9	WHO Child Growth Standards (CGS)

LIST OF TABLES

LIST OF FIGURES

LIST OF ACRONYMS

AO	Administrative Order
BHC	Barangay Health Center
BHS	Barangay Health Station
BNAP	Barangay Nutrition Action Plan
BNC	Barangay Nutrition Committee
C/MNAO	City/Municipal Nutrition Action Officer
C/MSWD	City/Municipal Social Welfare and Development
cm	centimeter
CRPD	Convention on the Rights of Persons with Disabilities
DOH	Department of Health
eOPT Plus	Electronic Operation Timbang Plus
FANTA	Food and Nutrition Technical Assistance III
FAO	Food and Agriculture Organization of the United Nations
GMP	Growth Monitoring and Promotion
HB	Height Board
kg	kilogram
L/HFA	Length/Height-for-Age
LFA	Length-for-Age
LGU	Local Government Unit
HFA	Height-for-Age
MW/MAM	Moderately Wasted/Moderate Acute Malnutrition
MUAC	Mid-Upper Arm Circumference
MUW	Moderately Underweight
MW	Moderately Wasted
NNC	National Nutrition Council
NNC CO	National Nutrition Council Central Office
PNAO	Provincial Nutrition Action Officer
RHU	Rural Health Unit
RNPC	Regional Nutrition Program Coordinator

SW/SAM	Severely Wasted/Severe Acute Malnutrition
SUW	Severely Underweight
SW	Severely Wasted
TWG-CGS	Technical Working Group on Child Growth Standards
WFA	Weight-for-Age
WFH	Weight-for-Height
WFL	Weight-for-Length
WHO	World Health Organization
WHO-CGS	World Health Organization Child Growth Standards

DEFINITION OF TERMS

Disability (Children with Special Conditions/ Needs)	Children affected by developmental, neurologic, and genetic conditions, which may affect growth patterns and overall growth potential.
Bilateral Pitting Edema	A clinical sign of severe malnutrition that makes feet, legs, arms, hands, and/or face appear swollen or puffed up due to severe muscle wasting.
eOPT Plus Tool	An excel-based tool used by nutrition workers in the community to help them consolidate and summarize the results of OPT Plus among 0-to-59-month-old children.
Length/Height-for-Age (LFA)	An indicator used to classify stunting among children 0-59 months old.
OPT Plus	Regular growth assessment of all 0-to-59-month-old children in the barangay, done by the trained team and involves planning, follow-up, and data quality check.
Tare	Adjusting/resetting the equipment's display to zero.
Triangle Chart	It is a simplified and comprehensive pictorial flow chart to identify the root causes of malnutrition in each child.
Weight-for-Age (WFA)	An indicator used to classify underweight among 0-to-59-monthold children.
Weight-for-Length/ Height (WFL/H)	An indicator used to classify wasting and overweight and obesity among 0-to-59-month-old children.
WHO Child Growth Standards (WHO-CGS)	Represents how children should grow through globally accepted prescriptive measures considered normal growth for all 0-to-59-month-old children.

Section 3.1

ROLES AND RESPONSIBILITIES OF OPT PLUS TEAM AND SUPERVISORS

One of the main activities in the OPT Plus is the collection of the anthropometric measurements of 0-to-59-month-old children. Anthropometric measurement is a non-invasive quantitative measurement of the body useful in assessing the nutritional status of both adults and children. Several activities are conducted during this activity including the actual measurements of weight, length, height, and MUAC, checking the severity of bilateral pitting edema, and referral for the identified malnourished children. Table 3.1 shows the roles and functions of the different persons involved during the OPT Plus Anthropometric Measurement.

Table 3.1. Roles and Functions of the different persons involved in the OPT Plus during the OPT Plus Anthropometric Measurement.

ORGANIZATION/ PERSON INVOLVED	FUNCTION
National Nutrition Council Central Office (NNC CO)	1. Monitor the status of the OPT Plus activity at the national level.
National Nutrition Council Regional Office/ Regional Nutrition Program Coordinator (RNPC)	1. Monitor the status of the OPT Plus activity. 2. Provide mentoring and supportive supervision visits to LGUs/ barangays needing assistance.
Provincial Nutrition Office/Provincial Nutrition Action Officer (PNAO)	1. Monitor the status of the OPT Plus activity. 2. Provide mentoring and supportive supervision visits to barangays needing assistance.
Supervisors: City/ Municipal Nutrition Action Officers, District/City/Municipal Coordinators, Midwives, Barangay Nutrition Action Officers among others as applicable	1. Check the master list of 0-to-59-month-old children. 2. Oversee the actual collection of length/height measurement and weighing. 3. Review the recording of the sex, date of birth, date of weighing, length/height, and weight. 4. Refer to the Barangay Health Centers (BHC)/Rural Health Unit (RHU) the children with bilateral pitting edema and obviously manifesting malnutrition (i.e., extreme thinness, hollow cheeks, edematous feet, and hands, etc.). 5. Feedback the child's nutritional status immediately after measurement using the weight-for-length/height index.
City/Municipal Nutrition Action Officer (C/ MNAO)	1. Monitor the status of the conduct of the OPT in each Barangay Health Station (BHS).
OPT Plus Team Members	1. Conduct monthly OPT Plus monitoring of MW/MAM, SW/SAM, underweight, and 0-to-23-month-old children, and quarterly monitoring of 24-to-59-month-old children. 2. Participate in the bi-annual review of the Barangay Nutrition Action Plan (BNAP) and present BNAP accomplishments during the Barangay General Assembly/Barangay Nutrition Committee (BNC) meetings.

ORGANIZATION/ PERSON INVOLVED	FUNCTION
OPT Plus Team Members	3. Prepare the master list of 0-to-59-month-old children. 4. Conduct the actual collection of length/height measurement and weighing with the supervisor, if feasible. 5. Record the sex, date of birth, date of weighing, length/height, and weight. 6. Refer to the BHC/ RHU the children with bilateral pitting edema and obviously manifest malnutrition if the supervisor is unavailable. 7. Provide feedback regarding the child's nutritional status immediately after measuring using the weight-for-length/height index through the Growth Chart in the ECCD Card or NNIS mobile app, if the supervisor is unavailable. 8. Coordinate with the DCW to measure the children who were not covered during the center-based OPT Plus or home visit.
Barangay Nutrition Committee (BNC)	1. Assist in the preparation of the master list of 0-to-59-month-old children. 2. Mobilize the under-five children to the weighing post. 3. Provide logistic support, i.e., transportation and food allowance to the OPT Plus Team. 4. Provide transportation and other services to the families with malnourished children to visit the BHC/RHU. 5. Assist coordination with the DCW for the children who have not been measured at home or center-based.

Section 3.2

OVERVIEW OF THE ANTHROPOMETRIC MEASUREMENTS

During this activity, the following measurements are taken to assess the nutritional status of 0-to-59-month-old children:

WEIGHT	The measurement of body mass or heaviness of a child.
LENGTH	The stature measurement of a child 0-to-23-month-old while in a recumbent or lying position.
HEIGHT	The stature measurement of a child 24-to-59-month-old collected while standing up.
MID-UPPER ARM CIRCUMFERENCE (MUAC)	The circumference of the left arm taken between the elbow and shoulder at the midpoint.
EDEMA	The presence of excess fluid in muscle tissues caused by severe muscle wasting.

Section 3.3

GENERAL REMINDERS IN TAKING ANTHROPOMETRIC MEASUREMENTS

Some Considerations When Planning to Take Anthropometric Measurements (FAO, 2021)

Before taking the child's weight and length or height measurements, a few things should be considered to ensure the collection of high-quality anthropometric data and the child's safety. Here are some reminders:

1. PLACEMENT OF TOOLS AND EQUIPMENT

a. Observe and select an appropriate spot in the household where the equipment may be set up.

b. Length boards and platform scales should be placed on a flat surface while hanging scales should be securely hung on a sturdy post, beam, or tree.

c. If possible, conduct the measurement outdoors during daylight hours.

d. Opt to measure indoors in case of bad weather or if there are too many people outdoors.

2. SAFETY AND PRECAUTION

a. Since children's bones are still developing, only apply gentle pressure on their limbs or legs when getting their measurements.

b. Do not leave the child alone with any measuring tools or equipment.

c. Make sure that the measurer's fingernails are trimmed, and accessories such as rings, bracelets, or watches are removed before handling a child to avoid accidentally hurting the child.

d. Do not hold a pencil or pen while weighing or measuring.

e. When using hanging scales, make sure they are hung on a sturdy pole, beam, or tree. Check the pants, straps, and hooks and make sure they are not broken or nearly broken.

f. When taking height measurements, use a leveling tool to make sure that the board is on level ground. For measuring recumbent length, place the board on top of a flat and sturdy platform or table.

g. Since children tend to be irritated or uncomfortable while taking their weight and height, make sure to be attentive and alert to avoid accidents.

3. DEALING WITH DISTRESSED CHILD

a. If the child is uncooperative or distressed, let the parent/caregiver calm the child first before proceeding with the measurement.

b. If measuring more than one child, measure another different child first and let the distressed child watch the process.

4. CHILDREN WITH SPECIAL CONDITIONS/ NEEDS

This will include suffering from chronic diseases, deformities, disfigurement and disabilities.

a. Children with special conditions may have to be weighed and measured differently.

b. If the child's height cannot be measured, get the length instead, and subtract 0.7 cm from the measurement.

c. If a child cannot form a vertical plane where the head, shoulders, buttocks, and heels are aligned, a minimum of two contact points (i.e., back of the head and buttocks or heels and buttocks) should touch the surface of the measuring tool. This may occur due to obesity.

d. If a child has leg length asymmetry, have the child stand on the longer leg and place a block or wedge on the shorter leg to align the hips before taking the height measurement. To get the length, hold the legs together and get the measurement to the heel of the longer leg.

e. In case weight and length/height cannot be measured, note the child's information (i.e., name of the child, name of parent/caregiver, address) in the recording form and note that the measurements cannot be taken due to the child's special condition.

Note: These guidelines were adapted from Alberta Health Services (2015)

5. TAKING CARE OF ANTHROPOMETRIC TOOLS

a. Clean and check the anthropometric tools regularly. If there are signs of damage that may affect the performance of the tool, report them immediately for a replacement.

b. Make sure that tools and equipment are regularly calibrated and verified.

c. Follow specific guidelines for taking care of the measurement tools.

d. When using digital scales, make sure that batteries are replaced and that extras are available.

6. RECORDING MEASUREMENTS

a. Use pencils instead of pens in recording to easily correct errors.

b. Make sure to record all relevant information for one child before moving on with the next to avoid confusion or mismatch of records.

c. Record weight in the nearest 0.1 kg

d. Record length, height, and MUAC in the nearest 0.1 cm.

e. Always validate and confirm the recorded weight, length, height, and MUAC after the assistant writes them down.

f. Record anthropometric data legibly in the OPT Plus Form 1A Recording Sheet.

NOTE 3.1. Reminders on Recording Data

1. Use OPT Plus Form 1A Recording Sheet in recording the sex, weight, length/height, date of birth, and date of measure (Annex 3.2).
2. Get all the measurements on the same day. The weight and length/height taken at different days will result in inaccurate data.

Writing numbers the correct way when recording anthropometric measurements

In recording anthropometric measurements, the following guidelines are to be followed to minimize errors:

1 Draw the number 1 as a single vertical line.
2, 3 Write the numbers 2 and 3 without extra loops.
4 Leave the number 4 open. A closed 4 can look like a 9.
5 Don't close the 5. A closed 5 can look like a 6.
6 Be careful that the circle on the 6 does not look like a 0.
7 Make a small horizontal line to cross the 7. This will distinguish it from a 1.
8 Make two circles to draw the number 8 so that it does not look like a 0. Don't separate the two circles.
9 Close the circle on the number 9 so that it does not look like a 4.
0 Put a diagonal line through the 0 so that it is easy to identify and does not look like a 6.

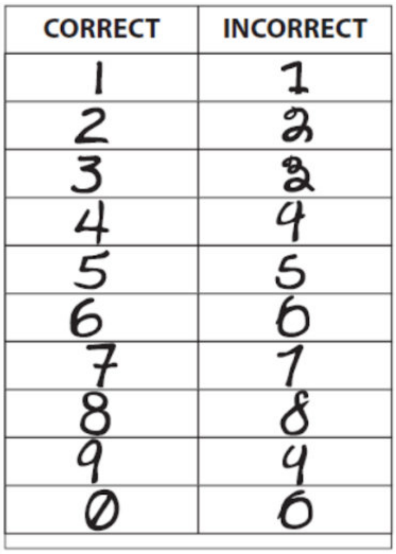

Figure 3.1. Correct and incorrect ways of writing numbers.

Getting ready to measure weight, length, height, and MUAC

Before you proceed to measure the child, make sure to do the following:

1. **Introduce yourself and briefly explain the purpose of your visit to the parent/ caregiver.** Mention that OPT Plus is the annual weighing and measurement of 0-to-59-month-old children and that it aims to determine the nutritional status of their child for referral to other local programs if needed.

2. **Provide the parent/caregiver with an overview of what will be done to the child during weighing and length/height measurement.** Also mention that the parent's/ caregiver's help will be needed during the activity.

3. **After explaining, get the consent of the parent/caregiver by using Annex 3.2.** Once the parent/caregiver agrees, request her to sign the OPT Plus Form 1A, then you may proceed. A sample consent form can be seen in Annex 3.3. Sample Data Privacy Notice and Informed Consent Form.

Section 3.4

INSTRUCTIONS ON HOW TO WEIGH THE 0-TO-59-MONTH-OLD CHILDREN

Using mechanical/digital hanging weighing scale (Cashin and Oot, 2018):

1. Set up the hanging scale.

a. Hang the scale using the hooks and a rope looped at a sturdy post, beam, or tree.

For mechanical scales: Make sure that the scale's dial is at the measurer's eye level.
For digital scales: Press the power button to turn the device on.

b. Tare the scale to "0."

For mechanical scales: Tare the scale to zero with the empty hanging pants attached to the scale. Adjustments can be made through taring knobs.
For digital scales: Press the tare button with the empty hanging pants attached to the scale.

c. Ensure that the set-up is not too low such that the foot of the child reaches the floor and not too high to avoid injuries in case the child falls.

2. Prepare the child for weighing.

a. Explain the reason for getting the child's weight to the parent/caregiver. Mention that it will be used to assess the child's growth and nutritional status.

b. To ensure that accurate weight measurement is obtained, ask the parent/caregiver to remove any heavy clothing from the child. If possible, the child should be weighed only in his/her underpants.

3. Weigh the child.

a. Remove the weighing pants from the scale and carefully direct the child's legs through the pants using the measurer's hand inserted through the leg holes from the lower part of the weighing pants. Make sure that the strap of the pants is in front of the child with his/her hands placed on the sides.

b. Once the weighing pants are securely in place, lift the child with the help of the trained assistant or parent/ caregiver and hang the straps onto the scale's hook. Ensure that straps are securely attached to the hook before carefully letting go of the child. DO NOT CARRY THE CHILD BY THE STRAP ONLY.

c. Before taking the measurement, make sure that the child's position is correct, that they are not holding anything, and that the child is hanging freely where their feet do not touch the ground. If the child is agitated, calm them down first to avoid inaccurate weight measurements.

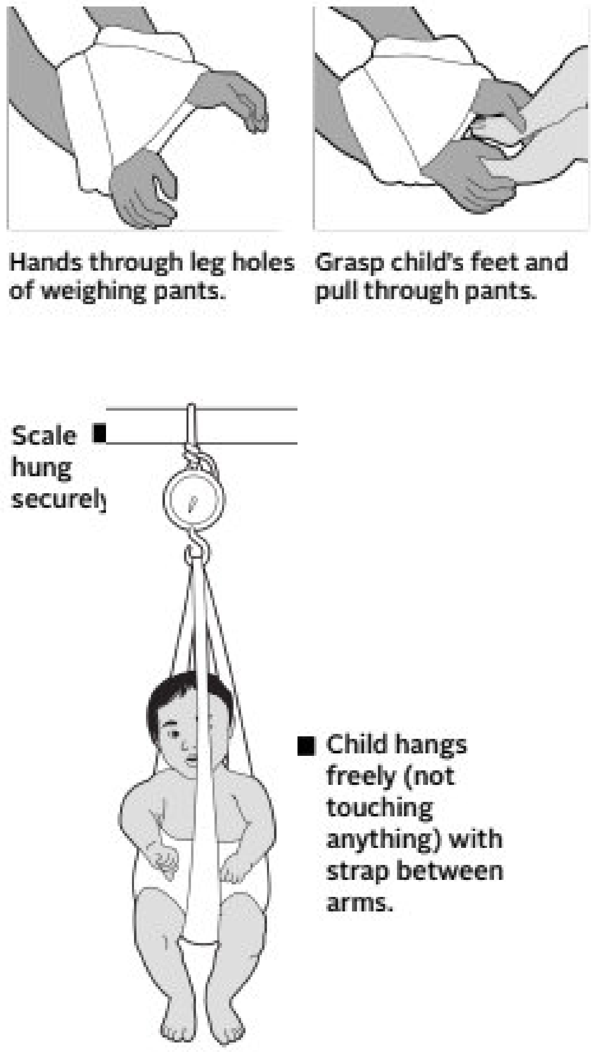

Figure 3.2. Illustration on proper positioning of the child in the hanging weighing scale.

4. Get the measurement.

For mechanical scales: The measurer should stand at eye level in front of the scale to ensure that the child is still, and the dial is not moving or fluctuating.

For digital scales: Weight will register on the digital display of the device or scale.

a. The measurer should read aloud the child's weight (in the nearest 0.1 kg), and the assistant should record it in the OPT Plus Form 1A Recording Sheet.

b. Lift the child by carrying their body and removing the pants with the help of the assistant or parent/caregiver.

c. Validate and confirm recorded weight.

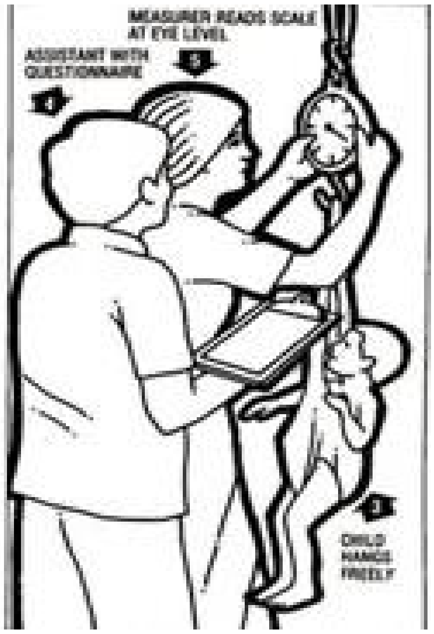

Figure 3.3. Illustration on how to read the weight using hanging weighing scale.

Using digital platform scales for children who can stand alone (Cashin and Oot, 2018):

1. Set up the scale.

a. Place the scale on a flat and even surface, preferably the floor.

b. Turn on the scale according to the instructions.

c. Tare the scale to "0.0." Note that some digital scales are automatically tared when turned on.

2. Prepare the child for weighing.

a. Explain the reason for getting the child's weight to the parent/caregiver. Mention that it will be used to assess the child's growth and nutritional status.

b. To ensure that accurate weight measurement is obtained, ask the parent/caregiver to remove any heavy clothing from the child and to make sure the diapers are empty.

3. Weigh the child.

a. Ask the child to stand in the middle of the scale.

b. Wait until the displayed weight remains fixed.

c. Read aloud the measurement and let the assistant record it in the OPT Plus Form 1A Recording Sheet.

d. Validate and confirm recorded weight.

Figure 3.4. Illustration on getting the weight of the child using digital platform.

Using digital platform scales to weigh children who cannot stand alone (Cashin and Oot, 2018):

Most recommended platform scales have a taring function which resets the scale to "0.0" while a person is standing on the scale. This function is useful when weighing infants and children who are unable to stand on their own with the help of their parents/caregivers. If your scale has no taring function, see Note 3.2. Manual Taring.

Here are the steps on using digital platform scales to weigh children who cannot stand alone:

1. Set up the scale.

a. Place the scale on a flat and even surface, preferably the floor.

b. Turn on the scale according to the instructions.

c. Tare the scale to "0." Note that some digital scales are automatically tared when turned on.

2. Prepare the child for weighing.

a. Explain the reason for getting the child's weight to the parent/caregiver. Mention that it will be used to assess the child's growth and nutritional status.

b. To ensure that accurate weight measurement is obtained, ask the parent/caregiver to remove any heavy clothing from the child and to make sure the diapers are empty. If possible, the child should be weighed only in his/her underpants or wrapped in a blanket.

3. Weigh the parent/caregiver.

a. Instruct the parent/caregiver to stand on the scale.

b. Once the weight registers, tare the digital scale to "0."

4. Get the child's weight.

a. Have the child placed in the parent/caregiver's arms.

b. Read the weight out loud to the nearest 0.1 kg and let the assistant record it in the OPT Plus Form 1A Recording Sheet.

5. Record and verify the weight.

Using a beam balance:

1. Set up the scale. Make sure that the scale is at 0.0 kg by placing all the weights on the left side.

2. Prepare the child for weighing.

a. Explain the reason for getting the child's weight to the parent/caregiver. Mention that it will be used to assess the child's growth and nutritional status.

b. To ensure that accurate weight measurement is obtained, ask the parent/caregiver to remove any heavy clothing from the child and to make sure the diapers are empty.

3. Weigh the child.

a. Ask the child to stand in the middle of the scale.

b. Wait until the displayed weight remains fixed.

c. Read aloud the measurement and let the assistant record it in the OPT Plus Form 1A Recording Sheet.

NOTE 3.2. Manual Taring

If the digital scale has no taring function, you may follow these steps:

1. Make sure that the scale is tared or in "0.0."
2. Have the parent/caregiver stand on the scale, then write down the weight.
3. Place the child on the parent's/caregiver's arms, then write down the combined weight.
4. Subtract the parent's/caregiver's weight from the combined weight to get the child's weight:

   Child's Weight = Combined Weight - Parent's/Caregiver's Weight
5. Verify and record.

NOTE:

Access the instructional video on anthropometric measurement at the NNC website: www.nnc.gov.ph.

Section 3.5

INSTRUCTIONS ON MEASURING THE LENGTH OF THE 0-TO-23-MONTH-OLD CHILDREN

There are different recommended length and height measuring tools (i.e., wooden height board (HB), aluminum-acrylic height board, stadiometer, microtoise, infantometer, and measuring mat) and each one has its own unique features. If available, one may refer to the User's Manual for the proper use and maintenance of the tool. If there is a need to fabricate the height boards, one should refer to the NNC's Governing Board Resolution No. 3, Series 2018 Approving the Guidelines on the Selection of Non-Wood Height and Length Measurement Tool (Annex 3.3) and Guidelines on the Fabrication, Verification, and Maintenance of Wooden Height Boards (Annex 3.4) before fabricating it. Below are the protocols you need to understand for getting the length or height of a child to get reliable, accurate, and precise OPT Plus data.

Using length board (Cashin and Oot, 2018):

1. **Prepare the set-up for measuring length.**

a. Place the board horizontally on a hard, flat, and durable surface such as the ground or sturdy table. If placed on a table or any elevated surface, make sure that it is sturdy to avoid accidents.

b. Explain to the parent/caregiver that the child will be measured using the board. In the absence of an assistant, mention that you will need their assistance to keep the 0-to-23-month-old baby while measuring. Also mention that two (2) or three (3) people will be measuring the length/height of the child.

c. Clean and sanitize the board before placing the child.

2. **Prepare the child for measuring recumbent length.**

a. Ask the parent/caregiver to remove any footwear or head/hair accessories that could interfere with the measurement.

3. **Measure the length.**

a. Ask the parent/caregiver to carefully place the child on the board.

b. Instruct the parent/caregiver to position the child's head such that the child is looking straight up while the top of the head is situated firmly on the headboard. The head of the assistant or parent/caregiver should be directly above the child's head while holding the correct position by gently placing their hands over the child's ears.

c. Stand at the side of the board, and check if the child is lying straight with the shoulders touching the board.

d. Gently hold down the child's legs with one hand and use the other hand to slide the footboard.

e. Apply pressure on the knees to straighten the legs as possible to get the child's length. Be careful not to exert too much force and cause injury to the child.

f. Check the child's position and make adjustments as needed.

g. Once the position is correct, slide the footboard firmly on the child's heels.

h. Read aloud the measurement (in the nearest 0.1 cm), and the assistant should record it in the OPT Plus Form 1A Recording Sheet.

i. Validate and confirm the recorded length.

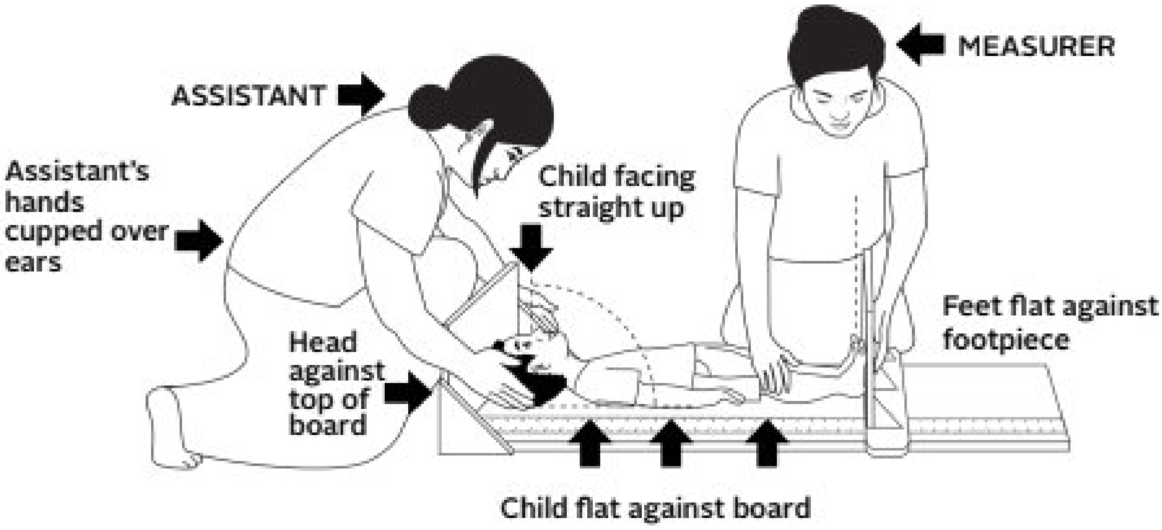

Figure 3.5. Illustration on how to position the child in the length board.

Section 3.6

INSTRUCTIONS ON MEASURING THE HEIGHT OF THE 24-TO-59-MONTH-OLD CHILDREN

Using height board or stadiometer (Cashin and Oot, 2018):

1. Prepare the set-up for measuring height.

a. Place the board vertically on a hard and flat surface against a wall or pillar.

b. A leveling tool may be used to ensure that the board stands at a 90° angle against the floor.

If using a stadiometer: Adjust the spacer found at the topmost part until the built-in leveling tool shows that the stadiometer is completely level or perpendicular to the floor.

c. Clean and sanitize the board before measuring.

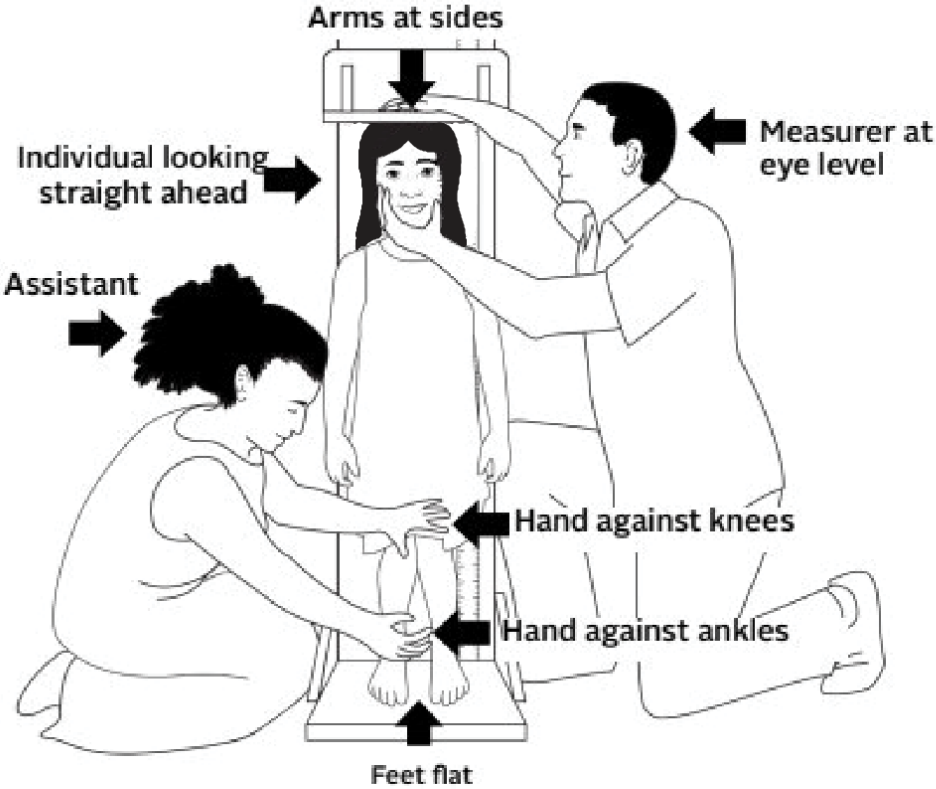

Figure 3.6. Illustration on how to position the child in the height board.

2. Prepare the child for measurement.

Remove any footwear and head/hair accessories that may interfere with the measurement.

3. Measure the child.

a. Kneel at the right side of the child and help them stand on the baseboard of the board with feet slightly apart and back against the board.

b. To help ensure that the child's position is correct, check if the tip of the shoulders is aligned to the position of the heel. It may help to check if the back of the head, shoulder blades, buttocks, calves, and heels touch the board.

c. Ask the assistant or parent/caregiver to hold the child's knees and legs in place.

c. Before getting the measurement, check the child's position and adjust as necessary. You may help a child who cannot stand straight by gently pushing on their stomach.

d. Slide the moveable headpiece down carefully while cupping the child's chin. Ensure that the line of sight is parallel to the ground. Make sure that the hair is pushed down before taking the measurement.

e. Read aloud the measurement (in the nearest 0.1 cm) and let the assistant record it in the OPT Plus Recording Sheet.

f. Validate and confirm recorded height.

Using microtoise:

1. Prepare the set-up.

a. Attach a plumb line on the wall by taping a long string with a rock at the end of the string. Make sure that the rock is hanging freely.

b. Place the microtoise on the floor and pull the measuring tape until the "0" mark is on the red line of the read-off area.

c. Make sure that the measuring tape is against the wall. Mark the end of the tape to indicate where to mount the microtoise.

d. Drill a hole exactly where the end of the measuring tape is and mount the tool using a screw.

e. Pick up the body of the tool from the floor and join it together with the end of the measuring tape

2. Take the child's height.

a. Ensure that the child is not wearing any footwear and has no hair ornaments that may interfere with the accurate measurement of height.

b. Have the child stand directly below the body of the tool while making sure that the back of the head, shoulder blades, buttocks, calves, and heels touch the wall.

c. Slide the body of the tool down until it reaches the child's head. Make sure that the hair is compressed when taking the measurement.

d. Read the measurement in the read-off area.

e. Read aloud the measurement (in the nearest 0.1 cm) and let the assistant record in the OPT Plus Recording Sheet.

f. Validate and confirm recorded height.

NOTE 3.3.
Reminder on Length and Height Measurement

Generally, 0-to-23-month-old children are measured lying down to get their recumbent length, while 24-to-59-month-old children are measured standing up. However, there are times this is not followed due to the child's inability to stand or refusal to lie down. In this case, the measurer will have to adjust the measurement accordingly:

- If a 0-to-23-month-old child was measured standing up, add 0.7 cm to the obtained measurement.
- If a 24-to-59-month-old child was measured lying down, subtract 0.7 cm to the obtained measurement.

Section 3.7

INSTRUCTIONS ON HOW TO MEASURE MUAC OF 6-TO-59-MONTH-OLD CHILDREN

The use of mid-upper arm circumference (MUAC) tape detects the possible problem of malnutrition among children immediately, especially during emergencies (See Annex 3.5. NNC Memorandum 2020-010 Interim Guidelines in the Conduct of OPT Plus, Nutrition Screening, Growth Monitoring and Promotion (GMP) Activities in the Context of the COVID-19 and other Disasters). Below are the instructions on how you are going to use the MUAC.

Using MUAC Tape:

1. **Position the child so that the measurer can work at eye level.**

2. **Locate the midpoint of the child's upper left arm. It is the area between the tip of the shoulder and the tip of the elbow.**

   a. Locate the shoulder and elbow tip and adjust the arm to make a right angle.

   b. Place one end of the MUAC tape on the tips of the shoulder and measure the length until the elbow tip.

   c. Ask the assistant to mark the midpoint with a pen. The midpoint is half of the length between shoulder and elbow tips.

3. **Straighten the arm and wrap the MUAC tape around the midpoint. Do not pull the end of the tape too tight.**

4. **Read the measurement through the window at the end of the tape and record.**

5. **Explain the result to the mother.**

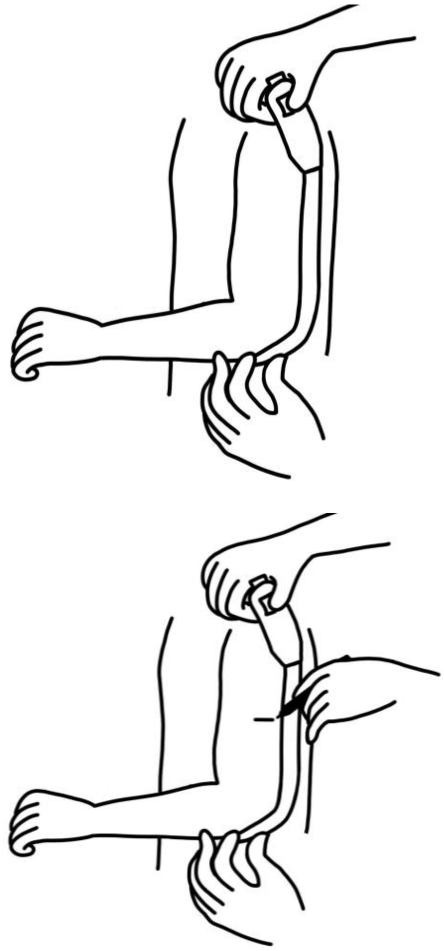

Figure 3.7. Illustration on how to get the midpoint between the shoulder and elbow.

Note 3.4. Use of MUAC

In the OPT Plus Protocol, MUAC can be used only during emergencies. If one barangay uses MUAC, all the barangays should use MUAC during OPT plus and follow-up activities.

Under the PIMAM Program, it is very important to note that MUAC is and continues to be used as a criteria for identifying, admission, and discharge of children with acute malnutrition or wasting. This is applicable for both regular and emergency contexts (see National Guidelines for the Management of Acute Malnutrition for Children under 5 years, DOH AO 2015-0055).

Section 3.8

INSTRUCTIONS ON HOW TO CHECK AND CLASSIFY SEVERITY OF BILATERAL PITTING EDEMA

Bilateral pitting edema is the presence of excess fluid in muscle tissues caused by severe muscle wasting. This condition is characterized by the puffiness or swelling of feet, hands, and/or face. A child with bilateral pitting edema is considered severely wasted or SW/SAM, at higher risk of mortality, and in need of urgent medical attention. Below are the steps for checking for presence of bilateral pitting edema.

Figure 3.8. Images of Bilateral Pitting Edema.

1. Prepare the child.

a. Explain to the parent/caregiver that the child will be checked for bilateral pitting edema and that their assistance may be needed during the procedure.

b. Ask the parent/caregiver to remove any footwear.

2. Check for bilateral pitting edema.

a. Hold the child's feet with your thumb on top of the foot.

b. Gently press your thumbs down and slowly count to three before lifting your thumb.

c. Check if there are indentations left by your thumbs.

d. Record the results based on severity. You may refer to Table 3.2 for the grades of bilateral pitting edema.

Table 3.2. Grades of bilateral pitting edema.

SEVERITY	BILATERAL PITTING EDEMA
+1	Both feet
+2	Both feet, legs, and may include hands, lower arms
+3	Both feet, legs, hands, lower arms, upper arms, face

Section 3.9

REFERRAL OF MALNOURISHED CHILDREN

What is Referral?

Referral is the process of linking identified malnourished children to appropriate interventions and services in the community and is one of the main objectives of the OPT Plus at the individual level. It is important to understand that OPT Plus is not just the annual weight and length/height measurement of 0-to-59-month-old children.

The information collected should be interpreted to identify which of the children are malnourished. Referral is an important process to utilize nutrition data and improve the nutritional status of children. The health and/or nutrition worker should fill out the OPT Plus Referral Form (Figure 3.9) if there is a child in need of referral..

Region
Province
City/Municipality
OPT Plus Referral Form

Nameof Child:_______________________________

Age:______________ Sex: _______________ Municipality:__________________________

Date of Referral: ____________________________ Severity of Bilateral Pitting Edema:

Weight: ____________ Height: ______________ ☐ None

MUAC: _____________ If yes: Grade ☐+1 ☐+2 ☐+3

WFH: ☐ WFA: ☐ HFA: ☐	SW/SAM: ☐ MUW: ☐ MSt: ☐	Normal: ☐ SUW: ☐ SSt: ☐	OW/Obese: ☐

Received by: Referred by:

______________________________ ______________________________
Name Name

______________________________ ______________________________
Designation Designation

- -

Return Slip

Name of Child:__________________________________

Date Referred:__________________________________

Nutritional Status:______________________________

____

Received by: Name/Signature

REMARKS

Figure 3.9. OPT Plus Referral Form.

NOTE:
The list of nutrition interventions can serve as guide for the OPT Plus Team where to refer the malnourished children.

Nutritional Status	Nutrition Interventions		
	0 to 5 months	6 to 23 months	24 to 59 months
Moderately wasted (MAM)/Severely wasted (SAM)	Refer to PIMAM		
Underweight	Counseling of mothers on the First 1000 Days	Dietary supplementation for children 6-23 months	Dietary supplementation in Child Development Centers and Supervised Neighborhood Plays
			School-Based Feeding Program
Overweight/Obese	Counseling of mothers on the First 1000 Days		Promotion of Healthy Lifestyle

For Child Who is Growing Well

A normal nutritional status suggests that the child is growing well. Parents/caregivers of these children should be commended by the OPT Plus Team during counseling. Moreover, normal growth should be sustained. This may be achieved by educating the parents/caregivers on the importance of proper nutrition and by giving feeding advice appropriate for the age of the child. It will also be helpful to remind the parents/caregivers that there are other interventions like immunization, deworming, dental check-ups, and other possible interventions that are accessible in the local health office.

For Malnourished Children

If there is a growth problem, the parent/caregiver should be interviewed and counseled to identify possible causes of the problem and to provide nutritional advice to improve the nutritional status of the child. It is very important to keep the discussion positive and to avoid accusing or judging the parent/caregiver. The parent's/caregiver's trust should be built to clearly communicate that he/she can help the child. Use clear, non-medical language as much as possible. If unfamiliar words such as "Obese" will be used, explain it to the mother. For example, being obese means being very heavy for one's height. **The triangle chart can be used in determining the causes of the child's condition.**

NOTE 3.5. Triangle Chart

The triangle chart was developed by the UPLB BIDANI Network Program (Jequinto et al., 1998).

The Triangle Chart is a simple and comprehensive pictorial flow chart used to identify the root causes of malnutrition of each child. It can also be used to plan possible interventions and monitor children and planned activities.

How to Use the Triangle Chart?

1. Identify the family situation of the child. This can be done by interviewing the parent/caregiver. Questions may include the child's past and present medical condition, feeding/caring practices, deworming, supplementation, immunization, etc. You may also directly observe the living condition of the household.
2. Together with the parent/caregiver, identify the problems that may have led or caused the child to be malnourished. Write each problem in big triangles.
3. Once the problems are identified, check the list of root causes and get the codes of causes that are applicable to the child's situation. Write it down in smaller triangles.

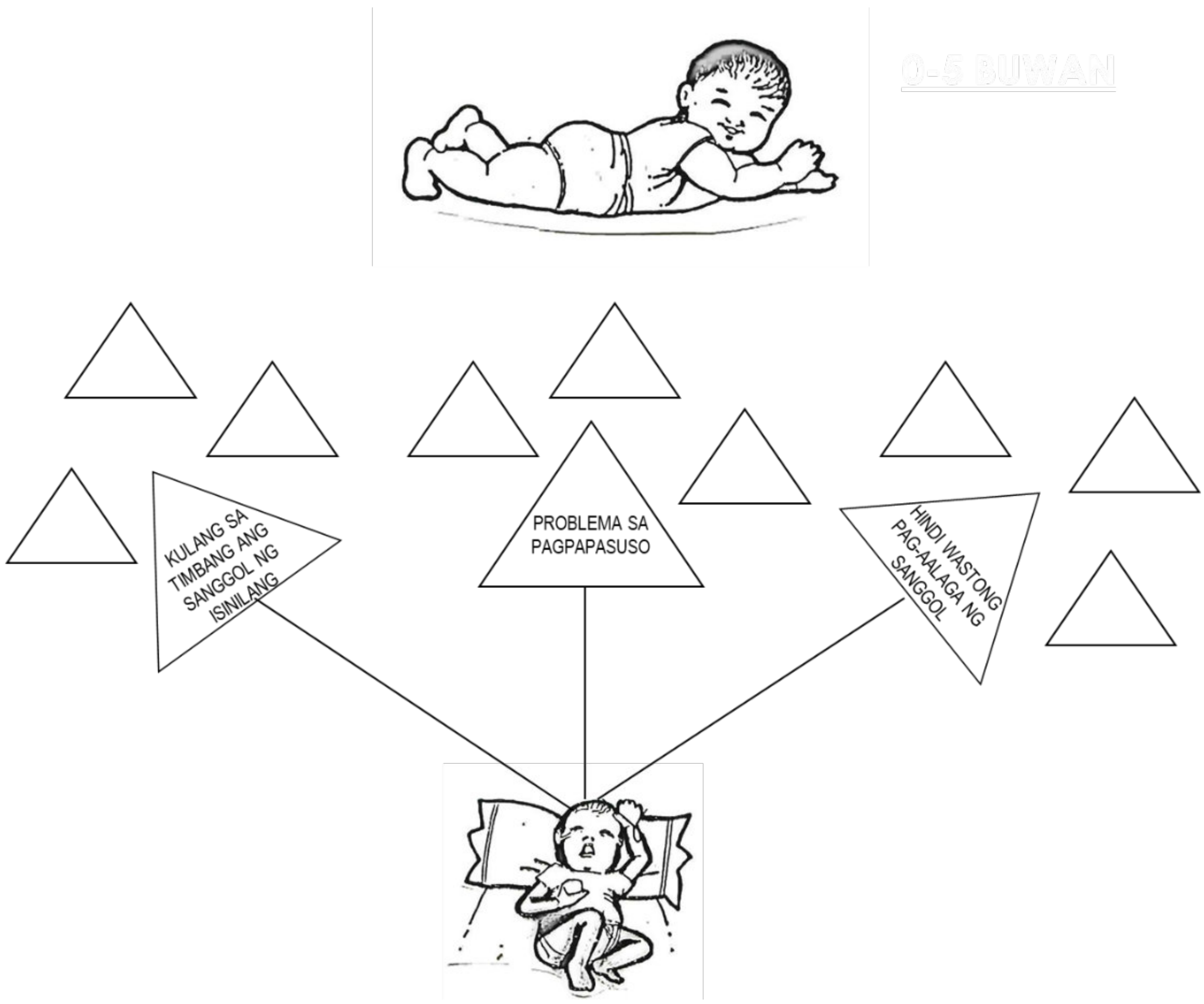

Figure 3.10. Triangle chart as an aid in identifying possible causes of malnutrition per child.

EXAMPLE 1:

A child's weight-for-age is between the -1 and -2 z-score lines, as it has been for the last three visits. The health worker shows the parent/caregiver the measurements and says, "Your child is lighter than most children of his age, but he is growing consistently. Notice how his growth line has stayed between these two lines. His weight has increased with his age, and this is good."

EXAMPLE 2:

The health worker shows the parent/caregiver the table and says, "You may have noticed that your child has become very thin. We will talk about what to do to help her gain weight. Do not worry; there are ways to help her grow."

However, if the health worker observed any of the following severe undernutrition problems, refer them immediately for urgent care:

1. Moderately wasted (MW/MAM)
2. Severely wasted (SW/SAM)

Whenever there is a need to refer a child, explain to the parent/caregiver the reasons for the referral and stress its importance.

Steps on how to explain the reasons for the referral:

1. Explain to the parent/caregiver the child's nutritional status is MAM or SAM.
2. Tell them the consequences if their child will not be referred immediately to the health center.
3. Inform them about the possible interventions that will be given to the child.

NOTE 3.6.
Assistance for Families of Malnourished Children

Sometimes the parents or caregivers refuse to bring their child to the health center because it is far or they lack the financial resources. The barangay can give a certification of indigency and ask financial support from the City/Municipal Social Welfare and Development (C/MSWD) while the barangay can help facilitate or arrange for transportation. Also, help arrange for care of other kids or family members.

For Children with Disability (Chronic conditions)

The Convention on the Rights of Persons with Disabilities (CRPD) defines people with disability as "those who have long-term physical, mental, intellectual or sensory impairments which in interaction with various barriers may hinder their full and effective participation in society on an equal basis with others." In case a child has a physical disability that makes him/her difficult or impossible to weigh and measure, the OPT Plus Team should still record the child's information whether the weight and length/height measurements were collected or not. Regardless of the type of disability, the OPT Plus Team should refer the child to the local health office and other specialized services.

Section 3.10

WHO CHILD GROWTH STANDARDS (CGS)

What is the WHO Child Growth Standards (CGS)?

The WHO Child Growth Standards (WHO-CGS) represents how children should grow. It provides a range of globally accepted prescriptive measures considered as normal growth for children less than five (5) years old. The standards were based on the WHO Multicentre Growth Reference Study (MGRS) where children from six (6) countries are receiving recommended feeding and care were included (WHO, 2008a).

NOTE 3.7.
Adoption of WHO-CGS in the Philipines

The country adopted the use of WHO-CGS in 2008. Since then, it has been used in determining the nutritional status of Filipino children. These standards are also being used in conducting OPT Plus and as the basis for the development of the eOPT Plus Tool. See Annex 3. NNC Governing Board Resolution No. 2, Series of 2008 Adoption of the New WHO Child Growth Standards for Use for Children 0-to-59-month-old in the Philippines.

The WHO-CGS currently has 12 indices used to determine the child's nutritional status. OPT Plus uses three of these indicators: length/height-for-age, weight-for-age, and weight-for-length/height.

1. **Length/Height-for-Age (L/HFA)** measures linear growth among children compared to their age. Children who are considered short for their age are classified as stunted. Stunting is a condition that manifests long-term or chronic malnutrition.
2. **Weight-for-Age (WFA)** measures the child's weight compared to their age. Unlike the HFA/LFA and WFH/WFL, this indicator does not distinguish between chronic or acute malnutrition. Children who are too light for their age are classified as underweight. Children can be underweight for their age because they are stunted, wasted, or both.
3. **Weight-for-Length/Height (WFL/H)** measures the child's weight compared to their length or height. Children who are too thin and of low stature can be classified as wasted. Wasting is a condition that reflects the acute or relatively recent onset of malnutrition. Overweight and Obese are defined as abnormal or excessive fat accumulation that may impair health (WHO, 2021).

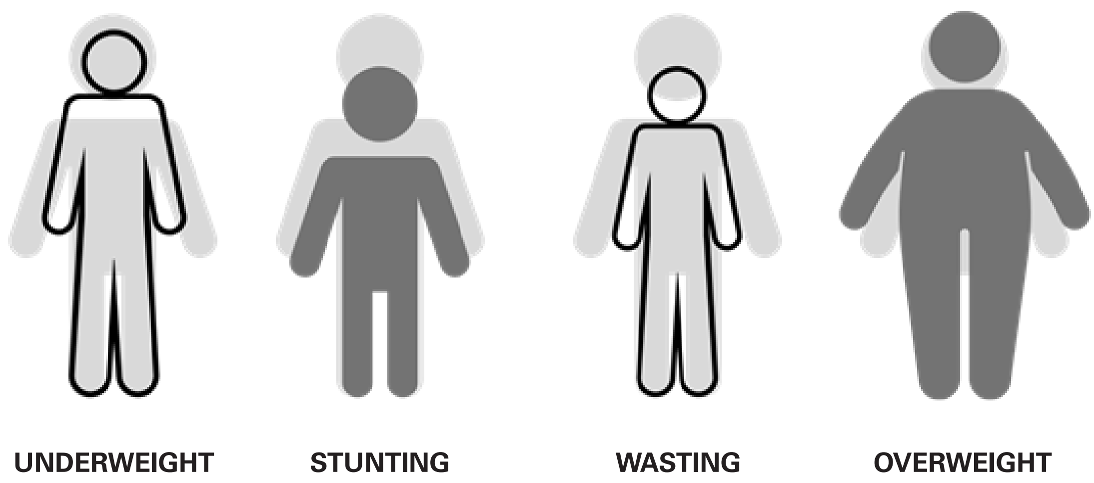

Figure 3.11. Illustrations of different forms of malnutrition.

How to determine the nutritional status of a child using the WHO-CGS?

The WHO-CGS uses anthropometric z-scores in determining the nutritional status of a child. A **Z-SCORE** compares the growth of a child through anthropometric measurements to the median value of the reference population having the same sex or age (FANTA, 2018). A child's z-score that falls below or beyond the normal range suggests a nutritional problem and the farther it is from the normal range, the more severe the problem is. A z-score can be determined using tables, charts, and/or computed with electronic tools or software.

Moreover, cutoffs are used to better make sense of a child's z-score. **CUTOFFS** are thresholds or ranges of values that correspond to classify the nutritional status of the child. WHO-CGS cutoffs for WFA, L/HFA, and WFL/H are presented in Table 3.3.

Table 3.3. Z-score cutoff points for classifying nutritional status using L/HFA, WFA, and WFL/H indicators (FANTA, 2018; WHO, 2008b).

INDICATOR	Z-SCORE CUT-OFF POINTS				
	<-3 SD	≥-3 TO <-2 SD	≥-2 TO ≤+2 SD	>+2 TO ≤+3 SD	>+3 SD
Length/Height-for-Age (L/HFA)	Severely Stunted	Moderately Stunted	Normal	Tall	
Weight-for-Age (WFA)	Severely Underweight	Moderately Underweight	Normal	Overweight	Obesity
Weight-for-Length/ Height (WFL/H)	Severely Wasted/SAM	Moderately Wasted/MAM	Normal	Overweight	Obesity

In determining the nutritional status of a child through the z-scores, age is a crucial data. Studies have shown that the mere estimation of age leads to inaccurate determination of nutritional status. During the previous years, tools such as the old eOPT Plus (v1) and the WHO-CGS Tables use the age in months to identify nutritional status. As part of the 2019 recommendations of WHO and UNICEF, age must be calculated by days for more precise values. Hence, age in days will be used in computing for nutritional status through length/height-for-age and weight-for-age.

By using any of these three tools that computes through age in days, nutritional status can be determined easily:

If with immediate access to a computer:	Electronic Operation Timbang (eOPT) Plus tool Version 2 (v2)
If with immediate access to a smartphone:	National Nutrition Information System (NNIS) mobile application
If without immediate access to a computer or mobile phone:	Growth Chart in the Early Childhood Care and Development (ECCD) Card

Section 4 will discuss the guidelines on the use of these tools. Due to the required precision when computing for age in days, digital tools must be used for efficiency and accuracy. The eOPT Plus Version 2 or the NNIS may be used for this.

MID-UPPER ARM CIRCUMFERENCE

MUAC is one of the common anthropometric measurements used in assessing the nutritional status of a child. It is a simple and fast way to determine whether the child is acutely malnourished.

SUPERVISOR'S BOX 3.1.

For more information on WHO-CGS, please visit this website: https://www.who.int/tools/child-growth-standards

For more information of WHO Multicentre Growth Reference Study (MGRS), please visit https://www.who.int/tools/child-growth-standards/who-multicentre-growth-reference-study

Table 3.4. MUAC cut-off points and classification.

AGE GROUP	NORMAL	WASTED	
		MODERATELY WASTED (MW)/MODERATE ACUTE MALNUTRITION (MAM)	SEVERELY WASTED (SW)/ SEVERE ACUTE MALNUTRITION (SAM)
6-59 months	≥ 12.5 cm	≥11.5 cm to <12.5 cm	<11.5 cm

Source: DOH-AO 2015-005

NOTE 3.8.
MUAC vs WFL/H in Assessing Wasting in 6-to-59-month-old Children

MUAC and WFL/H can both be used to identify wasted or acute malnutrition in children 6-to-59-month-old. A child that was initially classified as SW/SAM using MUAC should be monitored using MUAC after referring for treatment until discharge from the PIMAM Program.

BILATERAL PITTING EDEMA

Bilateral Pitting Edema also shows manifestations of malnutrition. When a child is severely wasted, signs of bilateral pitting edema or simply edema manifest physically. Edema is the presence of excess fluid in the child's tissues due to severe muscle wasting. It is characterized by a swollen and puffed-up appearance usually found at the child's feet that leaves a dimpled mark when pressed down. Children with edema are always classified as SAM. They are at higher risk of mortality, and in need of urgent medical attention. Edema may also be observed in other parts of the child's body such as the face and hands. The severity of edema can be classified into three (3) grades, as shown in Table 3.2.

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ANNEX 3.1. OPT Plus Form 1A Recording Form

Purok/ Area/ Block: Sampaguita	Barangay: Batong Malake	Municipality: LOS BAÑOS	Year of Last OPT Plus: 2022	Date of the Next OPT Plus Visit:
			IMPORTANT INFO ON DATE FORMAT	

Please add the names and details of new or unlisted children at the end of this list.

Note: For new children, just get the birthdate. Leave this space

OF PAGES FOR PRINTING: 0

Before printing, please read printing instructions (click on the link). PRINTING INSTRUCTIONS

Date printed: Dec 15, 2022

Child Seq. (Previous OPT)	Address (Purok or Location in the Barangay)	Name of Mother or Caregiver (SURNAME, FIRST NAME)	Child's Full Name (SURNAME, FIRST NAME)	Belongs to IP Group? YES/NO	Sex M/F	Date of Birth (MM/DD/YYYY)	Actual Date of Measurement (MM/DD/YYYY)	WEIGHT (kg)	HEIGHT (cm)	Age in Months *as of:*	Nutritional Status (Weight for L/Ht)	MUAC (cm)	Bilateral Pitting Edema NONE, +1, +2, +3	Disability (if any) YES/NO	Signature of Mother or Caregiver

ANNEX 3.2. Sample Data Privacy Notice and Informed Consent Form

SAMPLE DATA PRIVACY NOTICE AND CONSENT FORM

Privacy Notice

Information Collected
Collected information through the OPT Plus including name of child, name of parent/caregiver, address, birthdate, weight, height, and MUAC will be processed and stored for use of nutrition and health workers and Barangay Nutrition Committee.

Use of Information
Information collected will be stored and used by nutrition and health workers from government, non-government sectors, or any individual or organization. Collected information will primarily be used as baseline information on the nutritional status of children 0-to-59 months old in barangays. Information will also be used in local nutrition planning and prioritization of areas for nutrition interventions.

Information Sharing
Data collected will be collated and submitted from the barangay to the national level. This means that personnel at different government levels will have access to the data collected from you and your child. Other government agencies, the non-government sector, or any individual or organization may also be granted access upon approval of the National Nutrition Council (NNC).

Your Rights
You have the right to ask questions regarding the collection and use of information through the assigned health or nutrition worker who will be collecting the above mentioned information from you and your child.

Privacy Consent
I have understood the purpose of OPT Plus and the implication of agreeing with the process of collecting information from me and my child. The questions I have raised were also answered adequately. Hence, I hereby authorize collection and use of personal and anthropometric data from me and my child.

Signature over Printed Name Date

ANNEX 3.3. NNC Governing Board Resolution No. 3, Series 2018 Approving the Guidelines on the Selection of Non-Wood Height and Length Measurement Tool

Republic of the Philippines
Department of Health
NATIONAL NUTRITION COUNCIL

National Nutrition Council Governing Board
Resolution No. 3, Series 2018

APPROVING THE GUIDELINES ON THE SELECTION OF NON-WOOD HEIGHT AND LENGTH MEASURING TOOL

WHEREAS, the Philippines adopted the WHO Child Growth Standards (WHO-CGS) for determining the nutritional status of Filipino children under five years old;

WHEREAS, the WHO-CGS includes measurement of length and height of children to more accurately allow comprehensive child growth assessment;

WHEREAS, the NNC adopted the use of wooden height board in the country based on the recommendations of the World Health Organization where the first prototype of the wooden height board's specifications had been drawn and adapted;

WHEREAS, the NNC Governing Board on motion of the Department of Agriculture as Vice-Chair, instructed the Secretariat to explore other materials apart from wood for the fabrication of height board in view of the country's policy on total log ban;

WHEREAS, there is a need to field test height boards fabricated by various interest groups using other materials such as aluminum, acrylic, and plastics, a Field Trial of Alternative Height Measuring Equipment: Allenstick, Aluminum-Acrylic Height Board, Stadiometer vis-à-vis Wooden Height Board was conducted to provide evidence on accuracy, reliability, and ease of use of the other height board models; possibly replacing the wooden height board in the long-term in consideration of its weight and harm to the environment;

WHEREAS, the results of the field trial attested to the feasibility and practicability of the three (3) height boards tested made from other materials, i.e. aluminum-acrylic height board, fiberglass, and plastic, as acceptable measuring equipment for length/height of children in addition to the wooden height board with improvements as indicated in the study;

NOW THEREFORE, BE IT RESOLVED AS IT IS HEREBY RESOLVED, in consideration of the above premises, the National Nutrition Council Governing Board hereby approves the Guidelines on the Selection of Acceptable Non-Wood Height and Length Measuring Tool and use of height boards made of aluminum-acrylic, fiberglass, and plastic as acceptable measuring equipment in addition to the recommended wooden height board for assessing growth of children in the Philippines.

RESOLVED FURTHER, that the National Nutrition Council Secretariat shall ensure the dissemination of this Resolution and shall provide guidance to local government units and other agencies and organizations producing height boards, conducting length/height assessment of children and purchasing length/he24ight measuring equipment.

APPROVED this 24th day of April Two Thousand and Eighteen.

FRANCISCO T. DUQUE III, MD, MSc
Secretary of Health and Chairperson
National Nutrition Council Governing Board

Attested by:

Assistant Secretary of Health Maria-Bernardita T. Flores, CESO II
Council Secretary and Executive Director IV
National Nutrition Council

NNC Governing Board Resolution No. 3

Guidelines on the Selection of Acceptable Non-Wood Height and Length Measuring Tool

CONFORME:

EMMANUEL F. PIÑOL
Secretary of Agriculture
Vice-Chairperson, NNC Governing Board

EDUARDO M. AÑO
OIC-Secretary of the Interior and Local Government
Vice-Chairperson, NNC Governing Board

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BENJAMIN E. DIOKNO
Secretary of Budget and Management
Member, NNC Governing Board

LEONOR M. BRIONES
Secretary of Education
Member, NNC Governing Board

SILVESTRE H. BELLO III
Secretary of Labor and Employment
Member, NNC Governing Board

FORTUNATO T. DELA PEÑA
Secretary of Science and Technology
Member, NNC Governing Board

EMMANUEL A. LEYCO
OIC-Secretary of Social Welfare and Development
Member, NNC Governing Board

ERNESTO M. PERNIA
Secretary of Socio-Economic Planning and Director-General, National Economic and Development Authority
Member, NNC Governing Board

RAMON M. LOPEZ
Secretary of Trade and Industry
Member, NNC Governing Board

DR. AMADO R. PARAWAN
Health and Nutrition Advisor, Save the Children
Member, Philippine Coalition of Advocates for Nutrition Security
Member, NNC Governing Board

ROMEO C. DONGETO
Executive Director, Philippine Legislators' Committee on Population and Development
Member, NNC Governing Board

ANNEX 3.4. NNC Governing Board Resolution No. 3, Series 2018 Approving the Guidelines on the Selection of Non-Wood Height and Length Measurement Tool

Republic of the Philippines
Department of Health
NATIONAL NUTRITION COUNCIL

Guidelines on the Fabrication, Verification and Maintenance of Wooden Height Boards

Rationale

This guidepost is being issued to LGU functionaries, relevant government agencies and business entities with interest in fabrication, marketing, verification and maintenance of wooden height boards used in measuring the length of 0-23 months old and height of 24-71 months old children. This document serves as a guide in ensuring the integrity of all height boards for field use.

Coverage

This guidepost applies to wooden length/height boards, which is the length and height measuring equipment being recommended, by the Technical Working Group on Child Growth Standards (TWG-CGS), for use in both facility- and non-facility based child growth assessment. The TWG-CGS considered the WHO's recommendation on wooden height boards, an affordable measuring equipment, for which it has drawn up technical specifications for fabrication, and from which the TWG based its recommendation.

The guidepost is likewise provided for length boards for those who want to fabricate a separate length board for younger children. Illustrations of the length and height boards are attached for reference.

Materials

- Wood - Kiln dried, water-proofed light weight but sturdy wood, 1.5 cm thick with *Supplemental treatment using FPA Registered Wood Preservatives*
- Steel rule - At least 2.0 cm wide steel rule, 1.5 meter long, with engraved millimeter markings; major tick should be in centimeter and font of every 10- centimeter marks should be bigger and/or colored
- Wood screws for attaching the steel rule and pieces
- Rubber or equivalent material as stopper for the headpiece
- Handle
- Bag (optional)

Page 1 of 6

Additional note:

Advantages of properly kiln dried wood (15%-18% moisture) include:

a. Light in weight
b. Prevents dimensional changes and improves machining properties.
c. Provides good adhesion of glues and paint
d. Improves treatability and nail holding capacity
e. Resistant to decay but not to termites and beetles. Dry wood termites and other species of powder-post beetles prefer to attack wood materials even at the lowest MC of 10.0%.

Dimension (in centimeters)

<u>For height boards</u>

Assembled height of the height board: 146.5 cm

A. Head board (HHB) – movable; consists of 8 pieces of wood measuring as follows (*See illustration of head board*)

Label	**Quantity**	**Height** cm	**Length** cm	**Width** cm	**Diagonal/ Slant** **(approximate)**
HHB_A (back)	1	10.0	25.0	1.5	-
HHB_B [1/] (flat front)	1	25.0	25.0	1.5	-
HHB_C_1C_2 (outer side)	2	10.0 (at the back) 1.5 (in front)	6.5 (on top) 15.0 (at the bottom)	1.5	12.0
HHB_D_1D_2 (inner side)	2	8.5	3.5 (on top) 12.0 (at the bottom)	1.5	12.0
HHB_E [2/] (middle/handle)	1	8.5	3.5 (on top) 20.0 (at the bottom)	1.5	18.5
HHB_F (stopper)	1	3.5	15.0	2.5	-

1/ provide a thin slot measuring 4.5 cm x 0.5 cm, see illustration for the verification procedure
2/ provide a hole measuring 8 cm x 4cm which will serve as a handle (can be circular or rectangular in shape) of the headboard, see illustration

B. Back board (HBB) – consists of 3 pieces of wood measuring as follows
(*See illustration for back board*)

Label	Quantity	Height	Length	Width
HBB_A (backboard)	1	146.5	25.0	1.5
HBB_B (back top)	1	3.5	25.0	3.5
HBB_C (back bottom)	1	3.5	28.0	3.5

C. Foot board (HFB) – fixed; consists of 3 pieces of wood measuring as follows
(*See illustration of foot board*)

<table>
<tr><th>Label</th><th>Quantity</th><th>Height</th><th>Length</th><th>Width</th><th>Diagonal/ Slant</th></tr>
<tr><td>HFB_A (main foot board)</td><td>1</td><td>25.0</td><td>25.0</td><td>1.5</td><td>-</td></tr>
<tr><td rowspan="2">HFB_B_1B_2 (side)</td><td rowspan="2">2</td><td>15.0 (at the back)</td><td>6.0 (on top)</td><td rowspan="2">1.5</td><td rowspan="2">17</td></tr>
<tr><td>3.0 (in front)</td><td>18.0 (at the bottom)</td></tr>
</table>

For length boards

Assembled length of the length board: 116.5 cm

D. Head board (LHB) –fixed; consists of 3 pieces of wood measuring as follows
(*See illustration of head board*)

<table>
<tr><th>Label</th><th>Quantity</th><th>Height</th><th>Length</th><th>Width</th><th>Diagonal/ Slant</th></tr>
<tr><td>LHB_A (Head board)</td><td>1</td><td>20.0</td><td>20.0</td><td>1.5</td><td>-</td></tr>
<tr><td rowspan="2">LHB_B_1B_2 (side)</td><td rowspan="2">2</td><td>15.0 (at the back)</td><td>6.0 (on top)</td><td rowspan="2">1.5</td><td rowspan="2">17.0</td></tr>
<tr><td>3.0 (in front)</td><td>18.0 (at the bottom)</td></tr>
</table>

E. Back board (LBB) – consists of 3 pieces of wood measuring as follows
(*See illustration for back board*)

Label	Quantity	Height	Length	Width
LBB_A (backboard)	1	116.5	20.0	1.5
LBB_B (back top)	1	3.5	20.0	2.5
LBB_C	1	3.5	23.0	2.5

(back bottom)				

F. Foot board (LFB) – movable; consists of 8 pieces of wood measuring as follows (*See illustration of foot board*)

Label	Quantity	Height	Length	Width	Diagonal/ Slant (approximate)
LFB_A (back)	1	10.0	20.0	1.5	-
LFB_B [1/] (flat front)	1	20.0	20.0	1.5	-
$LFB\_C_1C_2$ (outer side)	2	10.0 (at the back)	4.5 (on top)	1.5	12.0
		1.5 (in front)	13.0 (at the bottom)		
$LFB\_D_1D_2$ (inner side)	2	8.5	1.5 (on top)	1.5	12.0
			10.0 (at the bottom)		
LFB_E [2/] (middle/handle)	1	8.5	1.5 (on top)	1.5	18.5
			18.0 (at the bottom)		
LFB_F (stopper)	1	3.5	15.0	2.5	-

1/ provide a thin slot measuring 4.5 cm x 0.5 cm, see illustration for the verification procedure

2/ provide a hole measuring 8 cm x 4 cm which will serve as a handle of the foot board, see illustration

Other specifications

1. Steel rule should be placed in the middle and a little bit embedded
2. Steel rule should be screwed to keep it in place
3. Headboard should have a slot (4.5 cm x 0.5 cm) through which the calibrated steel rule can be placed during the height board verification
4. Headpiece and footboard should be at 90^{o} angle with the back board
5. For models with detachable pieces (2-pc boards), care should be done that the pieces are properly attached and gaps in the joints are considered so as not to affect the accuracy of reading
6. Last reading should be at least 130.0 cm but not to exceed 1.0 cm for height boards.
7. Last reading should be at least 95.0 cm but not to exceed 1.0 cm for length boards.

Verification

Before a length or height board can be used, verification should be done by a DOST-trained personnel (passed the Competency Training on Verification). The points to be checked should include:

1. Correct placement and attachment of the steel rule
2. Zero placement of the steel rule at the base
3. Dimension of the board and parts
4. Quality of materials used
5. Perpendicularity of base/footboard and headpiece against the beam
6. Functionality and integrity of the unit
7. Correction factor, if any

A verified length/height board is one with a sticker of verification issued and posted at the upper right corner of the board. The certificate shall contain the date of verification, name and signature of person who verified and agency/institution, steel rule serial number (which will serve as unit number) and correction factor, if any.

Verification shall be repeated every 6 months by DOST-trained personnel.

Maintenance and Handling of the Height Board

Lifespan of wooden length/height board is 3-5 years, depending on the frequency of use and storage practices.

To maximize service life of the length/height board:

1. Use the length/height board properly according to its specific purpose.
2. Conduct regular tightness checks of all screws and bolts for safety.
3. Conduct regular visual inspection of the length/height board to check the board quality and avoid potential risk to children while in-service.
4. Do not drag during transport or transfer the board from one place to another.
5. Replace parts when necessary following the standard specifications in the fabrication of the length/height board.
6. Re-apply finishing when the length/height board in service needs supplemental finishes to retain its appearance. Excessive use can cause damage to both the finish and wood itself.
7. Let the kids stand on the board barefoot not only to obtain accurate measurements but also to avoid damage on the surface of the foot board.
8. Regularly clean with a soft, clean and dry cloth to keep the length/height board looking good.
9. Keep and store the length/height board in adequately dry area. Do not expose to water, excessive heat and direct sunlight.

The wooden board should be replaced if already showing warping or advanced deterioration. The rule should be replaced if the markings are already not legible.

Notes:

1. *Use of other materials in the fabrication of height board is still subject to review by the TWG-CGS and approval by the NNC Technical Committee.*

2. *The length/height boards with tape installed on the right side are acceptable only if these came from donations of UN organizations like UNICEF, WFP and Save the Children which were produced under the strictest guidance.*

ANNEX 3.5. NNC Memorandum No. 2020-010. INTERIM GUIDELINES IN THE CONDUCT OF OPT, GMP, AND NUTRITION SCREENING

Republika ng Pilipinas
KAGAWARAN NG KALUSUGAN
NATIONAL NUTRITION COUNCIL
Nutrition Building, 2332 Chino Roces Avenue Extension
Taguig City, Philippines

15 December 2020

MEMORANDUM 010

TO : **REGIONAL NUTRITION PROGRAM COORDINATORS OFFICERS-IN-CHARGE**

FROM : **AZUCENA M. DAYANGHIRANG, MD, MCH, CESO III**
Executive Director

SUBJECT : **INTERIM GUIDELINES IN THE CONDUCT OPT Plus, NUTRITION SCREENING, GROWTH MONITORING AND PROMOTION (GMP) ACTIVITIES IN THE CONTEXT OF COVID-19 PANDEMIC AND OTHER RELATED DISASTERS**

This is to provide an interim guidelines in the conduct of OPT, nutrition screening and growth monitoring and promotion activities in the context of COVID19 and other related disasters. The guidelines is in line with the provisions of DOH-Department Circular 2020-0167 Continuous Provision of Essential Health Services during the COVID-19 Epidemic and DOH-Department Memorandum 2020-0237 Interim Guidelines for the Delivery of Nutrition Services in the Context of COVID19 Pandemic under Growth and Development Monitoring and Promotion. The issuances provide for these activities to "still be done during health facility visits, community outreach and if the situation allows, strict observance of infection prevention and control (IPC) measures".

DOH-Department Memorandum 2020-0237 suggests the use of MUAC tapes instead of the weight and length/height measuring tools to adapt to a reduced physical contact approach of health care provider to infant or child. However, in the absence or insufficient MUAC tapes, the Global Nutrition Cluster cited key considerations for in-person interactions during data collection[1]:

1. Members of the OPT Team should thoroughly disinfect the weight and height/length measuring tools, MUAC tapes, and other frequently-touched surfaces and objects using diluted household bleach solutions or alcohol solutions with at least 70% alcohol after every use and between measurements.

[1] Global Nutrition Cluster (GNC) Nutrition Information Management, Surveillance and Monitoring in the Context of COVID-19. Brief No 1 (14 April 2020)

Batang Pinoy SANA TALL... Iwas stunting, SAMA ALL!
Iwas ALL din sa COVID-19!

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Makati City
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(63-2) 8843-5856 • 8843-5834 • 8816-4239
(63-2) 8818-7398 • 8892-4271
Fax No. (63-2) 8843-5818
www.nnc.gov.ph info@nnc.gov.ph www.facebook.com/nncofficial www.youtube.com/user/NNC1974

Wastong Nutrisyon: Alamin, Gawin at Palaganapin

2. between measurements. Refer to DOH-Department Memorandum 2020-0157 for the guidelines on cleaning and disinfection in various settings as an IPC measure against COVID-19.

3. Ensure physical safe distance between the mothers/caregivers and their children with the members of the OPT Team until their measurements can be taken. Train members of the OPT Team on how to practice IPC measures when taking measurements.

4. Ensure members of the OPT Team wear face masks, face shield and gloves when taking measurements. Mothers/caregivers should also wear face masks and face shields. Strictly follow the latest guidelines of COVID-19 Inter-Agency Task Force (IATF) for the Management of Emerging Infectious Diseases.

5. Re-consider positioning of measurer when reading the measurements, e.g., read from behind the individual for MUAC measurement to potentially reduce risk of droplet exposure

6. Members of the OPT Team can also consider using this opportunity to teach mothers/caregivers to assess nutritional status of their children using mid-upper arm circumference (MUAC); the use of MUAC tapes (color zones); potential outcomes and treatment; as well as how to assess for bilateral pitting edema. Use the video on how to measure MUAC downloadable from the NNC website: https://www.nnc.gov.ph/covid19/im/muac

Measurement schedule, recording and reporting of OPT Plus, nutrition screening, and growth monitoring

1. Conduct of OPT Plus using traditional weight and height measuring tools should be continued as basis to provide interventions to prevent deterioration of nutritional status. This should be completed within the first quarter of the year (January to March).

   Use the electronic OPT tool downloadable from the NNC Website https://www.nnc.gov.ph/?view=article&id=3461 to facilitate recording and reporting.

2. Process and submit results to NNC Central Office using the prescribed formats: Form 3 [for breakdown by municipality] and Form 4 [for breakdown by city].

3. If situation does not allow the use of traditional tools, use MUAC to assess nutritional status of children and recorded results in the e-OPT tool for Acute Malnutrition, downloadable from the NNC website: https://www.nnc.gov.ph/optacute.

4. The e-OPT Tool for Acute Malnutrition is similar to the structure and method of encoding, consolidation, and auto-generation of reports as the original e-OPT tool. Additional features of the tool include filterable list of MAM and SAM based

on MUAC and weight-for-length/height and recording presence of bilateral pitting edema.

5. Refer to **Annex 1**: Decision Tree for the LGUs on the conduct of OPT Plus, nutrition screening and growth monitoring and promotion activities to determine the method of measurement, tools for measurement, and recording and reporting tool.

6. Growth monitoring and promotion activities should also be continued and reported: monthly for children under 2 years old and those identified as malnourished based on OPT; and quarterly for children 2 to under 5 years old, regardless of nutritional status.

7. MUAC of children aged 6 to 23 months with normal nutritional status and no bilateral pitting edema should be measured monthly, and quarterly for 24 to 59 months old. If the child has normal MUAC but has bilateral pitting edema, immediately refer to the nearest barangay health center or rural health unit for further assessment and appropriate management.

8. While undergoing appropriate management, MUAC of children identified with severe acute malnutrition(SAM) or severely wasted shall be measured every 2 weeks, and moderate acute malnutrition (MAM) or moderately wasted without bilateral pitting edema every month until MUAC is greater than 12.5cm for two consecutive measurements. Monthly reports should be generated at the city/municipal level through the health officer and the nutrition action officer and reported to the local nutrition committee.

9. Immediately refer children identified as acutely malnourished (moderately of severely wasted) to the nearest barangay health center/rural health unit for further assessment, and appropriate and timely management. Validation of nutritional status by other health professional should not delay appropriate care and management, especially for children identified as SAM or severely wasted, as their condition can easily and quickly deteriorate. Moderately acutely malnourished children (moderately wasted) should be referred to targeted supplementary feeding programs in child development centers, health/nutrition centers, in schools, or where available.

10. Relay the measurements of the child and provide feedback and counselling to mothers and caregivers of the children measured. Reinforce good behaviors and respectfully correct any misconceptions or wrong practices.

11. Identify viable or preferred medium for communication with parents/caregivers for regular follow up of children, e.g., SMS, Facebook messenger and other available virtual platforms and during home visits.

Refer to **Annex 2** for the Scenarios at the Local Government Units and Nutrition Screening, Growth Monitoring and Promotion Activities.

For information and wide dissemination.

ANNEX 2 – Scenarios at the Local Government Units and Nutrition Screening, Growth Monitoring and Promotion Activities

<table>
<tr><td rowspan="3">Scenario (based on DOH-DM No. 2020-0237)</td><td colspan="5">Nutrition Screening, Growth Monitoring and Promotion Activities</td></tr>
<tr><td colspan="3">Normal Nutritional Status</td><td rowspan="2">SAM children</td><td rowspan="2">MAM children</td></tr>
<tr><td>0 to <6 months old</td><td>6 to 23 months old</td><td>24 to 59 months old</td></tr>
<tr><td rowspan="3">Scenario 1-2 during Recognition Phase Stage 1

Zero Cases or importation</td><td>Weight-for-length</td><td>Weight-for-height or MUAC

Bilateral pitting edema</td><td>Weight-for-height or MUAC

Bilateral pitting edema</td><td>• Weight-for-length for less than 6 months old
• Weight-for-height or MUAC for 6 to 59 months old</td><td>• Weight-for-length for less than 6 months old
• Weight-for-height or MUAC for 6 to 59 months old</td></tr>
<tr><td colspan="2">Monthly monitoring</td><td>Quarterly monitoring</td><td>Monitoring every 2 weeks</td><td>Monthly monitoring</td></tr>
<tr><td colspan="5">Follow-up through virtual platforms, home visits, and primary health care facilities</td></tr>
<tr><td rowspan="3">Scenario 3 Initiation Phase

Stage 2 Localized Transmission

Health and nutrition staff are engaged in COVID-19 response</td><td>Weight-for-length</td><td>MUAC

Bilateral pitting edema</td><td>MUAC

Bilateral pitting edema</td><td>• Weight-for-length for less than 6 months old
• MUAC for 6 to 59 months old
• Bilateral pitting edema</td><td>• Weight-for-length for less than 6 months old
• MUAC for 6 to 59 months old</td></tr>
<tr><td colspan="2">Monthly monitoring</td><td>Quarterly monitoring</td><td>Monitoring every 2 weeks</td><td>Monthly monitoring</td></tr>
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<tr><td rowspan="3">Scenario (based on DOH-DM No. 2020-0237)</td><td colspan="5">Nutrition Screening, Growth Monitoring and Promotion Activities</td></tr>
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<tr><td>0 to <6 months old</td><td>6 to 23 months old</td><td>24 to 59 months old</td></tr>
<tr><td rowspan="3">Scenario 4 During Acceleration Phase

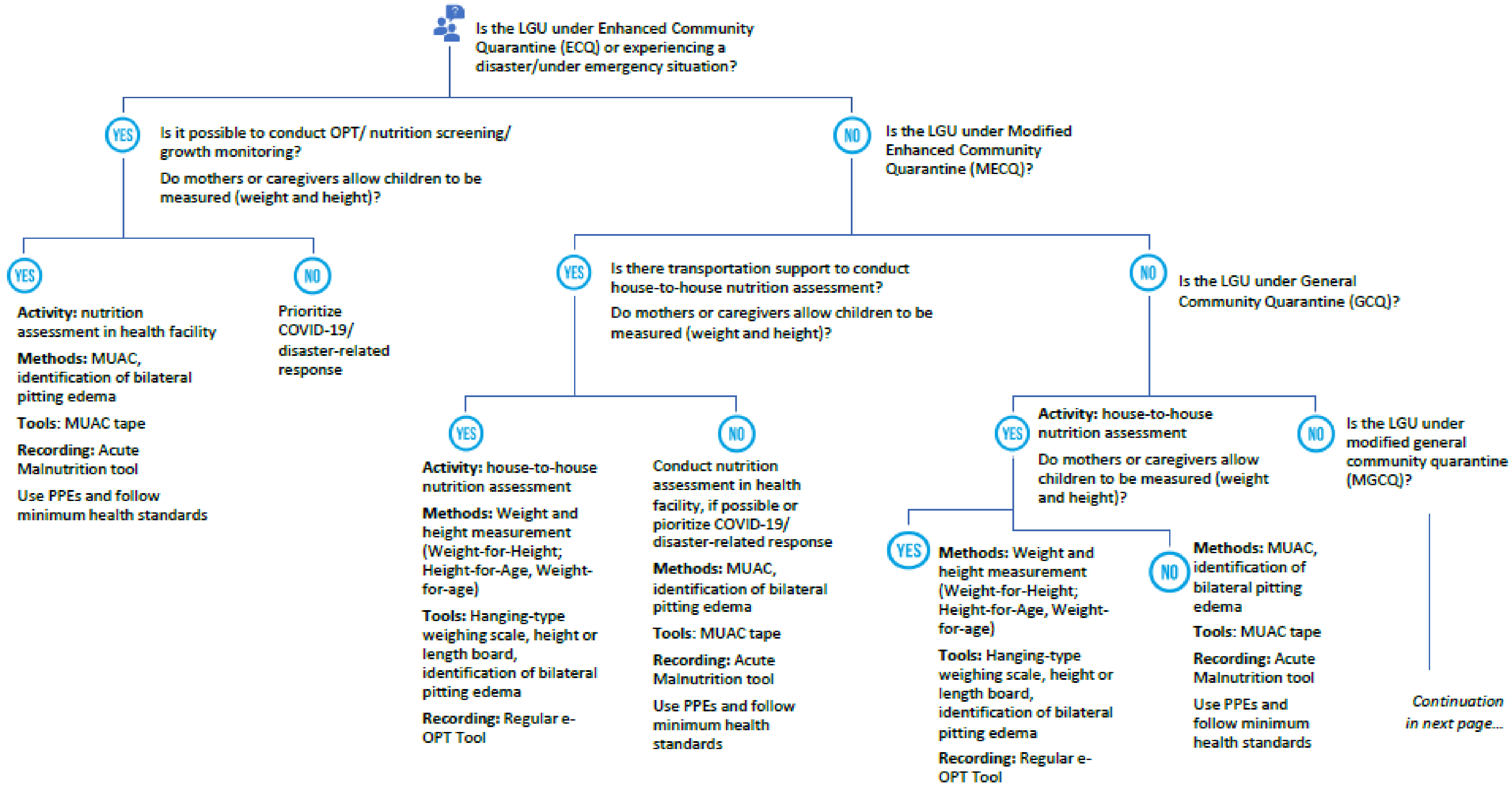
Stage 3 Community Transmission

All health and nutrition staff and other health office personnel are engaged in COVID-19 response</td><td>Weight-for-length</td><td>MUAC

Bilateral pitting edema</td><td>MUAC

Bilateral pitting edema</td><td>• Weight-for-length for less than 6 months old
• MUAC for 6 to 59 months old
• Bilateral pitting edema</td><td>• Weight-for-length for less than 6 months old
• MUAC for 6 to 59 months old</td></tr>
<tr><td colspan="2">Monthly monitoring</td><td>Quarterly monitoring</td><td>Monitoring every 2 weeks</td><td>Monthly monitoring</td></tr>
<tr><td colspan="5">• Visits to non-emergency concerns to primary health care facilities are suspended.
• Follow-up through virtual platforms and home visits</td></tr>
</table>

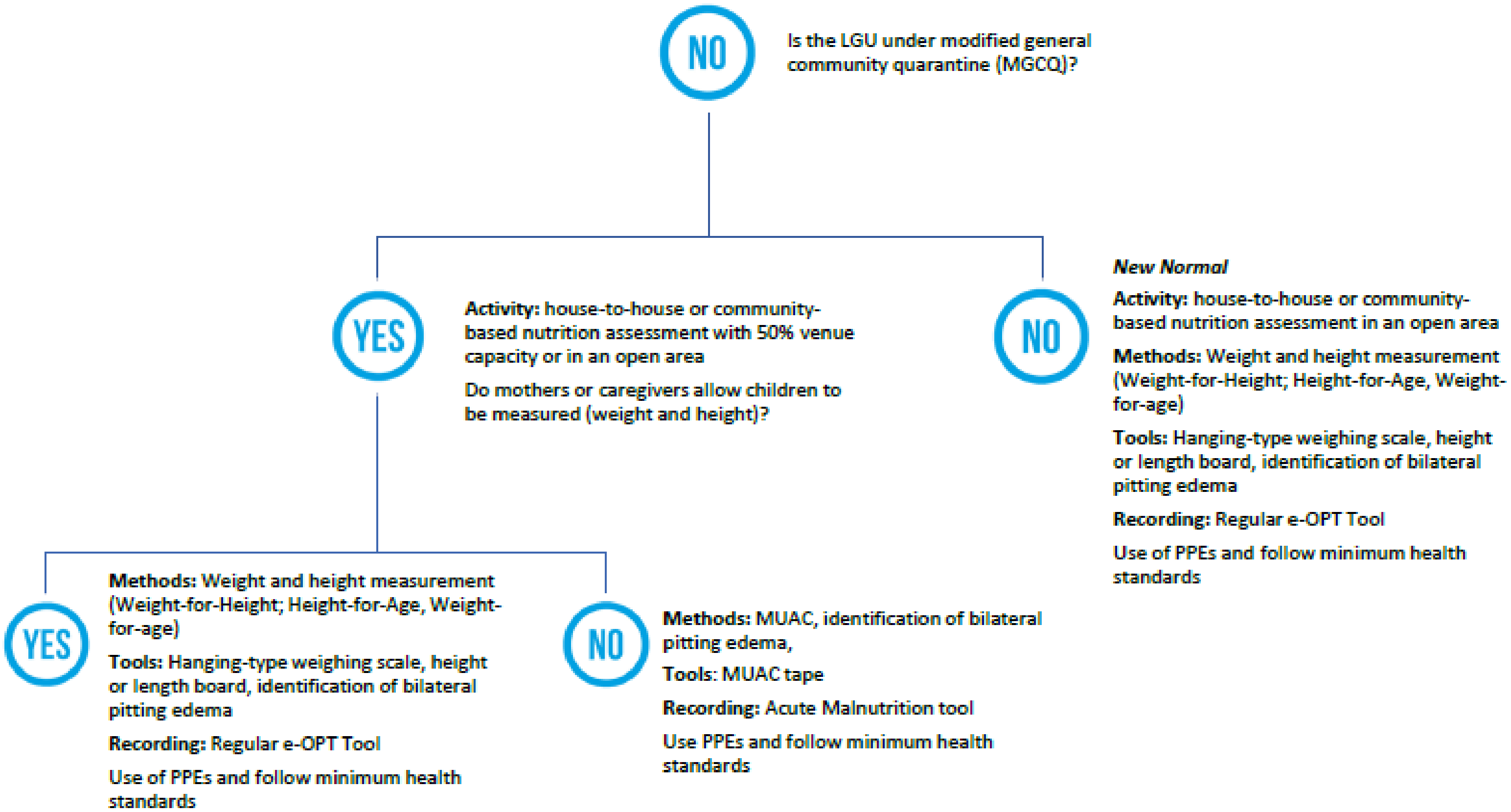
ANNEX 1 – Decision Tree for Local Government Units on the conduct of Operation Timbang, Nutrition Screening, Growth Monitoring and Promotion Activities in the Context of Pandemic and Other Related Disasters



Continuation in next page...

*Use only one assessment method for the whole barangay and municipality (either traditional weight and height, or MUAC)

...continuation of Annex 1



*Use only one assessment method for the whole barangay and municipality (either traditional weight and height, or MUAC)

OPT Plus TSEK

Operation Timbang Plus

Tamang Sukat, Eksaktong Kalidad!

Section Four

OPT PLUS PROTOCOL ON DATA PROCESSING, ENCODING, AND REPORT GENERATION

OPT PLUS PROTOCOL ON DATA PROCESSING, ENCODING, AND REPORT GENERATION

OVERVIEW

The general objective of Section 4 is to offer information on how to process the OPT Plus data. It also provides knowledge in determining the nutritional status of the children using the Electronic Operation Timbang Plus Tool (eOPT Plus Tool), using the National Nutrition Information System (NNIS) and using the World Health Organization Child Growth Standards (WHO-CGS) Growth Chart for rapid assessment.

After reading this section, one should be able to:

1. Enumerate the roles and responsibilities of each person involved in the OPT Plus;
2. Be familiar with the basic eOPT Plus Tool functions including encoding and generating reports; and
3. Determine the nutritional status using the eOPT, NNIS, and WHO-CGS Growth Chart.

This section covers the post-processing of anthropometric measurements, electronic and manual methods. Sections include the following:

Section 4.1	Roles of the OPT Plus Team and Supervisors
Section 4.2	The Electronic Operation Timbang Plus Tool (eOPT Plus Tool)
Section 4.3	Guidelines on Determination of Nutritional Status

LIST OF ACRONYMS

BNC	Barangay Nutrition Committee
C/MNAO	City/Municipal Nutrition Action Officer
eOPT Plus	Electronic Operation Timbang Plus
FAO	Food and Agriculture Organization of the United Nations
L/HFA	Length/Height-for-Age
LGU	Local Government Unit
M/C	Mother/caregiver
MAM	Moderate Acute Malnutrition
MSt	Moderately Stunted
MUAC	Mid-Upper Arm Circumference
MUW	Moderately Underweight
MW	Moderately Wasted
N	Normal
NNC	National Nutrition Council
NNC CO	National Nutrition Council Central Office
Ob	Obese
OW	Overweight
PNAO	Provincial Nutrition Action Officer
RNPC	Regional Nutrition Program Coordinator
SAM	Severe Acute Malnutrition
SSt	Severely Stunted
SUW	Severely Underweight
SW	Severely Wasted
UNICEF	United Nations Children's Fund
UPLB	University of the Philippines Los Baños
WFA	Weight-for-Age
WFL/H	Weight-for-Length/Height
WHO-CGS	World Health Organization Child Growth Standards
yyyy-mm-dd	Year-month-day

DEFINITION OF TERMS

eOPT Plus Tool	An Excel-based tool used by nutrition workers in the community to help them consolidate and summarize the results of OPT Plus among 0-to-59-months-old children.
Length/Height-for-Age (L/HFA)	Indicator used to classify stunting among 0-to-59-months-old children.
Weight-for-Age (WFA)	Indicator used to classify underweight among 0-to-59-months-old children.
Weight-for-Length/Height (WFL/H)	Indicator used to classify wasting, overweight, and obesity among 0-to-59-months-old children.
Child Growth Standards Table	Represents how children should grow through globally accepted prescriptive measures considered as normal growth of all 0-to-5-year-old children.

Section 4.1

ROLES AND RESPONSIBILITIES OF THE OPT PLUS TEAM AND SUPERVISORS

It is important to understand the roles and responsibilities of each person involved in implementing the OPT Plus encoding and report generation. Table 1 presents the roles and responsibilities of the persons involved in the activity.

Table 4.1. Roles and responsibilities of persons involved in the OPT Plus encoding and report generation.

ORGANIZATION/PERSON INVOLVED	FUNCTION
National Nutrition Council Central Office (NNC CO)	1. Provide technical assistance at the national level in using the eOPT Plus Tool. 2. Document issues and concerns encountered in the tool for future improvements.
National Nutrition Council Regional Office/ Regional Nutrition Program Coordinator (RNPC)	1. Provide technical assistance at the regional level in using the eOPT Plus Tool. 2. Submit reports to the NNC CO on issues and concerns encountered in the tool for future improvements.

ORGANIZATION/PERSON INVOLVED	FUNCTION
Provincial Nutrition Office/Provincial Nutrition Action Officer (PNAO)	1. Provide capacity building in using the eOPT Plus Tool.
Supervisors	1. Assist in encoding the anthropometric measurements in the eOPT Plus Tool. 2. Review the outputs of the BNS using the eOPT Plus. 3. Endorse the OPT Plus reports to the barangay. 4. Consolidate all barangay eOPT Plus into the municipal level eOPT Plus tool.
City/Municipal Nutrition Action Officer (C/MNAO)	1. Conduct a data quality check of the eOPT Plus Tool.
OPT Plus Team Members	1. If the team has access to a computer or laptop, use the eOPT Plus Tool to automatically compute children's age in days and nutritional status. If not possible, guide the encoder and review the encoded data. Make sure to get a soft copy of the eOPT Plus Tool from the official NNC website www.nnc.gov.ph. 2. Submit the OPT Plus reports to the C/MNAO and Midwife.
Barangay Nurition Committee (BNC)	1. Review the submitted OPT Plus reports and endorse them to the C/MNAO.

Section 4.2

THE ELECTRONIC OPT (EOPT) PLUS TOOL VERSION 2 (V2)

THE ELECTRONIC OPT TOOL (eOPT TOOL)

The eOPT Tool calculates the nutritional status of 0-to-59-month-old children using reference tables based on the WHO Child Growth Standards. It is an excel-based program created to improve efficiency and accuracy in determining nutritional status and reporting the OPT Plus data. With the use of the tool, barangays and cities/municipalities can better manage their data while ensuring accuracy, timeliness, and utilization of OPT Plus data in the communities. Some key features of the eOPT Tool include the following:

- Automated age calculation
- Automated classification of nutritional status
- Flagging of children with age >59 months
- Flagging of too low or too high encoded values for weight, length or height
- Error-checking features (i.e., duplicate entries, same birthdate)
- Simplifies data consolidation through data export
- Simplifies tabulation of results and report generation
- Print-ready forms

Overview of the Electronic OPT Plus Tool (eOPT Plus Tool) v2

The eOPT Plus Tool is used by nutrition workers in the community to help them record and summarize the results of OPT Plus among 0-59-month-old children. The eOPT Plus Tool runs on MS Excel version 2007 or later. Full functionality of the tool requires a licensed version of MS Excel.

The eOPT Plus tool is a spreadsheet used to automatically generate the nutritional status of children. The validity of the nutritional assessment depends on the accuracy and reliability of the encoded child's weight, length, height, mid-uppar arm circumference (MUAC), sex, date of birth, and measurement date. The nutritional status is determined using three indices: weight-for-age (WFA), height-for-age (HFA), and weight-for-length/height (WFL/H), or in the case where the city/municipality decided not to use the weight, length, and height, MUAC and edema can be used. The nutritional status is classified using the World Health Organization (WHO) Child Growth Standards (CGS) z-scores cut-off.

Aside from the automation features of the tool, it also lessens the chance of measurement errors which should be avoided to achieve reliable nutritional assessment results. The eOPT Plus Tool was developed specifically for barangay, city/municipal, provincial, and regional levels. It can be downloaded from the National Nutrition Council's (NNC) Website. The tables and reports that this tool provides are discussed in Section 5.

The previous version (v1) of the eOPT Plus tool was developed by the NNC together with the Food and Agriculture Organization of the United Nations (FAO). It was updated by the national project team composed of experts from the NNC, University of the Philippines Los Baños (UPLB), and United Nations Children's Fund (UNICEF) Philippines. The updating of the tool into the eOPT Plus v2 resulted to the use of age in days and the nearest 0.1 cm when computing the nutritional status for a more accurate and precise results, especially among those with borderline measurements and age in eOPT v1.

Change in Nutritional Status Computation

Unlike the previous versions of the eOPT Plus (v1) which computes for the nutritional status using age in months for weight-for-age and length/height-for-age; and using the nearest 0.5 cm for weight-for-length/height, the eOPT Plus v2 used a more accurate approach as stated in the comparison below:

	Version 1	Version 2
Weight for Age Length/Height for Age	Age-in-months	Age-in-days
Weight for Length/Height	nearest 0.5 cm	nearest 0.1 cm

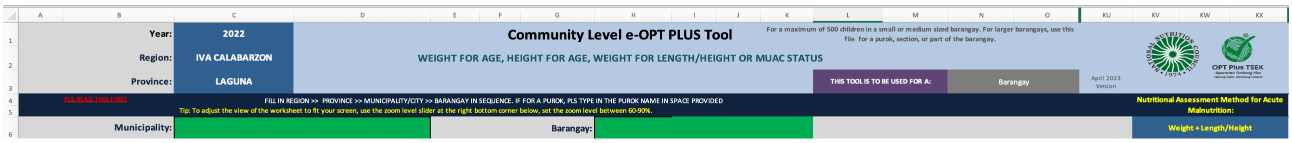
How to encode the OPT Plus data in the eOPT Plus Tool?

1. Fill the necessary information in the 'Nut_Statustool' headings such as the year, region, province, municipality, and barangay. The entry worksheet will not appear if the information is blank.

    - Type in the YEAR (cell C1). Choose first the PROVINCE/LGU, then the CITY/MUNICIPALITY, and lastly the name of the BARANGAY from the dropdown lists in each corresponding cell. Please fill these out in sequence.

    - If the message "Note: Large Barangay" appears in Cell L3, you should use this file for a Purok or smaller part of the barangay. Please type in the name of the Purok or sub-part of the barangay in Cell N6.

2. Choose from the dropdown list in Cell KV6 which Nutritional Assessment Method (Weight+Length/Height or MUAC) will be used during the OPT Plus. If Weight+Length/Height was chosen as a method, the 'MUAC' and 'MUAC Status' columns will be automatically blocked, and vice versa.



3. Refer to the paper forms (i.e., NNC Form 1A) that you used to record details of preschool children from your last OPT Plus activity. Type in all the required data for each child into the 'Nut_StatusTool' worksheet one at a time. One child, one row. Enter all the information for each child in the table. Do not skip rows - it will cause your OPT Plus Form 1A to not work properly.

    - You can then begin entering information for each child in the main table -- one child at a time – starting from the first blank row (Row # 10). Do not skip rows or leave any rows blank - this will cause your OPT Plus Form 1A recording sheet to not work properly.

4. For the 'Address or Location': make sure to specify the house street, purok/sitio or landmarks

5. For the 'Name of Mother or Caregiver': make sure to follow the format: SURNAME, FIRST NAME SUFFIX (e.g., DELA CRUZ, JUAN or DELA CRUZ, JUAN JR.)

6. For the 'Full Name of Child': make sure to follow the format: SURNAME, FIRST NAME SUFFIX (e.g., DELA CRUZ, JUANITO or DELA CRUZ, JUANITO JR.)

7. For the 'Belongs to the IP Group?': from the dropdown list, choose YES if the child belongs to an indigenous group; otherwise, choose NO.

8. For the 'Sex': from the dropdown list, choose M if the child is a male; otherwise, choose F.

9. For the 'Date of Birth' AND 'Date Measured': first determine the date format settings on your computer at the bottom right corner of your screen on your taskbar (or you can look at the date format in cell G7 in 'Nut_StatusTool'). For example, if your computer's date format is MM-DD-YYYY, "January 21, 2021" would appear as "01/21/2021"; if it is set as DD-MM-YYYY, it would appear as "21-01-2021".

10. For the 'Weight': make sure that it is in kilograms (kg) and the value is up to one decimal place only.

11. For the 'Height': make sure that it is in centimeters (cm) and the value is up to one decimal place only.

12. For the 'Bilateral Pitting Edema': from the dropdown list, choose 'NONE' if there was no bilateral pitting edema identified in the child, and if yes, indicate its severity (+1, +2 or +3)

13. For the 'Disability': from the dropdown list, choose 'NO' if the child has no disability; otherwise, choose 'YES'.

14. For the 'MUAC': make sure that it is in centimeters (cm) and the value is up to one decimal place only.

NOTE 4.1. Date Format

Important:

The way you type in the date format into the tool for the child's date of birth and the date of measurement must be the same as the way the date format in your computer's operating system appears (e.g., MM/DD/YYYY or DD/MM/YYYY). Always type in any date using the format consistent with your computer's settings. When you type in the numeric date, make sure that the month and the date are not interchanged.

Otherwise, this will lead to incorrect calculation of the child's age in days. After you have typed in the numeric date, it will appear in the cell as MM-DD-YYYY (for example, Jan-21-2021), regardless of your computer's date format settings. This is to help users read the correct dates after the date has been typed in. Please check that your computer's date and time settings are always correct.

NOTE 4.2. Data Entry

ENSURE ALL CELLS OR INFORMATION ARE FILLED OUT FOR EACH CHILD.

To auto-generate each child's nutritional status for all three indices as well as for the MUAC status, the following information must be complete: address, the name of the mother/caregiver, the child's name, IP status (whether the child belongs to an indigenous group), child's sex, date of birth, date measured, weight (in kg), height (in cm), presence or absence of bilateral pitting edema, presence or absence of disability, and MUAC (if MUAC is the nutritional assessment method to be used). It is important that the address, or the location in the barangay where the child lives should be entered to facilitate follow-up visits by nutrition workers. This can be the name of the purok, block, or part of the barangay.

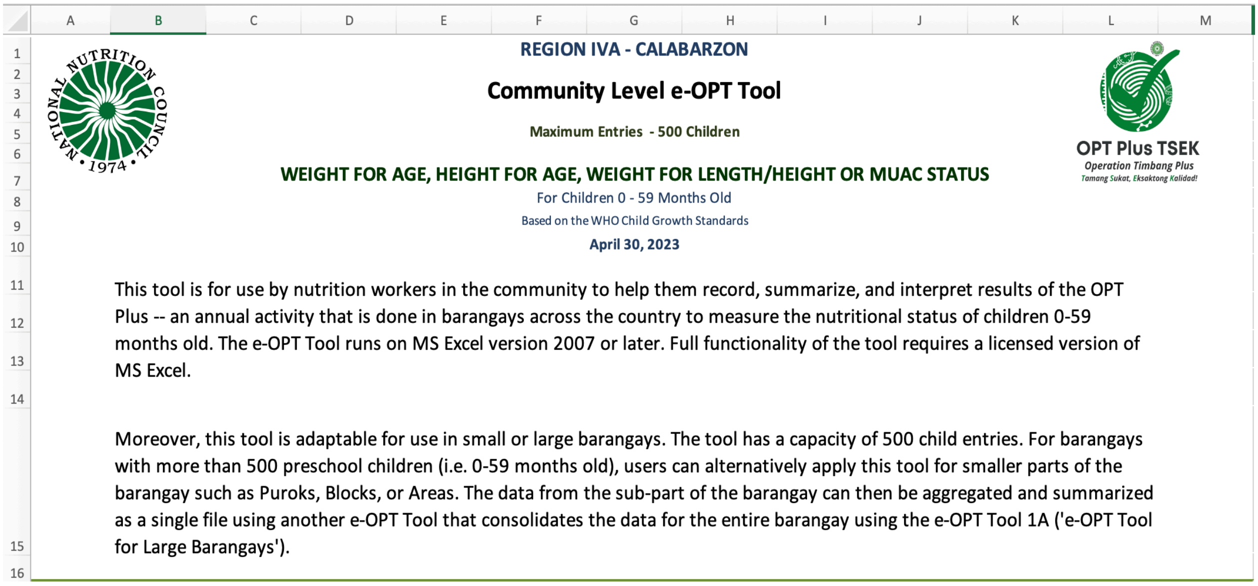
If the child is missing any essential information (e.g., name of mother/caregiver), type in "TBD," and make sure to obtain the needed information during the next house visit. Otherwise, if any details are left blank in the tool, one or all nutritional status indices for that child might not appear. To help users of the tool quickly see if a child has any missing information/measurement in the Nut_StatusTool, the cell for the corresponding missing information/measurement will be highlighted in yellow. Moreover, the cell opposite the child's name in Column A will be highlighted with a gray color.

How to update the eOPT Plus Tool in the next OPT Plus?

If there is already an encoded worksheet with eOPT data from previous use of the tool, batch copy the list of multiple children's names and details into the blank Nut_StatusTool using the "Copy>>Paste>>Values" function of MS Excel. Doing this avoids errors that might happen as underlying formulas and formats in cells from the source will also be copied if just "Paste" is used. Please just note that the children's names and other information from the source worksheet should be arranged in exactly the same way the columns in the tool are set up. Also make sure that the source data are pasted into the blank tool starting from Row #10 downwards.

What are the tables and reports generated in the eOPT Plus Tool?

1. **'Instructions'** tab guides the users on how to encode the OPT data and key features of the tool


REGION IVA - CALABARZON

Community Level e-OPT Tool

Maximum Entries - 500 Children

WEIGHT FOR AGE, HEIGHT FOR AGE, WEIGHT FOR LENGTH/HEIGHT OR MUAC STATUS

For Children 0 - 59 Months Old

Based on the WHO Child Growth Standards

April 30, 2023

OPT Plus TSEK
Operation Timbang Plus
Tamang Sukat, Eksaktong Kalidad!

This tool is for use by nutrition workers in the community to help them record, summarize, and interpret results of the OPT Plus -- an annual activity that is done in barangays across the country to measure the nutritional status of children 0-59 months old. The e-OPT Tool runs on MS Excel version 2007 or later. Full functionality of the tool requires a licensed version of MS Excel.

Moreover, this tool is adaptable for use in small or large barangays. The tool has a capacity of 500 child entries. For barangays with more than 500 preschool children (i.e. 0-59 months old), users can alternatively apply this tool for smaller parts of the barangay such as Puroks, Blocks, or Areas. The data from the sub-part of the barangay can then be aggregated and summarized as a single file using another e-OPT Tool that consolidates the data for the entire barangay using the e-OPT Tool 1A ('e-OPT Tool for Large Barangays').

2. **'Nut_Status Tool'** is the main data entry worksheet. The encoder can enter information of a child one at a time into each row, or through can do a batch entry for many children.

Year: 2022 **Region:** IVA CALABARZON **Province:** Laguna

Community Level e-OPT PLUS Tool

WEIGHT FOR AGE, HEIGHT FOR AGE, WEIGHT FOR LENGTH/HEIGHT OR MUAC STATUS

For a maximum of 500 children in a small or medium sized barangay. For larger barangays, use this file for a purok, section, or part of the barangay.

THIS TOOL IS TO BE USED FOR A: Barangay

April 2023 Version

OPT Plus TSEK

Tip: To adjust the view of the worksheet to fit your screen, use the zoom level slider at the right bottom corner below, set the zoom level between 60-90%.

Nutritional Assessment Method for Acute Malnutrition: Weight + Length/Height

CITY OF SANTA ROSA | Barangay: Ibaba

CHILD SEQ.	ADDRESS OR LOCATION OF CHILD'S RESIDENCE Purok, Block #, Area or Location in the Barangay	NAME OF MOTHER OR CAREGIVER (Surname, First Name Suffix)	FULL NAME OF CHILD (Surname, First Name Suffix)	Belongs to IP Group? YES/NO	SEX M/F	10/6/23 IMPORTANT: ALWAYS USE THIS DATE FORMAT WHEN TYPING IN DATES DATE OF BIRTH	DATE MEASURED	WEIGHT (kg)	HEIGHT (cm)	NO DATA ENTRY REQUIRED RESULTS WILL BE AUTO-FILLED Age in Months	Age in Days	Weight for Age Status	Length/ Height for Age Status	Weight for Length/ Height Status	Bilateral Pitting Edema NONE,+1,+2,+3	Disability YES/NO	MUAC (cm)	NO DATA ENTRY REQUIRED RESULTS WILL BE AUTO-FILLED MUAC Status
1	Purok 1	Mother 1	Child 1	NO	F	Aug-29-2020	Jan-10-2021	6.7	60.1	4	134	N	N	N		NO		
2	Purok 2	Mother 2	Child 2	YES	F	Oct-13-2019	Jan-10-2021	10.8	100.0	14	455	N	T	SW/SAM	NONE	NO		
3	Purok 3	Mother 3	Child 3	NO	F	Apr-19-2017	Jan-10-2021	14.0	97.0	44	1362	N	N	N	NONE	NO		
4	Purok 4	Mother 4	Child 4	NO	F	Sep-07-2020	Jan-10-2021	8.0	72.3	4	125	N	T	N		NO		
5	Purok 5	Mother 5	Child 5	NO	F	Mar-05-2019	Jan-10-2021	13.7	90.0	22	677	N	N	N	NONE	NO		
6	Purok 6	Mother 6	Child 6	NO	F	Apr-10-2019	Jan-10-2021	15.0	60.0	21	641	Use the WL/H column	SSt	Ob	NONE	NO		
7	Purok 7	Mother 7	Child 7	NO	F	May-04-2020	Jan-10-2021	7.0	70.0	8	251	N	N	N	NONE	NO		
8	Purok 8	Mother 8	Child 8	NO	F	Apr-14-2020	Jan-10-2021	9.0	73.0	8	271	N	N	N	NONE	NO		
9	Purok 9	Mother 9	Child 9	YES	F	Oct-16-2017	Jan-10-2021	15.0	99.0	38	1182	N	N	N	NONE	NO		

3. **'OPT Plus Form 1A'** serves as the pre-printed list of preschool children in the barangay, which can be used to find age-eligible children on the next rounds in the community. Names of preschoolers are alphabetically arranged to help match children and names. Make sure that the names of new or previously unlisted children are written down at the end of this list.

Barangay: Ibaba **City:** CITY OF SANTA ROSA **Year of Last OPT Plus:** 2022 Est. Date of Next OPT Plus Visit:

Province: Laguna

You must enter the estimated date of the next OPT Plus visit for the list to appear >>>>>

Please add the names and details of new or unlisted children at the end of this list.

NOTE: 0-23 mos old-measure length (lying down)

24 mos old & above-measure height (standing)

Note: For new children, just get the birthdate. Leave age in months blank.

OF PAGES FOR PRINTING: 0

Date printed: Oct 06, 2023

PRINTING INSTRUCTIONS

Child Seq. (Previous OPT)	Address Purok or Location in the Barangay	Name of Mother or Caregiver (SURNAME, FIRST NAME)	Child's Full Name (SURNAME, FIRST NAME)	Belongs to IP Group? YES/NO	Sex M/F	Date of Birth (MM/DD/YYYY)	Actual Date of Measurement (MM/DD/YYYY)	WEIGHT (kg)	HEIGHT (cm)	Age in Months *as of:* --	Nutritional Status (Weight for Lt/Ht)	MUAC (cm)	Bilateral Pitting Edema NONE, +1, +2, +3	Disability (if any) YES/NO	Signature of Mother or Caregiver

4. **'OPT Plus Form 1B'** is the consolidation of the results of the nutrition assessment.

- 0-to-59-month-old children who are Wasted and/or Stunted
- 24-to-59-month-old children who are Wasted and/or Stunted
- Number of children 0-59 months old who are overweight/obese
- Total number of children 0-23 months old
- Number of children 0-23 months old who are Wasted and/or Stunted
- Total number of children 0-29 months old
- Total number of children 30-59 months old
- Total number of children 24-49 months old
- Number of children 0-59 months old with Edema
- Number of children 0-59 months old with disability
- Total Number of mothers/ caregivers (M/Cs) of children 0-59 months old
- Number of M/Cs of 0-59 months old children affected by wasting and/or stunting
- Number of M/Cs of 0-59 months old children who are overweight/ obese
- Total number of M/Cs of children 0-23 months old
- Number of M/Cs of 0-23 months old children affected by Wasting and/or Stunting
- Number of Children with names and birthdate repeated
- Number of Children with missing information
- Number of Children with no name of parents/address
- Number of Children with no sex data
- Number of Children with no date of birth data
- Number of children older than 59 months
- Number of children with length/ height but no weight
- Number of children with weight but no length/height
- Number of children with no MUAC

SPACE FOR OFFICIAL SEAL OF LGU

Republic of the Philippines
Department of Health
NATIONAL NUTRITION COUNCIL
IVA CALABARZON

OPT Plus TSEK

OPT Plus Form B. Summary Sheet of the Nutritional Status of 0-59 Month-Old Children in a Barangay

Barangay:	Ibaba	**Province:**	Laguna
City:	CITY OF SANTA ROSA	**Total Popn Barangay:**	8,260
PSGC:	0403428007	**Estimated Popn of Children 0-59 mos in Barangay:**	750
		Prevalence Rate MUW & SUW (Last OPT Plus):	

OPT Plus Coverage: 1.20%
Source: DOH - EB 2021
Prevalence Rate MSt & SSt (Last OPT Plus):

Total # Indigenous Preschool Children Measured: 0-59 mos: 2

OPT PLUS 2022

Prevalence Rate MW/MAM & SW/SAM (Last OPT Plus):

ACRONYMS & ABBREVIATIONS	Total Boys:			Total Girls:		9	Total MUAC:			Total WFA:		9	Total HFA:		9	Total WFL/H:		9	Birth to 5 Years		F1K		# IP Children		
	0-5 Months			6-11 Months			12-23 Months			24-35 Months			36-47 Months			48-59 Months			0-59 Months		0-23 Months				
	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total	Total	Prev	Total	Prev	Boys	Girls	Total
WFA - Normal	0	2	2	0	2	2	0	2	2	0	0	0	0	2	2	0	0	0	8	88.9%	6	100.0%	0	2	2
WFA - Ob	No Obese/Overweight classification in the WFA. Following international standards, we use WL/HZ to classify overweight and obesity in children.																								
WFA - OW																									
WFA - MUW	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0%	0	0.0%	0	0	0
WFA - SUW	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0%	0	0.0%	0	0	0
HFA - Normal	0	1	1	0	2	2	0	1	1	0	0	0	0	2	2	0	0	0	6	66.7%	4	57.1%	0	1	1
HFA - Tall	0	1	1	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	2	22.2%	2	28.6%	0	1	1
HFA - MSt	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0%	0	0.0%	0	0	0
HFA - SSt	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	1	11.1%	1	14.3%	0	0	0
WFL/H - Normal	0	2	2	0	2	2	0	1	1	0	0	0	0	2	2	0	0	0	7	77.8%	5	71.4%	0	1	1
WFL/H - OW	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0%	0	0.0%	0	0	0
WFL/H - Ob	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	1	11.1%	1	14.3%	0	0	0
WFL/H - MW/MAM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0%	0	0.0%	0	0	0
WFL/H - SW/SAM	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	1	11.1%	1	14.3%	0	1	1
Total (WFA)	0	2	2	0	2	2	0	2	2	0	0	0	0	2	2	0	0	0							

Summary of Children covered by e-OPT Plus	
# Children 0-59 mos. who are Wasted and/or Stunted:	2
# Children 24-59 mos. who are Wasted and/or Stunted:	0
# Children 0-59 mos. who are Overweight/Obese:	1
Total Number of Children 0-23 mos. old:	7
# Children 0-23 mos. who are Wasted and/or Stunted:	2
Total Number of Children 0-29 mos. old:	7
Total Number of Children 30-59 mos. old:	2
Total Number of Children 24-59 mos. old:	2
# Children 0-59 mos. w/ Edema:	0
# Children 0-59 mos. w/ disability:	0

Mothers/Caregivers Summary	
Total Number of M/Cs of children 0-59 mos. old:	9
# M/Cs of 0-59 mos children affected by Wasting and/or Stunting:	2
# M/Cs of 0-59 mos. children who are Overweight/Obese:	1
Total Number of M/Cs of children 0-23 mos. old:	7
# M/Cs of 0-23 mos children affected by Wasting and/or Stunting:	2

Data Inaccuracy	
# Children with names and birthdate repeated:	0
# Children with missing information:	0
# Children with no name of parents/address:	0
# Children with no sex data:	0
# Children with no date of birth data:	0
# Children older than 59 months:	0
# Children with length/height but no weight:	0
# Children with weight but no length/height:	0
# Children with no MUAC:	
# Children with no data on edema:	0

Note: This is a printable worksheet. Before printing, please read printing instructions (click on the link).

PRINTING INSTRUCTIONS

Prepared by:	Checked:	Approved:
Name and Signature of Barangay Nutrition Scholar	Name and Signature of Midwife/Nurse/MNAO/MHO or District/City Nutrition Program Coordinator	Name and Signature of Barangay Captain, BNC Chairperson
Date:	Date:	Date:

5. **'OPT Plus Form 1C'** contains the list of Affected/At-risk 0-to-59-month-old preschool children.

PRINTING INSTRUCTIONS

SPACE FOR LGU SEAL

Republic of the Philippines
Department of Health
NATIONAL NUTRITION COUNCIL
IVA CALABARZON

NATIONAL NUTRITION COUNCIL 1974

OPT Plus TSEK

YEAR: 2022

OPT Plus Form C. List of Affected/At-risk 0-59 Month-Old Children

Pages for Printing: 1

Total # Children Affected/At-Risk: **2**

MUW= 0 MSt= 0 MW/MAM= 0

SUW= 0 **SSt= 1** **SW/SAM= 1**

OW= 0

Ob= 1

Barangay: Ibaba

City: CITY OF SANTA ROSA

Province: Laguna

Number of Children Affected by Undernutrition: **2**

Number of Children with Overweight or Obesity: **1**

PRINTING INSTRUCTIONS

Child Seq.	Address (Purok or Location in the Barangay)	Name of Mother or Caregiver	Full Name of Child	Sex	Age in Months	Nutritional Status: Weight for Age	Nutritional Status: Length/ Height for Age	Nutritional Status: Weight for Length/ Height	Nutritional Status: MUAC
2	Purok 2	Mother 2	Child 2	F	14	**N**	**T**	SW/SAM	
6	Purok 6	Mother 6	Child 6	F	21	**Use the WL/H column**	**SSt**	Ob	

6. **'NutStatusBrgy'** presents sex-aggregated summary tables of the nutritional status of 0-to-23-month and 0-to-59-month-old children which can be used for presentations.

SPACE FOR LGU SEAL

Barangay **Ibaba**
City: CITY OF SANTA ROSA
Province: Laguna

OPERATION TIMBANG PLUS 2022

NUTRITIONAL STATUS OF CHILDREN 0-23 and 0-59 MONTHS OLD

SEX-DISAGGREGATED SUMMARY TABLES FOR PRESENTATIONS

OPT Plus TSEK

WEIGHT FOR AGE

PRINTING INSTRUCTIONS

	0-23 Months (F1K)				0-59 Months			
	Boys	Girls	Total	Prev	Boys	Girls	Total	Prev
Normal	0	6	6	100.00%	0	8	8	88.89%
Overweight	No Obese/Overweight classification in the WFA. Following international standards, we use WL/HZ to classify overweight and obesity in children.							
Obese								
Moderately Underweight	0	0	0	0.00%	0	0	0	0.00%
Severely Underweight	0	0	0	0.00%	0	0	0	0.00%

HEIGHT FOR AGE

	0-23 Months (F1K)				0-59 Months			
	Boys	Girls	Total	Prev	Boys	Girls	Total	Prev
Normal	0	4	4	57.14%	0	6	6	66.67%
Tall	0	2	2	28.57%	0	2	2	22.22%
Moderately Stunted	0	0	0	0.00%	0	0	0	0.00%
Severely Stunted	0	1	1	14.29%	0	1	1	11.11%

WEIGHT FOR LENGTH/HEIGHT

	0-23 Months (F1K)				0-59 Months			
	Boys	Girls	Total	Prev	Boys	Girls	Total	Prev
Normal	0	5	5	71.43%	0	7	7	77.78%
Overweight	0	0	0	0.00%	0	0	0	0.00%
Obese	0	1	1	14.29%	0	1	1	11.11%
Moderately Wasted	0	0	0	0.00%	0	0	0	0.00%
Severely Wasted	0	1	1	14.29%	0	1	1	11.11%

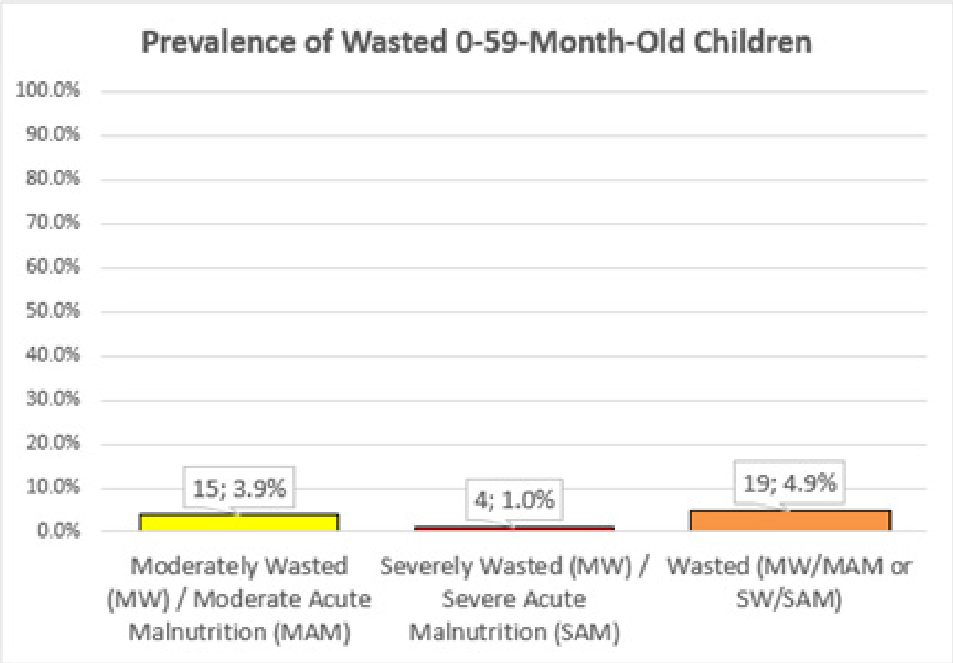
MUAC

	0-23 Months (F1K)				0-59 Months			
	Boys	Girls	Total	Prev	Boys	Girls	Total	Prev
Normal								
Moderate Acute Malnutrition								
Severe Acute Malnutrition								

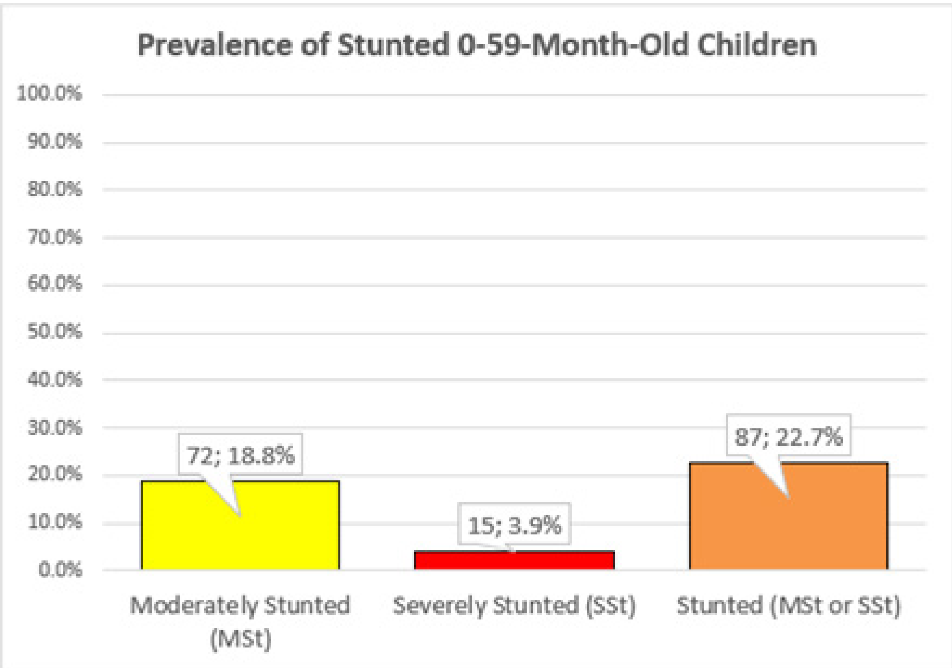
TOTAL NUMBER OF MOTHERS/CAREGIVERS OF CHILDREN (0-59 MOS OLD) AFFECTED BY UNDERNUTRITION	2	**TOTAL NUMBER OF MOTHERS/CAREGIVERS OF CHILDREN (0-23 MOS OLD) AFFECTED BY UNDERNUTRITION**	2

7. **'Graphs'** shows a graphical representation of the following prevalences in the community:

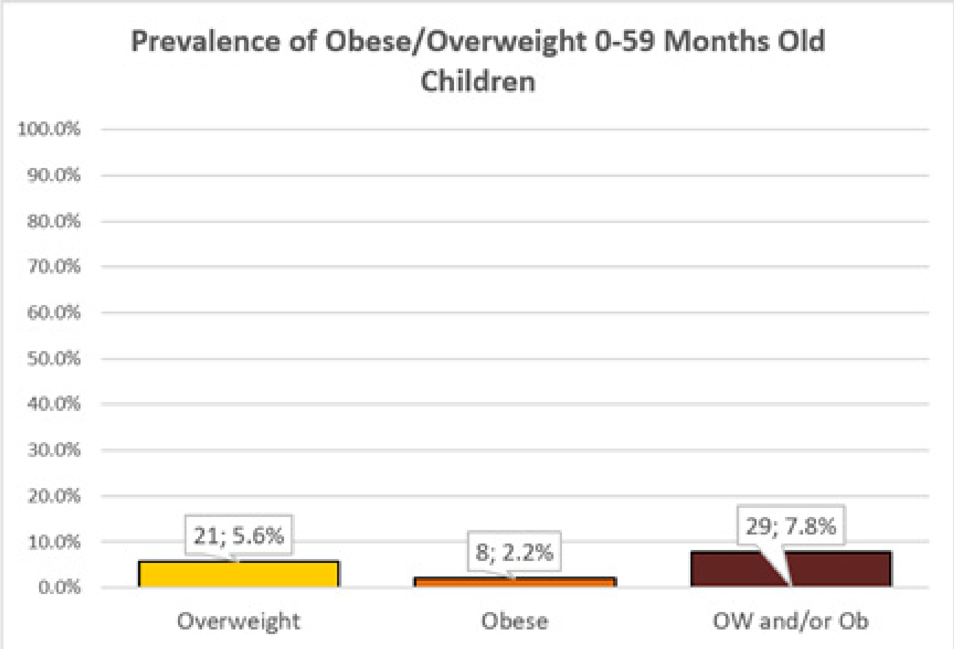
- Prevalence and Number of Wasted 0-to-59-Month-Old Children



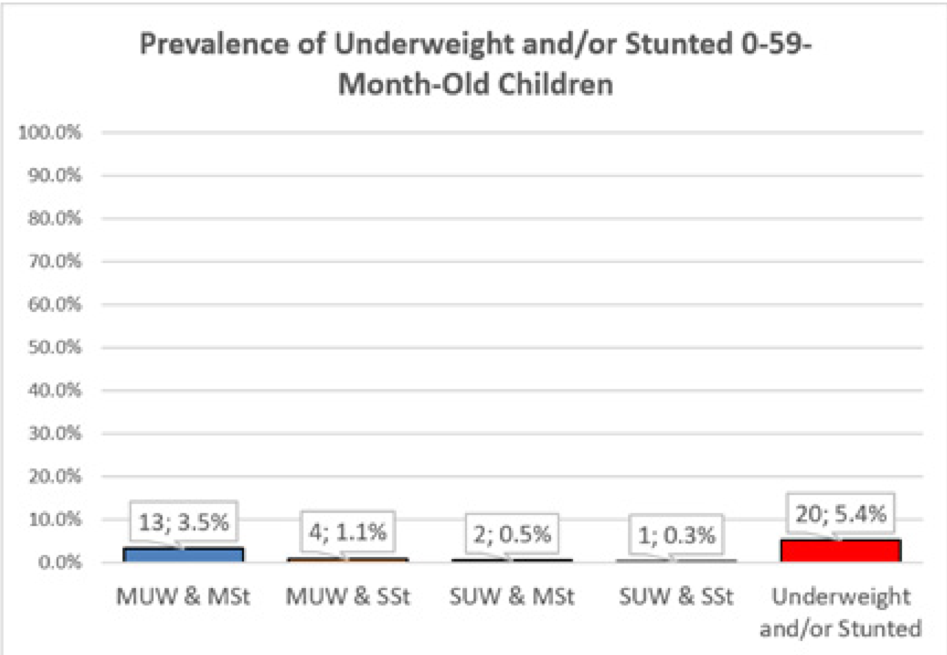
- Prevalence and Number of Stunted 0-to-59-Month-Old Children



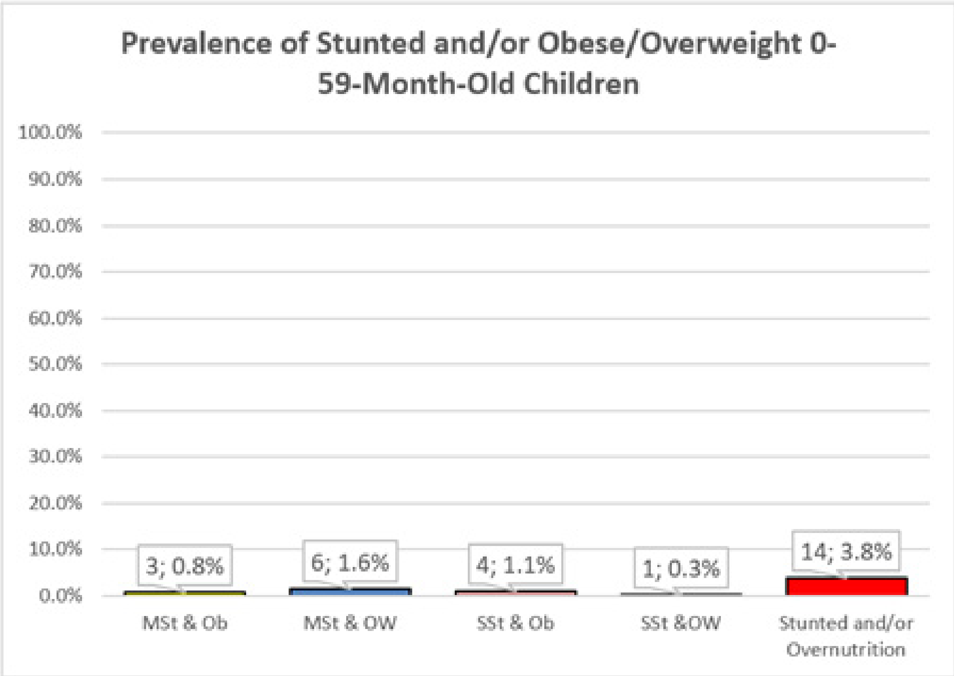
- Prevalence and Number of Obese/Overweight 0-to-59-month-old Children



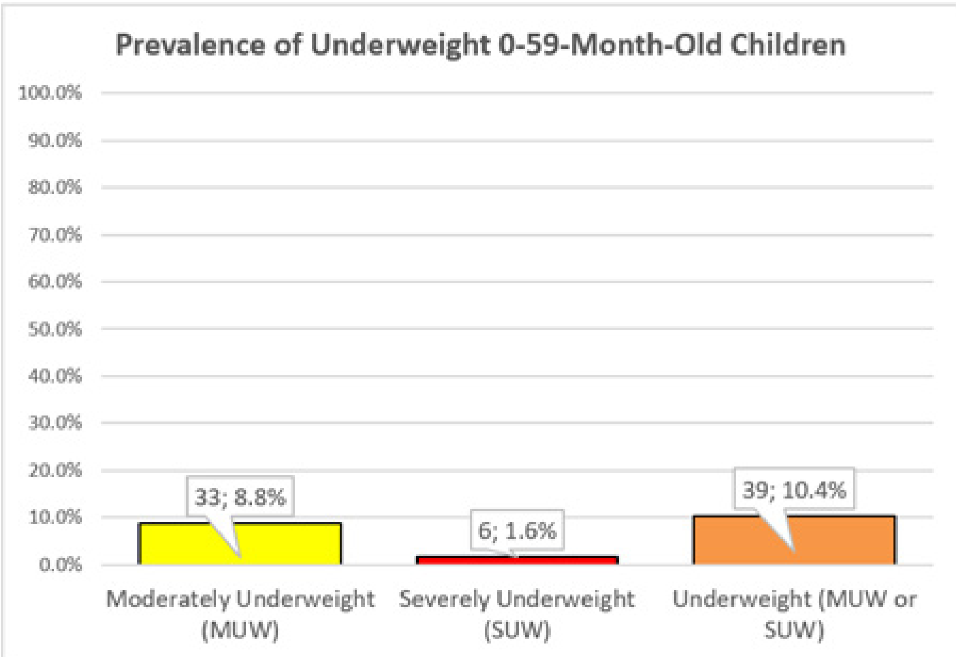
- Prevalence and Number of Underweight and/or Stunted 0-to-59-Month-Old Children



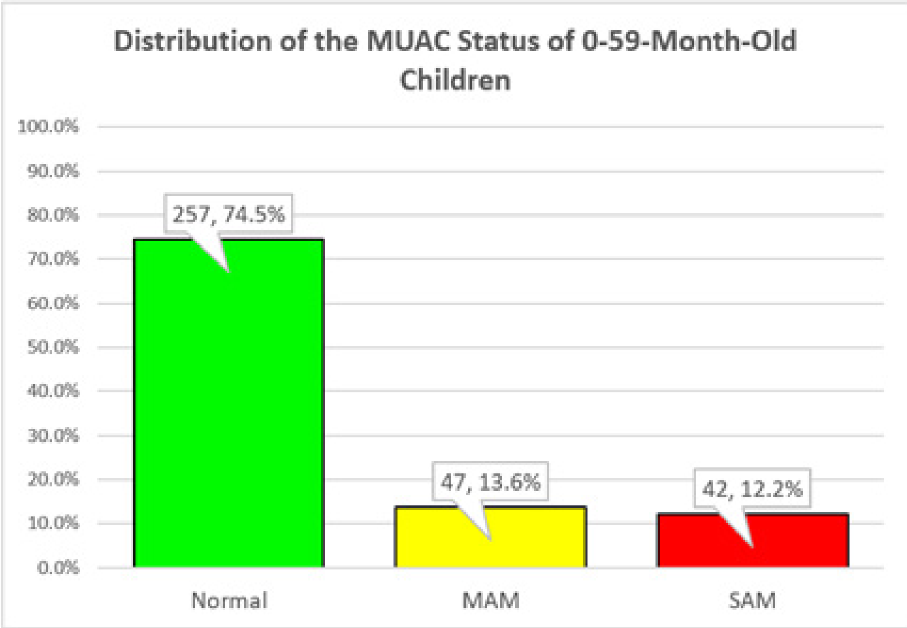
- Prevalence and Number of Stunted and/or Obese/Overweight 0-to-59-month-old Children



- Prevalence and Number of Underweight 0-to-59-month-old Children



- Distribution of the MUAC Status of 0-to-59-month-old Children



8. **'DQC Summary'** summarizes the needed information to be copied and pasted in the OPT Plus DQC Tool.

FOR DATA QUALITY CHECK (DQC)

OPT Plus TSEK
Operation Timbang Plus
Tamang Sukat, Eksaktong Kalidad!

	COMPLETENESS	
A	% Coverage (population of 0-59 months):	1.20
B	% Children measured with duplicate cases:	0.00
C	% Children with length/height but no weight:	0.00
D	% Children with weight but no length/height:	0.00
E	% Children with no date of birth data:	0.00
F	% Children with no sex data:	0.00
G	% Children with no name of parents/address:	0.00
H	% Children with no data on edema:	0.00

	ACCURACY	
A	% Children with flagged measurement based on z-scores:	11.11
B	Digit preference score for anthropometric data:	70
C	Skewness of weight-for-height/length z-score:	1.43
D	Kurtosis of weight-for-height/length z-score:	1.82
E	Poisson distribution (p-value) for WL/Hz (<-2):	

	RELIABILITY	
A	Standard deviation of weight-for-height/length z-score:	3.23

9. **'List of 0-23 mos'** is the list of all 0-to-23-month-old children in the community.

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MONITORING LIST FOR CHILDREN 0 - 23 MONTHS OLD

Barangay: **Ibaba** City: **CITY OF SANTA ROSA** Province: **Laguna**

Year: **2022**

# of Children: 7

Note: This list can be copied to create other customized monitoring worksheets.

PRINTING INSTRUCTIONS

pages for printing: 1

Child Seq.	Address: Purok or Location in the Barangay	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	Length/ Height (cm)	Weight (kg)	MUAC (cm)	WFA Status	Lt/HtFA Status	WFLt/Ht Status	MUAC Status	Follow-up Visits: Month # __	Month # __	Month # __	Month # __	Month # __	Month # __
1	Purok 1	Mother 1	Child 1	F	08/29/2020	60.1	6.7		N	N	N							
2	Purok 2	Mother 2	Child 2	F	10/13/2019	100.0	10.8		N	T	SW/SAM							
4	Purok 4	Mother 4	Child 4	F	09/07/2020	72.3	8.0		N	T	N							
5	Purok 5	Mother 5	Child 5	F	03/05/2019	90.0	13.7		N	N	N							
6	Purok 6	Mother 6	Child 6	F	04/10/2019	60.0	15.0		Use the WL/H column	SSt	Ob							
7	Purok 7	Mother 7	Child 7	F	05/04/2020	70.0	7.0		N	N	N							
8	Purok 8	Mother 8	Child 8	F	04/14/2020	73.0	9.0		N	N	N							

10. **'List_MW (MAM)'** is the list of moderately wasted/MAM children 0-to-59-months old.

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MONITORING LIST FOR MODERATELY WASTED (MAM) CHILDREN 0-59 MONTHS OLD

Barangay: **Ibaba** City: **CITY OF SANTA ROSA** Province: **Laguna**

Year: **2022**

PRINTING INSTRUCTIONS

# of Children: 1

Note: This list can be copied to create other customized monitoring worksheets.

pages for printing: 1

Child Seq.	Address (Purok or Location in the Barangay)	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	Length/ Height (cm)	Weight (kg)	MUAC (cm)	Follow-up Visits					
									Month # __	Month # __	Month # __	Month # __	Month # __	Month # __
3	Purok 3	Mother 3	Child 3	F	04/19/2017	97.0	12.0							

11. 'List_SW(SAM)' is the list of severely wasted/SAM children 0-to-59-months old.

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MONITORING LIST FOR SEVERELY WASTED (SAM) CHILDREN 0-59 MONTHS OLD

Barangay: **Ibaba** City: **CITY OF SANTA ROSA** Province: **Laguna**

Year: **2022**

PRINTING INSTRUCTIONS

# of Children: 1

Note: This list can be copied to create other customized monitoring worksheets.

pages for printing: 1

Child Seq.	Address	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	Length/ Height (cm)	Weight (kg)	MUAC (cm)	Follow-up Visits					
	Purok or Location in the Barangay								Month # __	Month # __	Month # __	Month # __	Month # __	Month # __
2	Purok 2	Mother 2	Child 2	F	10/13/2019	100.0	10.8							

12. 'List_MSt&SSt' is the list of moderately stunted and severely stunted children 0-to-59-months old.

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MONITORING LIST FOR MODERATELY OR SEVERELY STUNTED CHILDREN 0-59 MONTHS OLD

Barangay: **Ibaba** City: **CITY OF SANTA ROSA** Province: **Laguna**

Year: **2022**

PRINTING INSTRUCTIONS

# of Children: 1

Note: This list can be copied to create other customized monitoring worksheets.

pages for printing: 1

Child Seq.	Address (Purok or Location in the Barangay)	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	Length/ Height (cm)	Weight (kg)	Follow-up Visits					
								Month #__	Month #__	Month #__	Month #__	Month #__	Month #__
6	Purok 6	Mother 6	Child 6	F	04/10/2019	60.0	15.0						

13. **'List_OW&Ob'** is the list of overweight and obese 0-to-59-month-old children.

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MONITORING LIST FOR OVERWEIGHT OR OBESE CHILDREN 0-59 MONTHS OLD

Barangay: **Ibaba** City: **CITY OF SANTA ROSA** Province: **Laguna**

Year: **2022**

PRINTING INSTRUCTIONS

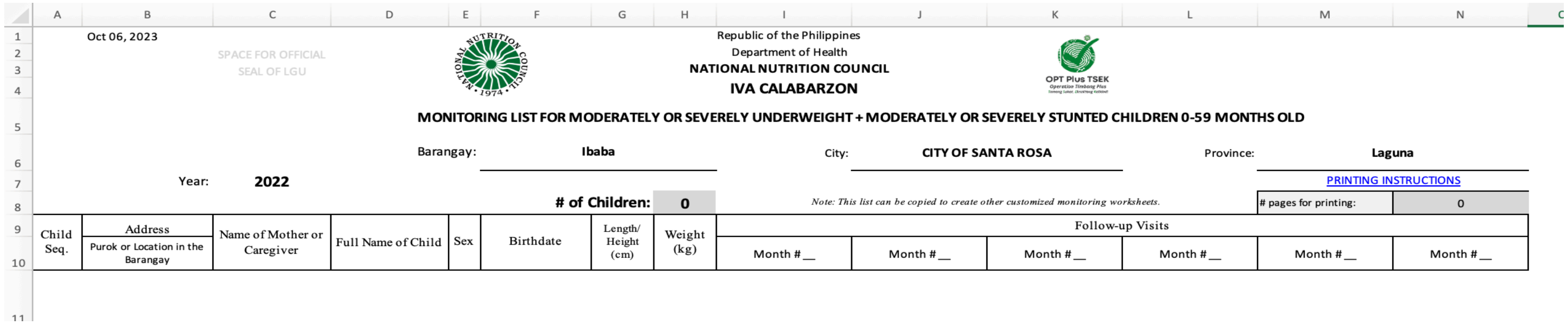
# of Children: **1**

Note: This list can be copied to create other customized monitoring worksheets.

pages for printing: 1

Child Seq.	Address Purok or Location in the Barangay	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	Length/ Height (cm)	Weight (kg)	Follow-up Visits Month #__	Month #__	Month #__	Month #__	Month #__	Month #__
6	Purok 6	Mother 6	Child 6	F	04/10/2019	60.0	15.0						

14. **‘List_MUW,SUW,MSt&SSt‘** is the list of underweight (moderately underweight or severely underweight) and stunted (moderately stunted or severely stunted) 0-to-59-month-old children.



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MONITORING LIST FOR MODERATELY OR SEVERELY UNDERWEIGHT + MODERATELY OR SEVERELY STUNTED CHILDREN 0-59 MONTHS OLD

Barangay: Ibaba City: CITY OF SANTA ROSA Province: Laguna

Year: 2022

PRINTING INSTRUCTIONS

# of Children: 0

Note: This list can be copied to create other customized monitoring worksheets.

pages for printing: 0

Child Seq.	Address Purok or Location in the Barangay	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	Length/ Height (cm)	Weight (kg)	Follow-up Visits					
								Month # __	Month # __	Month # __	Month # __	Month # __	Month # __

15. **'List_MSt,SSt,MW&SW'** is the list of stunted (moderately stunted or severely stunted) and wasted (moderately wasted and severely wasted) children 0-to-59-months old.

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OPT Plus TSEK

MONITORING LIST FOR MODERATELY OR SEVERELY STUNTED + MODERATELY OR SEVERELY WASTED CHILDREN 0-59 MONTHS OLD

Barangay: **Ibaba** City: **CITY OF SANTA ROSA** Province: **Laguna**

Year: **2022**

PRINTING INSTRUCTIONS

# of Children: 0

Note: This list can be copied to create other customized monitoring worksheets.

pages for printing: 0

Child Seq.	Address Purok or Location in the Barangay	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	Length/ Height (cm)	Weight (kg)	Follow-up Visits					
								Month # __	Month # __	Month # __	Month # __	Month # __	Month # __

16. **'List_MSt,SSt,OW&Ob'** is the list of stunted (moderately stunted or severely stunted), overweight, and obese 0-to-59-month-old children.

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MONITORING LIST FOR MODERATELY OR SEVERELY STUNTED + OVERWEIGHT OR OBESE CHILDREN 0-59 MONTHS OLD

Barangay: **Ibaba** City: **CITY OF SANTA ROSA** Province: **Laguna**

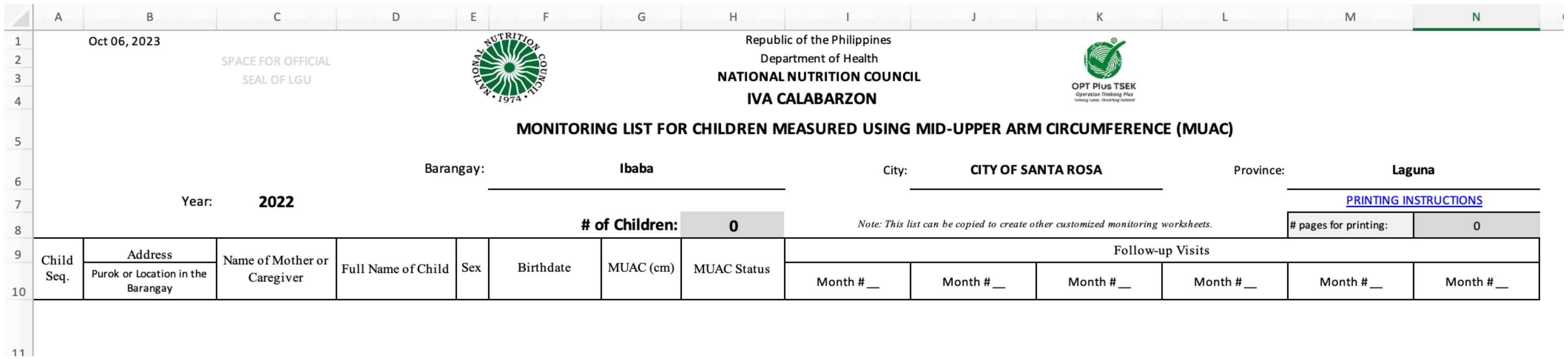
Year: **2022**

PRINTING INSTRUCTIONS

# of Children: 1 *Note: This list can be copied to create other customized monitoring worksheets.* # pages for printing: 1

Child Seq.	Address (Purok or Location in the Barangay)	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	Length/ Height (cm)	Weight (kg)	Follow-up Visits					
								Month #__	Month #__	Month #__	Month #__	Month #__	Month #__
6	Purok 6	Mother 6	Child 6	F	04/10/2019	60.0	15.0						

17. **'List_MUAC Status'** is the list of the MUAC status of 6-to-59-month-old children.



Oct 06, 2023

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OPT Plus TSEK

MONITORING LIST FOR CHILDREN MEASURED USING MID-UPPER ARM CIRCUMFERENCE (MUAC)

Barangay: **Ibaba** City: **CITY OF SANTA ROSA** Province: **Laguna**

Year: **2022**

PRINTING INSTRUCTIONS

# of Children: 0

Note: This list can be copied to create other customized monitoring worksheets.

pages for printing: 0

Child Seq.	Address Purok or Location in the Barangay	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	MUAC (cm)	MUAC Status	Follow-up Visits					
								Month #__	Month #__	Month #__	Month #__	Month #__	Month #__

18. **'Data Export'** contains all the data that are meant to be exported for consolidation to either a large barangay (in areas where the barangay preschool population exceeds 500 children) or to the municipal or city-level data.

Barangay	No. of Children w/ Valid WFA	No. of Children w/ Valid HFA	No. of Children w/ Valid WFL/H	No. of Children w/ Valid MUAC	Barangay Population	0-5 Months																
place the cursor and start copying from the green shaded cell below	0 - 59 mos			6-59 mos		N	SUW	MUW	N	SSt	MSt	T	N	OW	Ob	SW	MW	N	SUW	MUW	N	SSt
Ibaba	9	9	9	0	8,260	2	0	0	1	0	0	1	2	0	0	0	0	2	0	0	2	0

TO COPY THE ENTIRE ROW OF DATA, PLACE YOUR CURSOR ON THE GREEN SHADED CELL. THEN, USING YOUR KEYBOARD, **PRESS CTRL+Shift + Right Arrow.**

CELLS **B4 TO JA4** SHOULD BE HIGHLIGHTED AND COPIED TO ANOTHER TOOL CALLED 'e-OPT TOOL FOR MUNICIPALITIES OR CITIES.'

Section 4.3

GUIDELINES ON DETERMINATION OF NUTRITIONAL STATUS

By using any of these three tools that computes through age in days, nutritional status can be determined:

- **Electronic Operation Timbang Plus tool (eOPT) v2**
- **National Nutrition Information System mobile application**
- **Early Childhood Care and Development (ECCD) Card**

Electronic Operation Timbang Plus tool (eOPT) v2

As discussed in the previous section, the eOPT, when complete with all the required information of a child, can automatically compute nutritional status through the Child Growth Standards (CGS) z-scores.

Step 1. Open the eOPT

Step 2. At the 'Nut_StatusTool' worksheet, choose from the dropdown list in Cell KV6 which nutritional assessment method (Weight+Length/Height or MUAC) was used. If Weight+Length Height was chosen as a method, the 'MUAC' and 'MUAC Status' columns will be automatically blocked, and vice versa.

Step 3. Input all the required data of the child on the first row, ensure complete information. Refer to NOTE 4.1 for important information about the date format.

Step 4. See columns M-O for the nutritional status. Provide feedback to caregiver.

National Nutrition Information System mobile application

The National Nutrition Information System (NNIS) collects data for the OPT Plus. It was developed by the National Nutrition Council according to the section 6 of the Republic Act No. 11037 "Masustansyang Pagkain para sa Batang Pilipino Act" which states that the system shall harmonize all existing national and local nutrition databases from the national government agencies (NGAs), local government units (LGUs), and other relevant agencies of the government to identify individual groups, and localities that have the highest magnitude of hunger and undernutrition.

Upon completion of input of a child's anthropometric measurements and other relevant information needed in the OPT Plus, the NNIS mobile application automatically computes for the nutritional status of the child in real-time.

Refer to the NNIS User's Manual for step-by-step information in using the application.

Early Childhood Care and Development (ECCD) Card

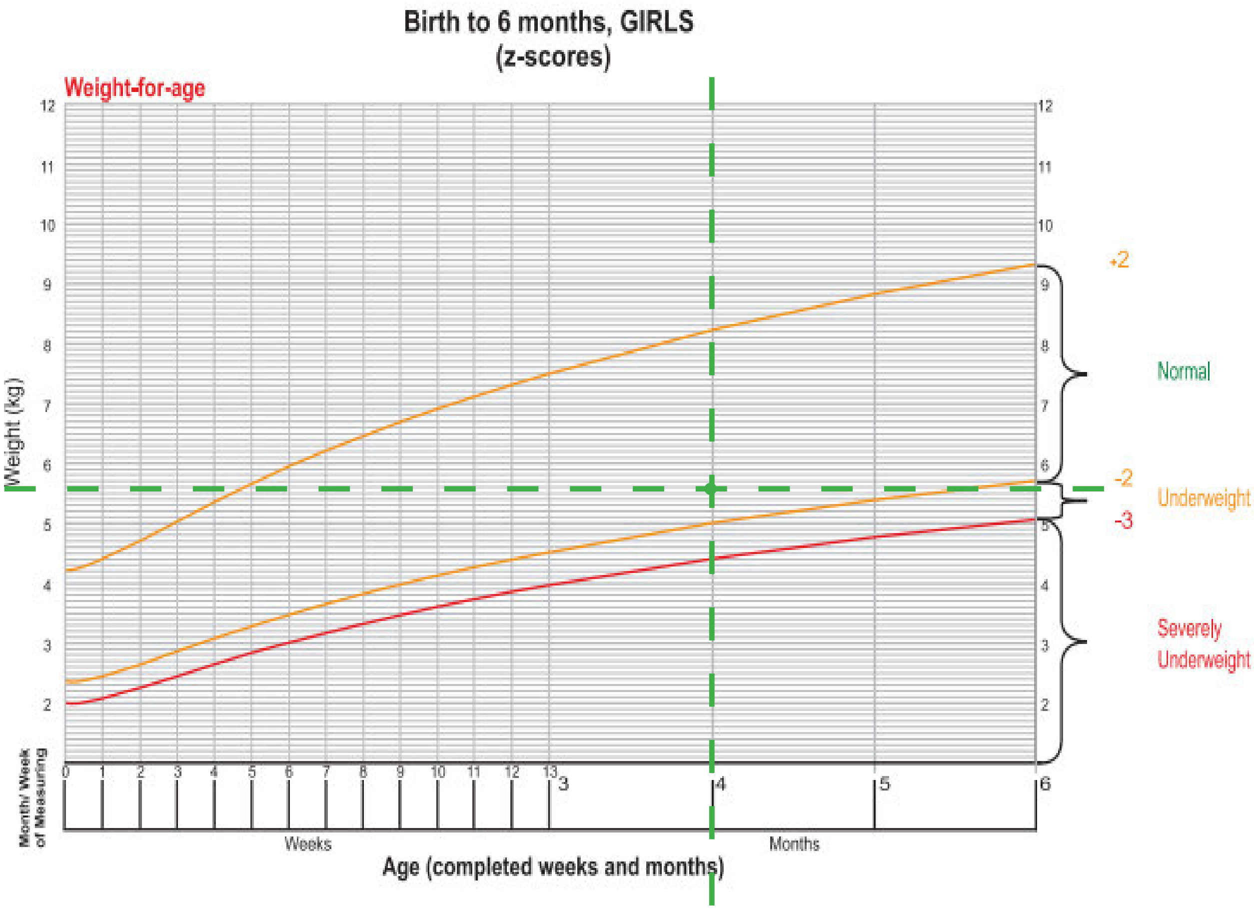
Unlike the two other tools, the ECCD Card does not specifically compute nutritional status, but instead plots the position of the measured indicators on the standard growth charts based on the Child Growth Standards (CGS) z-scores. In the table below is the matrix of growth indicators and applicable growth charts with the corresponding step-by-step procedures to identify nutritional status when used for immediate feedbacking to caregivers. Section 5.4 elaborates on the more detailed use of the ECCD Card.

Growth Indicator	Applicable CGS Charts	Steps to identify nutritional status of child
Weight-for-age	• Weight-for-age, birth to 6 months • Weight-for-age, 6 to 24 months • Weight-for-age, 24 to 71 months	1. Select the applicable growth chart. 2. Identify the exact vertical line corresponding to the age then identify the horizontal line corresponding to the weight. 3. Mark the point where these lines intersect. 4. Looking at the right side of the plotted point, refer to the label of the bracket to identify the nutritional status.
Length/height-for-age	• Length-for-age, birth to 6 months • Length-for-age, 6 to 24 months • Height-for-age, 24 to 71 months	1. Select the applicable growth chart. 2. Identify the exact vertical line corresponding to the age then identify the horizontal line corresponding to the length or height 3. Mark the point where these lines intersect. 4. Looking at the right side of the plotted point, refer to the label of the bracket to identify the nutritional status.
Weight-for-length/height	• Weight-for-length, birth to 24 months • Weight-for-height, 24 to 60 months	1. Select the applicable growth chart. 2. Identify the exact vertical line corresponding to the length or height (e.g., 75 cm, 78.5 cm), then identify the horizontal line corresponding to the weight. Note: *Due to the limited graphical representation, it is necessary to round the measurement to the nearest 0.5 centimeter (i.e., round down if 0.1 to 0.2 or 0.6 to 0.7 and round up if 0.3 to 0.4 or 0.8 to 0.9).* 3. Mark the point where these lines intersect, as precisely as possible. 4. Looking at the right side of the plotted point, refer to the label of the bracket to identify the nutritional status.

Weight-for-Age (WFA)

1. Select the applicable growth chart.
2. Identify the exact vertical line corresponding to the age then identify the horizontal line corresponding to the weight.
3. Mark the point where these lines intersect.
4. Looking at the right side of the plotted point, refer to the label of the bracket to identify the nutritional status.

Example: Maria is a 4-month old baby girl who weighs 5.6kg.

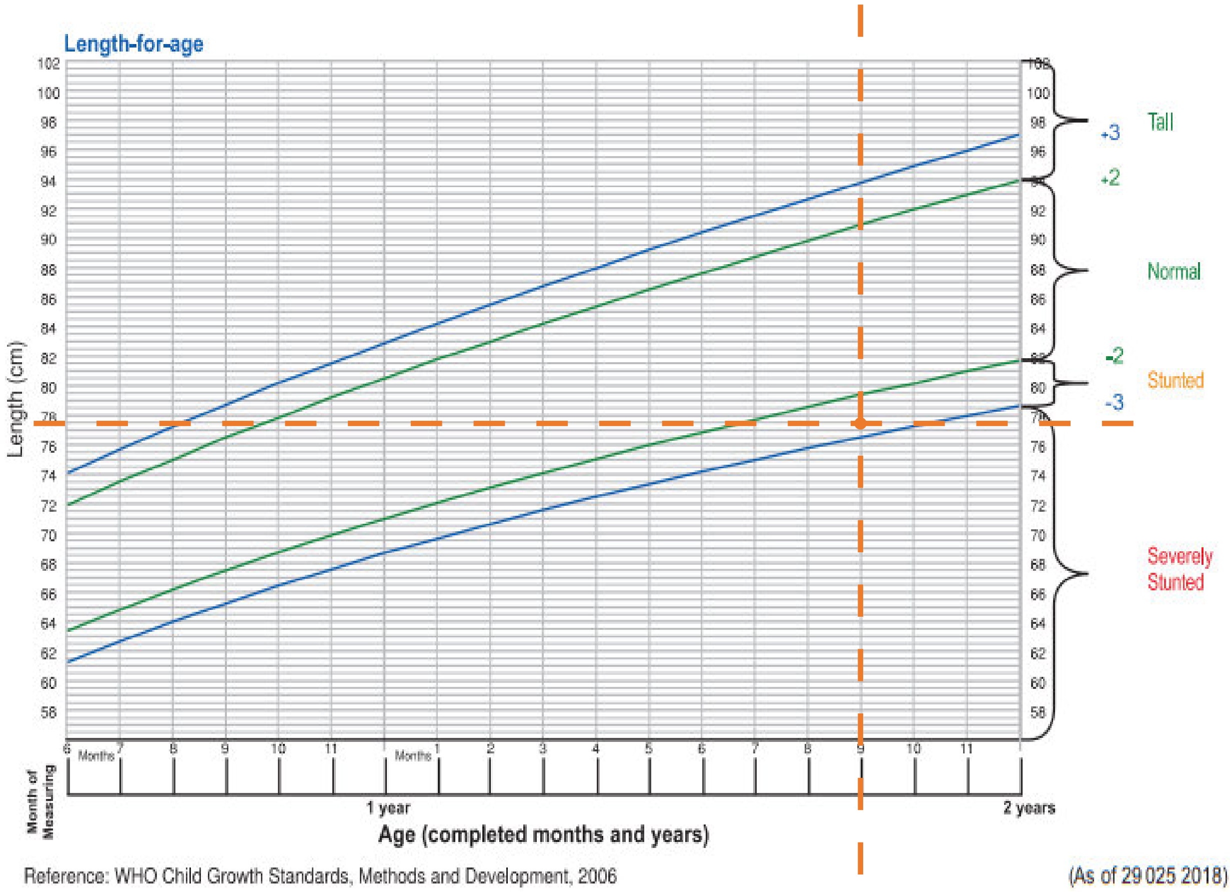


Using the ECCD Card's WFA Growth Chart for birth to 6 months for girls, we first identify the exact vertical line corresponding to the age then identify the horizontal line where her weight falls. Maria's age in months is 4, and since her weight falls between the two orange lines which is within the Normal bracket on the right, MARIA is classified as NORMAL.

Length/Height-for-Age (LFA/HFA)

1. Select the applicable growth chart.
2. Identify the exact vertical line corresponding to the age then identify the horizontal line corresponding to the length or height
3. Mark the point where these lines intersect.
4. Looking at the right side of the plotted point, refer to the label of the bracket to identify the nutritional status.

Example: Juan is a 9-month old boy whose measured length is 77.8cm.

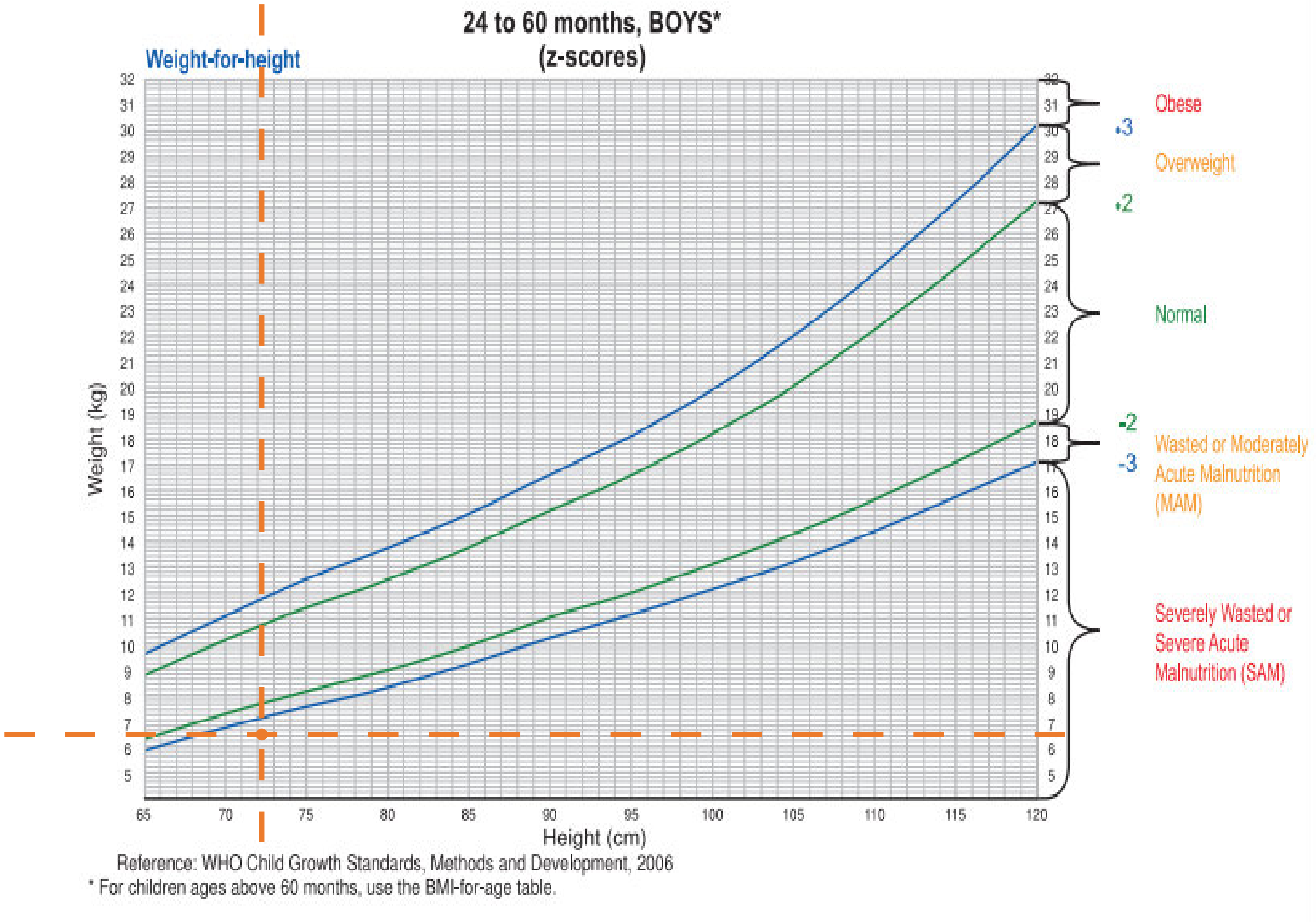


Using the ECCD Card's LFA Growth Chart for 6 to 24 months for boys, we first identify the exact vertical line corresponding to the age then identify the horizontal line where his length falls. Juan's age in months is 9, and since his length falls between the green and blue lines which is within the Stunted bracket on the right, JUAN is classified as MODERATELY STUNTED.

Weight-for-Length/Height (WFL/WFH)

1. Select the applicable growth chart.
2. Identify the exact vertical line corresponding to the length or height (e.g., 75 cm, 78.5 cm), then identify the horizontal line corresponding to the weight.
   Note: *Due to the limited graphical representation, it is necessary to round the measurement to the nearest 0.5 centimeter (i.e., round down if 0.1 to 0.2 or 0.6 to 0.7 and round up if 0.3 to 0.4 or 0.8 to 0.9).*
3. Mark the point where these lines intersect, as precisely as possible.
4. Looking at the right side of the plotted point, refer to the label of the bracket to identify the nutritional status.

Example: Pedro is a 33-month old boy who weighs 6.5kg and height is 72.2cm.



First, round off Pedro's weight to the nearest 0.5cm, making the height 72.0cm. Using the ECCD Card's WFH Growth Chart for 24 to 60 months for boys, identify the exact vertical line corresponding to the rounded off height then identify the horizontal line where his weight falls. Since Pedro's rounded off height is 72.0 cm and his weight falls below the blue line which is within the Severely Wasted bracket on the right, Pedro is classified as SEVERELY WASTED/SAM.

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Section Five

OPT PLUS PROTOCOL ON DATA INTERPRETATION AND ACTION PLANNING

OPT PLUS PROTOCOL ON DATA INTERPRETATION AND ACTION PLANNING

OVERVIEW

Data is the backbone of a strong action plan. Are all the available data utilized to make the right decisions? Section 5 discusses analyzing and interpreting the OPT Plus data in action planning.

After reading this section, one should be able to:

1. Enumerate the roles and responsibilities of the supervisors and OPT Plus Team;
2. Enumerate the uses of the OPT Plus results;
3. Describe the process of report generation via eOPT Plus;
4. Analyze and interpret the plotted points in the growth chart, and identify normal growth and growth problems; and
5. Understand how to utilize the OPT Plus results in action planning.

This section covers the interpretation of the OPT Plus results and guidance on how these data can be used for local action planning. Specific sections include the following:

Section 5.1	Roles and Responsibilities of OPT Plus Team and Supervisors
Section 5.2	Uses of OPT Plus Results
Section 5.3	Report Generation (eOPT Plus and Forms)
Section 5.4	Data Analysis and Interpretation
Section 5.5	Data Utilization and Action Planning

LIST OF TABLES

LIST OF FIGURES

LIST OF ACRONYMS

BNAP	Barangay Nutrition Action Plan
BNC	Barangay Nutrition Committee
BNS	Barangay Nutrition Scholar
C/MNAO	City/Municipal Nutrition Action Officer
C/MNAP	City/Municipal Nutrition Action Plan
C/MNC	City/Municipal Nutrition Committee
ECCD	Early Child Care and Development
eOPT Plus	Electronic Operation Timbang Plus
GB	Governinig Board
LNAP	Local Nutrition Action Plan
MW/MAM	Moderately Wasted/Moderate Acute Malnutrition
MUw	Moderately Underweight
MSt	Moderately Stunted
NNC	National Nutrition Council
NNC CO	National Nutrition Council Central Office
Ob	Obese
OW	Overweight
PNAO	Provincial Nutrition Action Officer
PNAP	Provincial Nutrition Action Plan
PNC	Provincial Nutrition Committee
PPAN	Philippine Plan of Action for Nutrition
RNAP	Regional Nutrition Action Plan
RNC	Regional Nutrition Committee
RNPC	Regional Nutrition Program Coordinator
RPAN	Regional Plan of Action for Nutrition
RSDC	Regional Social Development Committee
RTWG	Regional Technical Working Group
SD	Standard Deviation

SSt	Severely Stunted
SSt	Severely Stunted
St	Stunted
SUW	Severely underweight
SW/SAM	Severely Wasted/Severe Acute Malnutrition
UW	Underweight

DEFINITION OF TERMS

Barangay Nutrition Action Plan	Barangay-level version of the Philippine Plan of Action for Nutrition (PPAN), the country's directional framework for nutrition improvement.
eOPT Plus Tool	An Excel-based tool used by nutrition workers in the community to help them consolidate and summarize the results of OPT Plus among 0-to-59-month-old children.
OPT Plus	Regular growth assessment of all 0-to-59-month-old children in the barangays done by a trained team including planning, follow-up, and quality check activities.

Section 5.1

ROLES AND RESPONSIBILITIES OF THE OPT PLUS TEAM AND SUPERVISORS

The OPT Plus or mass measurement of children at the start of every year can help:

- Identify children's normal growth patterns and any growth problems from plotted points using appropriate growth charts or Early Child Care and Development (ECCD) card.
- Locate and identify preschoolers who are wasted, stunted, underweight, or overweight to be referred to appropriate nutrition and nutrition-related programs and interventions.
- Allows the community to determine the magnitude and prevalence of malnutrition and observe its trends that may be used to assess the effectiveness of local nutrition programs.
- Serve as a guide in setting prioritiesand targets, nutrition objectives, and preparing nutrition action plans for funding at the local level.

Table 5.1 presents the roles and responsibilities of the persons involved in the activity.

Table 5.1 Roles and Functions of the different persons involved in the OPT Plus during the OPT Plus Data Interpretation and Action Planning.

ORGANIZATION/PERSON INVOLVED	FUNCTION
National Nutrition Council Central Office (NNC CO)	1. Present the OPT Plus trend during National Barangay Nutrition Scholar (BNS) Conference and to the NNC Technical Committee, Technical Working Group, and other meetings. 2. Consolidate the OPT Plus results. 3. Include OPT Plus results in the Philippine Plan of Action for Nutrition specifically areas not covered by National Nutrition Survey to address nutrition problems.
Regional Nutrition Program Coordinator (RNPC)	1. Present the OPT Plus trend during BNS Conference and to the Regional Nutrition Council (RNC), Regional Technical Working Group (RTWG), and Regional Social Development Committee (RSDC) among other fora. 2. Consolidate the OPT Plus results. 3. Include OPT Plus results in the Regional Nutrition Action Plan (RNAP) to address nutrition problems.
Provincial Nutrition Action Officer (PNAO)	1. Present the OPT Plus trend to the Provincial Nutrition Committee (PNC) and Provincial Board (Sangguniang Panlalawigan). 2. Consolidate the OPT Plus results. 3. Include OPT Plus results in the Provincial Nutrition Action Plan (PNAP) to address nutrition problems. 4. Provide technical assistance in data analysis and interpretation.
Supervisor	1. Participate in the preparation of the Local Nutrition Action Plan (LNAP).

ORGANIZATION/PERSON INVOLVED	FUNCTION
City/Municipal Nutrition Action Officer (C/MNAO)	1. Organize the presentation of OPT PLUS results to City/Municipal Nutrition Committee (C/MNC). 2. Review and consolidate the updated Barangay Nutrition Action Plan (BNAP). 3. Facilitate the preparation of the City/Municipal Nutrition Action Plan (C/MNAP). 4. Present the results of the OPT Plus to the Sangguniang Panlungsod/ Bayan for the budget appropriation of the C/MNAP. 5. Facilitate approval of resolutions at the barangay, municipal, and city levels concerning OPT Plus activities.
OPT Plus Team members	1. Organize the presentation of OPT Plus results to BNC. 2. Facilitate the preparation and updating of the BNAP to the BNC. 3. Present the results of the OPT Plus to the Sangguniang Barangay for the budget appropriation of the BNAP. 4. Facilitate approval of resolutions at the barangay level concerning OPT Plus activities.
Barangay Nurition Committee (BNC)	1. Participate in the presentation of OPT Plus. 2. Participate in preparing and updating the BNAP. 3. Allocate funds to implement the programs/projects/ activities of the BNAP. 4. Review and approve resolutions concerning OPT Plus activities.

Section 5.2

USES OF OPT PLUS RESULTS

The OPT Plus or mass measurement of children at the start of every year can help:

- Identify children's growth problems from plotted points using a single indicator growth chart or ECCD card.
- Locate and identify preschoolers who are wasted, stunted, underweight, or overweight to be referred to appropriate nutrition and nutrition-related programs and interventions.
- Allows the community to determine the magnitude and prevalence of malnutrition and observe its trends that may be used to assess the effectiveness of local nutrition programs.
- Serve as a guide in setting priorities and targets, nutrition objectives, and preparing nutrition action plans for funding at the local level.

Section 5.3

REPORT GENERATION (EOPT PLUS AND FORMS)

The results of the nutritional assessment will be more useful if the data will be consolidated. The summary table can be generated automatically in the eOPT Plus Summary worksheet. If without access to a computer, tally the child's nutritional status using OPT Plus Form 1B.

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OPT Plus TSEK

OPT Plus Form B. Summary Sheet of the Nutritional Status of 0-59 Month-Old Children in a Barangay

Purok/ Area/ Block:	Sampaguita	**Province:**	Laguna	**Regn:**	IVA CALABARZON	**OPT Plus Coverage:**	24.06%
Barangay:	Batong Malake	**Total Popn Barangay:**	17,143	**Source:**	OH - EB 20		
Municipality:	LOS BAÑOS	**Estimated Popn of Children 0-59 mos in Barangay:**	1,546				
PSGC:	043411004	**Prevalence Rate MUW & SUW (Last OPT Plus):**		**Prevalence Rate MSt & SSt (Last OPT Plus):**		**Prevalence Rate MW/MAM & SW/SAM (Last OPT Plus):**	

Total # Indigenous Preschool Children Measured: 0-59 mos: 4

OPT PLUS **2022**

Note: Children with same names and birthdates found. Please validate on next visit. # of same name and birthdate: 1

'otal Boys: 214 | 'otal Girls: 168 | ›tal MUAC: | Total WFA: 375 | Total HFA: 373 | Total WFL/H: 372

																			Birth to 5 Years		**F1K**		**# IP Children**		
ACRONYMS & ABBREVIATIONS	0-5 Months			6-11 Months			12-23 Months			24-35 Months			36-47 Months			48-59 Months			**0-59 Months**		**0-23 Months**		Boys	Girls	Total
	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total	**Total**	**Prev**	**Total**	**Prev**			
WFA - Normal	14	12	26	18	14	32	35	31	66	29	32	61	36	32	68	51	28	79	**332**	**88.5%**	**124**	**86.7%**	1	0	1
WFA - Ob	**No Obese/Overweight classification in the WFA. Following international standards, we use WL/HZ to classify overweight and obesity in children.**																								
WFA - OW																									
WFA - MUW	1	1	2	7	2	9	4	1	5	4	1	5	4	1	5	4	3	7	**33**	**8.8%**	**16**	**11.2%**	0	1	1
WFA - SUW	0	0	0	1	1	2	1	0	1	1	0	1	0	2	2	0	0	0	**6**	**1.6%**	**3**	**2.1%**	0	0	0
HFA - Normal	**11**	**8**	**19**	**18**	**14**	**32**	**26**	**27**	**53**	**24**	**24**	**48**	**29**	**30**	**59**	**47**	**26**	**73**	**284**	**76.1%**	**104**	**71.7%**	0	1	1
HFA - Tall	**1**	**1**	**2**	**0**	**0**	**0**	**1**	**0**	**1**	**1**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**4**	**1.1%**	**3**	**2.1%**	0	0	0
HFA - MSt	**2**	**2**	**4**	**7**	**3**	**10**	**10**	**6**	**16**	**7**	**8**	**15**	**9**	**5**	**14**	**7**	**6**	**13**	**72**	**19.3%**	**30**	**20.7%**	0	0	0
HFA - SSt	**1**	**2**	**3**	**1**	**1**	**2**	**2**	**1**	**3**	**2**	**1**	**3**	**2**	**0**	**2**	**0**	**0**	**0**	**13**	**3.5%**	**8**	**5.5%**	0	0	0
WFL/H - Normal	12	8	20	22	15	37	34	27	61	27	31	58	37	34	71	50	30	80	**327**	**87.9%**	**118**	**81.9%**	0	0	0
WFL/H - OW	2	2	4	1	0	1	4	3	7	2	2	4	1	0	1	3	1	4	**21**	**5.6%**	**12**	**8.3%**	0	0	0
WFL/H - Ob	1	2	3	0	1	1	1	1	2	0	0	0	2	0	2	0	0	0	**8**	**2.2%**	**6**	**4.2%**	0	0	0
WFL/H - MW/MAM	0	1	1	2	0	2	0	2	2	5	0	5	0	0	0	1	1	2	**12**	**3.2%**	**5**	**3.5%**	0	0	0
WFL/H - SW/SAM	0	0	0	1	2	3	0	0	0	0	0	0	0	1	1	0	0	0	**4**	**1.1%**	**3**	**2.1%**	0	1	1
MUAC - Normal																									
MUAC - MW/MAM																									
MUAC - SW/SAM																									
Total (WFA)	**15**	**13**	**28**	**26**	**17**	**43**	**40**	**32**	**72**	**34**	**33**	**67**	**40**	**35**	**75**	**55**	**31**	**86**							

Summary of Children covered by e-OPT Plus		**Mothers/Caregivers Summary**		**Data Inaccuracy**	
ldren 0-59 mos. who are Wasted and/or Stunted :	100	Total Number of M/Cs of children 0-59 mos. old:	316	en with names and birthdate repeated:	1
dren 24-59 mos. who are Wasted and/or Stunted:	54	# M/Cs of 0-59 mos children affected by Wasting and/or Stunting:	91	# Children with missing information:	7
Children 0-59 mos. who are Overweight/Obesity:	29	# M/Cs of 0-59 mos. children who are Overweight/Obesity:	28	dren with no name of parents/address:	2
Total Number of Children 0-23 mos. old:	153	Total Number of M/Cs of children 0-23 mos. old:	146	# Children with no sex data:	1
ildren 0-23 mos. who are Wasted and/or Stunted:	46	# M/Cs of 0-23 mos children affected by Wasting and/or Stunting:	45	# Children with no date of birth data:	1
Total Number of Children 0-29 mos. old:	185			# Children older than 59 months:	0
Total Number of Children 30-59 mos. old:	198			dren with length/height but no weight:	2
Total Number of Children 24-59 mos. old:	230			dren with weight but no length/height:	3
# Children 0-59 mos. w/ Edema:	3			# Children with no MUAC:	4
# Children 0-59 mos. w/ disability:	2				

Figure 5.1. Summary Sheet generated from the eOPT Plus (OPT Plus Form 1B).

Section 5.4

DATA ANALYSIS AND INTERPRETATION

Data needs to be organized and visualized so that the reader can understand the information for its purpose. Organization also allows proper data analysis and interpretation and thus derives meaning. Data analysis and interpretation can be made on individual and community levels.

Individual Level

Using obtained information and accurate measurements, these are then plotted as points using appropriate growth charts which can also be found in the Early Childhood Care and Development Card. The following growth indicators are plotted to create line graphs that indicate the rank of the child's measurement. The charts can be seen in Annexes 5.1a and 5.1b.

Plotting length/height-for-age

PURPOSE

- Monitor growth in length or height of the child's age at a given visit.
- Identify children who are stunted (short) due to prolonged undernutrition or repeated illness.
- Identify children who are tall for their age.

PLOTTING POINTS

1. Select the growth chart for a girl or boy.
2. Identify the age then plot on the corresponding vertical line (not between vertical lines).
3. Plot on or between the horizontal lines as precisely as possible.
4. When points are plotted for two or more visits, connect adjacent points with a straight line to better observe the trend.

Plotting weight-for-age

PURPOSE

- Monitor weight relative to the child's age at the time of visit.
- Assess whether a child is moderately or severely underweight.

Note: The WFA indicator is NOT USED to classify a child as overweight or obese. THIS CAN NOT BE RELIED UPON situations where the child's age cannot be accurately determined

PLOTTING POINTS

1. Select the growth chart for a girl or boy.
2. Identify the age then plot on the corresponding vertical line (not between vertical lines).
3. Plot on or between the horizontal lines to show weight measurement to 0.1 kg, e.g., 7.8 kg.
4. When points are plotted for two or more visits, connect adjacent points with a straight line to better observe trends.

Plotting length/height-for-age

PURPOSE

- Monitor weight in proportion to attained growth in length or height.
- Identify children with low weight-for-height who may be moderately wasted or severely wasted, especially when the children's ages are unknown.
- Identify wasting caused by a recent illness or food shortage that causes acute and severe weight loss, although chronic undernutrition or illness can also cause this condition.
- Identify children with high weight-for-length/height who may be at risk of becoming overweight or obese.

PLOTTING POINTS

1. Select the growth chart for a girl or boy.
2. Identify the height and plot the length or height on the corresponding vertical line (e.g., 75 cm, 78 cm).

   Note: It is necessary to round the measurement to the nearest whole centimeter (i.e., round down if 0.1 to 0.4 and round up if 0.5 to 0.9).
3. Plot weight on the corresponding horizontal line as precisely as possible.
4. When points are plotted for two or more visits, connect adjacent points with a straight line to observe the trend better.

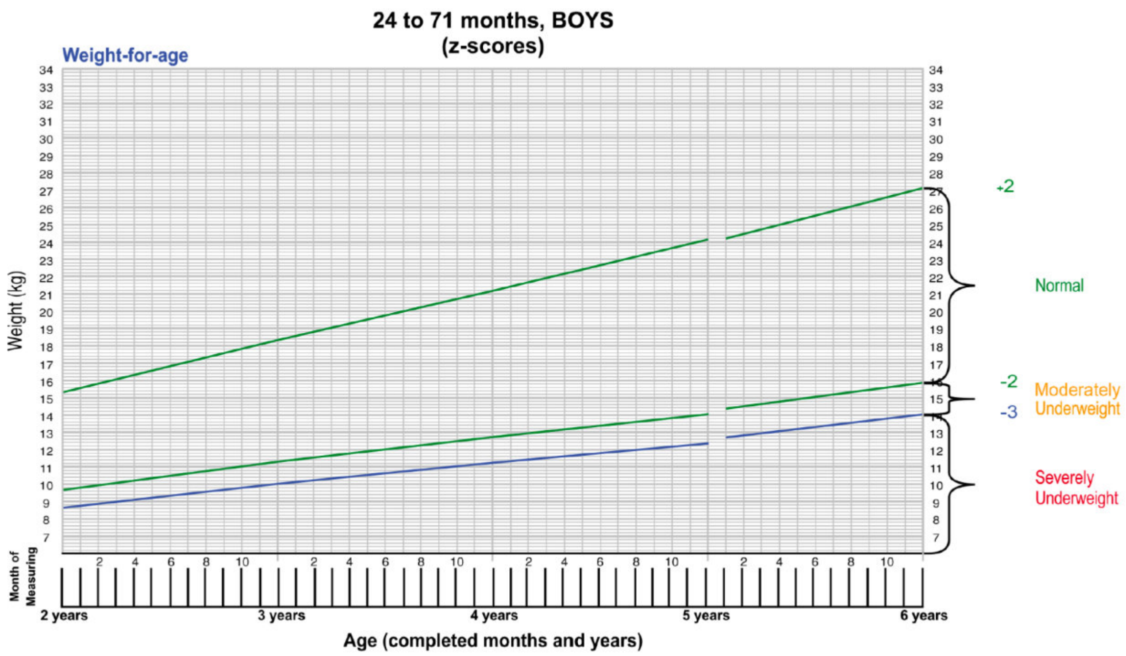
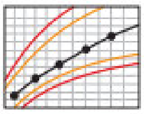


Figure 5.2. Sample ECCD Growth Chart for plotting points.

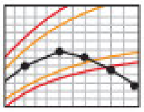
Interpreting plotted points for growth indicators

HOW TO INTERPRET GROWTH CHARTS

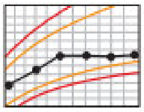
It is important to consider the child's whole situation when interpreting trends on the growth chart. How to interpret trends of growth on charts:



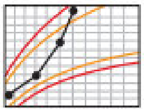
A child who is growing well has a trend line that tracks fairly close to the same Z-score line and will be on or between the -2 and +2 Z-score lines. The caregiver should be encouraged to continue giving proper care to the child.



A normal or undernourished child with a sharp decline on the growth line indicates growth problem to be investigated and remedied.



A child with a flat growth curve is in danger of developing malnutrition. The child should be referred to the trained health worker for appropriate advice and/or intervention.



A child who is already overweight should not have a rising growth line. Counseling to increase the child's physical activity while maintaining a healthy diet should be done.

To determine correctly if a child is obese, refer to the Weight-for-length or Weight-for-height charts.

Using the symbols below, record relevant information or condition of the child at the time of visit to the health center/clinic:

- • - weight
- B - breastfeeding
- CF - complementary feeding
- A - therapeutic vitamin A
- EB - exclusive breastfeeding
- F - fever
- C - cold/cough
- CP - cleft palate
- D - diarrhea
- H - hospitalized
- I - injury
- O - others

Table 5.2. Growth indicators according to Z-score.

<table>
<tr><th rowspan="2">Z-SCORE</th><th colspan="4">GROWTH INDICATOR</th></tr>
<tr><th>LENGTH/HEIGHT FOR AGE</th><th>WEIGHT FOR AGE</th><th>WEIGHT FOR LENGTH</th><th>WEIGHT FOR HEIGHT</th></tr>
<tr><td>>+3SD</td><td rowspan="3">Tall</td><td rowspan="3">Overweight</td><td>Obese</td><td>Obese</td></tr>
<tr><td>+3SD</td><td rowspan="2">Overweight</td><td rowspan="2">Overweight</td></tr>
<tr><td>>+2SD</td></tr>
<tr><td>+2SD</td><td rowspan="2">Normal</td><td rowspan="2">Normal</td><td rowspan="2">Normal</td><td rowspan="2">Normal</td></tr>
<tr><td>-2SD</td></tr>
<tr><td><-2SD</td><td rowspan="2">Moderately Stunted</td><td rowspan="2">Moderately Underweight</td><td rowspan="2">Moderately Wasted</td><td rowspan="2">Moderately Wasted</td></tr>
<tr><td>-3SD</td></tr>
<tr><td><-3SD</td><td>Severely Stunted</td><td>Severely Underweight</td><td>Severely Wasted</td><td>Severely Wasted</td></tr>
</table>

Community Level

1. **PRESENT.** Select, organize, and group ideas and evidence in a logical way. Below are some ways to present the OPT Plus results.

   The BNS may present the overall prevalence of underweight, wasted, and stunted in the barangay by sex, age group, and purok using the OPT Plus Form 1B table (Figure 5.3) and graphs from the eOPT tool and with the aid of a spot map.

Republic of the Philippines
Department of Health
NATIONAL NUTRITION COUNCIL
IVA CALABARZON

OPT Plus TSEK

OPT Plus Form B. Summary Sheet of the Nutritional Status of 0-59 Month-Old Children in a Barangay

Purok/ Area/ Block:	Sampaguita	**Province:**	Laguna	**Regn:**	IVA CALABARZON	**OPT Plus Coverage:**	24.06%
Barangay:	Batong Malake	**Total Popn Barangay:**	17,143	**Source:**	OH - EB 20		
Municipality:	LOS BAÑOS	**Estimated Popn of Children 0-59 mos in Barangay:**	1,546				
PSGC:	043411004	**Prevalence Rate MUW & SUW (Last OPT Plus):**		**Prevalence Rate MSt & SSt (Last OPT Plus):**		**Prevalence Rate MW/MAM & SW/SAM (Last OPT Plus):**	

Total # Indigenous Preschool Children Measured: 0-59 mos: 4

OPT PLUS 2022

Note: Children with same names and birthdates found. Please validate on next visit. # of same name and birthdate: 1

otal Boys: 214 | otal Girls: 168 | tal MUAC: | Total WFA: 375 | Total HFA: 373 | Total WFL/H: 372 | **Birth to 5 Years** | **F1K** | **# IP Children**

ACRONYMS & ABBREVIATIONS	0-5 Months			6-11 Months			12-23 Months			24-35 Months			36-47 Months			48-59 Months			**0-59 Months**		**0-23 Months**				
	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total	**Total**	**Prev**	**Total**	**Prev**	Boys	Girls	Total
WFA - Normal	14	12	26	18	14	32	35	31	66	29	32	61	36	32	68	51	28	79	332	88.5%	124	86.7%	1	0	1
WFA - Ob	**No Obese/Overweight classification in the WFA. Following international standards, we use WL/HZ to classify overweight and obesity in children.**																								
WFA - OW																									
WFA - MUW	1	1	2	7	2	9	4	1	5	4	1	5	4	1	5	4	3	7	33	8.8%	16	11.2%	0	1	1
WFA - SUW	0	0	0	1	1	2	1	0	1	1	0	1	0	2	2	0	0	0	6	1.6%	3	2.1%	0	0	0
HFA - Normal	11	8	19	18	14	32	26	27	53	24	24	48	29	30	59	47	26	73	284	76.1%	104	71.7%	0	1	1
HFA - Tall	1	1	2	0	0	0	1	0	1	1	0	1	0	0	0	0	0	0	4	1.1%	3	2.1%	0	0	0
HFA - MSt	2	2	4	7	3	10	10	6	16	7	8	15	9	5	14	7	6	13	72	19.3%	30	20.7%	0	0	0
HFA - SSt	1	2	3	1	1	2	2	1	3	2	1	3	2	0	2	0	0	0	13	3.5%	8	5.5%	0	0	0
WFL/H - Normal	12	8	20	22	15	37	34	27	61	27	31	58	37	34	71	50	30	80	327	87.9%	118	81.9%	0	0	0
WFL/H - OW	2	2	4	1	0	1	4	3	7	2	2	4	1	0	1	3	1	4	21	5.6%	12	8.3%	0	0	0
WFL/H - Ob	1	2	3	0	1	1	1	1	2	0	0	0	2	0	2	0	0	0	8	2.2%	6	4.2%	0	0	0
WFL/H - MW/MAM	0	1	1	2	0	2	0	2	2	5	0	5	0	0	0	1	1	2	12	3.2%	5	3.5%	0	0	0
WFL/H - SW/SAM	0	0	0	1	2	3	0	0	0	0	0	0	0	1	1	0	0	0	4	1.1%	3	2.1%	0	1	1
MUAC - Normal																									
MUAC - MW/MAM																									
MUAC - SW/SAM																									
Total (WFA)	15	13	28	26	17	43	40	32	72	34	33	67	40	35	75	55	31	86							

Summary of Children covered by e-OPT Plus		Mothers/Caregivers Summary		Data Inaccuracy	
ldren 0-59 mos. who are Wasted and/or Stunted :	100	Total Number of M/Cs of children 0-59 mos. old:	316	en with names and birthdate repeated:	1
dren 24-59 mos. who are Wasted and/or Stunted:	54	# M/Cs of 0-59 mos children affected by Wasting and/or Stunting:	91	# Children with missing information:	7
Children 0-59 mos. who are Overweight/Obesity:	29	# M/Cs of 0-59 mos. children who are Overweight/Obesity:	28	dren with no name of parents/address:	2
Total Number of Children 0-23 mos. old:	153	Total Number of M/Cs of children 0-23 mos. old:	146	# Children with no sex data:	1
ildren 0-23 mos. who are Wasted and/or Stunted:	46	# M/Cs of 0-23 mos children affected by Wasting and/or Stunting:	45	# Children with no date of birth data:	1
Total Number of Children 0-29 mos. old:	185			# Children older than 59 months:	0
Total Number of Children 30-59 mos. old:	198			dren with length/height but no weight:	2
Total Number of Children 24-59 mos. old:	230			dren with weight but no length/height:	3
# Children 0-59 mos. w/ Edema:	3			# Children with no MUAC:	4
# Children 0-59 mos. w/ disability:	2				

Figure 5.3. Summary sheet generated from the eOPT Plus.

2. **ANALYZE.** Examine the data for patterns and initial observations and then explore any data limitations. Before beginning data analysis, it is important to discuss any limitations in the data. Make sure not only the correct measurements are done but as well as the correct encoding and identification of nutritional status. There are also data quality checks that can be used in Section 7.

Different Ways to Analyze the OPT Plus results:

a. Present the previous OPT results and compare them with the current year with the MNAO/OPT Team. Table 5.2 is the sample matrix. Discuss the data if there is a decrease or increase in the prevalence in the different indices. Take note of the questions raised by the BNC.

Table 5.3. Template for recording yearly results of nutritional status.

INDICES	NUTRITIONAL STATUS	YEAR/PREVALENCE		
Weight-for-Age	Normal			
	MUW			
	SUW			
Heigh-for-Age	Normal			
	Tall			
	MSt			
	SSt			
Weight-for-Length/Height	Normal			
	OW			
	Ob			
	MW			
	SW			

b. Use prevalence thresholds for wasted, overweight, and stunted (Table 5.3) in 0-59-month-old children.
   - Thresholds are cut-off values used for interpreting measurements of individual children.
   - The different labels used for the Interpretation of Anthropometry are 'low,' 'medium,' 'high,' and 'very high.' A category labeled 'very low' has no public health concern.
   - The labels can be used for descriptive purposes to map communities according to severity levels, identify priority communities for action, and by governments to trigger action and target programs to achieve 'low' or 'very low' levels.

Table 5.4. Prevalence thresholds for wasting, overweight, and stunting in children under five years (de Onis et al., 2019).

WASTING PREVALENCE THRESHOLDS (%)	LABELS	OVERWEIGHT PREVALENCE THRESHOLDS (%)	LABELS	STUNTING PREVALENCE THRESHOLDS (%)	LABELS
<2.5	Very Low	<2.5	Very Low	<2.5	Very Low
2.5 - <5	Low	2.5 - <5	Low	2.5 - <10	Low
5 - <10	Medium	5 - <10	Medium	10 - <20	Medium
10 - <15	High	10 - <15	High	20 - <30	High
≥15	Very High	≥15	Very High	≥30	Very High

c. Use the spot map (Figure 5.4) to visualize which purok/s have the highest malnourished children.

Figure 5.4. Sample spot-map showing the location of malnourished children.

3. **INTERPRET.** Understanding the findings and why they are occurring can help address them and focus on the action plan.

   - A high prevalence of stunting in a community indicates the presence of a more severe and chronic form of undernutrition, which may be due to long-term food deprivation and overall poor health. Use the graphs (Figures 5.5 and 5.6) generated from the eOPT Plus tool to present the OPT Plus results.

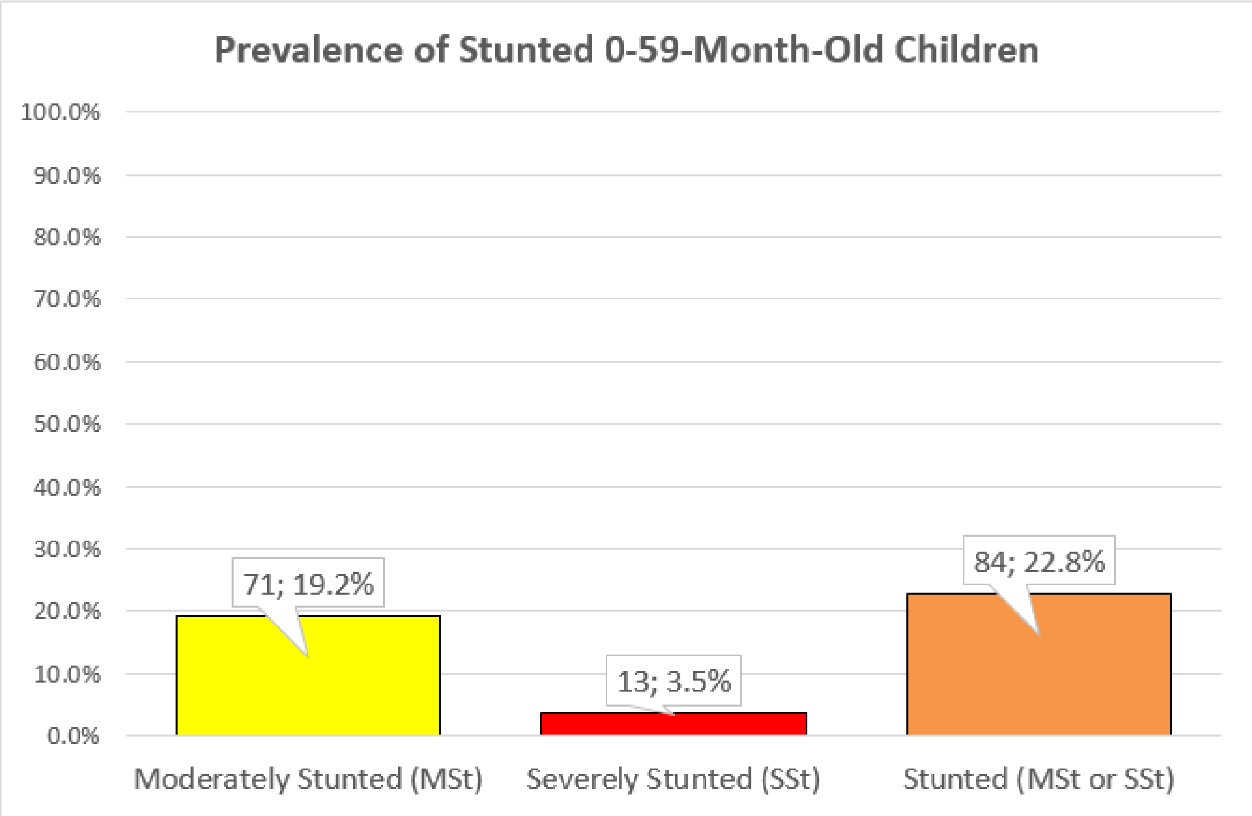


Figure 5.5. Graph on the prevalence of stunted 0-to-59-month-old children generated from the eOPT Plus.

- A high prevalence of wasting in a community may indicate a short-term/ acute food shortage as in the cases of calamities and/or widespread infections in the immediate past. If not addressed immediately, wasting may increase stunting in the community in the long term.

- Several data points for wasting can allow analysis of seasonality patterns and trends of food shortages and certain illnesses. It will enable more sensitive forecasting and targeting of children selected for supplemental feeding to prevent further undernutrition.

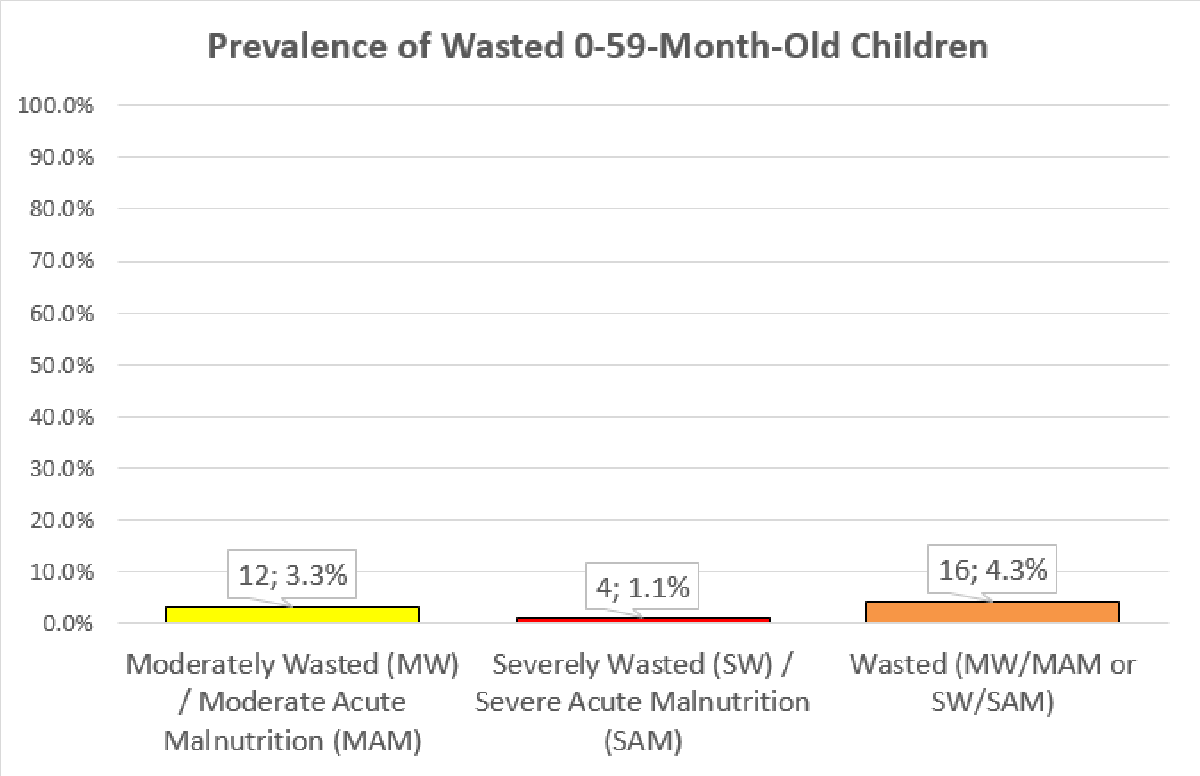


Figure 5.6. Graph on the prevalence of wasted 0-to-59-month-old children generated from the eOPT Plus.

Section 5.5

DATA UTILIZATION AND ACTION PLANNING

The activity does not end with submitting and presenting results. The BNC should use the data generated from OPT Plus for action planning. The National Nutrition Council (NNC) issued Governing Board (GB) Resolution No. 3, S. 2014 Approving and Adopting the Guidelines on Local Nutrition Planning. Using the OPT Plus results, household profile, and other data available in the barangay, the following guide questions will guide you in ensuring that nutrition problems in your community are properly addressed:

1. What forms of malnutrition were identified in the OPT Plus?
2. How many are malnourished children?
3. Where are they located?
4. What are the causes of malnutrition in the area? Or the child?
5. What nutrition interventions should be implemented?
6. What is the cost of implementing the nutrition interventions?
7. Where can we get the resource?

NOTE 5.1.

Use the OPT Plus data in the targeting of the number of children in the BNAP. The OPT Plus Form 1B already has the number of 0-to-59-month-old children, 0-to-23-months, and children UW, SUW, MW/MAM, SW/SAM, St, OW, and Ob summarized. Use this data in the Annual Target number in the different nutrition activities.

Make a table on where data will be utilized (from OPT Plus to other sectors and how they can utilize data) in the preparation of the BNAP. The BNAP is a local version of the Philippine Plan of Action for Nutrition (PPAN), the country's directional framework for nutrition improvement. Below is the BNAP Outline which can be found in the Barangay Nutrition Program Management Handbook.

Similarly, the consolidated results of the OPT Plus must be used at the municipal/city and provincial level in the targeting of the number of children in the municipal/city/provincial nutrition action plan. The OPT Plus results may also be used at the regional level for targeting of the Regional Plan of Action for Nutrition (RPAN).

REFERENCES

de Onis, M., Borghi, E., Arimond, M., Webb, P., Croft, T., Saha, K., De-Regil, L. M., Thuita, F., Heidkamp, R., Krasevec, J., Hayashi, C., & Flores-Ayala, R. (2019). Prevalence thresholds for wasting, overweight and stunting in children under 5 years. Public health nutrition, 22(1), 175–179. https://doi.org/10.1017/S1368980018002434

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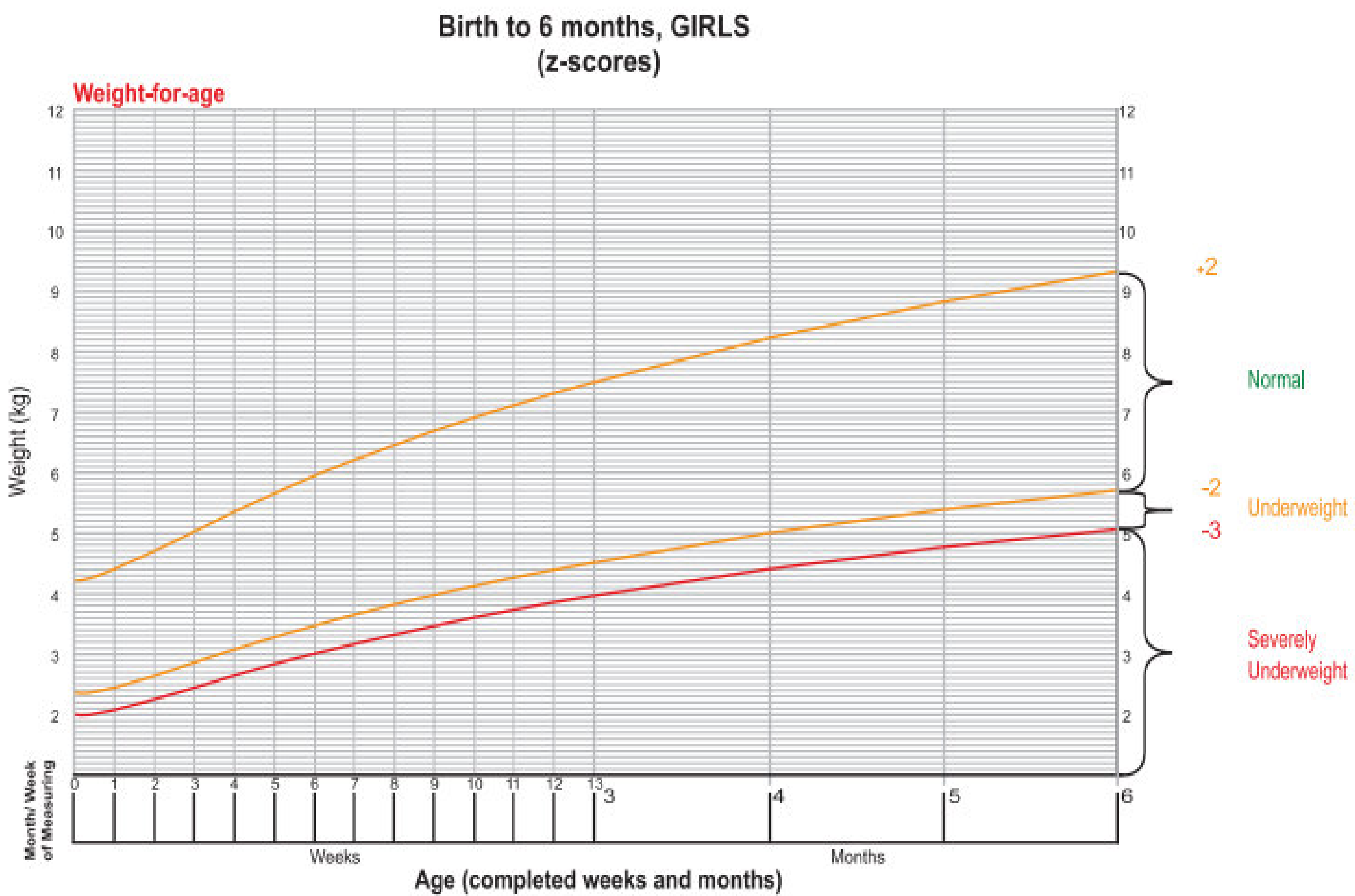
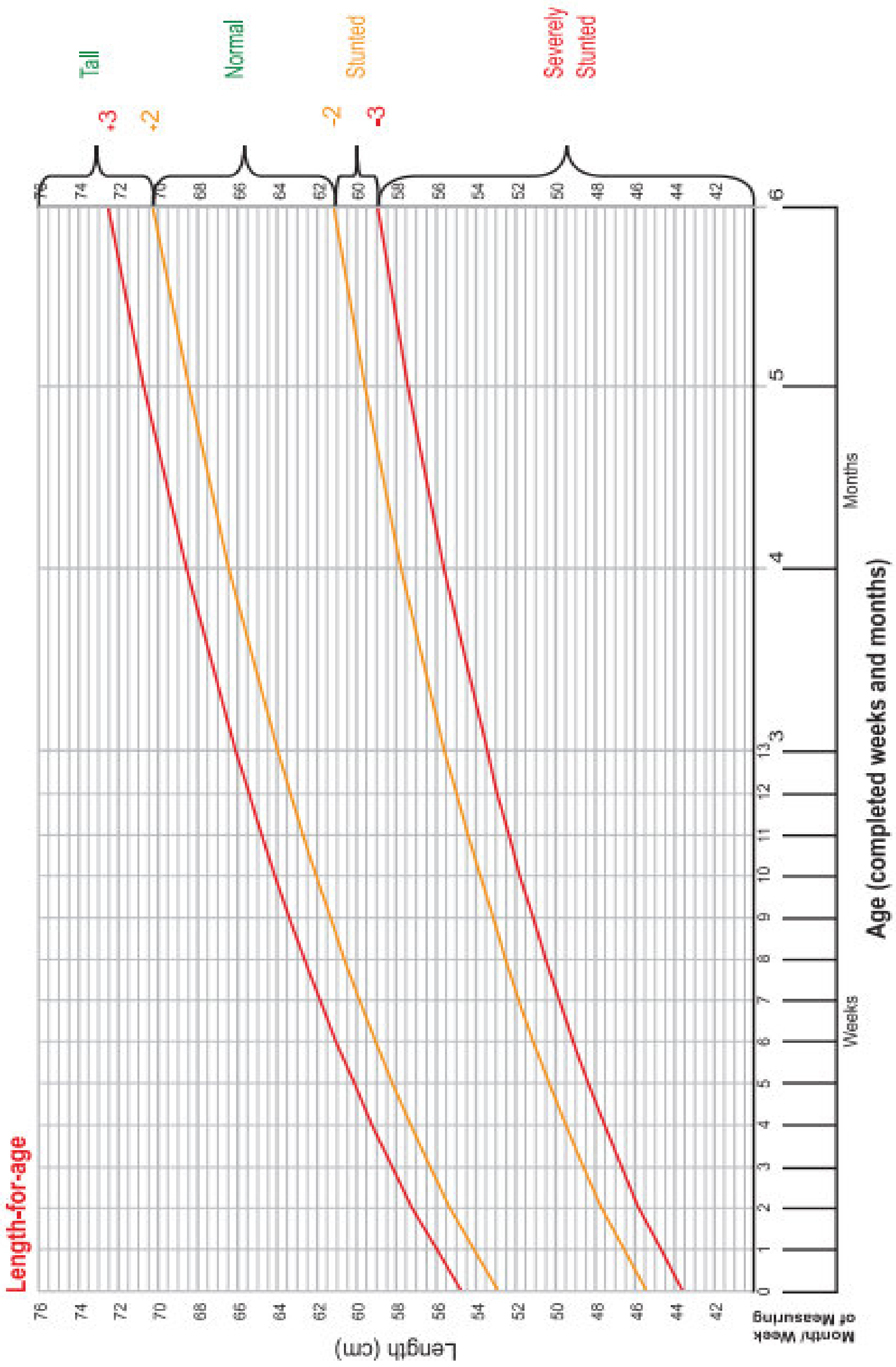
Lalap, B.A, Maneja, M.C.P., Gonzales, P.G., Abuyog, A.T., Castaneda, A.B., Garma, G.F., Felix, A. dR. BIDANI Network Program. Participative Nutrition Enhancement Approach Manual. Unpublished. 2021

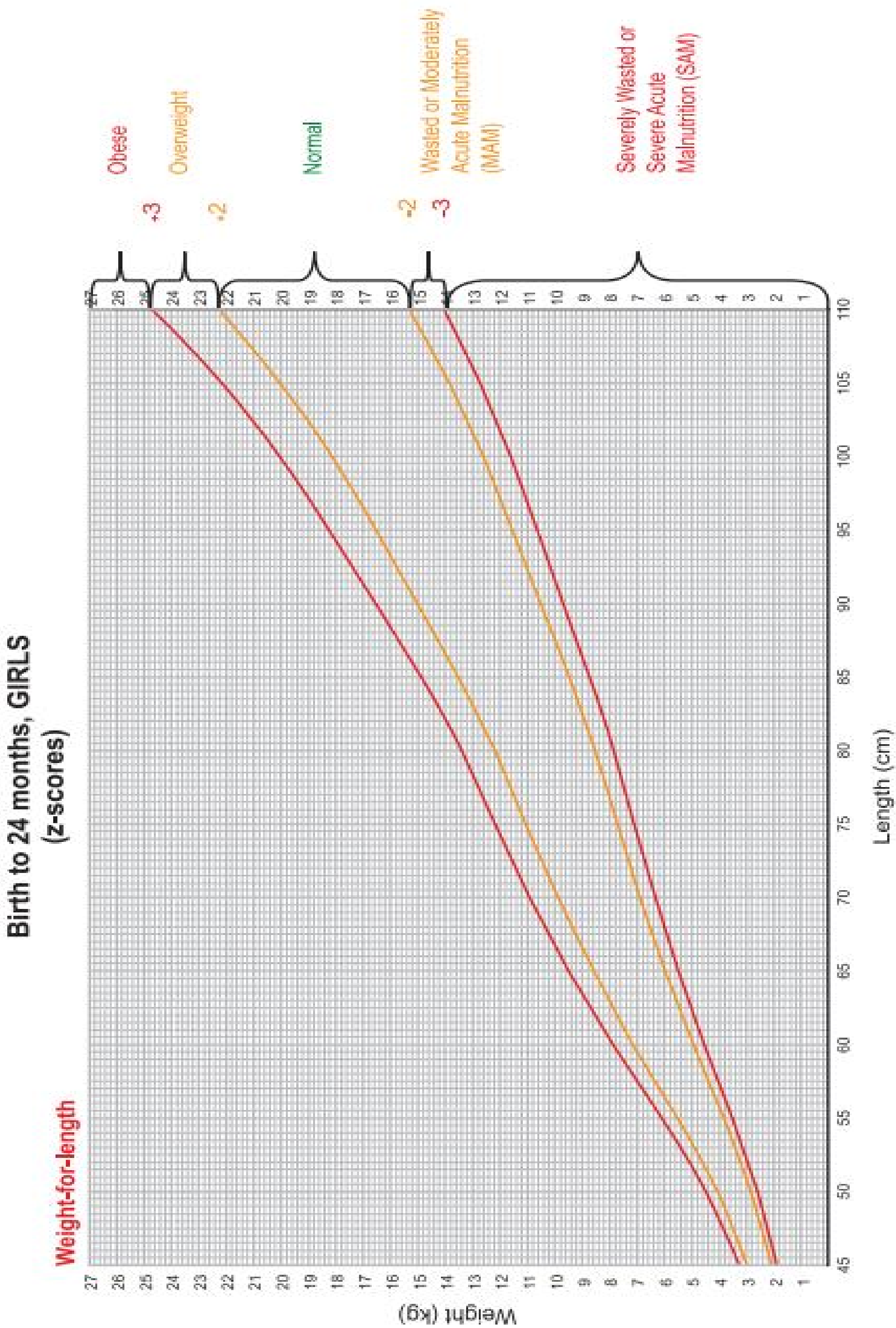
LIST OF ANNEXES

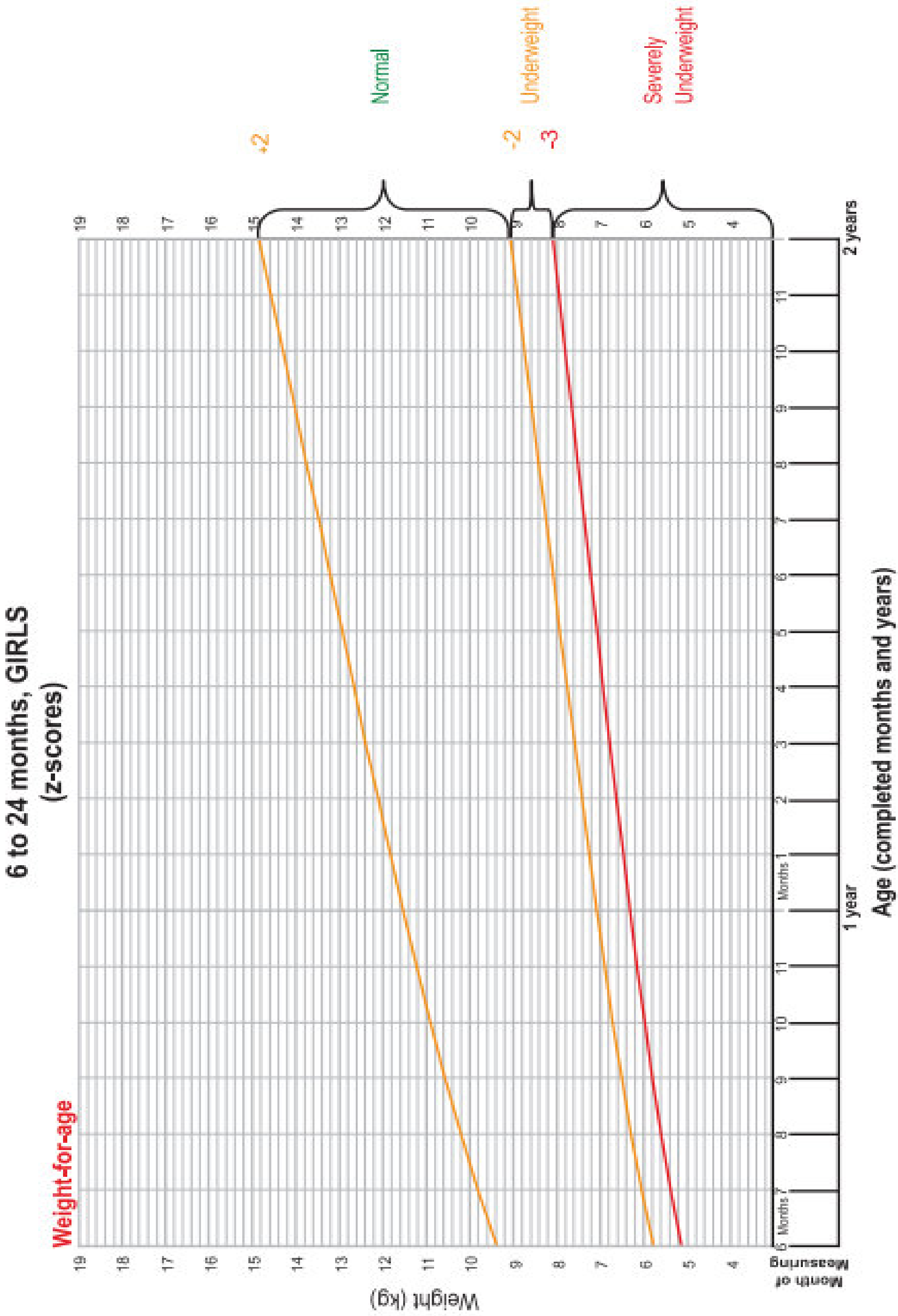
ANNEX 5.1a. Growth chart for girls by age group

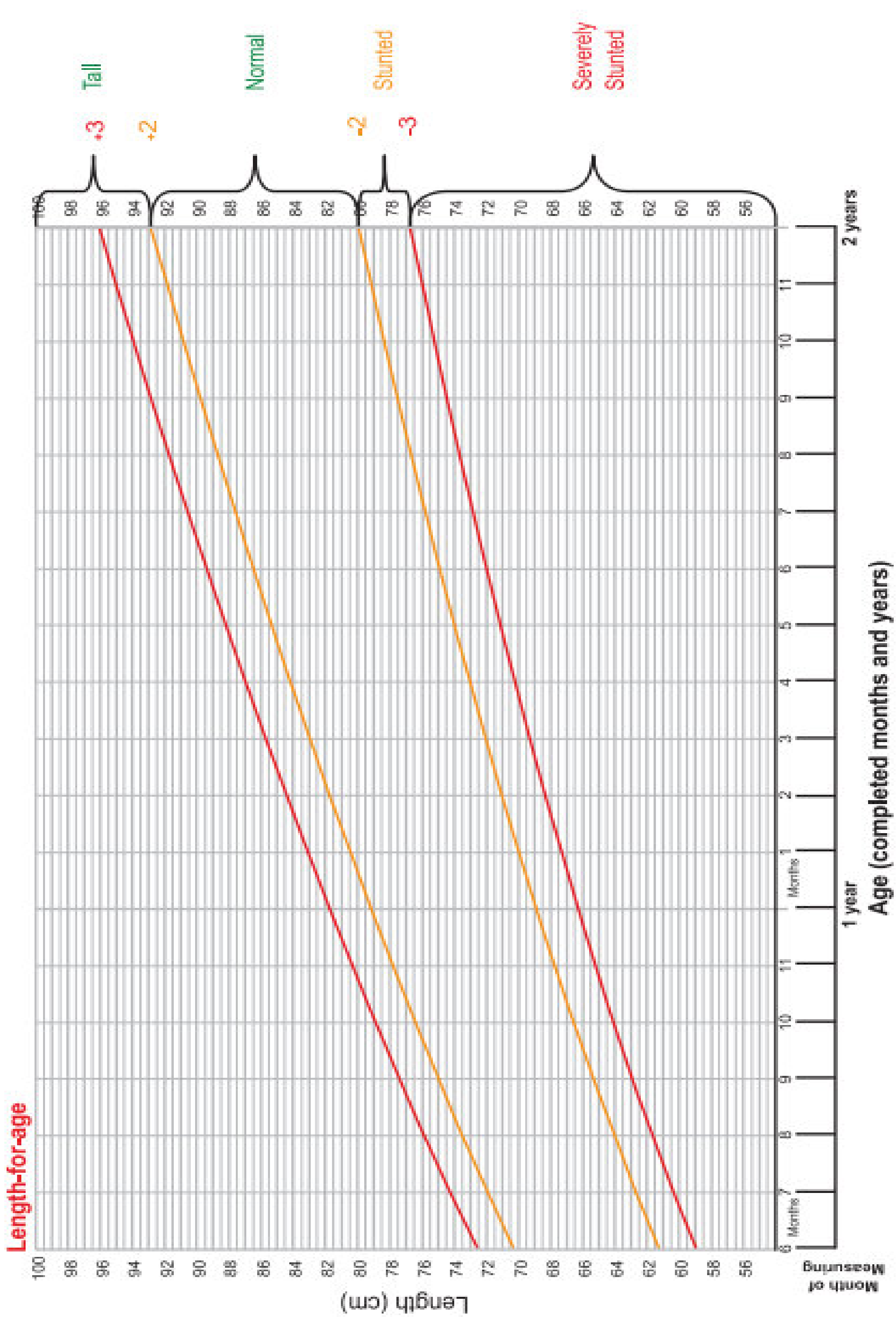




Reference: WHO Child Growth Standards, Methods and Development, 2006


Weight-for-length
Birth to 24 months, GIRLS
(z-scores)
Weight (kg)
Length (cm)
+3
+2
-2
-3
Obese
Overweight
Normal
Wasted or Moderately Acute Malnutrition (MAM)
Severely Wasted or Severe Acute Malnutrition (SAM)


Weight-for-age
6 to 24 months, GIRLS
(z-scores)
Weight (kg)
Month of Measuring
Age (completed months and years)
6 Months
1 year
2 years
+2
-2
-3
Normal
Underweight
Severely Underweight

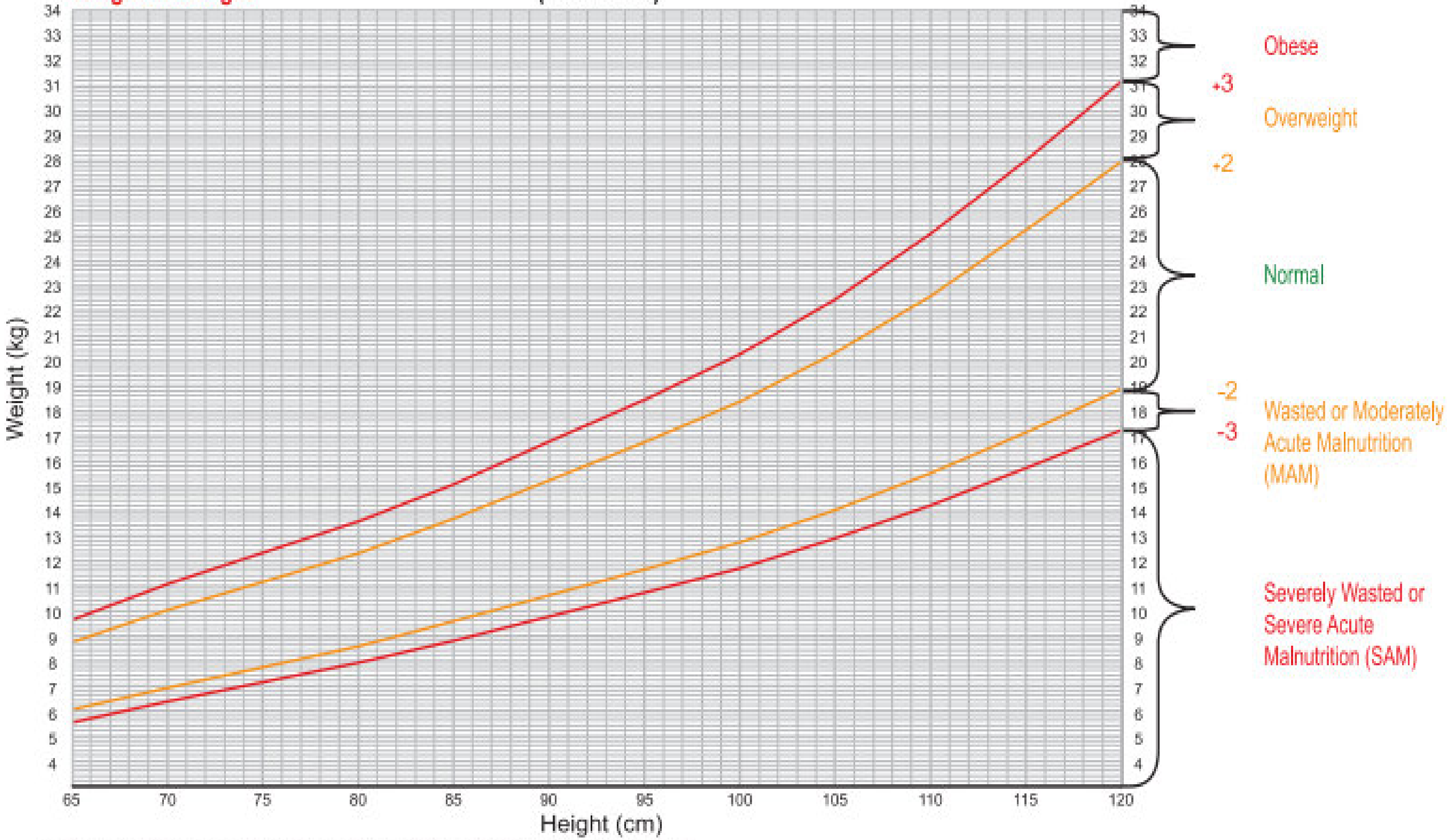


Reference: WHO Child Growth Standards, Methods and Development, 2006

(As of 29 025 2018)

24 to 60 months, GIRLS*
(z-scores)

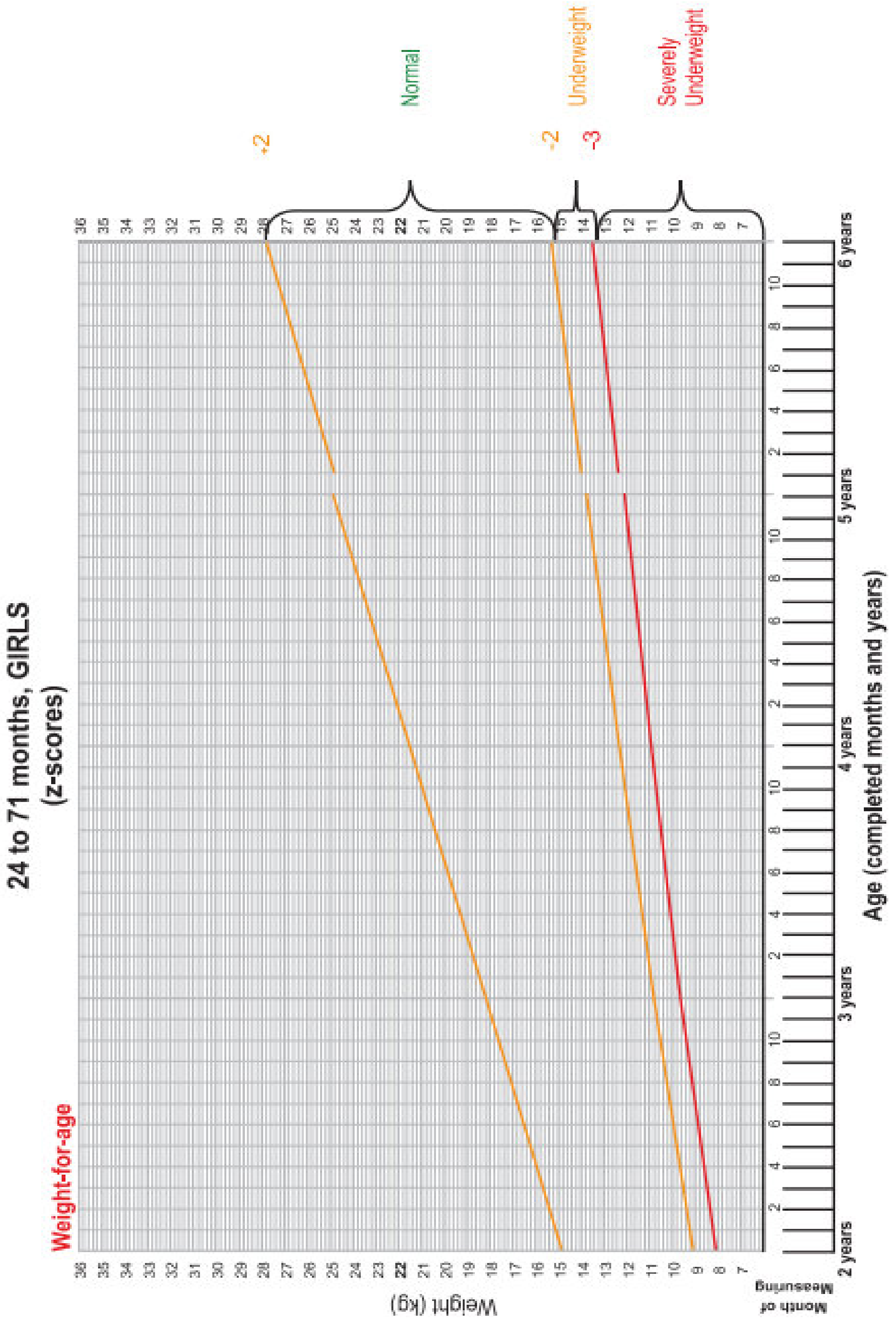
Weight-for-height

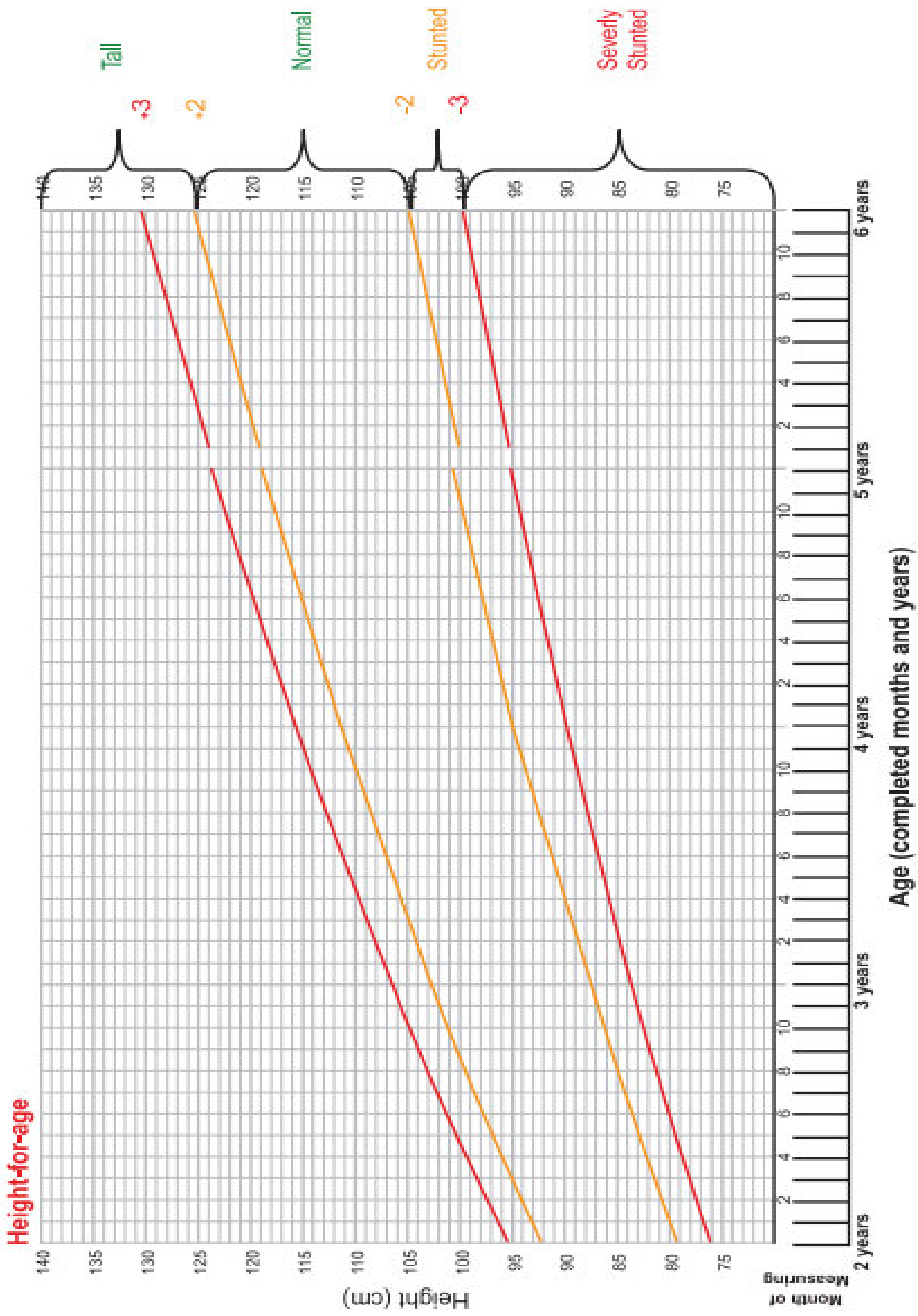


Reference: WHO Child Growth Standards, Methods and Development, 2006

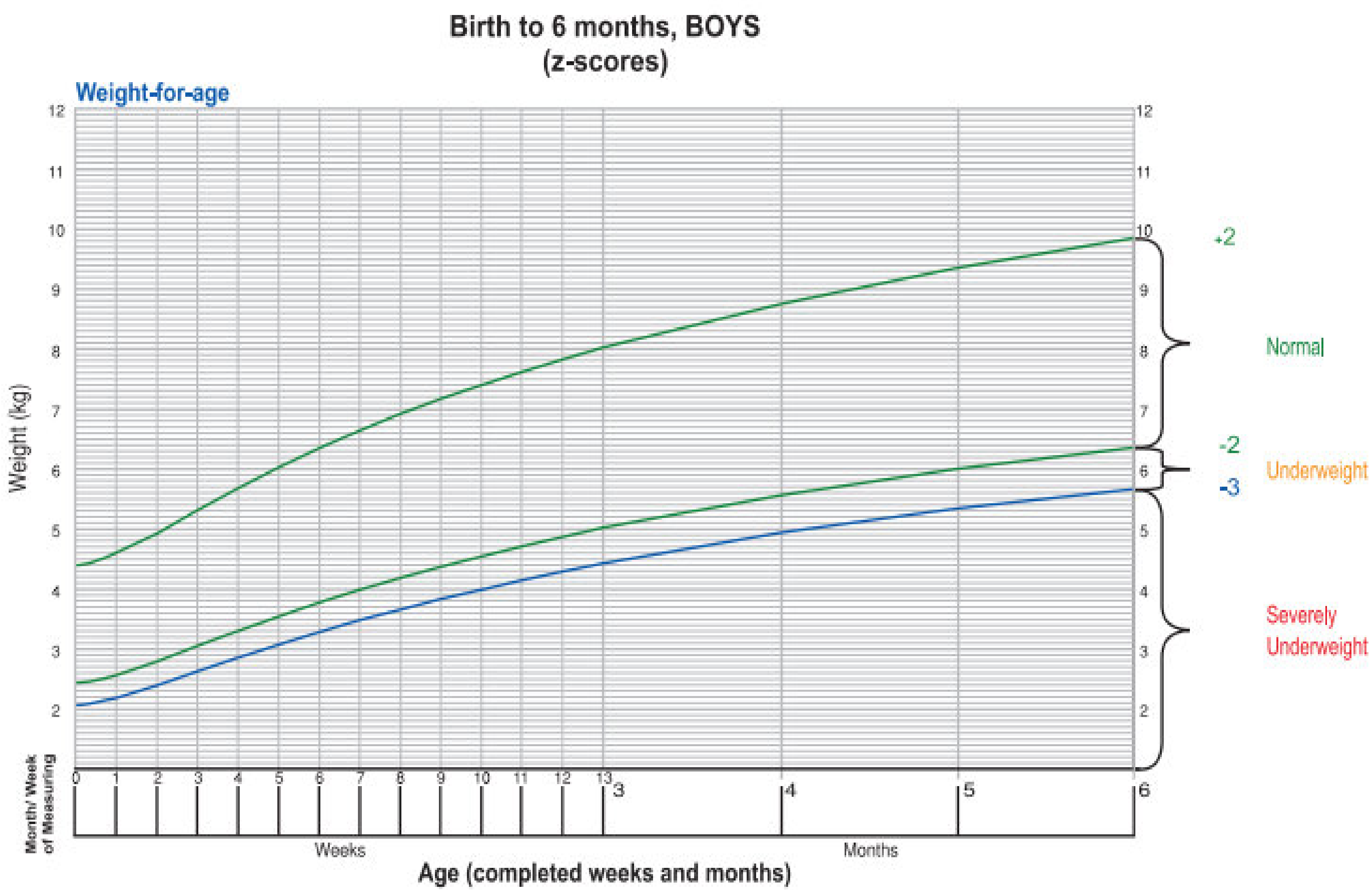
* For children ages above 60 months, use the BMI-for-age table.

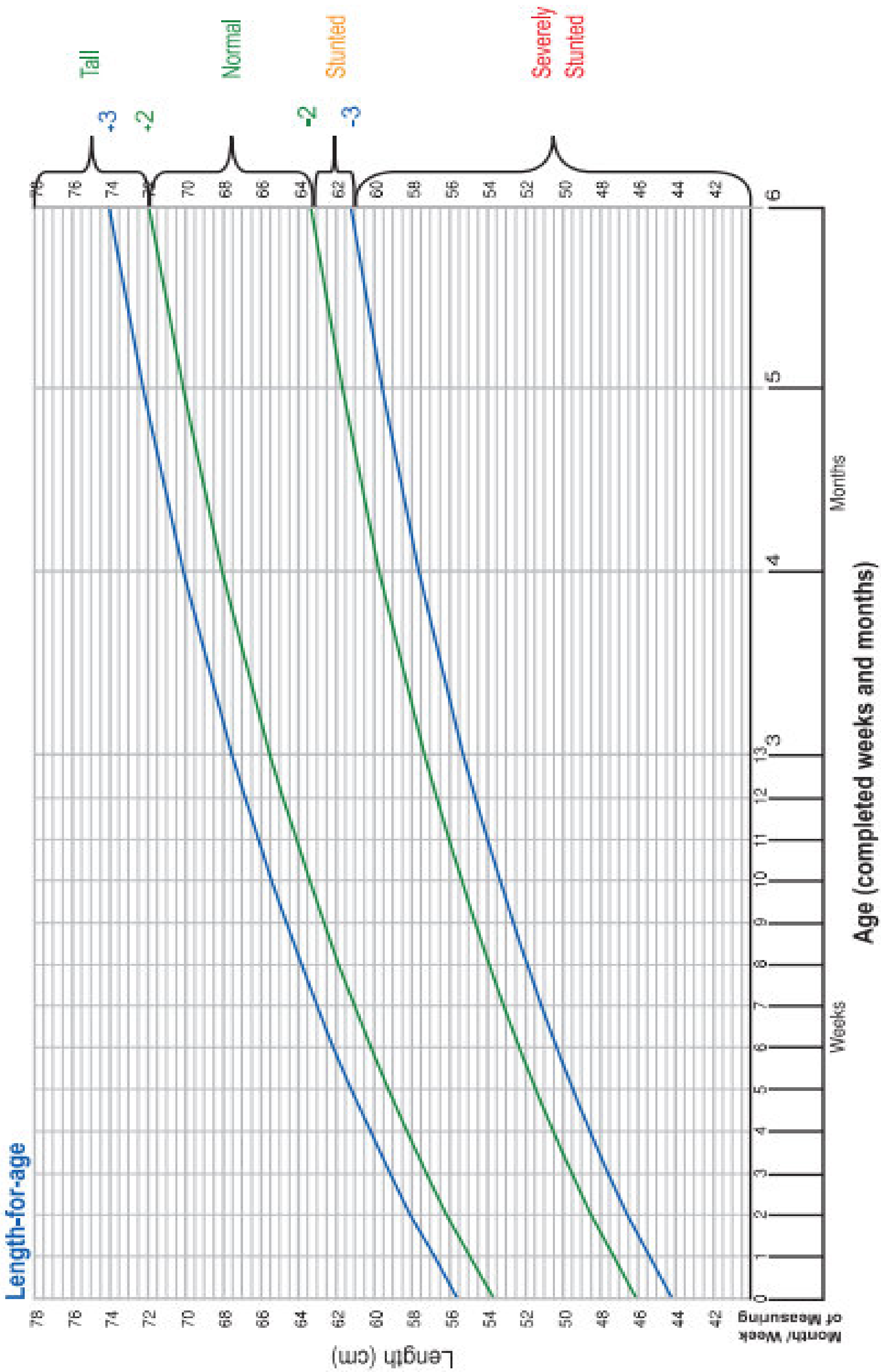
(As of 29 025 2018)


Weight-for-age
24 to 71 months, GIRLS
(z-scores)
Weight (kg)
Month of Measuring
Age (completed months and years)
2 years
3 years
4 years
5 years
6 years
+2
-2
-3
Normal
Underweight
Severely Underweight

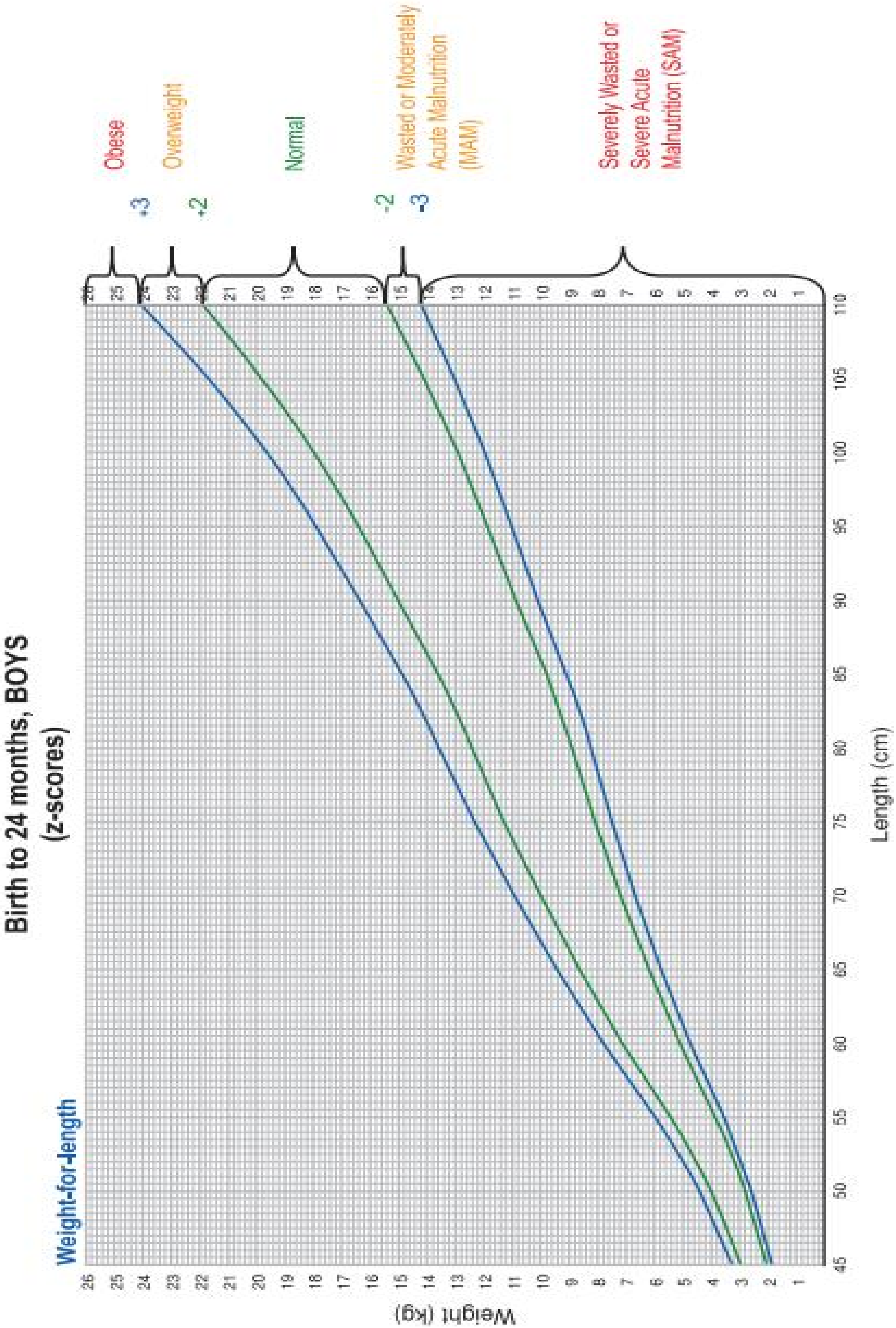

Height-for-age
Height (cm)
Month of Measuring
Age (completed months and years)
2 years
3 years
4 years
5 years
6 years
Tall
Normal
Stunted
Severly Stunted
+3
+2
-2
-3
140
135
130
125
120
115
110
105
100
95
90
85
80
75

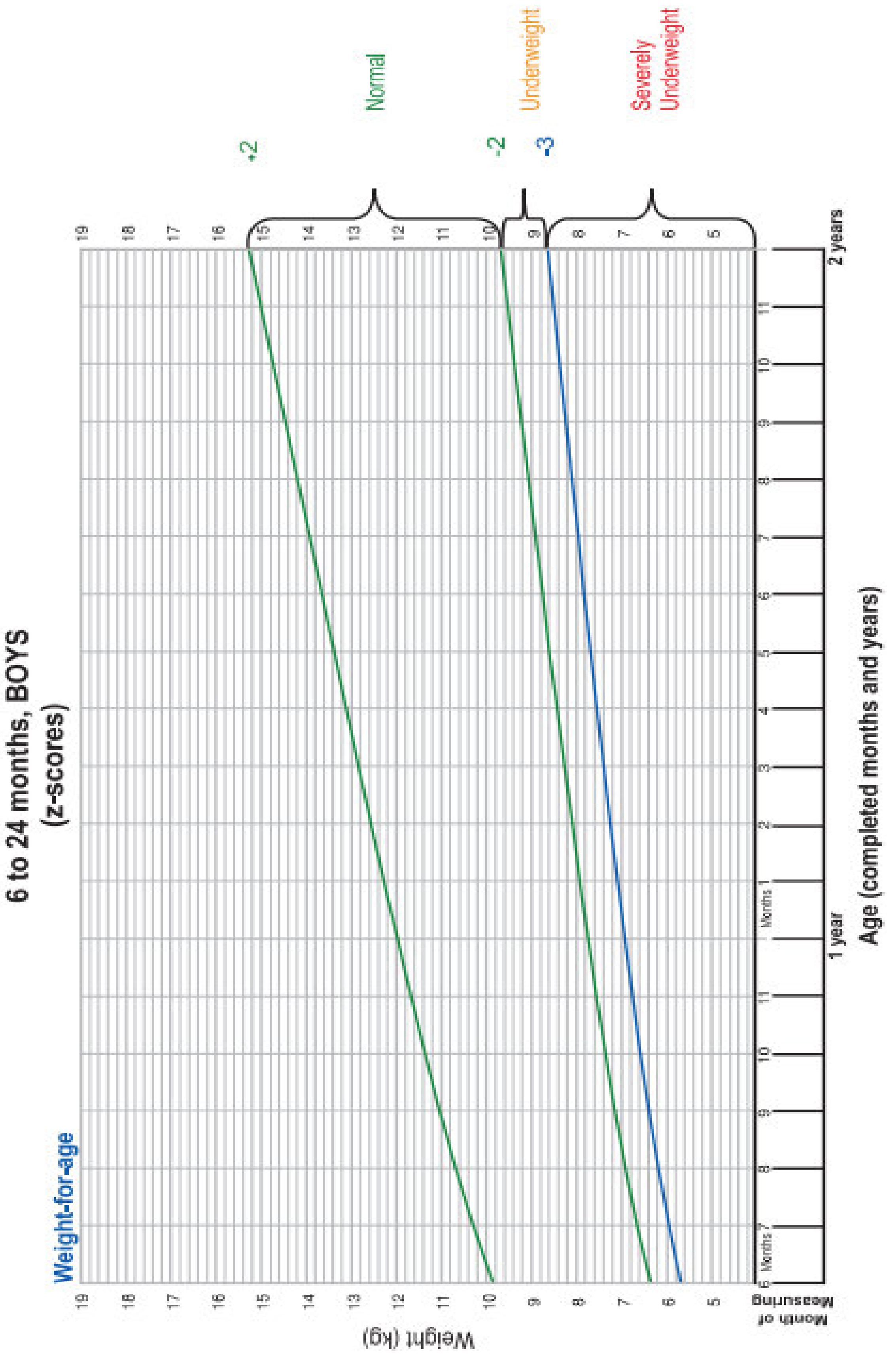
ANNEX 5.1b. Growth chart for boys by age group

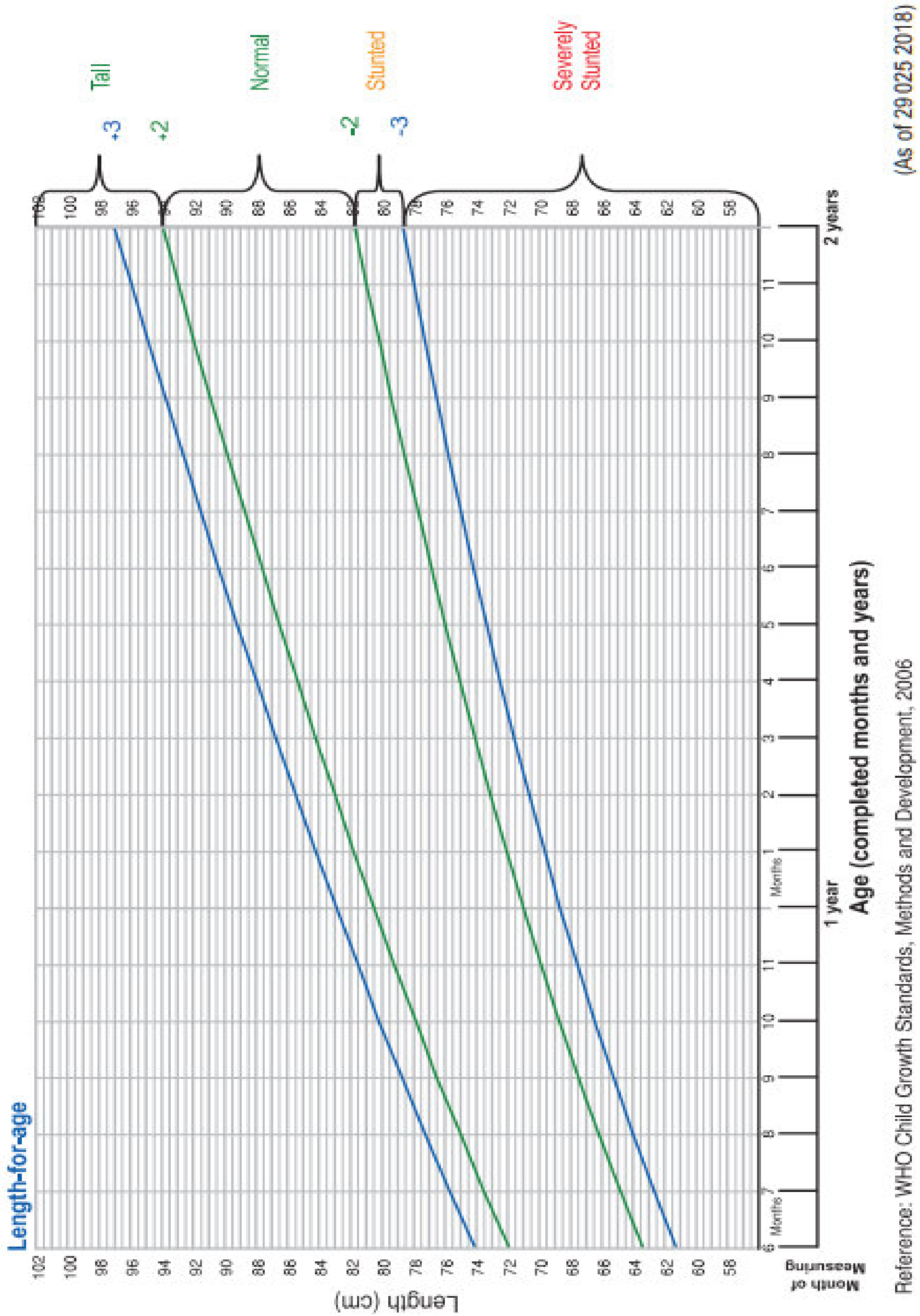




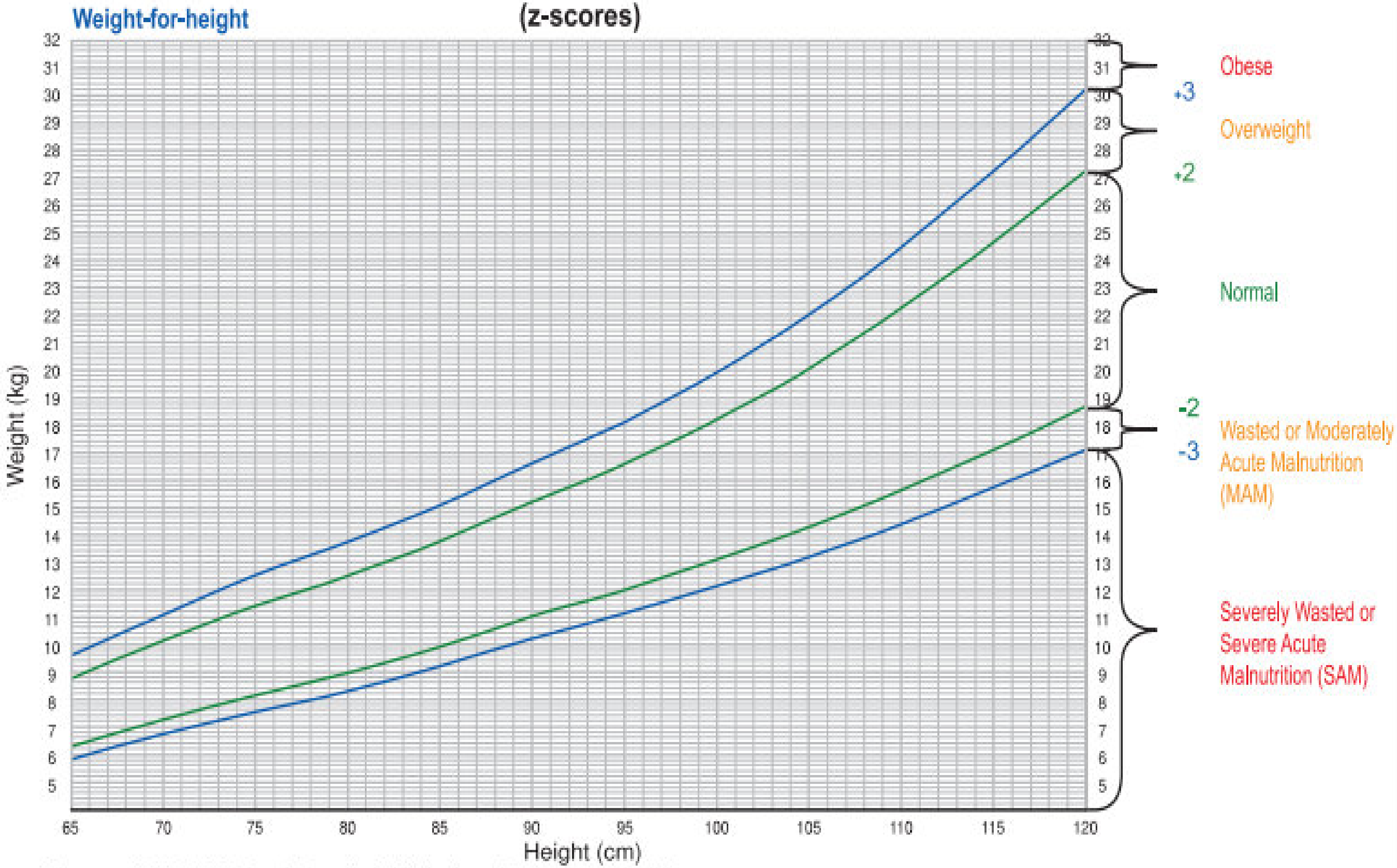
Reference: WHO Child Growth Standards, Methods and Development, 2006


Weight-for-length
Birth to 24 months, BOYS
(z-scores)
Weight (kg)
Length (cm)
+3
+2
-2
-3
Obese
Overweight
Normal
Wasted or Moderately Acute Malnutrition (MAM)
Severely Wasted or Severe Acute Malnutrition (SAM)


Weight-for-age
6 to 24 months, BOYS
(z-scores)
Weight (kg)
Month of Measuring
Age (completed months and years)
1 year
2 years
Months
+2
-2
-3
Normal
Underweight
Severely Underweight

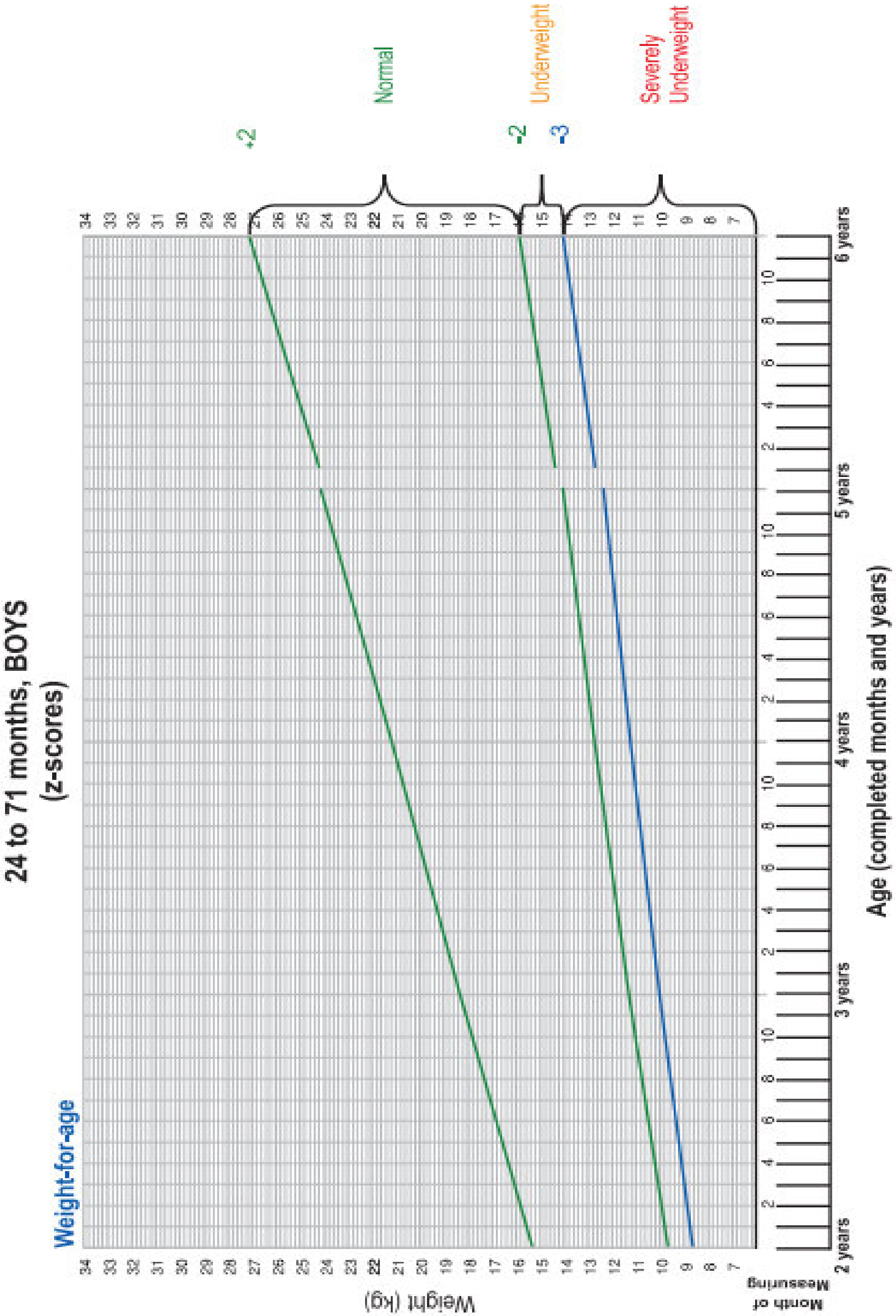

Length-for-age
Length (cm)
Month of Measuring
Months
1 year
2 years
Age (completed months and years)
Tall
Normal
Stunted
Severely Stunted
+3
+2
-2
-3
Reference: WHO Child Growth Standards, Methods and Development, 2006
(As of 29 025 2018)

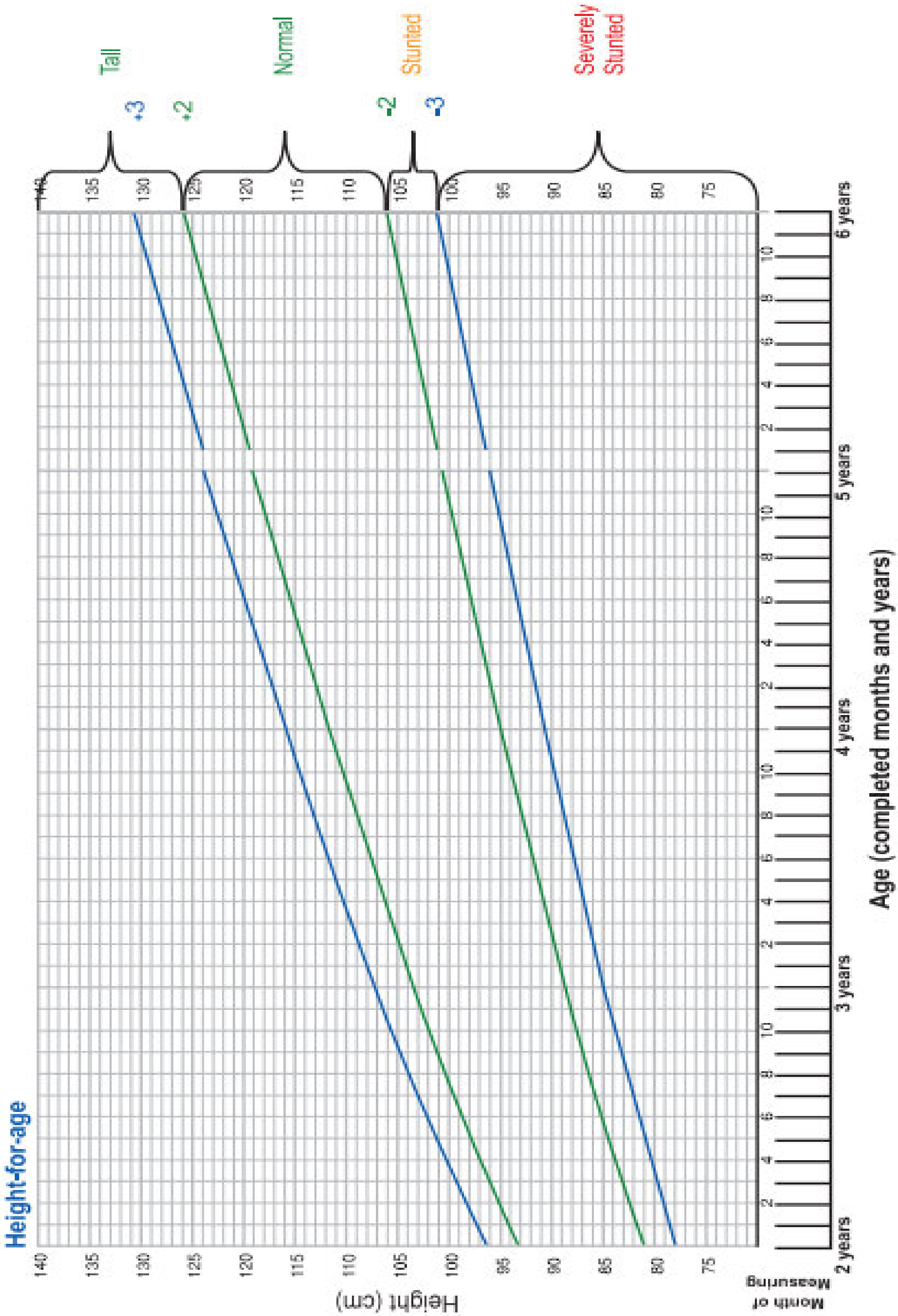
24 to 60 months, BOYS*
(z-scores)



Reference: WHO Child Growth Standards, Methods and Development, 2006

* For children ages above 60 months, use the BMI-for-age table.


Weight-for-age
24 to 71 months, BOYS
(z-scores)
Weight (kg)
Age (completed months and years)
Month of Measuring
2 years
3 years
4 years
5 years
6 years
+2
-2
-3
Normal
Underweight
Severely Underweight


Height-for-age
Height (cm)
Age (completed months and years)
Month of Measuring
2 years
3 years
4 years
5 years
6 years
+3
+2
-2
-3
Tall
Normal
Stunted
Severely Stunted

ANNEX 5.2. BNAP Template

Barangay Nutrition Action Plan
(Year)
Barangay, Municipality, or City

A. Introduction (Panimula)

B. Nutrition Situation (Ang Kalagayang pang-nutrisyon sa Barangay)

- What forms of malnutrition were identified in the OPT Plus? (Anu-anong uri ng malnutrisyon?)
- How many are malnourished children? (Ilan ang bilang ng malnutrisyon?)
- Who are malnourished? (Sinu-sino ang mga mayroong malnutrisyon?)
- Where are the malnourished located? (Saan nakatira ang mga mayroong malnutrisyon?)
- What are the causes of malnutrition in the area? Or the child? (Anu-ano ang mga sanhi ng malnutrisyon?)
- What are the steps to address malnutrition (Anu-ano ang mga hakbang na ginawa para matugunan ang malnutrisyon?)
- How effective are the interventions? (Gaano ito ka-epektibo?)
- What are the resources available to address malnutrition? (Anu-anong resources ang mayroon para sapag tugon sa malnutrisyon?)
- What are the possible hindrances that will affect the implementation of the nutrition programs? (Anu-ano ang mga balakid na maaaring makaapekto sa implementasyon ng mga programang pang-nutrisyon?)

C. Objectives

D. Work / Operational Plan

E. Monitoring and Evaluation (Pagsubaybay at Pagsusuri)

F. Budget / Financial Strategy (Badyet /Stratehiyang Pinansyal)

G. Attachment: Nutrition in Emergency Work Plan (Response and Recovery)

NOTE:
Refer to the NNC Barangay Nutrition Program Management Handbook.

OPT Plus TSEK

Operation Timbang Plus

Tamang Sukat, Eksaktong Kalidad!

Section Six

OPT PLUS PROTOCOL ON FOLLOW THROUGH ACTIVITIES

OPT PLUS PROTOCOL ON FOLLOW THROUGH ACTIVITIES

OVERVIEW

The follow through activities is an important component of the OPT Plus monitoring and evaluation process which aims to enhance continuity and implementation. Section 6 covers practical guidelines for monitoring and evaluating the OPT Plus activities.

After reading this section, one should be able to:

1. Enumerate the roles and responsibilities of the nutrition offices, supervisors, and OPT Plus team members;
2. Define the importance of conducting follow through activities;
3. Identify the steps in conducting child growth monitoring and active search; and
4. Be familiar with the form used for BNAP monitoring.

This section covers the process of conducting OPT Plus follow through activities. Specific sections include the following:

LIST OF ACRONYMS

BNAP	Barangay Nutrition Action Plan
BNC	Barangay Nutrition Committee
BNS	Barangay Nutrition Scholar
C/MNAO	City/Municipal Nutrition Action Officer
CMAM	Community Management of Acute Malnutrition
eOPT Plus	Electronic Operation Timbang Plus
IDA	Iron Deficiency Anemia
IGP	Income Generating Project
IYCF	Infant and Young Child Feeding
MAM	Moderate Acute Malnutrition
MNP	Micronutrient Powder
MSt	Moderately Stunted
MUW	Moderately Underweight
MW	Moderately Wasted
MW/MAM	Moderately Wasted/Moderate Acute Malnutrition
NAP	Nutrition Action Plan
NNC CO	National Nutrition Council Central Office
Ob	Obese
OW	Overweight
P/C/MNAO	Provincial/City/Municipal Nutrition Action Officer
PPAN	Philippine Plan of Action for Nutrition
RNPC	Regional Nutrition Program Coordinator
SAM	Severe Acute Malnutrition
SPS	Sangkap Pinoy Seal
SSt	Severely Stunted
St	Stunted
SUW	Severely Underweight
SW	Severely Wasted
SW/SAM	Severely Wasted/Severe Acute Malnutrition

DEFINITION OF TERMS

Active Search	A more aggressive approach in locating the target children than passive search, which relies on when the child is brought to the health center.
Barangay Nutrition Action Plan	Barangay-level version of the Philippine Plan of Action for Nutrition (PPAN); plan of a village to address its malnutrition problems and their causes, it includes the nutrition situation, list of nutrition interventions, monitoring and evaluation process, and budget for BNAP implementation.
Barangay Nutrition Committee	Serves as the advisory committee to the Sangguniang Barangay in the planning and implementation of health and nutrition programs in the barangay.
Interventions	Actions taken to improve the nutritional status of children.
Monitoring and evaluation	Ensures that implementation of OPT Plus continues after the measurement of children and that the information collected is being utilized in the area.
Stunting	A form of malnutrition that results from chronic or long-term undernutrition and can be classified using the length/height-for-age indicator. It is a general term that refers to both moderate and severe stunting.
Underweight	A form of malnutrition classified using the weight-for-age indicator. It is also a general term referring to both moderately and severely underweight.
Wasting	A form of malnutrition that results from acute onset of undernutrition classified using the weight-for-length/height indicator or MUAC. It is also the term referring to both moderately and severely wasted.

Section 6.1

ROLES AND RESPONSIBILITIES OF THE SUPERVISORS AND OPT PLUS TEAM IN THE MONITORING AND EVALUATION

The OPT Plus Monitoring and Evaluation is a two-pronged approach that aims to monitor the growth of children and the Nutrition Action Plans (NAP) of the barangay and city/ municipality. Monitoring and evaluation is an important part to:

- Ensure that implementation of OPT Plus continues after the initial measurement of the children, and
- The information collected is being utilized in the community.

Table 6.1 enumerates the roles and responsibilities of the nutrition offices, supervisors, and OPT Plus members.

Table 6.1. Roles and Functions of the different persons involved in the OPT Plus during the OPT Plus Protocol on Follow Through Activities.

ORGANIZATION/PERSON INVOLVED	FUNCTION
National Nutrition Council Central Office (NNC CO)	1. Conduct mid-year and end-year meetings within the Nutrition Surveillance Division staff regarding the conduct of OPT Plus. 2. Conduct national activities related to data quality check.
Regional Nutrition Program Coordinator (RNPC)	1. Conduct end-year meetings with P/C/MNAOs to discuss the implemented programs and projects and updates on the nutritional status of children. 2. Conduct regional activities related to data quality check.
Provincial Nutrition Action Officer (PNAO)	1. Conduct mid-year meetings with C/MNAOs to discuss the implemented programs and projects and updates on the nutritional status of children. 2. Conduct provincial activities related to data quality check.
Supervisor	1. Assist in reviewing the monthly and quarterly OPT Plus monitoring coverage if all the children weighed and measured in the first quarter were followed up. 2. Participate in the bi-annual review of the BNAP and assist in presenting BNAP accomplishments during the Barangay General Assembly.
City/Municipal Nutrition Action Officer (C/MNAO)	1. Remind the BNC of the schedules of the monthly OPT Plus monitoring of SW/MAM and MW/MAM, underweight and 0-to-23-month-old, and quarterly monitoring of 24-to-59-month-old children. 2. Review the monthly and quarterly OPT Plus monitoring coverage if all the children weighed and measured in the first quarter were all followed up OR Assist in reviewing the monthly and quarterly OPT Plus monitoring coverage if all the children measured were covered during child growth monitoring. 3. Conduct a bi-annual review of the BNAP and present BNAP accomplishments during the Barangay General Assembly. 4. Follow-up to the Barangay Nutrition Scholar (BNS) on the status of the referred malnourished children. 5. Conduct city/municipal activities related to data quality check.

ORGANIZATION/PERSON INVOLVED	FUNCTION
OPT Plus Team members	1. Remind the BNC of the schedules of the monthly OPT Plus monitoring of SW/MAM and MW/MAM, underweight and 0-to-23-month-old, and quarterly monitoring of 24-to-59-month-old children. 2. Review the monthly and quarterly OPT Plus monitoring coverage if all the children weighed in the first quarter were all weighed. 3. Organize a bi-annual review of the BNAP and assist in presenting BNAP accomplishments during the Barangay General Assembly.
Barangay Nurition Committee (BNC)	1. Assist in conducting monthly OPT Plus monitoring of SW/MAM and MW/MAM, underweight and 0-to-23-month-old, and quarterly monitoring of 24-to-59-month-old children. 2. Participate in the bi-annual review of the BNAP and present BNAP accomplishments during the Barangay General Assembly. 3. Conduct barangay activities related to data quality check.

Section 6.2

CHILD GROWTH MONITORING AND ACTIVE SEARCH

After weighing and measuring, the nutritional status of 0-to-59-month-old children will be determined. While some will be classified as normal, a percentage of these children could be identified with at least one form of undernutrition. The target of OPT Plus child growth monitoring and conducting an active search is 0-to-59-month-old children classified as underweight and/or wasted (MW/MAM and SW/SAM) along with all 0-to-23-month-old infants. These children will be weighed monthly to monitor their growth and ensure that they are referred to appropriate services and interventions. Instead of the center- or facility-based search mode, the child growth monitoring will employ active cases search. Active Search is a more aggressive approach to locating the target children than passive search, which relies on instances when the child is brought to the health center. Use the forms for OPT Plus Monitoring, which can be generated from the eOPT Plus.

Instructions

The OPT Plus Monitoring Forms 1 to 8 (eOPT Plus Tool Worksheets) have a similar format for all, varying only on the identified nutritional status of the children. The columns on the left provide the information of the child which includes the address, name of the child, name of mother or caregiver, sex, birth date, and the initial length/height (cm), and weight (kg). This is automatically generated when information is encoded in the OPT Plus Form 1A. Moreover, the columns after the personal information are allotted for recording the weight and length/height during the follow-up visits.

eOPT Plus v2 Worksheet: List of 0-23 mos is the list of 0-23-month-old children

Oct 06, 2023

SPACE FOR OFFICIAL SEAL OF LGU

Republic of the Philippines
Department of Health
NATIONAL NUTRITION COUNCIL
IVA CALABARZON

MONITORING LIST FOR CHILDREN 0 - 23 MONTHS OLD

Barangay: **Ibaba** City: **CITY OF SANTA ROSA** Province: **Laguna**

Year: **2022**

# of Children: 7

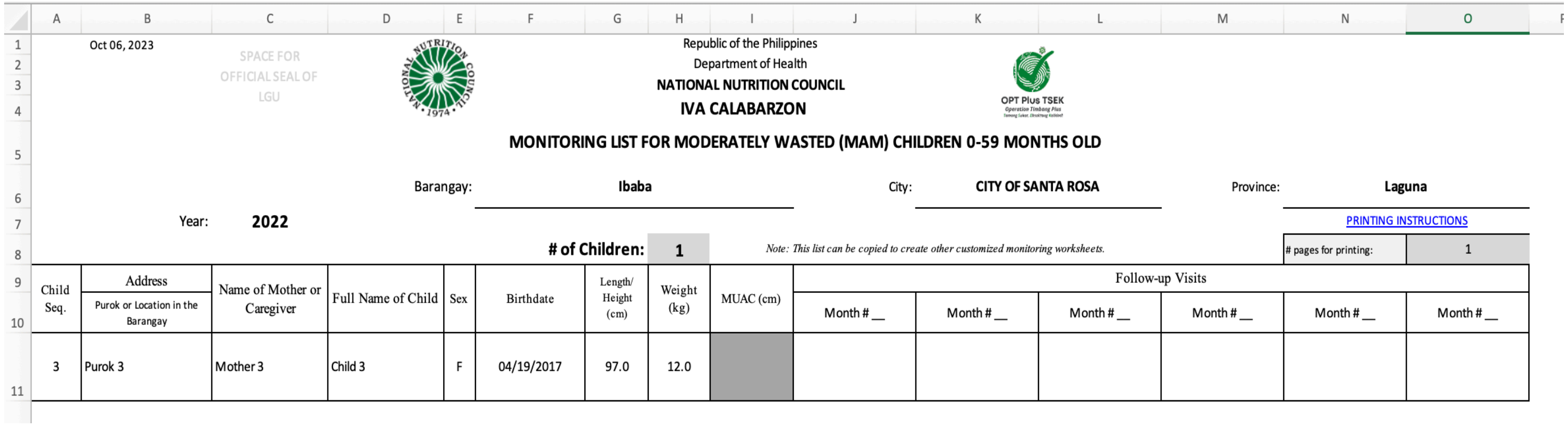
Note: This list can be copied to create other customized monitoring worksheets.

PRINTING INSTRUCTIONS

pages for printing: 1

Child Seq.	Address: Purok or Location in the Barangay	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	Length/ Height (cm)	Weight (kg)	MUAC (cm)	WFA Status	Lt/HtFA Status	WFLt/Ht Status	MUAC Status	Follow-up Visits: Month #__	Month #__	Month #__	Month #__	Month #__	Month #__
1	Purok 1	Mother 1	Child 1	F	08/29/2020	60.1	6.7		N	N	N							
2	Purok 2	Mother 2	Child 2	F	10/13/2019	100.0	10.8		N	T	SW/SAM							
4	Purok 4	Mother 4	Child 4	F	09/07/2020	72.3	8.0		N	T	N							
5	Purok 5	Mother 5	Child 5	F	03/05/2019	90.0	13.7		N	N	N							
6	Purok 6	Mother 6	Child 6	F	04/10/2019	60.0	15.0		Use the WL/H column	SSt	Ob							
7	Purok 7	Mother 7	Child 7	F	05/04/2020	70.0	7.0		N	N	N							
8	Purok 8	Mother 8	Child 8	F	04/14/2020	73.0	9.0		N	N	N							

eOPT Plus v2 Worksheet: List_MW(MAM) is the list of 0-59-month-old children who are moderately wasted/moderate acute malnutrition



Oct 06, 2023

SPACE FOR OFFICIAL SEAL OF LGU

Republic of the Philippines
Department of Health
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IVA CALABARZON

MONITORING LIST FOR MODERATELY WASTED (MAM) CHILDREN 0-59 MONTHS OLD

Barangay: Ibaba City: CITY OF SANTA ROSA Province: Laguna

Year: 2022

PRINTING INSTRUCTIONS

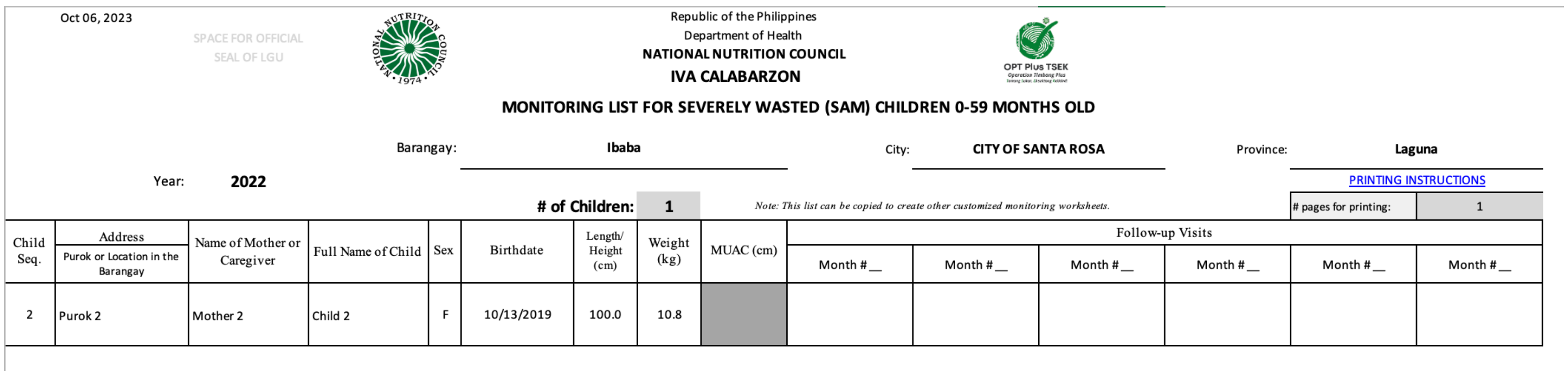
# of Children: 1

Note: This list can be copied to create other customized monitoring worksheets.

pages for printing: 1

Child Seq.	Address Purok or Location in the Barangay	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	Length/ Height (cm)	Weight (kg)	MUAC (cm)	Follow-up Visits					
									Month # __	Month # __	Month # __	Month # __	Month # __	Month # __
3	Purok 3	Mother 3	Child 3	F	04/19/2017	97.0	12.0							

eOPT Plus v2 Worksheet: List_SW(SAM) is the list of 0-59-month-old children who are severely wasted/severe acute malnutrition



Oct 06, 2023

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MONITORING LIST FOR SEVERELY WASTED (SAM) CHILDREN 0-59 MONTHS OLD

Barangay: Ibaba City: CITY OF SANTA ROSA Province: Laguna

Year: 2022

PRINTING INSTRUCTIONS

# of Children: 1

Note: This list can be copied to create other customized monitoring worksheets.

pages for printing: 1

Child Seq.	Address Purok or Location in the Barangay	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	Length/ Height (cm)	Weight (kg)	MUAC (cm)	Follow-up Visits					
									Month # __	Month # __	Month # __	Month # __	Month # __	Month # __
2	Purok 2	Mother 2	Child 2	F	10/13/2019	100.0	10.8							

eOPT Plus v2 Worksheet List_MSt&SSt is the list of 0-59-month-old children who are moderately stunted/ severely stunted

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OPT Plus TSEK

MONITORING LIST FOR MODERATELY OR SEVERELY STUNTED CHILDREN 0-59 MONTHS OLD

Barangay: **Ibaba** City: **CITY OF SANTA ROSA** Province: **Laguna**

Year: **2022**

PRINTING INSTRUCTIONS

# of Children: 1

Note: This list can be copied to create other customized monitoring worksheets.

pages for printing: 1

Child Seq.	Address (Purok or Location in the Barangay)	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	Length/ Height (cm)	Weight (kg)	Follow-up Visits					
								Month #__	Month #__	Month #__	Month #__	Month #__	Month #__
6	Purok 6	Mother 6	Child 6	F	04/10/2019	60.0	15.0						

eOPT Plus v2 Worksheet: List_OW&Ob is the list of 0-59-month-old children who are overweight and obese

Oct 06, 2023

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OPT Plus TSEK

MONITORING LIST FOR OVERWEIGHT OR OBESE CHILDREN 0-59 MONTHS OLD

Barangay: **Ibaba** City: **CITY OF SANTA ROSA** Province: **Laguna**

Year: **2022**

PRINTING INSTRUCTIONS

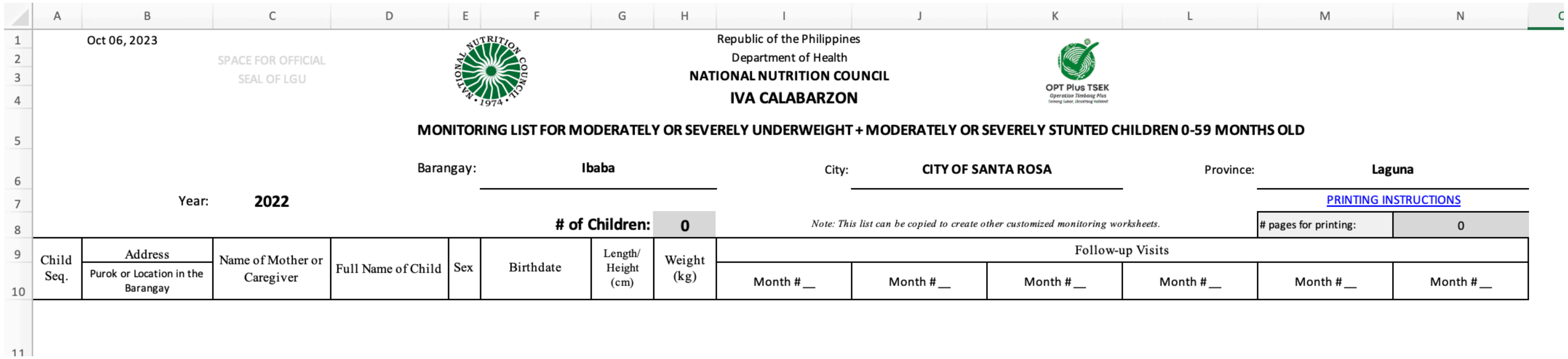
# of Children: 1

Note: This list can be copied to create other customized monitoring worksheets.

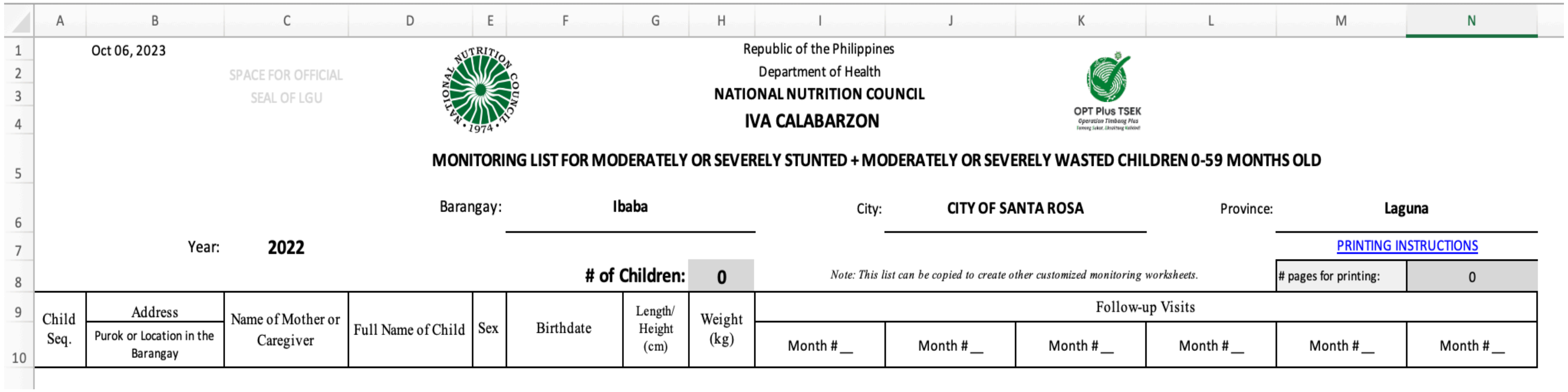
pages for printing: 1

Child Seq.	Address (Purok or Location in the Barangay)	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	Length/ Height (cm)	Weight (kg)	Follow-up Visits					
								Month #__	Month #__	Month #__	Month #__	Month #__	Month #__
6	Purok 6	Mother 6	Child 6	F	04/10/2019	60.0	15.0						

eOPT Plus v2 Worksheet: List_MUW, SUW,MSt&SSt is the list of 0-59-month-old children who are both stunted and underweight



eOPT Plus v2 Worksheet: List_MSt,SSt,MW&SW is the list of 0-59-month-old children who are both stunted and wasted



eOPT Plus v2 Worksheet: List_MSt,SSt,OW&Ob is the list of 0-59-month-old children who are both stunted and overweight/obese

Oct 06, 2023

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OPT Plus TSEK

MONITORING LIST FOR MODERATELY OR SEVERELY STUNTED + OVERWEIGHT OR OBESE CHILDREN 0-59 MONTHS OLD

Barangay: **Ibaba** City: **CITY OF SANTA ROSA** Province: **Laguna**

Year: **2022**

PRINTING INSTRUCTIONS

# of Children: 1

Note: This list can be copied to create other customized monitoring worksheets.

pages for printing: 1

Child Seq.	Address (Purok or Location in the Barangay)	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	Length/ Height (cm)	Weight (kg)	Follow-up Visits: Month #__	Month #__	Month #__	Month #__	Month #__	Month #__
6	Purok 6	Mother 6	Child 6	F	04/10/2019	60.0	15.0						

eOPT Plus v2 Worksheet: List_MUAC Status is the list of 6-59-month-old children by MUAC status

Oct 06, 2023

SPACE FOR OFFICIAL SEAL OF LGU

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OPT Plus TSEK

MONITORING LIST FOR CHILDREN MEASURED USING MID-UPPER ARM CIRCUMFERENCE (MUAC)

Barangay: **Ibaba** City: **CITY OF SANTA ROSA** Province: **Laguna**

Year: **2022**

PRINTING INSTRUCTIONS

# of Children: 0

Note: This list can be copied to create other customized monitoring worksheets.

pages for printing: 0

Child Seq.	Address (Purok or Location in the Barangay)	Name of Mother or Caregiver	Full Name of Child	Sex	Birthdate	MUAC (cm)	MUAC Status	Follow-up Visits: Month #__	Month #__	Month #__	Month #__	Month #__	Month #__

Section 6.3

BNAP MONITORING

During the BNC Meeting every quarter, report the updates on the nutritional status of the target groups for monthly weighing using the OPT Plus Monitoring Form 1A to 1G. Monitor the BNC Action Plan implementation status by answering the BNC Quarterly Accomplishment Report (Annex 6.1). Worksheet 1 is the sample template and instructions.

Worksheet 1. BNC Quarterly Accomplishment Report.

ACTIVITY (COLUMN 1)	TARGET GROUP (COLUMN 2)	TARGET NUMBER (COLUMN 3)	QUARTERLY OUTREACH (COLUMN 4)							
			1ST QUARTER		2ND QUARTER		3RD QUARTER		4TH QUARTER	
			NO.	%	NO.	%	NO.	%	NO.	%
A. ACTIVITY 1										
1.										
2.										

Instructions

- **First Column (Activity):** identify the interventions listed in the BNAP,
- **Second Column (Target group):** list the target groups (i.e. 0-59-month-old children, 0-23-month-old children, stunted, wasted, underweight, etc.).
- **Third Column (Target number):** write down the target number per age group or nutritional status classification based on the OPT Plus results.
- **Fourth Column (Quarterly Outreach):** update the number of children that were involved or referred to the listed activities in the first column. This should be done every quarter of the year (March, June, September, and December).

REFERENCES

Department of Health (DOH). (2010). Manual of Operation on Growth Monitoring and Promotion of Children in the Philippines using the WHO Child Growth Standards.

ANNEX 6.1 BNC Quarterly Accomplishment Report

Activity	Target Group	Target Number	Quarterly Outreach							
			1st Quarter		2nd Quarter		3rd Quarter		4th Quarter	
			No.	%	No.	%	No.	%	No.	%
A. Weighing Activities										
1. Full Weighing (OPT+)	All 0-59-month old children									
2. OPT Plus Monitoring	0-23-months									
	Underweight (UW)									
	Severely UW (SUW)									
	Moderately Wasted (MW/ MAM)									
	Severely Wasted (SW/SAM)									
	Stunted (St)									
	Overweight (OW)									
	Obese (Ob)									
	24-59-month-old									
B. Nutrition Specific Programs										
1. Infant and Young Child Feeding										
a. Referrals for breastfeeding counseling	Pregnant									
	Lactating									
b. Conduct breastfeeding/ complementary feeding counseling	Pregnant									
	Lactating									
	Mothers with MW/MAM or SW/ SAM children									
c. Other activities										
2. Integrated Management of Acute Malnutrition										
a. Referral of MW/MAM and SW/SAM children	SW (SAM) 6- to 59- months									
	MW (MAM) 6- to 59- months									
b. Referrals for ITC and OTC	SW (SAM) 6- to 59- months									
	MW (MAM) 6- to 59- months									
c. Assistance in OTC										
3. National Dietary Supplemental Feeding										
a. Referrals to/for: Dietary Supplementation	MUW and SUW 6- to 23- months									
	MUW and SUW 24-to 59-months									
	Nutritionally At-risk Pregnant Mothers									
	Infants 6- to 23- months									
	Pregnant									
	Lactating									

Activity		Target Group	Target Number	Quarterly Outreach							
				1st Quarter		2nd Quarter		3rd Quarter		4th Quarter	
				No.	%	No.	%	No.	%	No.	%
a.	Assistance for Moderate Acute Malnutrition (MW/ MAM) Supplementary Feeding	Families of MW (MAM)									
4.	**National Promotion for Behavior Change**										
a.	Conduct of Nutrition Education, Education Classes (Pabasa sa Nutrisyon, Idol ko si Nanay, etc)	Mothers of MUW (children)									
		Mothers of SUW children									
		Mothers of MW (MAM) children									
		Mothers of SW (SAM) children									
		Mothers of Stunted children									
		Mothers of OW children									
		Pregnant									
		Lactating									
		Mothers with infants 0- to 23-months									
		Mothers with infants 24- to 59-months									
b.	Home visits/follow up/ counseling	Mothers/ caregivers of 0- to 23- months									
		Mothers of MUW									
		Mothers of SUW									
		Mothers of MW (MAM)									
		Mothers of SW (SAM)									
		Mothers of St									
		Mothers of OW									
		Fathers or other male family members of 0- to 23- months									
c.	Distribution of Information, Education and Communication materials (IECM)	Mothers of MUW									
		Mothers of SUW									
		Mothers of MW (MAM)									
		Mothers of SW (SAM)									
		Pregnant									
		Lactating									
		Mothers with infants 0- to 23-months									
		Mothers with infants 24- to 59-months									

Activity	Target Group	Target Number	Quarterly Outreach							
			1st Quarter		2nd Quarter		3rd Quarter		4th Quarter	
			No.	%	No.	%	No.	%	No.	%
5. Micronutrient Supplementation										
a. Vitamin A Supplementation	All 6- to 11-month old									
	All 12- to 59- month old									
	Pregnant									
	Post-partum women									
b. Iron-folic acid supplementation	Pregnant									
	Lactating									
	All 6- to 11-month old									
	All 12- to 59- month old									
	5-9 years old with IDA									
	13-18 years old adolescent girls									
c. Zinc Supplementation	0-59 years old with diarrhea									
d. Provision of MNP	6- to 23-months									
e. Provision of iodized oil capsules	Pregnant and lactating mothers in endemic areas									
f. Provision of calcium supplements	Pregnant									
6. Mandatory Food Fortification										
Assist on: a. Promotion of Sangkap Pinoy Seal (SPS)	Families of MUW									
	Families of SUW									
	Families of MW (MAM)									
	Families of Severely SW (SAM)									
	Families of St									
	Others									
b. Promoting the use of iodized salt	All families									
c. Promotion of the use of fortified rice/flour/cooking oil	All Families									
7. Nutrition-in-emergencies										
Referral on a. Assessment	0- to 59- months									
b. IYCF Counselling	0- to 59- months									
c. Distribution of Vitamin A capsules	0- to 59- months									
d. Community management of Acute Malnutrition (CMAM) e.g. referral of cases	0- to 59- months									

Activity	Target Group	Target Number	Quarterly Outreach							
			1st Quarter		2nd Quarter		3rd Quarter		4th Quarter	
			No.	%	No.	%	No.	%	No.	%
8. Overweight and Obesity Management and Prevention Program										
Assist on a. Conduct weight management referral for overweight and obese individual	OW and Ob children									
b. Conduct of weight management intervention for overweight and obese individual	OW and Ob children									
C. Nutrition-Sensitive Programs										
1. Home, school, and community food production										
Assist on: a. Seeds/seedlings distribution home gardening	Families of MUW									
	Families of SUW									
	Families of MW (MAM)									
	Families of Severely SW (SAM)									
	Families of Pregnant									
	Families with 0 to 59 months									
	Other families									
b. Establishment of b1. home yard gardens	Families of MUW									
	Families of SUW									
	Families of MW (MAM)									
	Families of Severely SW (SAM)									
	Families with 0-23 months old									
b2. community gardens	Barangays									
b3. school gardens	Schools									
c. small animal dispersal	Families of MUW									
	Families of SUW									
	Families of MW (MAM)									
	Families of Severely SW (SAM)									
	Other families									
2. Livelihood Assistance										
Referral on a. Credit or lending institutions for credit assistance	Families of MUW									
	Families of SUW									
	Families of MW (MAM)									
	Families of Severely SW (SAM)									
	Other families									

Activity	Target Group	Target Number	Quarterly Outreach							
			1st Quarter		2nd Quarter		3rd Quarter		4th Quarter	
			No.	%	No.	%	No.	%	No.	%
a. Agencies providing material support for IGP	Families of MUW									
	Families of SUW									
	Families of MW (MAM)									
	Families of Severely SW (SAM)									
	Other families									
c. IGP training supports IGP	Families of MUW									
	Families of SUW									
	Families of MW (MAM)									
	Families of Severely SW (SAM)									
	Other families									

OPT Plus TSEK

Operation Timbang Plus

Tamang Sukat, Eksaktong Kalidad!

Section Seven

DATA QUALITY CHECK (DQC)

DATA QUALITY CHECK (DQC)

OVERVIEW

The goal of OPT Plus is not just to produce the nutritional status data in the community but also to ensure that the data are produced through quality procedures. Section 7 discusses the basic principles of data quality and the OPT Plus Data Quality Check (DQC).

After reading this section, one should be able to:

1. Enumerate the basic concepts and principles of data quality;
2. Describe the data quality dimensions of OPT Plus; and
3. List the Data Quality Check Protocol at different levels.

Section 7 covers the data quality assessment components of OPT Plus. Specific sections include the following:

LIST OF ACRONYMS

AIP	Annual Investment Plan
BNAO	Barangay Nutrition Action Officer
BNAP	Barangay Nutrition Action Plan
BNC	Barangay Nutrition Committee
BNS	Barangay Nutrition Scholar
BNSAP	Barangay Nutrition Scholar Action Plan
C/MNAO	City/Municipal Nutrition Action Officer
C/MNC	City/Municipal Nutrition Committee
DQC	Data Quality Check
eOPT	Electronic Operation Timbang
L/HFA	Length/Height-for-Age
LGU	Local Government Unit
MAM	Moderate Acute Malnutrition
MSt	Moderately Stunted
MUW	Moderately Underweight
MW/MAM	Moderately Wasted/Moderate Acute Malnutrition
NNC CO	National Nutrition Council Central Office
NNC RO	National Nutrition Council Regional Office
NSD	Nutrition Surveillance Division
Ob	Obese
OPT Plus	Operation Timbang Plus
OW	Overweight
PIR	Program Implementation Review
PNAO	Provincial Nutrition Action Officer
PNC	Provincial Nutrition Committee
RNPC	Regional Nutrition Program Coordinator
SAM	Severe Acute Malnutrition

SSt	Severely Stunted
SUW	Severely Underweight
SW/SAM	Severely Wasted/Severe Acute Malnutrition
UNICEF	United Nations Children's Fund
WFA	Weight-for-Age
WFL/H	Weight-for-Length/Height
WHO	World Health Organization

DEFINITION OF TERMS

Accuracy	Aims to determine the quality of measurements and the quality of the tools used for measurement.
Completeness	Refers to the extent to which the data collection covers the target population.
Integrity	Refers to when the system used to generate them is protected from deliberate bias or manipulation for political or personal reasons.
Reliability	Concerned with producing data using standard protocols or procedures that can be consistently repeated regardless of user or frequency of the conduct.
Timeliness	Refers to the review of OPT Plus data and reports submitted to key personnel during the period specified in the protocol.
Usefulness	Refers to the provision of nutrition services to at-risk 0-59-month-old children.
Z-score	Indicates how much a given value differs from the standard deviation. The child's z-score determines his/her nutritional status.

Section 7.1

CONCEPTS, PRINCIPLES, AND DIMENSIONS OF DATA QUALITY

Quality data is a prerequisite for better information needed in decision-making and policy making. In nutrition, quality anthropometric data and quality socio-demographic information such as sex and age are vital in order to have accurate and reliable nutritional status assessment. This information will be the primary basis of policymakers, program managers, researchers, and advocates of nutrition in devising their nutrition programs and interventions.

Anthropometry data quality poses implications in understanding the prevalence and burden of malnutrition. Threats to data quality can arise at various stages of program implementation, from initial development, training, and actual fieldwork to data recording, cleaning, and analysis (Corsi et al., 2017).

Factors affecting anthropometry

Inaccuracies and deficiencies in anthropometry data quality frequently exist especially in large-scale survey implementation, thus posing implications in understanding the prevalence and burden of malnutrition. Threats to data quality can arise at the various stages of program implementation, from initial development, training, and actual fieldwork to data recording, cleaning, and analysis (Corsi et al., 2017). Below is a descriptive summary of the various factors that adversely affect anthropometry data quality as identified by the studies and literature reviewed for this desk research.

Improper age reporting and calculation

Age is considered to be a crucial data in processing and interpreting anthropometry measurements since it is primarily used in determining the presence of stunting and underweight. Studies (Grellety & Golden, 2018; Finaret & Hutchinson, 2018; Figueroa & Kurdi, 2019) report that data errors are usually found in age estimation and misreporting. Due to poor date of birth information, discontinuities in the estimates of mean height-for-age (HAZ) by month of birth (MOB) may occur (Perumal et al., 2020). Likewise, data errors introduced by merely estimating the age, rather than computing it, may overestimate the prevalence of nutrition indicators (Leidman, Mwirigi, Maina-Gathigi, Wamae, Imbwaga, & Bilukha, 2018).

Tools such as the eOPT Plus, SMART ENA, and WHO AnthroPlus allow for the calculation of age in months. For more precise values, the WHO Anthro Survey Analyser, on the other hand uses age calculation in days, as part of the 2019 recommendations of WHO and UNICEF. In their report, WHO and UNICEF (2019) noted that if only the month and year of birth are provided, it must be imputed as missing day of birth. For standard analysis, however, the 15th day of the month is recommended to be used.

Imprecise anthropometry measurements

Ensuring the accuracy and precision of anthropometry measurements are of great importance in generating a reliable base of information for growth monitoring and nutrition interventions. This can be owed to the fact that anthropometry data quality is directly related to the evaluation of nutrition prevalence and status (Leidman et al., 2018). Adversely, imprecise and inaccurate anthropometry measurements pose a significant threat to overall data quality. As emphasized by Leidman et al. (2018), imprecise measures of weight and height can result in wider values of standard deviation, suggesting an overestimated value for the prevalence nutrition indicators.

Imprecise measures can also lead to statistically and biologically implausible values, which in some cases are flagged and corrected to fit the accepted range of growth standards. Such practice may generate data that are within the acceptable range but are of poor quality (Corsi et al., 2017). Few studies (Finaret & Hutchinson, 2018; Manzoni et al., 2019), on the other hand, reported that implausible values are rather tagged as missing data than having them flagged or corrected.

Furthermore, inaccurate and imprecise anthropometry measurements can be attributed to some factors including inconsistencies in measuring techniques and protocols (Phan et al., 2020), use of uncalibrated and nonstandard measuring equipment, as well as human error itself or the competence of measurers. The latter factor is further elaborated in a succeeding subsection.

The use of automated tools, such as the ones mentioned in the previous section are deemed useful in preventing entry of data that are inaccurate and precise. This is possible through the inclusion of plausibility check and range check features. However, it must be noted that data are susceptible to manipulation or correction in order to fit the standard plausible range or limits (Corsi et al., 2017).

Incomplete anthropometry data

Missing anthropometry data was reported to be another hindrance in ensuring data quality. Completeness of data from the child and mother's basic information down to the actual anthropometry measurements must be reviewed in order to generate reliable results.

For instance, the evaluation report published by the Innovations for Poverty Action(2019) showed that there was a significant percentage of children with missing data in the 2017 and 2018 OPT, suggesting that despite being eligible, a number of children were not measured. Similarly, an international study conducted by Grijalva-Eternod et al. (2017) revealed problematic anthropometry data quality among infants due to the large proportion of missing anthropometry data.

Bilukha, Couture, McCain, and Leidman (2020) highlighted in their methodology that missing and incomplete data has the potential to bias nutritional indicators, thus the completeness of weight, height, age, and sex variables must be initially evaluated.Incomplete data are easily detected and imputed with the use of DQA tools similar to the ones previously mentioned. WHO and UNICEF's recommendations, in particular, foregrounds data completeness as one of the primary measures of data quality.

Incompetence of measurers

Ramirez, Viajar, and Azaña (2019) suggested that human error can affect the accuracy of data measurements, calculation of age, data encoding, and validation. All of which are considered to be significant in maintaining the accuracy of OPT Plus's database. Moreover, data error may occur with the incorrect use of the eOPT Plus Tool especially with untrained BNS or health workers as end users. This is comparable with the study of Grellety and Golden (2018) in which they excluded from their analysis the surveys that were administered by inexperienced persons who lack performance testing and adequate supervision, since such surveys are indicative of data with questionable quality. Given such, anthropometry training and standardization tests are among the recommended activities to be implemented prior to actual anthropometry measurements in order to ensure that measurers are capable of collecting and reporting data of high quality.

Data manipulation

Studies also reported instances of data fabrication and manipulation which can adversely affect data integrity. In their analysis of malnutrition surveys administered across 55 countries, Grellety and Golden (2018) excluded the data that had been fabricated and replicated from one survey to another in order to maintain quality within the level of their analysis.

In a local setting, the Innovations for Poverty Action (2019) suggested in their exploratory analysis that OPT is susceptible to data manipulation. In their evaluation, the HAZ and weight-for-age (WAZ) distribution in the specified study areas did not resemble a normal curve. This was found to be indicative of data manipulation deliberately done to underreport stunting and underweight. The manipulation was presumably executed by increasing the actual height and weight measurements by a small margin to fit the normal weight and height criteria. The discontinuities found in the analysis may also be due to the assumption that undernourished children were taken of their best measurements outside of the OPT period or right after the LGU's implementation of micronutrient supplementation and feeding program.

Reviewed literature on DQA tools, however, were silent on the likelihood and mitigation of data manipulation. Hence, the strengthening of monitoring and supervision on the part of field officials and data collectors.

Contextual factors

Other factors that may directly or indirectly affect data quality were mentioned in a few literature. For instance, Perumal et al. (2020) emphasized that variations in the quality of anthropometry survey may be attributed to contextual factors that may hinder successful program implementation such as conflict, political instability, and geographical barriers.

Furthermore, WHO and UNICEF (2020) listed inadequate research infrastructure, low investment, lack of data collection, and data processing technology as limiting factors in the conduct of nutrition surveys, especially in the context of low-income countries.

Considerations to address anthropometry problems

The **use of a well-established method** or tool in child anthropometry and data quality assessment is evidently a good practice in data management. Tools that have the ability to clean criteria and feedback through plausibility checks (Grellety & Golden, 2018) can mark some improvements in data quality. The tools developed by SMART, WHO, and UNICEF are considered to be especially useful in ensuring data quality given the fact that these references are based on international standards and valid research.

Dimensions of Data Quality

As for the Operation Timbang Data Quality Check (OPT DQC), the following dimensions were identified for evaluation:

1. **ACCURACY** is concerned with the correctness of data such that it measures what it intends to measure (WHO, 2008). This dimension aims to determine the quality of measurements and the tools used for measurement. Indicators to be included directly affect the quality of data collected (weight and length/height measurement) and the generated child's nutritional status through OPT Plus.
2. **RELIABILITY** is concerned with producing data using standard protocols or procedures that can be consistently repeated regardless of user or frequency of the conduct. Indicators that will be used to measure reliability include training, standardization, and the ability of the OPT Plus team to follow standard protocols and procedures to produce consistent results. The reliability of the data on weight, height/length, date of birth, and sex are also essential in this dimension.
3. **COMPLETENESS** refers to the extent to which the data collection covers the target population.
4. **TIMELINESS** refers to the review of OPT Plus data and reports submitted to key personnel during the period specified in the protocol.
5. **USEFULNESS** refers to providing nutrition services to at-risk 0-59-month-old children

The aforementioned dimensions can be completely achieved if done with Integrity - protected from deliberate bias or manipulation for political or personal reasons.

Section 7.2

GUIDELINES IN ASSURING DATA QUALITY IN KEY AREAS OF OPT PLUS PROTOCOL

A. DQC Tools

The data quality assessment of the OPT will be instituted at the different LGU levels. There are four DQC tools in excel worksheet including 1) barangay, 2) large barangay, 3) city/municipality, and 4) province. The details per level are described as follows.

1. BARANGAY LEVEL

The barangay DQC tool has eight worksheets including the form for: 1) Welcome; 2) DQC Guide; 3) Basic Information (Form 1.1); 4) Supportive Supervision (Form 1.2A); 5) Action Plan resulting from Supportive Supervision (Form 1.2B); 6) DQC dimension; 7) DQC Radial diagram; and 8) OPT Plus Action Plan. The following are the step-by-step instructions:

a. Welcome

WELCOME

OPERATION TIMBANG PLUS

DATA QUALITY CHECK AT THE BARANGAY LEVEL

This Operation Timbang Plus Data Quality Check for BARANGAY has the following forms:

Sheet #	Sheet Title	Person-In-charge
Sheet 1	Welcome Page	
Sheet 2	DQC Guide	
Sheet 3	Brgy. OPT DQC FORM 1.1: BARANGAY OPT PLUS DATA QUALITY CHECK BASIC INFORMATION	Head of the OPT Team/ NAO or D/CNPC
Sheet 4	Brgy. OPT DQC FORM 1.2: BARANGAY OPT PLUS DATA QUALITY CHECK: SUPERVISOR'S SUPPORTIVE SUPERVISION	NAO, District/City Nutrition Coordinator
Sheet 5	Brgy. OPT DQC FORM 1.3: BARANGAY OPT PLUS DATA QUALITY CHECK BY DIMENSION	
Sheet 6	Brgy. OPT PLUS DQC FORM 1.4: BARANGAY OPT PLUS DQC RADIAL DIAGRAM	
Sheet 7	Brgy. OPT DQC FORM 1.5: BARANGAY OPT PLUS DQC ACTION SHEET TO IMPROVE QUALITY	Municipality

b. OPT DQC Form 1.1: Barangay OPT Plus Data Quality Check Basic Information

Instructions:

Form 1.1. A.

1. Write all the names of the OPT Plus Team members.
2. Encode the latest OPT training information.

Form 1.1. B.

1. Encode the type of weighing scale and the length/height equipment used and assess its condition.
2. Encode the availability of mid-upper arm circumference (MUAC) tape and Child Growth Standards (CGS) charts, and record the OPT budgetary allocation.

Form 1.1. C.

1. Encode the availability of computer/s for eOPT and write its condition.
2. Specify the eOPT version used.
3. Indicate if there is an available barangay data encoder and write how many encoders are available.
4. Encode the date when data was completed and encoded and write how many days it was finished.
5. Encode the date when the data was checked and signed by the Brgy. Captain.
6. Indicate the date when the data was presented to the Barangay Nutrition Committee (BNC)/BDC, Brgy. Assembly.
7. Indicate the date when the data was submitted to the municipality.

Form 1.1. D.

1. Refer to the eOPT Plus tool and encode the number of children measured based on their sex and age and the total number of children measured.
2. Encode the total number of children based on the latest population projection used in the eOPT Plus tool.
3. Encode the total number of children based on the actual Population based on household profile.
4. Refer to the printed Brgy. eOPT - OPT Plus Form 1A used and encode the total NS feedbacked to the parents.

Form 1.1. E.

1. Encode the number of children identified as underweight/stunted/ overweight/ obese.
2. Encode the number of identified Moderate Acute Malnutrition (MAM) & Severe Acute Malnutrition (SAM).
3. Encode the number and percentage of MAM & SAM referred.
4. Encode the number and percentage of MAM & SAM that was rehabilitated.
5. Encode the date when nutrition planning was conducted.
6. Encode the percentage and number of children targeted for nutrition programs.
7. Encode the percentage and number of nutrition programs with budget based according to the AIP.
8. Encode the interventions provided to the identified malnourished children.
9. Encode the interventions provided for the first 1000 days.

OPT PLUS DQC FORM 1.1: BARANGAY OPT PLUS DATA QUALITY CHECK BASIC INFORMATION

For the Head of the OPT Team/ NAO or D/CNFC

		YEAR OF OPT PL	2022		
BARANGAY:	Batong Malake	Y/MUNICIPALI	Los Baños	PROVINCE:	Laguna

OPT Plus Barangay Team M 4

1. Reanne	7.	13.
2. Kat	8.	14.
3. Danica	9.	15.
4. Gelay	10.	16.
5.	11	17.
6.	12.	18.

A. LATEST OPT TRAINING (INPUT DATA BEFORE THE CONDUCT OF OPT PLUS)					
INCLUSIVE DA		VENUE:		# OF TRAINED TEAM	
TRAINING ORGANIZATION(S)/				# OF CERTIFIED TEAM	

B. AVAILABLE RESOURCES (EQUIPMENT, TOOLS, BUDGET, ETC) add colors per action needed

WEIGHING SCALE (check for verification sticker for the condition)	TYPE	CONDITION	LENGTH MEASUREMENT (check for verification sticker for the condition)	TYPE	CONDITION	HEIGHT MEASUREMENT (check for verification sticker for the condition)	TYPE	CONDITION
	MECHANICAL HANGING WEIGHING SCALE	CALIBRATED & VERIFIED			NOT CALIBRATED W/IN 3 YEARS			CALIBRATED & VERIFIED
	Actions Needed	NONE (Fit for use)		Actions Needed	CALIBRATE		Actions Needed	NONE (Fit for use)
	AVAILABILITY	BARANGAY-OWNED		AVAILABILITY			AVAILABILITY	

UNSTRETCHABLE MUAC TAPE	AVAILABILITY	CONDITION	AGE COMPUTATION TABLE (OPTIONAL)	AVAILABILITY	CONDITION	OPT BUDGETARY ALLOCATION IN LGU AIP	₱300,000.00
		NOT IN GOOD CONDITION		OT AVAILABL			
	Action Point	REPLACE		Action Point	ACQUIRE		

CGS CHART FOR WEIGHT-FOR-AGE	AVAILABILITY	CONDITION	CGS CHART FOR LENGTH/HEIGHT-FOR-AGE	AVAILABILITY	CONDITION	CGS CHART FOR WEIGHT-FOR-LENGTH/HEIGHT	AVAILABILITY	CONDITION
	AVAILABLE	IN GOOD CONDITION		OT AVAILABL			AVAILABLE	NOT IN GOOD CONDITIO
	Action Point	MAINTAIN		Action Point	ACQUIRE		Action Point	REPLACE

C. DATA PROCESSING

COMPUTER AVAILABLE FOR E-OPT	AVAILABILITY	CONDITION	e-OPT VERSION USED	eOPT version		BARANGAY DATA ENCODER	AVAILABILITY	NUMBER
	AVAILABLE	NOT IN GOOD CONDITION		December 2022-January 2023			OT AVAILABL	
# OF DAYS OPT DATA WERE COLLECTED	TE COMPLETE	# OF DAY(S)	# OF DAYS DATA WERE ENCODED	TE COMPLET	# OF DAY(S)	DATE WHEN OPT REPORT WAS SIGNED BY		
DATE PRESENTED TO BNC/BDC			DATE PRESENTED TO BRGY ASSEMBLY			DATE SUBMITTED TO MUNICIPALI		

D. BASIC INFORMATION (0-59 MONTHS OLD CHILDREN)							
NUMBER OF CHILDREN MEASURED	OTAL NUMBER OF CHILDRE		0-29 MONTHS	30-59 MONTH	9-23 MONTHS	24-59 MONTH	TAL NUMBER OF CHILDR
	BOYS	GIRLS					
Projected population							
Actual Population based on Household							
Number of children whose NS was immediately feedbacked to parents/caretaker after measurement:							

E. USES OF E-OPT

ASSESSMENT OF NUTRITIONAL STATUS

UNDERWEIGHT			STUNTED			OVERWEIGHT/OBESE		
MODERATE	SEVERE	TOTAL	MODERATE	SEVERE	TOTAL	OVERWEIGHT	OBESE	TOTAL

MW/MAM & SW/SAM (WASTED) Identified (SOURCE: e-OPT Plus Data)	MAM/SAM	COUNTS	MW/MAM & SW/SAM referred (SOURCE:REFERRAL FORM)	COUNTS	% (# of MAM and SAM referred/total # of children with MAM & SAM)	MW/MAM & SW/SAM IMPROVED/REHABILITATED (SOURCE: opt LIST OF MALNOURISHED CHILDREN)	COUNTS	%
	MAM							
	SAM							
	TOTAL							
DATE OF NUTRITION PLANNING			TARGETED MALOURISHED CHILDREN (SOURCE: BNAP)	COUNTS	% (# of children targeted for nutrition programs/total number of malnourished)	PROGRAMS/ PROJECT WITH BUDGET (SOURCE: AIP)	COUNTS	% (# of programs with budget/total projects in BNAP)

INTERVENTIONS IDENTIFIED		
INTERVENTIONS FOR MALNOURISHED		
UNDERWEIGHT	STUNTED	OBESE
Feeding Program	Feeding Program	Diet Program

INTERVENTIONS FOR FIRST 1000 DAYS			
0-5 MONTHS		6-23 MONTHS	
Feeding Program		Feeding Program	

c. OPT DQC Form 1.2A: Barangay OPT Plus Data Quality Check: Supervisor's Supportive Supervision

Instructions:

1. Encode the date, venue where OPT Plus was conducted, and name of the OPT Plus Team member monitored.
2. Check the anthropometric equipment of the OPT Plus Team. Record 1 if the statement/ practice is observed otherwise record 0.
3. Check whether the proper practice in taking anthropometric measurements were followed by the OPT Plus team. Record 1 if the statement/practice is observed otherwise record 0.

Note that if it was indicated in OPT DQC FORM 1.1: BARANGAY OPT PLUS DATA QUALITY CHECK BASIC INFORMATION that there were no available equipment or the equipment used were not calibrated/verified or damaged, the checklist for the said equipment will be blocked.

OPT PLUS DQC FORM 1.2A: BARANGAY OPT PLUS DATA QUALITY CHECK: SUPERVISOR'S SUPPORTIVE SUPERVISION

For Nutrition Action Officer, District/City Nutrition Coordinator

(Use this form to check if the anthropometric protocol is being followed)

YEAR OF OPT PLUS: 2022

BARANGAY: Batong Malake | CITY/MUNICIPALITY: Los Baños | PROVINCE: Laguna

OPT Plus Barangay Team Members: 4

1. Reanne	7.	13.
2. Kat	8.	14.
3. Danica	9.	15.
4. Gelay	10.	16.
5.	11.	17.
6.	12.	18.

OPT PLUS SUPPORTIVE SUPERVISION					
INCLUSIVE DATES		PLACE/VENUE		OPT TEAM MONITORED AND	
INCLUSIVE DATES		PLACE/VENUE		OPT TEAM SUPERVISED	
INCLUSIVE DATES		PLACE/VENUE		OPT TEAM SUPERVISED	
INCLUSIVE DATES		PLACE/VENUE		OPT TEAM SUPERVISED	

*CHECK the anthropometric equipment of the OPT team. Record **1 if the statement/practice is observed otherwise record 0**.*

MEASUREMENT CHECKLIST	SCORE	REMARKS
1.1. The OPT Team used the recommended/accepted weighing scale according to the MOP.		
1.2. The weighing scale was available and owned by the barangay.		
1.3. The weighing scale was calibrated within the last three years, verified within the year & fit for use.		
1.4. The measurer assured that the child is wearing light clothes or ask the mother/caregiver to remove unnecessary clothing including shoes, slippers, socks, extra clothes and diaper; the child's pockets were checked and emptied.	1	---
1.5. All weight measurements were recorded/written correctly and in the nearest 0.1 kg.	1	---
1.6. The OPT Team's measurer and recorder/assistant conducted the OPT Plus in the barangay	1	---
1.7. The child was measured twice and if the replicate measurement had a difference greater than or equal to 0.3 kg, a third weight measurement was obtained.	1	---

Observe the anthropometric measurement of the OPT team. Record 1 if the statement/practice is observed otherwise record 0.

CHECKLIST ON WEIGHING USING MECHANICAL HANGING WEIGHING SCALE	SCORE	REMARKS
1.8. The weighing scale was hung from a stable and sturdy branch, ceiling beam, or pole and the ropes used for hanging the scale was sturdy.	1	---
1.9. The weighing scale dial was at eye level of the measurer.	1	---
1.10. The measurer was assisted by another person while putting the child in the weighing pants to be measured.	1	---
1.11. The strap of the pants were placed in front of the child and the child's feet were not touching the ground during weight measurement.	1	---
1.12. The child was measured twice and if the replicate measurement had a difference greater than or equal to 0.3 kg, a third weight measurement was obtained.	1	---
1.13. The team performed ZERO adjustment procedure (pulling the sling seat/weighing pants downwards until half of the scale's capacity 3 times.)	1	---

CHECKLIST ON WEIGHING USING DIGITAL HANGING WEIGHING SCALE	SCORE	REMARKS
1.14. The weighing scale was hung from a stable and sturdy branch, ceiling beam, or pole and the ropes used for hanging the scale was sturdy.		
1.15. The weighing scale dial was at eye level of the measurer.		
1.16. The measurer was assisted by another person while putting the child in the weighing pants to be measured.		
1.17. The strap of the pants was placed in front of the child and the child's feet were not touching the ground during weight measurement.		
1.18. The child was measured twice and if the replicate measurement had a difference greater than or equal to 0.3 kg, a third weight measurement was obtained.		
1.19. The team performed ZERO adjustment procedure (pulling the sling seat/weighing pants downwards until half of the scale's capacity 3 times.)		

CHECKLIST ON WEIGHT MEASUREMENT: USING DIGITAL FLAT SCALE	SCORE	REMARKS
1.20. The scale was placed on a flat surface.		
1.21. For scale without taring: the mother/caregiver was measured and her weight was recorded then the child was given to the mother/caregiver on the weighing scale and their weight was taken. The weight of the mother was deducted from the total weight of mother/caregiver and child.		
1.22. For scale with taring: the mother/caregiver was asked to be on the weighing scale before taring the scale; the child was given to the mother/caregiver on the weighing scale and the child's weight was taken.		
1.23. The child was measured twice and if the replicate measurement had a difference greater than or equal to 0.3 kg, a third weight measurement was obtained.		

CHECKLIST ON WEIGHT MEASUREMENT: MECHANICAL COLUMN SCALE	SCORE	REMARKS
1.24. The scale was placed on a flat surface.		
1.25. The child was measured twice and if the replicate measurement had a difference greater than or equal to 0.3 kg, a third weight measurement was obtained.		

CHECKLIST ON LENGTH/HEIGHT MEASUREMENT	SCORE	REMARKS
2.1. The length/height board was available and owned by the barangay.		
2.2. The length/height board was verified within the year and was fit for use.		
2.3. The measurer requested the mother/caregiver to remove headgear, braids, etc. and shoes, slippers, and socks, etc. before the measurement.		
2.4.The length/height was read and called out by the measurer to the nearest 0.1 cm.		

CHECKLIST ON LENGTH MEASUREMENT	SCORE	REMARKS
2.5. The measuring flatboard was placed on a hard flat surface on the floor or on the table.		
2.6. If placed on the floor, the assistant was kneeling beside the headboard. If placed on the table, the assistant was standing beside the headboard		
2.7. The measurer was on the right side of the child while holding the foot piece using the right hand.		
2.8. The measurer was helped by the mother/caregiver in placing the child on the board and instructed the mother/caregiver to position at the opposite end of the board to help calm the child.		
2.9. The assistant cupped his/her hands over the child's ear to position the child's head ensuring that the child's sight was perpendicular to the ground and the assistant was at eye level with the child.		
2.10. The measurer made sure that the child was lying flat in the center of the board and checked the position of the child before getting the measurement		
2.11. The measurer pressed the child's shins or knees against the board using his/her left hand while maneuvering the foot piece firmly on the child's heels using their right hand.		

CHECKLIST ON HEIGHT MEASUREMENT	SCORE	REMARKS
2.12. The measuring board was placed on a hard flat surface against a wall, table, tree, staircase, etc. ensuring that the board was not moving.		
2.13. The measurer knelt with his/her right knee on the right side of the child while the assistant placed the OPT_Printout and pencil on the ground and knelt on both knees on the right side of the child.		
2.14. The assistant placed the child's feet flat and together at the center and against the back and base of the board/wall.		
2.15. The assistant placed his/her hand right above the child's ankles, his/her left hand on the child's knees and pushed it against the board/wall; the assistant then informed the measurer that he/she has already positioned the child's feet and legs accordingly.		
2.16. The measurer instructed the mother/caregiver to stand in front of the child and for the child to look straight ahead to the mother/caregiver so that the child's line of sight was level with the ground when the height was being measured.		
2.17. The measurer's left hand was placed under the child's chin and gradually closed his/her hand.		
2.18. The child's shoulders were level and the measurer pushed the child's tummy a little bit before measuring.		
2.19. The measurer ensured that the headpiece was pushed through the child's hair.		

CHECKLIST ON LENGTH/HEIGHT MEASUREMENT	SCORE	REMARKS
2.20. The measurer used the appropriate procedure for length (0-23mos - recumbent) and height (23-59 mos - standing) measurement and if length or height was measured due to any reason, this was documented under remarks		
2.21. The length/height was recorded by the assistant in the nearest 0.1 cm and showed to the measurer who checked the accuracy and legibility.		
2.22. The child was measured twice and in case the replicate measurements had a difference greater than or equal to 0.5 cm, a third length/height measurement was obtained; the two closest measurements were used.		
2.23. The length/height was read, called out and recorded by the assistant in the nearest 0.1 cm and showed to the measurer who checked the accuracy and legibility.		

CHECKLIST ON MUAC MEASUREMENT	SCORE	REMARKS
3.1. The measurer informed the mother/caregiver that the Child's MUAC will be measured and asked him/her to position the child such that the left upper arm is exposed (any clothing that may cover the child's upper arm is removed).		
3.2. The MUAC was measured in the left upper arm.		
3.3. Midpoint between the elbow and the tips of the shoulder was properly located.		
3.4. The arm was at a right angle and MUAC was measured at the midpoint.		
3.5. The MUAC was measured not too loose or too tight.		

CHECKLIST FOR OBSERVING SIGNS OF BILATERAL PITTING EDEMA	SCORE	REMARKS
4.1. The measurer informed the mother/caregiver that signs of edema will be checked.		
4.2. The measurer pressed both feet gently for three (3) seconds and if positive, assessed the severity of edema.		
4.3. The measurer recorded the severity of edema.		

CHECKLIST ON OPT PLUS UTILIZATION	SCORE	REMARKS
5.1. The OPT team immediately feedbacked the nutritional status of all children to the parents/caregiver immediately after measurement.		
5.2. The OPT team immediately referred all MAM/SAM cases to the RHU.		

OVERALL TOTAL SCORE:	4
SUPPORTIVE SUPERVISOR SCORE (%):	13%

d. OPT DQC Form 1.2B: Barangay OPT Plus Data Quality Check: Supervisor's Supportive Supervision (Practices that Needs Action)

This sheet will automatically reflect all actions that needs to be improved based on the OPT DQC FORM 1.2A: BARANGAY OPT PLUS DATA QUALITY CHECK: SUPERVISOR'S SUPPORTIVE SUPERVISION

OPT PLUS DQC FORM 1.2B: BARANGAY OPT PLUS DATA QUALITY CHECK: SUPERVISOR'S SUPPORTIVE SUPERVISION (PRACTICES THAT NEEDS ACTION)

YEAR OF OPT PLUS: 2022

BARANGAY: Batong Malake | CITY/MUNICIPALITY: Los Baños | PROVINCE: Laguna

OPT Plus Barangay Team Members: 4

1. Reanne	7.	13.
2. Kat	8.	14.
3. Danica	9.	15.
4. Gelay	10.	16.
5.	11.	17.
6.	12.	18.

OPT SUPPORTIVE SUPERVISION					
INCLUSIVE DATES		PLACE/VENUE		OPT TEAM MONITORED AND	
INCLUSIVE DATES		PLACE/VENUE		OPT TEAM SUPERVISED	
INCLUSIVE DATES		PLACE/VENUE		OPT TEAM SUPERVISED	
INCLUSIVE DATES		PLACE/VENUE		OPT TEAM SUPERVISED	

PRACTICES THAT NEEDS ACTION

e. OPT DQC Form 1.3: Barangay OPT Plus Data Quality Check by Dimension

Instructions:

COMPLETENESS

1. Refer to eOPT ‘DQC Summary' sheet and copy column I, Rows 8-15 and paste (paste special -> values) to Brgy OPT DQC FORM 1.3 column L, rows 21-28.

ACCURACY

2. Refer to eOPT ‘DQC Summary' sheet and copy column I, Rows 17-21 and paste (paste special -> values) to Brgy OPT DQC FORM 1.3 column L, rows 35-39.

RELIABILITY

3. Refer to eOPT ‘DQC Summary' sheet and copy column I, row 24 and paste (paste special -> values) to Brgy OPT DQC FORM 1.3 column L, row 46.
4. Refer to Brgy OPT DQC Form 1.1 and encode the actual performance scores on rows 48-53 in column L based on the performance level.

TIMELINESS

5. Refer to eOPT Plus submission forms and check whether the OPT Plus data was completed on or before March 31. Encode the actual performance score in column L, row 59 based on the performance level.
6. Refer to receiving copy of eOPT Plus report and check when the report was submitted. Encode the actual performance score in column L, row 60 based on the performance level.
7. Refer to Brgy OPT DQC Form 1.1 (Columns I-J, Row 66) & Brgy OPT DQC Form 1.2A (Column L, Row 119)2 and check the percentage of children's NS feedbacked to parents and encode the actual performance score in column L, row 61 based on the performance level.
8. Refer to Brgy OPT DQC Form 1.1 (Column G, Rows 78-80) & Brgy OPT DQC Form 1.2A (Column L, Row 120) and encode the actual performance score in column L, row 62 based on the performance level.

USEFULNESS

9. Refer to Brgy_OPT_DQC Form 1.1 (Column G, Rows 78-80) & receiving copy of the eOPT plus report and encode the actual performance score in column L, row 68 based on the performance level.
10. Refer to Brgy_OPT_DQC Form 1.1 (Columns C-D and F-G, Rows 56-58) and check the date OPT data was presented to BNC. Encode the actual performance score in column L, row 69 based on the performance level.
11. Refer to Barangay Nutrition Action Plan (BNAP) and Barangay Nutrition Scholar Action Plan (BNSAP) targeting and encode the actual performance score in column L, row 70 based on the performance level
12. Refer to the Program Implementation Review (PIR) and checked whether the OPT Plus results were discussed during the PIR. Encode the actual performance score in column L, row 71 based on the performance level.

I. COMPLETENESS

OPT PLUS DQC Form 1.3: BARANGAY OPT PLUS DATA QUALITY CHECK BY DIMENSION

YEAR OF OPT PLUS: 2022

Name of Barangay: Batong Malake

Name of Municipality: Los Baños

Province: Laguna

Region: Region IV-A (CALABARZON)

Legend

- Automatically filled up
- To be filled up

I. COMPLETENESS DPCB Memo: Scorecard results ; projected popn

	INDICATOR (UNIT)	TARGET (Reference)	PERFORMANCE LEVEL*				DOCUMENT SOURCE	Weight	ACTUAL DATA	NO DATA ENTRY REQUIRED RESULTS WILL BE		**REMARK
			0	2	4	10				SCORE (0-10)	% RATING	
A	% Coverage (population of 0-59 months) [%]	80-110% (NNC, 2012 OPT+ IG)	< 60	60-69	70-79	80-110	E-OPT Plus (DQC Summary)	32%	23.80	0	0.00	
B	Duplicate cases: % of eligible children measured with duplicate cases	0	>5	3-5	0.01-2.9	0	E-OPT Plus (DQC Summary)	20%	0.26	4	8.00	
C	With missing information: % of children with length/height but no weight	≤10	>20	15.01-20	10.01-15	≤10	E-OPT Plus (DQC Summary)	8%	0.52	10	10.00	
D	With missing information: % of children with weight but no length or height data	≤10	>20	15.01-20	10.01-15	≤10	E-OPT Plus (DQC Summary)	8%	0.78	10	10.00	
E	With missing information: % of children with no date of birth data	0	>5	3-5	0.01-2.9	0	E-OPT Plus (DQC Summary)	8%	0.26	4	4.00	
F	With missing information: % of Children with no sex data	0	>5	3-5	0.01-2.9	0	E-OPT Plus (DQC Summary)	8%	0.26	4	4.00	
G	With missing information: % of children with no name of	≤10	>20	15.01-20	10.01-15	≤10	E-OPT Plus (DQC Summary)	8%	0.78	10	10.00	
H	With missing information: % of children with no data on edema	≤10	>20	15.01-20	10.01-15	≤10	E-OPT Plus (DQC Summary)	8%	0.52	10	10.00	
								TOTAL		52	56.00	

SOURCE: WHO

II. ACCURACY

	INDICATOR [UNIT]	TARGET (Reference)	PERFORMANCE LEVEL				DOCUMENT SOURCE	% Distribution	ACTUAL PERFORMANCE	NO DATA ENTRY REQUIRED RESULTS WILL BE		REMARKS/ EVIDENCE
			0	2	4	10				SCORE (0-10)	% RATING	
A	# of children with flagged measurement based on Z-score (>	< 2.5 (ENA, 2020)	> 7.5	> 5 – 7.5	> 2.5 – 5	0 – 2.5	E-OPT Plus (DQC Summary)	30%	2.45	10	30.00	
B	average percentage of digit preference score (for weight & length/height or MUAC data)	< 8 (ENA, 2020)	>20	13-20	8-12	<8	E-OPT Plus (DQC Summary)	25%	28	0	0.00	
C	Skewness of weight-for-height Z-score (WHZ)	[-0.2, 0.2] (ENA, 2020)	<-0.6 or >0.6	[-0.6,-0.41] or [0.41,0.6]	[-0.4,0.-21] or [0.21,0.4]	[-0.2,0.2]	E-OPT Plus (DQC Summary)	15%	1.3	0	0.00	
D	Kurtosis of weight-for-height Z-score (WHZ)	[-0.2, 0.2] (ENA, 2020)	<-0.6 or >0.6	[-0.6,-0.41] or [0.41,0.6]	[-0.4,0.-21] or [0.21,0.4]	[-0.2,0.2]	E-OPT Plus (DQC Summary)	15%	5.1	0	0.00	
E	Poisson distribution of WHZ-score	>0 .05 (ENA, 2020)	<= 0.001	[0.002,0.009]	[0.01, 0.049]	>0.05	E-OPT Plus (DQC Summary)	15%	0.012564196	4	6.00	
								TOTAL		14	36.00	

*target, scoring and performance corridors are patterned on plausibility report of the SMART-ENA software (ENA, 2020)

III. RELIABILITY

	INDICATOR [UNIT]	TARGET (Reference)	PERFORMANCE LEVEL 0	2	4	10	DOCUMENT SOURCE	% Distribution	ACTUAL PERFORMANCE	NO DATA ENTRY REQUIRED RESULTS WILL BE AUTO-FILLED: SCORE (0-10)	% RATING	**REMARKS/ EVIDENCE
A	Standard Deviation of WHZ/MUAC (z-score)	> 0.9 to < 1.1	>= 1.20 OR <= 0.80	0.81 to 0.84 or 1.16 to 1.19	0.85 to 0.9 or 1.1 to 1.15	>0.9 to <1.1	E-OPT Plus (DQC Summary	15%	1.46	0	0.00	
B	Supportive Supervisor Score (%)	≥ 90%	< 70	70-79	80-89	≥ 90	Brgy OPT DQC Form 1.2A	25%	13	0	0.00	
C	Training of OPT Team (number of team member trained)	All team members are certified on accuracy and precision	no training or not all team members are trained	all members trained but no certified	at least 2 members certified	All team members certified	Brgy OPT DQC Form 1.1 (Column I, Rows 28-29)	15%	10	10	15.00	annual / certification / retraining specify inclusive dates
D	Recency of training & certification of OPT Team (date of re-training)	Training in the past 6 months	no training at all or only few were trained more than 1 year ago	training beyond 1 year ago for all team members	last 12 months for some members	last 12 months for all members	Brgy OPT DQC Form 1.1 (Column I, Rows 30-31)	5%	10	10	5.00	
E	Acceptable weighing equipment	Use of Calibrated & Acceptable Weighing Scale with all parts in good condition	Use of not recommended weighing scale	Use of recommended weighing scale but not in good condition and not calibrated	Use of recommended weighing scale and in good condition but not calibrated	Use of calibrated & recommended weighing scale with all parts in good condition	Brgy OPT DQC Form 1.1 (Columns C-D, Row 36)	10%	10	10	10.00	
F	Acceptable height equipment - height for 24-59	Use of verified & acceptable equipment for measuring height with all parts in good condition	Use of not recommended equipment for measuring height, e.g. metal stick, tape measure, improvised wall chart	Use of recommended equipment for measuring height but some parts are not in good condition	Use of recommended equipment for measuring height with all parts in good condition but not verified	Use of verified and recommended equipment for measuring height with all parts in good condition	Brgy OPT DQC Form 1.1 (Columns I-J, Row 36)	10%	10	10	10.00	
G	Acceptable length equipment - children below 24 months	Use of verified & acceptable equipment for measuring length with all parts in good condition	Use of not recommended equipment for measuring length, e.g. metal stick, tape measure, improvised wall chart	Use of recommended equipment for measuring length but some parts are not in good condition	Use of recommended equipment for measuring length with all parts in good condition but not verified	Use of verified & recommended equipment for measuring length with all parts in good condition	Brgy OPT DQC Form 1.1 (Columns F-G, Row 36)	10%	10	10	10.00	
H	Use of e-OPT plus	Complete, properly labeled and updated e-OPT plus file at the municipal level in the 1st quarter of the year	E-opt plus not used/ incomplete, not propoerly labeled and not updated e-OPT plus file at the brgy level			Complete, propoerly labeled and updated e-OPT plus file at the municipal level in the 1st quarter of the year	Brgy OPT DQC Form 1.1 (Column C & F, Rows 54-55) & file submitted to municipality	10%	10	10	10.00	
								TOTAL		60	60.00	

**You can add narrative explanation for the basis of scoring

IV.A. TIMELINESS

	INDICATOR [UNIT]	TARGET (Reference)	PERFORMANCE LEVEL				DOCUMENT SOURCE	% Distribution	ACTUAL PERFORMANCE	NO DATA ENTRY REQUIRED RESULTS WILL BE AUTO-FILLED		REMARKS/ EVIDENCE
			0	2	4	10				SCORE (0-10)	% RATING	
A	Conduct & completion of OPT plus data collection (inclusive period)	started in January and completed on or before 31 March	Completed in June and beyond	Completed in May	completed in April	started in January and completed on or before 31 March	E-OPT Plus Submission Forms	15%	10	10	15.00	
B	Submission of OPT plus report (Time)	Submitted in the 1st quarter of the year	Submitted in June and beyond	Submitted in May	Submitted in April	Submitted in the 1st quarter of the year	Receiving copy of the e-OPT plus report	15%	10	10	15.00	
C	Immediate feedback to parents of NS (%)	>90% of children's NS were feedbacked to parents	< 50	50-69	70-89	≥ 90	Brgy OPT DQC Form 1.1 (Columns I-J, Row 66) & Brgy OPT DQC Form 1.2A (Column L, Row 119)	15%	10	10	15.00	
D	immediate referral of MAM/SAM to RHU	all MAM/SAM children reported to RHU immediately	SAM & MAM cases are not reffered to RHU	some (not all) SAM & MAM cases are reported to RHU	all SAM & MAM cases are reported but not immediate	all children with MAM or SAM are immediately reported to RHU	Brgy OPT DQC Form 1.1 (Column G, Rows 78-80) & Brgy OPT DQC Form 1.2A (Column L, Row 120)	15%	10	10	15.00	

IV.A. USEFULNESS

	INDICATOR [UNIT]	TARGET (Reference)	PERFORMANCE LEVEL				DOCUMENT SOURCE	% Distribution	ACTUAL PERFORMANCE	NO DATA ENTRY REQUIRED RESULTS WILL BE AUTO-FILLED		**REMARKS/ EVIDENCE
			0	2	4	10				SCORE (0-10)	% RATING	
A	Referral of all malnourished children for appropriate intervention	all malnourished children were referred for appropriate intervention in the 1st quarter of the year	all malnourished children were not referred for appropritate intervention in the 1st quarter of the year	all malnourished children were referred for appropriate intervention in the 4th quarter of the year	all malnourished children were refered for appropriate intervention the 3rd quarter of the year	all malnourished children were referred for appropriate intervention in the 2nd quarter of the year	Brgy_OPT_DQC Form 1.1 (Column J, Rows 78-80) & Receiving copy of the e-OPT plus report	10%	10	10	10.00	
B	Reporting of OPT plus results to Barangay Nutrition Committee/Barangay Development Committee (BNC/BDC) and/or General assembly	Reported to BNC/BDC and/or Brgy General Assembly in the 2nd quarter of the year	OPT plus results not reported to BNC/BDC and/or General Assembly within the year	Reported to BNC/BDC and/or Brgy General Assembly in the 4th quarter of the year	Reported to BNC/BDC and/or Brgy General Assembly in the 3rd quarter of the year	Reported to BNC/BDC and/or Brgy General Assembly in the 2nd quarter of the year	Brgy_OPT_DQC Form 1.1 (Columns C-D and F-G, Rows 56-58)	10%	10	10	10.00	
C	Targeting of households with malnourished children	Household with malnourished children targetted for nutrition interventions	information on NS was not used in targeting participants in nutrition intervention	All household with malnourished children were targetted in BNAP but lacks nutrition action	Only limited number of households with malnourished children were targetted, participated and gain benefits from implemented nutrition interventions	All households with malnourished children were targetted, participated and gain benefits from implemented nutrition interventions	BNAP and BNSAP targeting	10%	10	10	10.00	
D	Presentation of OPT Plus results in the program implementation review (PIR)	OPT Plus results discussed in the PIR	OPT Plus results was reported to PIR			OPR Plus results were discussed in the PIR	PIR	10%	10	10	10.00	
								TOTAL		80	100.00	
								AVERAGE		52	63.00	

** You can add narrative explanation for the basis of scoring

f. OPT DQC Form 1.4: Barangay OPT Plus DQC Radial Diagram

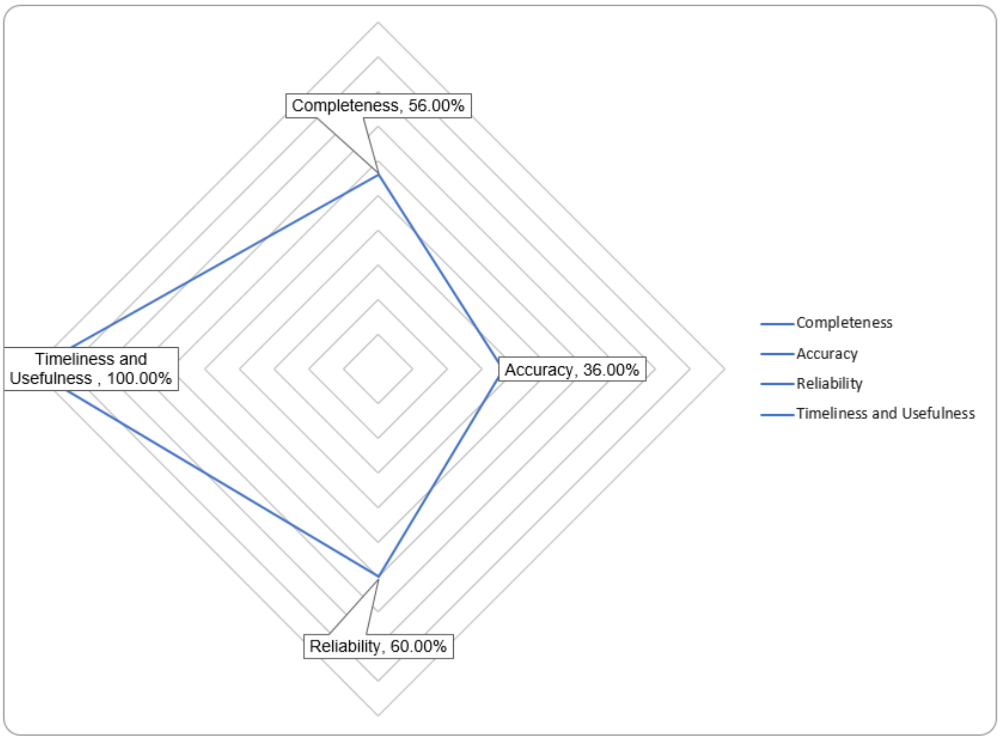
All results in this sheet including the radial diagram will automatically be encoded based on the data provided in the BRGY OPT DQC FORM 1.3.

OPT PLUS DQC FORM 1.4: BARANGAY OPT PLUS DQC RADIAL DIAGRAM

YEAR OF OPT PLUS:	2022	BARANGAY:	Batong Malake
MUNICIPALITY:	Los Baños	PROVINCE/REGION:	Laguna / Region IV-A (CALABARZON)

NUTRITION WORKERS MANAGEMENT FUNCTIONS		TARGET RATING	PERFORMANCE RATING (NO DATA ENTRY REQUIRED)
A.	Completeness	100%	56.00%
B.	Accuracy	100%	36.00%
C.	Reliability	100%	60.00%
D.	Timeliness and Usefulness	100%	100.00%
	AVERAGE	100%	63%
	PERFORMANCE LEVEL	FAIR	

Barangay OPT DQC Monitoring Radial Diagram



g. OPT DQC Form 1.4: Barangay OPT Plus DQC Action Sheet to Improve Quality

Answer all questions based on experience of the OPT Plus Team from the recent conduct of OPT Plus including action points to address the gaps identified in the DQC.

OPT DQC FORM 1.5: BARANGAY OPT PLUS DQC ACTION SHEET TO IMPROVE QUALITY

YEAR OF OPT PLUS:	2022	BARANGAY:	Batong Malake
MUNICIPALITY/CITY:	Los Baños	PROVINCE/REGION:	Laguna / Region IV-A (CALABARZON)

Answer the following questions based on experience of the OPT Plus Team from the recent conduct of OPT Plus.
Note: To be accomplished by the Municipality

Parameters	What is going well and why?	What are the problems and issues? Why do these problems and issues persist?	How did the LGUs manage these OPT Plus issues and problems?
A. Completeness			
B. Accuracy			
C. Reliability			
D. Timeliness and Usefulness			
E. Others			

Parameters	Recommendation for the Nutrition Worker	Recommendation for the Local Nutrition Committee	Recommendation for the National Nutrition Council
A. Completeness			
B. Accuracy			
C. Reliability			
D. Timeliness and Usefulness			
E. Others			

Name of Team Member	Designation and Office	Signature	Date Signed

Received: Date:

2. CITY/MUNICIPAL LEVEL

The city/municipal DQC tool has six worksheets including the form for: 1) Welcome; 2) Capacity Mapping of Human Resources (Form A); 3) Capacity Mapping of Anthropometric Equipments and Tools (Form B); 4) Data Quality Check Encoding (Form 2.1); 5) Data Quality Check Summary (Form 2.2); and 6) OPT Plus Action Plan. The following are the step-by-step instructions:

a. Welcome

WELCOME

OPERATION TIMBANG PLUS
DATA QUALITY QUALITY CHECK AT THE CITY/MUNICIPAL LEVEL

NATIONAL NUTRITION COUNCIL • 1974 •

OPT Plus TSEK
Operation Timbang Plus
Tamang Sukat, Eksaktong Kalidad!

This Operation Timbang Plus Data Quality Check for MUNICIPALITY/CITY has the following forms:

Sheet #	Sheet Title	Person-In-charge
Sheet 1	Welcome Page	
Sheet 2	Municipal OPT DQC FORM A: CITY/MUNICIPAL OPT PLUS CAPACITY MAPPING OF HUMAN RESOURCES	
Sheet 3	Municipal OPT DQC FORM B: CITY/MUNICIPAL OPT PLUS CAPACITY MAPPING OF ANTHROPOMETRIC EQUIPMENT AND TOOLS	
Sheet 4	Municipal OPT DQC FORM 2.1: CITY/MUNICIPAL OPT PLUS DATA QUALITY CHECK ENCODING	
Sheet 5	Municipal OPT DQC FORM 2.2: CITY/MUNICIPAL OPT PLUS DATA QUALITY CHECK SUMMARY	
Sheet 6	Municipal OPT DQC FORM 2.3: CITY/MUNICIPAL OPT DQC ACTION SHEET TO IMPROVE QUALITY	

b. OPT DQC Form A. City/Municipality OPT Plus Capacity Mapping of Human Resources

Instructions:

1. Print or copy this form in a city/municipal OPT logbook.
2. Indicate the municipality, province, and region and number of barangays in the municipality.
3. Write the dates when the supportive supervision was conducted.

Capacity Mapping of Human Resources

First Column	List down the barangays in the first column.
Second Column	List down all members of the OPT Plus team per barangay. Add rows if necessary.
Third Column	Write the designation of the members of the OPT Plus team.
Fourth Column	Write their contact information for each OPT Plus team member.
Fifth Column	Indicate if the OPT Plus team member was 1) trained only; 2) oriented only; 3. trained and passed the standardization test; or 0) not trained yet.
Sixth Column	If the OPT Plus team member was trained, indicate the classification (i.e., Level 1, 2, 3, 4 or 5).
Seventh Column	Indicate remarks such as newly appointed, resigned, with illness, and other problematic situations.

OPT PLUS DQC FORM A: CITY/MUNICIPAL OPT PLUS CAPACITY MAPPING OF HUMAN RESOURCES

For the period: April to June 2021

CITY/MUNICIPALITY: Los Baños **PROVINCE:** Laguna **REGION:** Region IV-A (CALABARZON)

TOTAL NUMBER OF BARANGAYS: 50 **Dates:** 21-Jan-21 - 1-Feb-21

BARANGAY	CAPACITY MAPPING OF HUMAN RESOURCES					
	Name (Surname, Given name, Middle Name)	Designation	Contact Details	Status: Trained/Oriented/ Passed the Standardization Test	If trained and Passed, indicate classification (Level 1, 2, 3, 4 or 5)	REMARKS
1 Batong Malake						
2 Maahas						
3						
4						
5						
6						
7						
8						
9						
10						

c. OPT DQC Form B. City/Municipality OPT Plus Capacity Mapping of Anthropometric Equipment and Tools

Instructions:

1. Print or copy this form in a city/municipal OPT logbook.
2. Indicate the municipality, province, and region and number of barangays in the municipality.
3. Write the dates when the supportive supervision was conducted.

Capacity Mapping of Anthropometric Equipment and Tools

First Column	Indicate the barangay.
Second Column	Indicate the type and the number of weighing scale available in the barangay.
Third Column	Indicate the type and number of measuring equipment for length available in the barangay.
Fourth Column	Indicate the type and number of measuring equipment for height available in the barangay.
Fifth Column	Indicate the number of available non-stretchable MUAC tape in the barangay.
Sixth Column	Indicate if the CGS tables for weight-for-age (WFA); length/height-for-age (L/HFA) and weight-for-length/height (WFL/H) for male and female are all available in the barangay.
Seventh Column	Indicate remarks such wrong unit of measurement in the equipment, bad condition, and other problematic conditions of the equipment and tools.

OPT PLUS DQC FORM B: CITY/MUNICIPAL OPT PLUS CAPACITY MAPPING OF ANTHROPOMETRIC EQUIPMENT AND TOOLS

For the period: April to June 2021

CITY/MUNICIPALITY: Los Baños | **PROVINCE:** Laguna | **REGION:** Region IV-A (CALABARZON)

TOTAL NUMBER OF BARANGAYS: 50 | **Suportive Supervision Inclusive Dates:** 21-Jan-21 - 01-Feb-21

BARANGAY	CAPACITY MAPPING OF Anthropometric Equipment and Tools												
	Weighing Equipment*		Equipment for measuring length		Equipment for measuring height		Non-stretchable MUAC tape		CGS tables				REMARKS (wrong unit of measurement, bad condition, etc)
	Type	#	Type	#	Type	#	Availability	#	WFL	WFH	WFA	LFH	
**Checklist													
1 Batong Malake													
2 Maahas													
3													
4													
5													
6													
7													
8													

d. OPT DQC Form 2.1: City/Municipality OPT Plus Capacity Mapping of Anthropometric Equipment and Tools

Instructions:

1. Indicate the municipality, province, and region and number of barangays in the municipality.
2. Indicate the total number of children, total number of male children, total number of female children, total number of 0-to-29-month-old children, and total number of 30-to-59-month old children measured during the latest OPT Plus.
3. Indicate when the OPT Plus data were submitted to the province.
4. If the eOPT Plus data was used in LNAP and was presented in the MNC, choose YES. If not, choose NO.

Data Quality Dimensions per barangay

First Column	List down all barangays in the municipality.
Second to Sixth Column	Refer to BRGY OPT DQC Form 1.4. Copy the performance rating in column F, row 19-22 and paste on the corresponding columns.
Seventh Column	Refer to BRGY OPT DQC Form 1.4. Copy the average performance rating in column F, row 24 and paste on the corresponding column.
Eighth Column	This will be auto-filled once the barangay's data quality dimensions are input.

OPT PLUS DQC FORM 2.1: CITY/MUNICIPAL OPT PLUS DATA QUALITY CHECK ENCODING

For the period: April to June 2021

CITY/MUNICIPALITY	Los Baños	**PROVINCE:**	Laguna	**REGION:**	Region IV-A (CALABARZON)
TOTAL NUMBER OF BARANGAYS IN THE MUNICIPALITY:	10	**TOTAL NO. OF CHILDREN:**		**TOTAL NO. OF MALE CHILDREN:**	
				TOTAL NO. OF FEMALE CHILDREN:	
TOTAL NO. OF 0-29 MONTHS CHILDREN:		**TOTAL NO. OF 30-59 MONTHS CHILDREN:**		**DATE SUBMITTED TO PROVINCE:**	On or before May 15

The eOPT Plus data was used in LNAP and was presented in the MNC? YES

BARANGAY		DATA QUALITY DIMENSION PER BARANGAY					
		COMPLETENESS (%)	ACCURACY (%)	RELIABILITY (%)	TIMELINESS & USEFULNESS (%)	AVERAGE RATING (%) [NO DATA ENTRY REQUIRED]	Remarks
1	Batong Malake	68.00	59.00	30.50	67.00	56.13	NEEDS IMPROVEMENT
2	Maahas	87.00	90.00	95.00	67.50	84.88	GOOD
3							
4							
5							
6							
7							
8							
9							
10							

e. OPT DQC Form 2.2: City/Municipal OPT Plus Data Quality Audit Summary

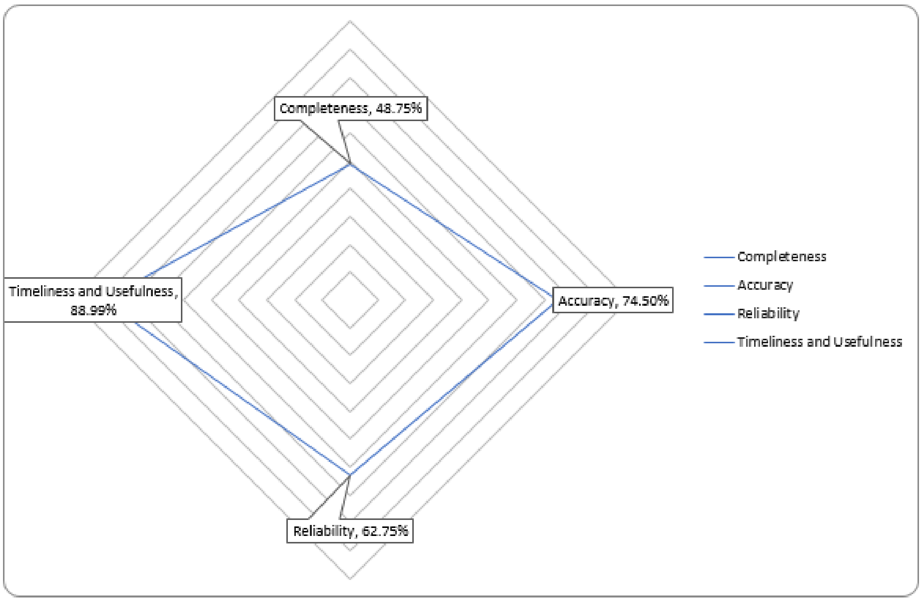
All results in this sheet including the radial diagram will automatically be encoded based on the data provided in the OPT DQC FORM 2.1: MUNICIPAL/CITY OPT PLUS DATA QUALITY CHECK ENCODING.

OPT PLUS DQC FORM 2.2: CITY/MUNICIPAL OPT PLUS DATA QUALITY CHECK SUMMARY

FOR THE PERIOD:	April to June 2021	REGION:	Region IV-A (CALABARZON)
CITY/MUNICIPALITY	Los Baños	PROVINCE:	Laguna

NUTRITION WORKERS MANAGEMENT FUNCTIONS		TARGET RATING	PERFORMANCE RATING (NO DATA ENTRY REQUIRED)
A.	Completeness	100%	48.75%
B.	Accuracy	100%	74.50%
C.	Reliability	100%	62.75%
D.	Timeliness and Usefulness	100%	88.99%
	AVERAGE	100%	68.75%
	PERFORMANCE LEVEL	FAIR	

Municipal/City OPT Plus DQC Monitoring Radial Diagram



f. OPT DQC Form 2.3: City/Municipal OPT DQC Action Sheet to Improve Quality

Answer all questions based on experience of the OPT Plus Team from the recent conduct of OPT Plus.

OPT PLUS DQC FORM 2.3: CITY/MUNICIPAL OPT PLUS DQC ACTION SHEET TO IMPROVE QUALITY

FOR THE PERIOD:	April to June 2021	REGION:	Region IV-A (CALABARZON)
CITY/MUNICIPALITY:	Los Baños	PROVINCE:	Laguna

Parameters	What is going well WITH OPT PLUS and why?	What are the problems and issues on OPT Plus? Why do these problems and issues persist?	How did the LGUs manage the these issues and problems?
A. Completeness			
B. Accuracy			
C. Reliability			
D. Timeliness and Usefulness			
E. Others			

Parameters	Recommendation for the OPT Plus Team	Recommendation for the Local Nutrition Committee	Recommendation for the National Nutrition Council
A. Completeness			
B. Accuracy			
C. Reliability			
D. Timeliness and Usefulness			
E. Others			

Name of Team Member	Designation and Office	Signature	Date Signed

Received: ____________________ Date: ________________________________

3. PROVINCIAL LEVEL

The province DQC tool has four worksheets including the form for: 1) Welcome; 2) Data Quality Check Encoding (Form 4.1); 5) Data Quality Check Summary (Form 4.2); and 6) OPT Plus Action Plan (Form 4.3). The following are the step-by-step instructions:

a. Welcome

WELCOME

OPERATION TIMBANG PLUS

DATA QUALITY CHECK AT THE PROVINCIAL LEVEL

OPT Plus TSEK

Operation Timbang Plus

Tamang Sukat, Eksaktong Kalidad!

This Operation Timbang Plus Data Quality Check for BARANGAY has the following forms:

Sheet #	Sheet Title	Person-in-Charge
Sheet 1	Welcome Page	
Sheet 2	Provincial OPT DQC Form 4.1 : PROVINCIAL OPT PLUS DATA QUALITY CHECK ENCODING	
Sheet 3	Provincial OPT DQC Form 4.2: PROVINCIAL OPT PLUS DATA QUALITY CHECK SUMMARY	
Sheet 4	Provincial OPT DQC Form 4.3: PROVINCIAL OPT DQC ACTION SHEET TO IMPROVE QUALITY	

b. Provincial OPT DQC Form 4.1. Provincial OPT Plus Data Quality Check Encoding

Instructions:

1. Indicate the period of the conduct of DQC.
2. Indicate the province, region, and number of cities/municipalities in the province.
3. Encode the total number of children measured, total number of 0-to-29-month-old and 30-to-59-month-old children.
4. Encode the total number of male and female children measured.

Data Quality Dimension per City/Municipality

First Column	List down the city/municipality in the first column.
Second to Fifth Column	Refer to city/municipality DQC form 2.2 and encode the performance ratings per functions.
Sixth to Seventh Column	The average rating and remarks will automatically be computed per city/ municipality.

PROVINCIAL OPT PLUS DQC FORM 4.1: PROVINCIAL OPT PLUS DATA QUALITY CHECK ENCODING

For the period: Jan-June 2021

PROVINCE:	Laguna	**REGION:**	Region IV-A (CALABARZON)		
NUMBER OF CITIES/MUNICIPALITIES:	10	**TOTAL NO. OF CHILDREN:**		**TOTAL NO. OF MALE CHILDREN:**	
TOTAL NO. OF 0-29 MONTHS CHILDREN:		**TOTAL NO. OF 30-59 MONTHS CHILDREN:**		**TOTAL NO. OF FEMALE CHILDREN:**	

CITY/MUNICIPALITY	DATA QUALITY DIMENSION PER MUNICIPALITY/CITY					
	COMPLETENESS (%)	ACCURACY (%)	RELIABILITY (%)	TIMELINESS & USEfULNESS (%)	AVERAGE RATING (%) [NO DATA ENTRY REQUIRED]	Remarks
Los Baños	100.00	92.00	97.00	80.00	92.25	GOOD
Maahas	96.00	65.00	58.00	95.00	78.50	FAIR

c. Provincial OPT DQC Form 4.2: Provincial OPT Plus Data Quality Check Summary

All results in this sheet including the radial diagram will automatically be encoded based on the data provided in the OPT DQC FORM 4.1: PROVINCIAL OPT PLUS DATA QUALITY CHECK ENCODING.

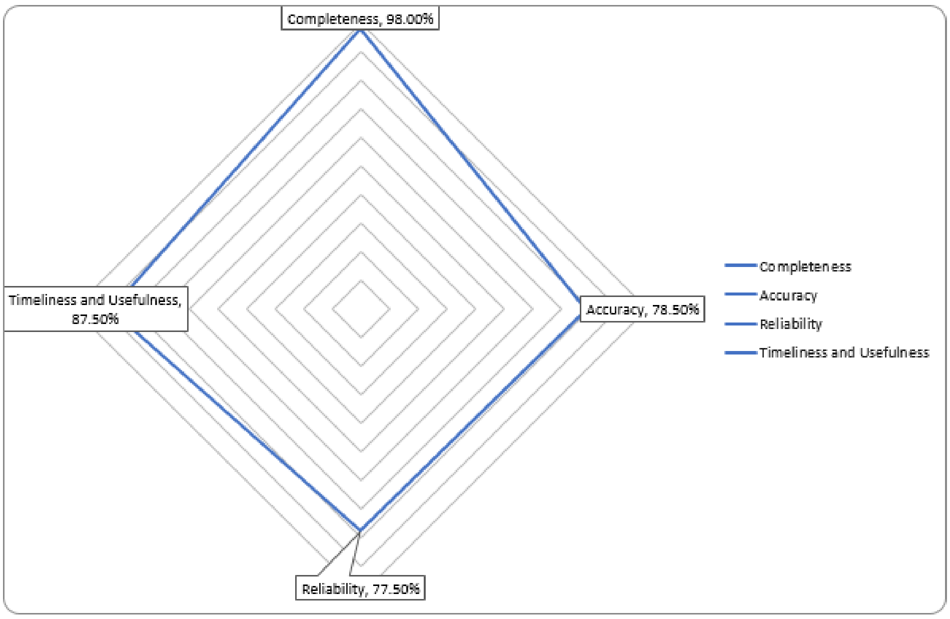
NATIONAL NUTRITION COUNCIL • 1974 •

PROVINCIAL OPT PLUS DQC FORM 4.2: PROVINCIAL OPT PLUS DATA QUALITY CHECK SUMMARY

FOR THE PERIOD:	Jan-June 2021		
PROVINCE:	Laguna	REGION:	Region IV-A (CALABARZON)

NUTRITION WORKERS MANAGEMENT FUNCTIONS		TARGET RATING	PERFORMANCE RATING (NO DATA ENTRY REQUIRED)
A.	Completeness	100%	98.00%
B.	Accuracy	100%	78.50%
C.	Reliability	100%	77.50%
D.	Timeliness and Usefulness	100%	87.50%
AVERAGE		100%	85.38%
PERFORMANCE LEVEL		GOOD	

Provincial OPT PLUS DQC Monitoring Radial Diagram



d. Provincial OPT DQC Form 4.3: Provincial OPT DQC Action Sheet to Improve Quality

Answer all questions based on experience of the OPT Plus Team from the recent conduct of OPT Plus.

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PROVINCIAL OPT PLUS DQC FORM 4.3: PROVINCIAL PLUS OPT DQC ACTION SHEET TO IMPROVE QUALITY

FOR THE PERIOD: Jan-June 2021

PROVINCE: Laguna | **REGION:** Region IV-A (CALABARZON)

Parameters	What is going well WITH OPT PLUS and why?	What are the problems and issues on OPT Plus? Why do these problems and issues persist?	How did the LGUs manage the these issues and problems?
A. Completeness			
B. Accuracy			
C. Reliability			
D. Timeliness and Usefulness			
E. Others			

Parameters	Recommendation for the OPT Plus Team	Recommendation for the Local Nutrition Committee	Recommendation for the National Nutrition Council
A. Completeness			
B. Accuracy			
C. Reliability			
D. Timeliness and Usefulness			
E. Others			

Name of Team Member	Designation and Office	Signature	Date Signed

Received: ____________________ Date: ________________________________

B. DQC Activity Cycle

The Data Quality Check (DQC) protocol for OPT Plus was installed to ensure that the five data quality dimensions are met. Four major DQC activities are 1) OPT Plus Measurement and Processing; 2) OPT Plus Monitoring; 3) Resource Evaluation; and 4) Capacity Development. Figure 1 illustrates the DQC activity cycle.

The initial implementation of the DQC starts with the Resources Evaluation on Human Resources and Equipment and Tools in August of 2023. The supervisors will use Municipal OPT DQC Form A - Capacity Mapping of Human Resources and Form B - Capacity Mapping of Anthropometric Equipment and Tools to plan for Capacity Development, including OPT Training, Planning for procurement, and Calibration and Verification of Equipment and Tools. It is done to prepare the OPT Plus Team members for conducting OPT Plus at the start of 2024 and data processing. These activities would like to check the accuracy, reliability, completeness, and timeliness of the OPT Plus data at this stage. The next DQC activity is OPT Plus monitoring to check the usefulness of the OPT data. It will document the intervention provided to the at-risk children and the improvement in their nutritional status. The next cycle will start again in August while the OPT Plus monitoring continues until December.

Figure 7.1 shows that the DQC Activity Cycle aligns with the OPT Plus activities enumerated in Table 7.1. The table also listed the forms to be used for the DQC Activities. These forms are discussed under the DQC tools topic.

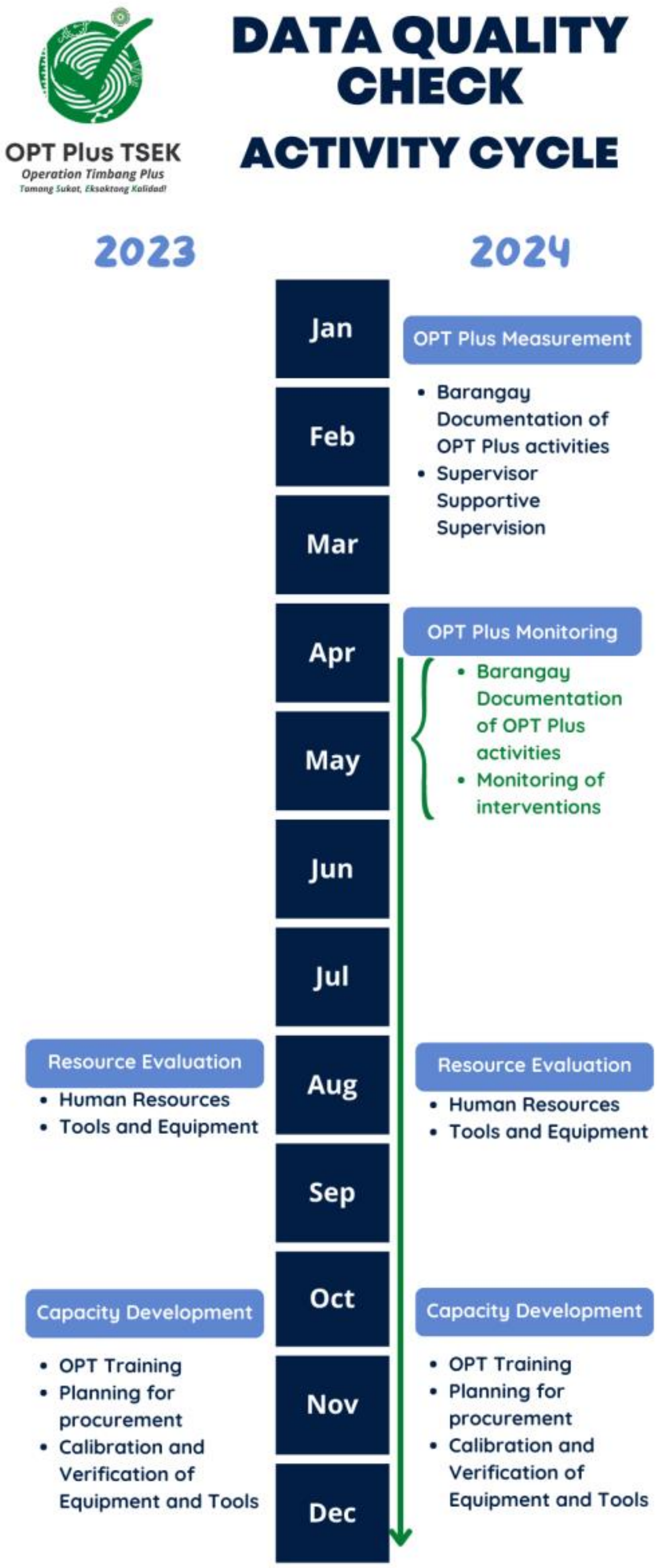

Figure 7.1. The DQC Activity Cycle.

MONTH	FUNCTION			DQC ACTIVITY CYCLE			
	Activities	Forms and Source	Submission of OPT Plus	2023	2024	Forms and Source	Submission of OPT TPlus DQC
January	1. Conduct OPT Plus 2. Data Processing	1. OPT Plus Form 1A			1. Barangay Documentation of OPT Plus activities 2. Supervisor Supportive Supervision	Brgy OPT DQC 1. OPT DQC Form 1.1: Barangay OPT Plus Documentation 2. OPT DQC Form 1.2: Barangay OPT Plus Data Quality Check: Supervisor's Supportive Supervision (Checklist)	
February	1. Conduct OPT Plus 2. Data Processing 3. Monthly monitoring of 0-to-23-month-old and malnourished children	OPT Plus: 2. List of 0-23 mos; 3. List_MW(MAM) 4. List_SW(SAM) 5. List_MSt&SSt 6. List_OW&Ob 7. List_MUW, SUW,MSt&SSt 8. List_MSt,SSt, MW&SW 9. List_MSt,SSt, OW&Ob 10. List_MSt,SSt, OW&Ob					

MONTH	FUNCTION			DQC ACTIVITY CYCLE			
	Activities	Forms and Source	Submission of OPT Plus	2023	2024	Forms and Source	Submission of OPT TPlus DQC
March	1. Presentation to BNC 2. Preparation and updating of BNAP 3. Monthly monitoring of 0-to-23-month-old and malnourished children	OPT Plus: Forms 2 to 10 11. NutStatusBrgy 12. Graphs	March 25: Barangay Nutrition Scholar (BNS) to Midwife/Barangay Nutrition Action Officer(BNAO) March 31: Midwife/BNAO to Barangay Nutrition Committee (BNC)				
April	Monthly monitoring of 0-to-23-month-old and malnourished children	Forms 2 to 10	April 16: BNC to City/Municipal Nutrition Action Officer (C/MNAO)		1. Barangay Documentation of OPT Plus activities 2. Monitoring of interventions (April to December)	OPT DQC Form 1.1: Barangay OPT Plus Data Quality Check Basic Information	
May	Monthly monitoring of 2-23-month-old and malnourished children	Forms 2 to 10	May 5: C/MNAO to City/Municipal Nutrition Committee (C/(MNC) May 15: C/MNC to Provincial Nutrition Action Officer (PNAO)				

MONTH	FUNCTION			DQC ACTIVITY CYCLE			
	Activities	Forms and Source	Submission of OPT Plus	2023	2024	Forms and Source	Submission of OPT TPlus DQC
June	Monthly monitoring of 0-to-23-month-old and malnourished children	OPT Plus: Forms 2 to 10 11. NutStatusBrgy 12. Graphs	June 5: PNAO to Provincial Nutrition Committee (PNC) June 15: PNC to Regional Nutrition Program Coordinator - NNC Regional Office				
July	Monthly monitoring of 0-to-23-month-old and malnourished children	Forms 2 to 10	July 5: RNPC-NNC Regional Office to NNC Central Office - Nutrition Surveillance Division				
August	1. Semestral monitoring of 0-to-59-month-old children 2. Inventory of resources 3. Organization of OPT Plus Team 4. Preparation of equipment and materials 5. Calibration/ verification of equipment/tools	OPT Plus Form 1A		Resource Evaluation	Resource Evaluation	Municipal OPT DQC 6. Form A. Capacity Mapping of Human Resources 7. Form B. Capacity Mapping of Anthropometric Equipment and Tools	August 31: BNC to C/MNC

MONTH	FUNCTION			DQC ACTIVITY CYCLE			
	Activities	Forms and Source	Submission of OPT Plus	2023	2024	Forms and Source	Submission of OPT TPlus DQC
September	1. Monthly monitoring of 0-to-23-month-old and malnourished children 2. BNAP Preparation 3. Community Mobilization	Forms 2 to 10 Household Profile	September 30/ 3rd Quarter NNC Technical Committee Meeting: NNC CO-NSD to submit and present to the NNC Technical Committee				September 30: C/ MNC to PNC
October	1. Monthly monitoring of 0-to-23-month-old and malnourished children 2. Community Mobilization	Forms 2 to 10		1. OPT Training 2. Planning for procurement 3. Calibration and Verification of Equipment and Tools	1. OPT Training 2. Planning for procurement 3. Calibration and Verification of Equipment and Tools		October 30: PNC to RNPC/NNC-RO
November	1. Monthly monitoring of 0-to-23-month-old and malnourished children 2. Community Mobilization	Forms 2 to 10					November 29: RNPC/NNC-RO to NNC CO
December	1. Monthly monitoring of 0-to-23-month-old and malnourished children 2. Community Mobilization						

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OPT Plus TSEK
Operation Timbang Plus
Tamang Sukat, Eksaktong Kalidad!

Section Eight

SUPERVISOR’S GUIDE: SUPPORTIVE SUPERVISION

SUPERVISOR'S GUIDE: SUPPORTIVE SUPERVISION

OVERVIEW

OPT Plus Supportive Supervision is the process of helping the OPT Plus Team to continuously improve their work performance and maintain quality data. Section 8 will discuss the importance of supportive supervision in implementing OPT Plus.

After reading this section, one should be able to:

1. Define supportive supervision;
2. Enumerate the roles and responsibilities of supervisors; and
3. Be familiar with the data quality check supportive supervision checklist.

This section covers guidelines specifically for OPT Plus supervisors. Specific sections include the following:

LIST OF ACRONYMS

BGA	Barangay General Assembly
BHC	Barangay Health Center
BHS	Barangay Health Station
BNAP	Barangay Nutrition Action Plan
BNC	Barangay Nutrition Committee
BNS	Barangay Nutrition Scholar
C/MNAP	City/Municipal Nutrition Action Plan
CNAO	City Nutrition Action Officer
CGS	Child Growth Standards
CHO	City Health Office
CNC	City Nutrition Committee
D/CNPC	District/City Nutrition Program Coordinator
DQC	Data Quality Check
ECCD	Early Child Care and Development
eOPT Plus	Electronic Operation Timbang Plus
KAP	Knowledge, Attitudes and Practices
L/HFA	Length/Height-for-Age
LCE	Local Chief Executive
MAM	Moderate Acute Malnutrition
MHO	Municipal Health Office
MNAO	Municipal Nutrition Action Officer
MNC	Municipal Nutrition Committee
MNPC	Municipal Nutrition Program Coordinator
MUAC	Mid-upper Arm Circumference
MW/MAM	Moderately Wasted/Moderate Acute Malnutrition
OPT Plus	Operation Timbang Plus
RHM	Rural Health Midwife
RHU	Rural Health Unit
SAM	Severe Acute Malnutrition

SW/SAM	Severely Wasted/Severe Acute Malnutrition
WFA	Weight-for-Age
WFH	Weight-for-Height
WFL	Weight-for-Length
WFL/H	Weight-for-Length/Height
WHO	World Health Organization
WHO-CGS	World Health Organization Child Growth Standards

Section 8.1

ROLES AND RESPONSIBILITIES OF SUPERVISORS

What is supportive supervision?

Supportive Supervision can be defined as a process of guiding, monitoring, and coaching workers to promote compliance with standards of practice and to ensure the delivery of quality health services. This process permits supervisors and supervisees the opportunity to work as a team to meet common goals and objectives (UNICEF, 2019).

Who are the supervisors?

At the city level, City Nutrition Action Officer (CNAO), the District/City Nutrition Program Coordinator (D/CNPC), or the Rural Health Midwives (RHM) will supervise and monitor the conduct of OPT Plus in the barangays.

At the municipal level, the Municipal Nutrition Action Officer (MNAO), Municipal Nutrition Program Coordinator (MNPC), nutritionist-in-charge, or midwife will supervise and monitor the conduct of Operation Timbang (OPT) Plus in the barangays.

What are the roles of the supervisors?

Supervisors are guides and trainers of the OPT Plus Team to ensure the proper and successful conduct of the OPT Plus. These can be achieved by developing the skills of the OPT Plus Team in communication, accurate, precise, and reliable anthropometric measurements, presentation, advocacy, documentation, reporting and recording, organizing, time management, and coordination. Likewise, during supportive supervision, the supervisor helps the members of the OPT Plus team make things work, identify mistakes/shortcomings, and correct them at the same time by demonstrating the correct procedure and developing the skills of the team members. The roles of supervisors and the C/MNAO are enumerated in Table 8.1.

Table 8.1. Roles and responsibilities of the supervisors and C/MNAO according to OPT Plus activities.

ACTIVITY	SUPERVISORS	CITY/MUNICIPAL NUTRITION ACTION OFFICER (C/MNAO)
Activity 1: Pre-OPT Plus Protocol	1. Assist in organizing the Training and Standardization and Refresher Training on the OPT Plus. 2. Facilitate the organization of the OPT Plus Team in the assigned cluster. 3. Check whether the weighing scales are calibrated and if length/height boards are verified. If not, initiate or lead the calibration or verification of the tools and equipment. 4. For highly urbanized cities, the City Health Officer (CHO) will prepare a letter of communication to be signed by the Local Chief Executive (LCE) to allow the conduct of OPT Plus in exclusive subdivisions and villages.	1. Organize the OPT Plus Training and Standardization Exercise and Refresher Training. 2. Remind the barangay OPT Plus Team of the scheduled conduct of the OPT Plus every January to March of the current year. 3. Oversee the conduct of family profiling and spot mapping. 4. Allocate funds for the purchase of OPT Plus.

ACTIVITY	SUPERVISORS	CITY/MUNICIPAL NUTRITION ACTION OFFICER (C/MNAO)
Activity 2: **OPT Plus Anthropometric Measurement**	1. Check the master list of under-five children. 2. Oversee the actual collection of length/height measurement and weighing. 3. Review the recording of the sex, date of birth, date of weighing, length/height, and weight. 4. Refer to the Barangay Health Centers (BHC)/Rural Health Unit (RHU) the children with bilateral pitting edema and/or obvious manifestation of malnutrition (i.e., extreme thinness, hollow cheeks, edematous feet and hands, etc.). 5. Provide feedback on the child's nutritional status immediately after measuring using the weight-for-length/height index.	Monitor the status of the conduct of the OPT Plus in each Barangay Health Stations (BHS).
Activity 3: **OPT Plus Nutritional Status Assessment**	1. If possible, assist in encoding the anthropometric measurements in the Electronic Operation Timbang Tool (eOPT Plus Tool). 2. Review the outputs of the Barangay Nutrition Scholars (BNS) using the WHO Child Growth Standards (WHO-CGS) Table. 3. Endorse the OPT Plus reports to the barangay. 4. Consolidate all barangay eOPT Plus into the municipal level eOPT Plus tool.	Conduct data quality check of the OPT Plus.

ACTIVITY	SUPERVISORS	CITY/MUNICIPAL NUTRITION ACTION OFFICER (C/MNAO)
Activity 4: OPT Plus Data Interpretation and Action Planning	Participate in the preparation of the LNAP.	1. Organize the presentation of OPT PLUS results to the City/Municipal Nutrition Committee (C/MNC). 2. Review and consolidate the updated Barangay Nutrition Action Plan (BNAP). 3. Facilitate the preparation of the C/MNAP. 4. Present the results of the OPT Plus to the Sangguniang Panlungsod/ Bayan for the budget appropriation of the C/MNAP. 5. Facilitate approval of resolutions at the barangay and municipal/city levels concerning OPT Plus activities.
Activity 5: OPT Plus Monitoring and Evaluation	1. Assist in reviewing the monthly and quarterly OPT Plus monitoring coverage if all the children weighed and measured in the first quarter were followed up. 2. Participate in the bi-annual review of the BNAP and assist in presenting BNAP accomplishments during the Barangay General Assembly (BGA).	1. Remind the Barangay Nutrition Committee (BNC) of the schedules of the monthly OPT Plus monitoring of SW/ MAM and MW/MAM, underweight and 0-to-23-month-old, and quarterly monitoring of 24-to-59-month-old children. 2. Review the monthly and quarterly OPT Plus monitoring coverage if all the children weighed and measured in the first quarter were all followed up OR Assist in reviewing the monthly and quarterly OPT Plus monitoring coverage to verify if all the children measured were covered during child growth monitoring. 3. Conduct a bi-annual review of the BNAP and present BNAP accomplishments during the BGA. 4. Follow-up the BNS on the status of the referred malnourished children. 5. Conduct city/municipal activities related to data quality check.

Section 8.2

PRACTICAL GUIDE IN CONDUCTING SUPPORTIVE SUPERVISION

Activity 1. Pre-OPT Plus Protocol

During this activity, the supervisor should determine the availability of trained OPT Plus measurers (OPT DQC Mun Form A and B) as well as calibrated/verified measuring equipment, tools, and forms. This documentation of the OPT Plus human resources, equipment, and supplies serves as an input to the Capacity Mapping Tool for Human Resources and Supplies Worksheets, which should be updated every year. The capacity mapping of human resources (Worksheet 1) provides information on trained or untrained OPT Plus Team Members while the capacity mapping of anthropometric equipment and tools provides information on the availability and condition of anthropometric tools, other supplies, and forms (Worksheet 2).

The capacity mapping will determine how much equipment should be calibrated, verified, or purchased and which barangays need OPT Plus training and certified measures. It is why pre-OPT Plus activities should start in August to give ample time to prepare the equipment and the OPT Plus members.

Worksheet 1. Capacity mapping of human resources.

OPT PLUS DQC FORM A: CITY/MUNICIPAL OPT PLUS CAPACITY MAPPING OF HUMAN RESOURCES

For the period: April to June 2021

CITY/MUNICIPALITY: Los Baños **PROVINCE:** Laguna **REGION:** Region IV-A (CALABARZON)

TOTAL NUMBER OF BARANGAYS: 50 **Dates:** 21-Jan-21 - 1-Feb-21

BARANGAY	CAPACITY MAPPING OF HUMAN RESOURCES					
	Name (Surname, Given name, Middle Name)	Designation	Contact Details	Status: Trained/Oriented/ Passed the Standardization Test	If trained and Passed, indicate classification (Level 1, 2, 3, 4 or 5)	REMARKS
1 Batong Malake						
2 Maahas						
3						
4						
5						
6						
7						
8						
9						
10						

Instructions:

1. Print or copy this form in a city/municipal OPT Plus logbook.
2. Indicate the municipality, province, and region and number of barangays in the municipality.
3. Write the dates when the supportive supervision was conducted.

Capacity Mapping of Human Resources

First Column List down the barangays in the first column.

Second Column List down all members of the OPT Plus team per barangay. Add rows if necessary.

Third Column Write the designation of the members of the OPT Plus team.

Fourth Column For each OPT Plus team member, write their contact information.

Fifth Column	Indicate if the OPT Plus team member was 1.) trained only; 2.) oriented only; 3.) trained and passed the standardization test; or 0) not trained yet.
Sixth Column	If the OPT Plus team member was trained, indicate the classification (Level 1, 2, 3, 4 or 5).
Seventh Column	Indicate remarks such as newly appointed, resigned, with illness, and other problematic situations.

Worksheet 2. Capacity mapping of anthropometric equipment and tools.

OPT PLUS DQC FORM B: CITY/MUNICIPAL OPT PLUS CAPACITY MAPPING OF ANTHROPOMETRIC EQUIPMENT AND TOOLS

For the period: April to June 2021

CITY/MUNICIPALITY: Los Baños **PROVINCE:** Laguna **REGION:** Region IV-A (CALABARZON)

TOTAL NUMBER OF BARANGAYS: 50 **Suportive Supervision Inclusive Dates:** 21-Jan-21 - 01-Feb-21

BARANGAY	CAPACITY MAPPING OF Anthropometric Equipment and Tools												
	Weighing Equipment*		Equipment for measuring length		Equipment for measuring height		Non-stretchable MUAC tape		CGS tables				REMARKS (wrong unit of measurement, bad condition, etc)
	Type	#	Type	#	Type	#	Availability	#	WFL	WFH	WFA	LFH	
**Checklist													
1 Batong Malake													
2 Maahas													
3													
4													
5													
6													
7													
8													

**Weighing Equipment: 1) Hanging weighing scale; 2) Digital Hanging scale; 3) Digital Platform Scale; 4) Others, specify*

Instructions:

1. Print or copy this form in a city/municipal OPT Plus logbook.
2. Indicate the municipality, province, and region and number of barangays in the municipality.
3. Write the dates when the supportive supervision was conducted.

Capacity Mapping of Anthropometric Equipment and Tools

First Column	Indicate the barangay.
Second Column	Indicate the type and the number of weighing scale available in the barangay.
Third Column	Indicate the type and number of measuring equipment for length available in the barangay.
Fourth Column	Indicate the type and number of measuring equipment for height available in the barangay.
Fifth Column	Indicate the number of available non-stretchable MUAC tape in the barangay.
Sixth Column	Indicate if the CGS tables for weight-for-age (WFA); length/height-for-age (L/HFA) and Weight-for-length/height (WFL/H) for male and female are all available in the barangay.
Seventh Column	Indicate remarks such wrong unit of measurement in the equipment, bad condition and other problematic conditions of the equipment and tools.

Activity 2. OPT Plus Anthropometric Measurement

The supervisors in this activity are expected to oversee the actual measurement of the 0-to-59-month-old children. The following are some guidelines on how to initiate supportive supervision and mentoring:

1. Schedule a date and time for a visit with the OPT Plus Team in advance.
2. Use the OPT DQC Form 1.2a: Barangay OPT Plus Data Quality Check: Supervisor's Supportive Supervision. The checklist will help the supervisors identify what knowledge and skills of the OPT Plus team need to be strengthened to generate accurate and reliable OPT data.
3. Make sure that the nutritional status of the children is immediately provided to parents/ caregivers and let them plot the weight and length/height of the child in the growth chart. It is usually included in the baby book or Early Child Care and Development (ECCD) card. Annexes 5.1a and 5.1b provides the growth chart for the three indices if it is not available. Please take note the growth charts for girls and boys are different.
4. Take photos if possible to help provide feedback in proper/improper measurement techniques.
5. As supervisors, children with bilateral pitting edema, obvious manifestations of undernutrition [moderately wasted or moderate acute malnutrition (MW/MAM) or severely wasted or severe acute malnutrition (SW/SAM)] should be immediately referred to the health center.
6. At the end of the supervisor's visit, discuss the observations with the OPT Plus team and decide on a doable action the team can do to improve the conduct of OPT Plus before the next mentoring visit.

Activity 3. OPT Plus Nutritional Status Assessment

As supervisors, make sure that the OPT Plus team, especially the BNS, is trained on using the eOPT Plus and having access to computers. Train them to use the computers to slowly shift them from the manual to automatic computation of nutritional status.

If the OPT Plus team has no access to computers and C/MNAO or the supervisors will encode the data, please do not instruct the BNS to compute the nutritional status manually. It will be an additional time and workload that might delay the submission of the OPT Plus data. The weight-for-length/height feedback to the parents/caregivers is enough. If they need to be practiced on manual computation, it can be done during refresher training or small-dosage training.

To lessen the burden in encoding, follow the instructions in the eOPT Tool on how to update it and use the BNS printout form for recording the weight and length/height.

Activity 4. OPT Plus Data Interpretation and Action Planning

The supportive supervision of the OPT Plus team on communicating and presenting the results of the OPT Plus to the local chief executives, and advocating to fund nutrition programs, projects, and activities should be done. Successful communication is one that brings the desired changes in actions (NNC and UPLB, 2011). The following are supervisors' responsibilities during OPT Plus data interpretation and action planning:

1. Assist the BNC in preparing the nutrition situation by answering the following guide questions:
    a) What forms of malnutrition?
    b) How many are malnourished?
    c) Who is malnourished?
    d) Where are the malnourished?
    e) What are the causes of malnutrition?
    f) What has been done to address malnutrition?
    g) What are the resources available to address malnutrition?
    h) What constraints could affect the effective implementation of interventions?
2. Practice the OPT Plus team by showing and explaining the OPT Plus results.
3. Guide the BNC in preparing the Barangay Nutrition Action Plan (BNAP) and make sure that the nutritionally at-risk families and children are targeted in the plan.

Activity 5. OPT Plus Monitoring and Evaluation

The expected supportive supervision in the activity involves monitoring monthly OPT plus follow-up and BNAP monitoring. A simple recording of activities is shown in Worksheet 3 for the supervisors. The supervisors should also record the actions done to share or correct any knowledge, attitudes, and practices (KAP) of the BNS and OPT Plus team. Worksheet 4 is the DQC checklist is the evaluation tool to determine the efficiency and effectiveness of the OPT Plus team in maintaining the data quality of the OPT.

Worksheet 3. Sample Logbook for monitoring.

Supervisor/Mentor: ______________________

Names of OPT Plus Team:

1. ______________________________ 4. ______________________________
2. ______________________________ 5. ______________________________
3. ______________________________ 6. ______________________________

Nutrition worker: ____

Date	Activity observed	Actions to be taken	Follow up (next visit schedule)

Worksheet 4. Sample Supportive Supervision Checklist.

OPT PLUS DQC FORM 1.2A: BARANGAY OPT PLUS DATA QUALITY CHECK: SUPERVISOR'S SUPPORTIVE SUPERVISION

For Nutrition Action Officer, District/City Nutrition Coordinator

(Use this form to check if the anthropometric protocol is being followed)

YEAR OF OPT PLUS: 2022

BARANGAY: Batong Malake

CITY/MUNICIPALITY: Los Baños

PROVINCE: Laguna

OPT Plus Barangay Team Members: 4

1. Reanne
2. Kat
3. Danica
4. Gelay
5.
6.
7.
8.
9.
10.
11.
12.
13.
14.
15.
16.
17.
18.

OPT PLUS SUPPORTIVE SUPERVISION					
INCLUSIVE DATES		PLACE/VENUE		OPT TEAM MONITORED AND	
INCLUSIVE DATES		PLACE/VENUE		OPT TEAM SUPERVISED	
INCLUSIVE DATES		PLACE/VENUE		OPT TEAM SUPERVISED	
INCLUSIVE DATES		PLACE/VENUE		OPT TEAM SUPERVISED	

CHECK the anthropometric equipment of the OPT team. Record ***1 if the statement/practice is observed otherwise record 0****.*

MEASUREMENT CHECKLIST	SCORE	REMARKS
1.1. The OPT Team used the recommended/accepted weighing scale according to the MOP.		
1.2. The weighing scale was available and owned by the barangay.		
1.3. The weighing scale was calibrated within the last three years, verified within the year & fit for use.		
1.4. The measurer assured that the child is wearing light clothes or ask the mother/caregiver to remove unnecessary clothing including shoes, slippers, socks, extra clothes and diaper; the child's pockets were checked and emptied.	1	---
1.5. All weight measurements were recorded/written correctly and in the nearest 0.1 kg.	1	---
1.6. The OPT Team's measurer and recorder/assistant conducted the OPT Plus in the barangay	1	---
1.7. The child was measured twice and if the replicate measurement had a difference greater than or equal to 0.3 kg, a third weight measurement was obtained.	1	---

Observe the anthropometric measurement of the OPT team. Record 1 if the statement/practice is observed otherwise record 0.

CHECKLIST ON WEIGHING USING MECHANICAL HANGING WEIGHING SCALE	SCORE	REMARKS
1.8. The weighing scale was hung from a stable and sturdy branch, ceiling beam, or pole and the ropes used for hanging the scale was sturdy.	1	---
1.9. The weighing scale dial was at eye level of the measurer.	1	---
1.10. The measurer was assisted by another person while putting the child in the weighing pants to be measured.	1	---
1.11. The strap of the pants were placed in front of the child and the child's feet were not touching the ground during weight measurement.	1	---
1.12. The child was measured twice and if the replicate measurement had a difference greater than or equal to 0.3 kg, a third weight measurement was obtained.	1	---
1.13. The team performed ZERO adjustment procedure (pulling the sling seat/weighing pants downwards until half of the scale's capacity 3 times.)	1	---

CHECKLIST ON WEIGHING USING DIGITAL HANGING WEIGHING SCALE	SCORE	REMARKS
1.14. The weighing scale was hung from a stable and sturdy branch, ceiling beam, or pole and the ropes used for hanging the scale was sturdy.		
1.15. The weighing scale dial was at eye level of the measurer.		
1.16. The measurer was assisted by another person while putting the child in the weighing pants to be measured.		
1.17. The strap of the pants was placed in front of the child and the child's feet were not touching the ground during weight measurement.		
1.18. The child was measured twice and if the replicate measurement had a difference greater than or equal to 0.3 kg, a third weight measurement was obtained.		
1.19. The team performed ZERO adjustment procedure (pulling the sling seat/weighing pants downwards until half of the scale's capacity 3 times.)		

CHECKLIST ON WEIGHT MEASUREMENT: USING DIGITAL FLAT SCALE	SCORE	REMARKS
1.20. The scale was placed on a flat surface.		
1.21. For scale without taring: the mother/caregiver was measured and her weight was recorded then the child was given to the mother/caregiver on the weighing scale and their weight was taken. The weight of the mother was deducted from the total weight of mother/caregiver and child.		
1.22. For scale with taring: the mother/caregiver was asked to be on the weighing scale before taring the scale; the child was given to the mother/caregiver on the weighing scale and the child's weight was taken.		
1.23. The child was measured twice and if the replicate measurement had a difference greater than or equal to 0.3 kg, a third weight measurement was obtained.		

CHECKLIST ON WEIGHT MEASUREMENT: MECHANICAL COLUMN SCALE	SCORE	REMARKS
1.24. The scale was placed on a flat surface.		
1.25. The child was measured twice and if the replicate measurement had a difference greater than or equal to 0.3 kg, a third weight measurement was obtained.		

CHECKLIST ON LENGTH/HEIGHT MEASUREMENT	SCORE	REMARKS
2.1. The length/height board was available and owned by the barangay.		
2.2. The length/height board was verified within the year and was fit for use.		
2.3. The measurer requested the mother/caregiver to remove headgear, braids, etc. and shoes, slippers, and socks, etc. before the measurement.		
2.4.The length/height was read and called out by the measurer to the nearest 0.1 cm.		

CHECKLIST ON LENGTH MEASUREMENT	SCORE	REMARKS
2.5. The measuring flatboard was placed on a hard flat surface on the floor or on the table.		
2.6. If placed on the floor, the assistant was kneeling beside the headboard. If placed on the table, the assistant was standing beside the headboard		
2.7. The measurer was on the right side of the child while holding the foot piece using the right hand.		
2.8. The measurer was helped by the mother/caregiver in placing the child on the board and instructed the mother/caregiver to position at the opposite end of the board to help calm the child.		
2.9. The assistant cupped his/her hands over the child's ear to position the child's head ensuring that the child's sight was perpendicular to the ground and the assistant was at eye level with the child.		
2.10. The measurer made sure that the child was lying flat in the center of the board and checked the position of the child before getting the measurement		
2.11. The measurer pressed the child's shins or knees against the board using his/her left hand while maneuvering the foot piece firmly on the child's heels using their right hand.		

CHECKLIST ON HEIGHT MEASUREMENT	SCORE	REMARKS
2.12. The measuring board was placed on a hard flat surface against a wall, table, tree, staircase, etc. ensuring that the board was not moving.		
2.13. The measurer knelt with his/her right knee on the right side of the child while the assistant placed the OPT_Printout and pencil on the ground and knelt on both knees on the right side of the child.		
2.14. The assistant placed the child's feet flat and together at the center and against the back and base of the board/wall.		
2.15. The assistant placed his/her hand right above the child's ankles, his/her left hand on the child's knees and pushed it against the board/wall; the assistant then informed the measurer that he/she has already positioned the child's feet and legs accordingly.		
2.16. The measurer instructed the mother/caregiver to stand in front of the child and for the child to look straight ahead to the mother/caregiver so that the child's line of sight was level with the ground when the height was being measured.		
2.17. The measurer's left hand was placed under the child's chin and gradually closed his/her hand.		
2.18. The child's shoulders were level and the measurer pushed the child's tummy a little bit before measuring.		
2.19. The measurer ensured that the headpiece was pushed through the child's hair.		

CHECKLIST ON LENGTH/HEIGHT MEASUREMENT	SCORE	REMARKS
2.20. The measurer used the appropriate procedure for length (0-23mos - recumbent) and height (23-59 mos - standing) measurement and if length or height was measured due to any reason, this was documented under remarks		
2.21. The length/height was recorded by the assistant in the nearest 0.1 cm and showed to the measurer who checked the accuracy and legibility.		
2.22. The child was measured twice and in case the replicate measurements had a difference greater than or equal to 0.5 cm, a third length/height measurement was obtained; the two closest measurements were used.		
2.23. The length/height was read, called out and recorded by the assistant in the nearest 0.1 cm and showed to the measurer who checked the accuracy and legibility.		

CHECKLIST ON MUAC MEASUREMENT	SCORE	REMARKS
3.1. The measurer informed the mother/caregiver that the Child's MUAC will be measured and asked him/her to position the child such that the left upper arm is exposed (any clothing that may cover the child's upper arm is removed).		
3.2. The MUAC was measured in the left upper arm.		
3.3. Midpoint between the elbow and the tips of the shoulder was properly located.		
3.4. The arm was at a right angle and MUAC was measured at the midpoint.		
3.5. The MUAC was measured not too loose or too tight.		

CHECKLIST FOR OBSERVING SIGNS OF BILATERAL PITTING EDEMA	SCORE	REMARKS
4.1. The measurer informed the mother/caregiver that signs of edema will be checked.		
4.2. The measurer pressed both feet gently for three (3) seconds and if positive, assessed the severity of edema.		
4.3. The measurer recorded the severity of edema.		

CHECKLIST ON OPT PLUS UTILIZATION	SCORE	REMARKS
5.1. The OPT team immediately feedbacked the nutritional status of all children to the parents/caregiver immediately after measurement.		
5.2. The OPT team immediately referred all MAM/SAM cases to the RHU.		

OVERALL TOTAL SCORE:	4
SUPPORTIVE SUPERVISOR SCORE (%):	13%

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